May 19, 2026

Colonel Chris Ellis, PhD
Senior Fellow and Director of Research
Homeland Defense Institute
United States Air Force Academy

Subject: Transmittal of Final Report for *Private-Sector Lifelines in the Arctic: Predicting and Assessing Risk to Critical Infrastructure*

Dear Colonel Ellis,

It is with both professional pride and personal gratitude that I transmit the final report for *Private-Sector Lifelines in the Arctic: Predicting and Assessing Risk to Critical Infrastructure*, prepared by the Private Sector Emergency Management Association (PSEMA) under the Homeland Defense Institute's 2026 research portfolio in support of NORAD and USNORTHCOM Arctic mission assurance priorities.

The report is attached, and I want to take a moment, before the findings stand on their own, to acknowledge how this work came to be. It did not begin with a research proposal. It began with a conversation.

Three years ago, at the Region 6 Homeland Defense Symposium, I spoke up during a session on critical infrastructure and made what I thought at the time was a fairly straightforward observation: that the private sector is not adjacent to homeland defense, it is part of it, and that the systems sustaining communities and installations alike are largely owned, operated, and maintained by industry. Colonel Richard Ball, then serving as the Region 6 Defense Coordinating Officer, approached me afterward. Rather than letting the comment pass, he invited PSEMA into the conversation. He invited us to subsequent events, asked hard and generous questions about how the private sector actually shows up during an incident, and over time turned what could have been a single hallway exchange into a sustained working dialogue.

That dialogue is the soil this project grew in. The framing of business-critical infrastructure as distinct from, but inseparable from, the federal critical-infrastructure sectors; the recognition that lifeline failure in the Arctic is not a logistics problem but a mission-assurance problem; the willingness to look honestly at where plans assume capacity that does not exist, these threads can all be traced back to that 2023 conversation and the ones it made possible. I am grateful to Colonel Ball for the doorway, and to HDI for opening the room behind it.

As I send this report forward, three things are on my mind.

First, I am genuinely eager to present these findings and recommendations at the Homeland Defense Awareness Symposium in June. The dual-theater scenario, West Coast port degradation concurrent with Arctic escalation, surfaced patterns that I think will resonate with the audience: the difference between paper redundancy and real redundancy; the operational reality that aviation cannot substitute for sealift at sustainment scale, even when *Operation Vittles* is invoked as precedent; the in-state refining picture and what it means when the closest replacement barrel is days away by tanker; and the way communications

fragility functions as a force degrader long before it functions as a tactical problem. These are findings I want decision-makers to sit with, not skim past.

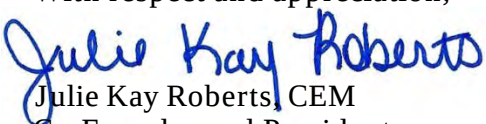
Second, the research makes clear that what we need next is more conversation, not less. The most consistent theme across the dependency mapping, the risk analysis, and the stakeholder engagement is that private-sector operators, DoD, and federal and state government partners are working from different mental models of what the others can do, will do, and are authorized to do under disruption. Those gaps are not malicious and they are not new, but they are operationally consequential, and they will not close on their own. The recommendations in the report deliberately call for sustained, structured, pre-event engagement, not just exercises, not just post-incident after-action reviews, but a standing rhythm of conversation where the assumptions on each side can be tested before they have to be relied upon.

Third, I hope PSEMA can serve as the conduit for the next steps. The recommendations in this report are not a stopping point. They are a map of where the harder work begins, in deeper modeling of specific lifeline systems, in expanded engagement with tribal and Alaska Native partners on terms that respect sovereignty and reciprocity, in piloting the cross-sector coordination mechanisms the report identifies, and in carrying these findings into the planning communities where they can actually shape decisions. PSEMA is positioned to convene that work because we sit in the seam between industry, government, and the operational community without being captured by any one of them. We would welcome the opportunity to continue, and we are ready to do so.

A note on the structure of this submission. Alongside the final report itself, PSEMA is transmitting two companion deliverables that are intended to be read with it rather than after it. The Executive Summary is designed as a walk-through of the report's analytical arc, scenario, dependency map, risk findings, and recommendations, written so that a reader who has the time only for the summary will still finish with the operationally consequential picture intact, and so that a reader proceeding into the full report will arrive there already oriented. The comprehensive Appendices stand as a separate document because the supporting material, historical precedents, lifeline-system detail, dependency tables, methodology notes, and the reference list, exceeded what could be integrated into the body of the report without obscuring the analytical line through it. Together the three documents are the deliverable; each is calibrated to a different reading need, and any one of them is incomplete without the others in reach.

My thanks to you, to Natasha, and to the broader HDI team for the trust placed in PSEMA on this project. Thank you as well to the stakeholders, named and unnamed, who gave their time and candor to the engagement track; their input is the reason this report has the texture it does. Please do not hesitate to reach out if any clarification would be useful in advance of the June symposium. I look forward to the briefing and to the conversations it will open.

With respect and appreciation,



Julie Kay Roberts, CEM
Co-Founder and President
Private Sector Emergency Management Association

P S E M A · H O M E L A N D D E F E N S E I N S T I T U T E

Private-Sector Lifelines in the Arctic

Predicting and Assessing Risk to Critical Infrastructure

Executive Summary

This publication was sponsored by the Homeland Defense Institute. The views expressed in this publication do not necessarily represent the views of the United States Air Force Academy, North American Aerospace Defense Command and United States Northern Command, the Department of Defense, or the United States Government.

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Purpose

This Executive Summary is a standalone, 25-page companion to the *Private-Sector Lifelines in the Arctic: Predicting and Assessing Risk to Critical Infrastructure* Final Report. It is designed for decision-maker, NORAD/USNORTHCOM leadership, the Homeland Defense Institute, FEMA, CISA, Alaska DHS&EM, and private-sector lifeline operators, who require an operationally precise synthesis of the research without engaging the full Final Report on first read.

Every section of this document points back to the corresponding section of the Final Report by section title. The findings, recommendations, and analytical framing presented here are drawn directly from the Final Report; no new analysis is introduced. Where the body of the Final Report provides the operational specificity, the Executive Summary provides the decision frame.

Target audience: senior decision-makers operating under time constraints who need to know what was found, what it means for Arctic homeland defense, and what should be done, in that order.

Final Report Reference: Executive Summary (full development of all material summarized here).

Research at a Glance

A one-page orientation for the reader. Four questions, in plain language, with a pointer back to the Final Report for full development.

What is this research?

A federally commissioned study examining how privately operated lifeline systems, fuel, food, transportation, communications, medical supply, and supply-chain distribution, sustain Arctic homeland defense under contested conditions. It was conducted by the Private Sector Emergency Management Association (PSEMA) in support of the Homeland Defense Institute and NORAD/USNORTHCOM Arctic mission assurance priorities. The research is analytically distinct from its companion study, Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense, which addresses the human-systems dimensions of the same disruption environment.

What did it set out to do?

Strengthen the resilience and adaptability of the Arctic homeland defense ecosystem by identifying interdependencies and capability gaps across critical infrastructure assets, systems, and networks, public and private. The work is directly responsive to *HDAS Research Focus Area 1: Predicting and Assessing Risk to Critical Infrastructure*, and to the NSM-22 and DoDD 3020.40 frameworks that recognize military readiness as inseparable from civilian critical infrastructure.

What did it find?

Arctic homeland defense depends on commercial systems that were not designed for the stress conditions a dual-theater disruption would impose. They fail faster than planners assume, cascade more broadly than frameworks anticipate, and leave decision-makers operating without adequate situational awareness during the Days 0–14 window when choices are most consequential. The twenty-two key findings developed in the Final Report consolidate into six operationally consequential conclusions, presented in Section 6 of this Executive Summary.

What is this research?

Layered investment across overlapping time horizons, pre-positioned reserves and pre-staged regulatory authority in the near term, distributed energy and coordination architecture maturity in the mid-term, and structural infrastructure and codified doctrine in the long term, operating concurrently, not sequentially. Section 8 presents the recommendation set in compact form.

Final Report Reference: Executive Summary and Strategic Implications

Strategic Context: Why This Research Matters

This section establishes a strategic frame. It draws directly from the Final Report's Introduction and Strategic Implications sections.

The Contested Homeland

The homeland is no longer a sanctuary. Adversaries can obstruct and disrupt power projection and sustainment operations from within and against the homeland through a combination of kinetic action, gray-zone activity, and exploitation of legally permitted but strategically harmful conduct. NORAD and USNORTHCOM's Arctic mission depends on a resilient homeland for force protection, power projection, and the sustained national will that enables both. The Arctic theater compresses these dependencies: the same lifeline systems that sustain remote civilian communities are the systems on which installation operations, family readiness, and contractor workforce availability all depend.

Defense Critical Infrastructure Is Privately Operated

DoDD 3020.40 (Mission Assurance) and NSM-22 (Critical Infrastructure Security and Resilience) both recognize that military readiness is inseparable from civilian critical infrastructure, most of it privately owned. This dependence intensifies in the Arctic, where lifeline systems are sparse, fragile, geographically constrained, and concentrated across a small number of operators. The implication is operational: *Arctic mission assurance cannot be planned, exercised, or sustained without sustained engagement with the private-sector entities that own and run the systems on which the mission depends.*

The Arctic Operating Environment

Limited deep-water ports and icebreakers; weather-restricted airfields; long distances between communities; communications fragility in high latitudes; limited redundancy in energy and fuel; cross-border dependencies with Canada; medical evacuation distance and weather constraints; tribal and local governance with varying authorities. Each is documented in the Final Report; each appears as a binding constraint in the dependency mapping, risk assessment, and substitution analysis that follow. The Arctic operating environment is not an austere version of CONUS infrastructure, it is a different infrastructure category, with different failure modes, recovery dynamics, and substitution limits.

Why Now

The 2025 National Defense Strategy identifies simultaneity, the prospect of multiple, overlapping disruptions, as a primary planning challenge. The dual-theater scenario this research models is structurally consistent with that planning challenge, calibrated to Arctic geographic and seasonal realities. The Days 0–14 decision window, during which planning assumptions fail and consequential decisions must be made under uncertainty, is the period for which preparatory investment yields the highest operational return. It is also the period for which preparatory investment must be made before disruption occurs; once the window opens, the options narrow with each passing day.

Final Report Reference: Strategic Implications and Introduction

Scenario Framework: The Dual-Theater Disruption

This section summarizes the scenario that grounds the research. It is a planning stress test, not a prediction.

The Scenario in One Sentence

An adversary disables major West Coast port operations (Days 0–14) concurrent with an Arctic escalation requiring rapid mobilization through limited northern corridors (Days 10–45).

Why This Scenario

West Coast ports, principally the Seattle/Tacoma gateway complex and the Los Angeles/Long Beach complex, are the maritime gateways through which most Alaska-bound freight, fuel, and population-supporting supply flows. An adversary acting to disable or degrade those gateways simultaneously affects the maritime supply that sustains Alaska's commercial inventory architecture and the strategic flow that would otherwise support an Arctic mobilization. Arctic escalation forces the system to draw on the same constrained logistics architecture that the port degradation has already strained. Days 0–14 is the period when planning assumptions fail and consequential decisions must be made under uncertainty. The scenario is calibrated to be operationally realistic,

consistent with the threat picture in the published, open-source defense literature, not a worst-case theoretical construction.

Arctic Sustainment Modes Examined

Five sustainment modes are examined for their operational reality under stress: maritime resupply (primary, and the only mode capable of moving freight at scale into Alaska); aviation sustainment (constrained by weather, fuel, runway, and crew); ALCAN Highway overland movement (constrained by Canadian sovereignty, weather, and chokepoints); pre-positioned stocks (limited, seasonal, and unevenly distributed); and substitution across modes (constrained at every level by the binding limits of each mode in isolation). The Substitution Inventory developed in the Final Report documents what can and cannot substitute for what, under what conditions, and at what capacity.

Seasonal Asymmetry

The vulnerability is not uniform across the year. Alaska's downstream distribution network serving western and northern communities is seasonal and ice-constrained. A disruption between October and April operates against already-depleted reserve margins with no barge resupply option, compressing time-to-shortage from weeks to days. The same disruption in June or July, with the barge season active and reserves replenished, presents a substantially different operational picture. Mission assurance planning that does not differentiate seasonally underestimates winter vulnerability by an operationally significant margin.

Historical Analog: The Berlin Airlift

Operation Vittles, the 1948–49 Berlin Airlift, is the most analytically relevant historical case for sustained, aviation-only sustainment of a contested theater. Its operational record, the volume of cargo, the aircrew requirements, the fuel and maintenance burden, the airfield-throughput limits, and the duration of the effort, provides the empirical baseline against which aviation-based Arctic sustainment must be evaluated. The Final Report treats the analog in detail. The summary conclusion: aviation-only sustainment of an Arctic theater of the scale this research examines is not feasible at the cargo volumes Arctic operations require and was not feasible in the Berlin case either except under conditions (short distance, high aircraft density, dedicated airfields, single-mission focus) that do not obtain in the Arctic.

Final Report Reference: Scenario Development (full framework, stressors, and Operations Vittles analog)

Methodology in Brief

This section summarizes the analytical approach, validation anchors, and limits. It is drawn from the Final Report's Research Foundation and Methodology sections.

Five-Deliverable Architecture

The research was structured as five integrated deliverables, each building on the prior:

- *Scenario Development:* establishes the dual-theater disruption environment and the seasonal, geographic, and operational parameters within which the analysis proceeds.
- *Targeted Dependency Mapping:* identifies what is connected to what across three layers of Arctic infrastructure (physical, organizational, and coordination).
- *Focused Risk and Vulnerability Assessment:* establishes where and when connections break, how failure cascades, and where the operational window for intervention closes.
- *Preliminary Mitigation and Cost–Benefit Prioritization:* evaluates high-impact resilience options against realistic parameters, with explicit treatment of which options yield Days 0–14 benefit and which yield longer-horizon resilience.
- *Stakeholder Engagement:* validates findings through structured engagement across the seven lifeline sectors examined; non-attributed, aggregated.

Analytical Posture

Arguments lead; research corroborates. The analytical posture throughout is operational, not theoretical. Vulnerabilities are framed in operational terms: what fails first, how fast, with what cascading effect, and at what point intervention is no longer feasible. The framing is deliberate. Arctic mission assurance is not advanced by cataloguing risks in the abstract; it is advanced by identifying the specific operational thresholds at which intervention becomes possible, becomes urgent, or becomes too late.

Validation Anchors

Findings are corroborated against a body of independent, published, validated work including: the RAND Plan Blue 21 wargame; the FEMA Supply Chain Resilience and Public-Private Partnership maturity frameworks; the CNA Supply Chain Operational Engagement methodology; Dewberry infrastructure dependency research; Alaska Task Force and Alaska Federal Senior Operational Group operational records; NSM-22 and DoDD 3020.40; Air Force Petroleum Center processing-capacity data; Caton's (2024) contested-logistics framework; and Fox and Shaffer-Miles tribal and Arctic Native scholarship. The convergence of these independent sources on the same operational conclusions is one of the principal indicators of analytical robustness.

Limits and Boundaries

The research was conducted within a three-month performance period. The analytical window is bounded at Days 0–14 of the dual-theater disruption; extended sustainment beyond Day 14, the recovery dynamics, structural reconstitution, and multi-month sustainment trajectories, is identified as the single most consequential follow-on research

priority and is not delivered within this engagement. Stakeholder engagement is non-attributed; no proprietary, sensitive, or classified data was requested or used. Conclusions are operationally specific where the evidence supports operational specificity and are deliberately framed in conditional language where the evidence does not.

Final Report Reference: Research Foundation, Analytical Framework, and Methodological Validation Anchors.

Six Key Findings

The Final Report develops twenty-two key findings in full, organized into eight analytical clusters that move from the master dependency (fuel), through the lifeline architecture, into the cascade dynamics that connect them, and finally to the governance and coordination conditions that determine whether intact infrastructure can be coordinated under stress. The six findings presented here are the consolidated, operationally most consequential conclusions, each absorbing several of the underlying twenty-two. Each finding follows the same structure: headline statement, why it matters, what the evidence shows, and a reference to the Final Report section where it is developed in full. Readers who require the full twenty-two findings in their cluster groupings, with the underlying analytical record, should consult the Key Findings section of the Final Report.

Finding 1: Fuel Is the Master Dependency

Headline

Fuel is not one vulnerability among several. It is the foundational dependency whose degradation initiates cascades across the entire Arctic lifeline architecture.

Why it matters

Fuel disruption does not stay in the energy sector. It simultaneously disables transportation, heating, power generation, and aviation, all of which depend on imported fuel products with limited regional storage and no rapid resupply alternative when maritime supply is degraded. Every other failure mode identified in this research is accelerated when fuel availability is constrained. The cascade from fuel scarcity to medical evacuation limits to workforce degradation to mission impact is not theoretical; it is the operationally expected trajectory under the scenario.

What the evidence shows

Alaska's fuel architecture is structurally concentrated, seasonally constrained, and dependent on a small number of West Coast and in-state pathways. In-state refining capacity, approximately 164,200 barrels per day, is distributed across a small number of facilities operating under thin margins and tightly coupled to a single crude source; there is no parallel refining architecture in-state that can absorb the loss of a single facility, and any disruption to refined product output cascades immediately into aviation, ground transportation, heating, and power generation. The maritime fuel chain that feeds western and northern Alaskan communities depends on a sole or near-sole commercial avgas importer and a small number of fuel and freight carriers; the underlying market

structure is consolidated, the assets are specialized (ice-class, shallow-draft, fuel-rated), and there is no substitution path that does not require the same vessels under a different flag. Stockpile depth across the system is measured in days, not weeks. A West Coast port disruption affecting refined-product flow into Anchorage compounds against in-state refining concentration and against the seasonal closure of downstream barge resupply to compress the operational window to a point well inside the Days 0–14 analytical horizon.

This finding consolidates three of the twenty-two underlying findings, fuel as the master dependency, in-state refining concentration, and maritime fuel resupply concentration.

Final Report Reference: Key Findings Cluster 1 (Fuel as the Master Dependency); Targeted Dependency Mapping (Layers 1–2); Focused Risk and Vulnerability Assessment.

Finding 2: Sustainment Modes Cannot Substitute at the Volumes the Mission Requires

Headline

West Coast maritime gateways are no-fail nodes with no rapid substitute; aviation cannot replace maritime sustainment at scale; and the Jones Act waiver architecture adds friction to the one substitution path that is structurally available.

Why it matters

The contingency planning literature consistently treats aviation as the alternative when maritime routes are disrupted. The operational record contradicts that assumption at the cargo volumes Arctic sustainment requires. Maritime sustainment is the only mode capable of moving freight at scale into Alaska, and the West Coast gateway complex through which that maritime flow originates is concentrated, congestion-sensitive, and without a rapid rerouting alternative within a fourteen-day window. The Jones Act waiver process, designed for incident-by-incident administrative review, was not designed for the compressed timelines of an Arctic-Pacific dual-theater event. Each sustainment mode examined in this research has a binding constraint that either eliminates it as a Days 0–14 option or reduces its capacity to a fraction of what mission requirements demand.

What the evidence shows

The Port of Alaska is the single intermodal hub through which most of Alaska's freight, fuel, and population-supporting supply flows from the Pacific Northwest gateway complex. Peer-reviewed computable general equilibrium modeling of a compound disruption at the twin ports of Los Angeles and Long Beach finds GDP losses propagating well beyond the port region through commercial supply-chain transmission, with no rerouting alternative available at sufficient scale within the operational window; those national-level figures exclude Alaska entirely, so Arctic mission-assurance consequences are additive to the modeled losses, not captured within them. The RAND Plan Blue 21 wargame documents directly that the aviation-as-substitute-for-maritime assumption is operationally wrong at scale, and the *Operation Vittles* historical record confirms that air-only sustainment requires conditions (short distance, high aircraft density, dedicated airfields, single-mission focus) that do not obtain in the Arctic. Replacing one week of

Port of Alaska cargo throughput by air would require the equivalent of seven hundred wide-body 747 cargo landings, a figure that does not yet account for fuel, de-icing, crew, or runway capacity, and that makes the substitution argument operationally implausible on its face. Cabotage requirements under the Jones Act constrain the universe of vessels available for emergency Alaska resupply to U.S.-flagged, U.S.-built, U.S.-crewed bottoms; waivers are administratively available but are time-limited, require demonstration of national defense necessity, and were not designed to activate within the compressed timeline of a dual-theater event.

This finding consolidates three of the twenty-two underlying findings, West Coast gateways as no-fail nodes, aviation as inadequate maritime substitute, and the Jones Act friction point.

Final Report Reference: Key Findings Cluster 2 (Maritime and Aviation Sustainment Limits); Arctic Sustainment Modes and Lifeline Constraints; Scenario Development (*Operation Vittles* analog).

Finding 3: Cascades Are Fast, Non-Linear, and Compressed by Seasonal Asymmetry

Headline

Failure pathways are non-linear, fast-cascading, and self-reinforcing. The window for effective intervention closes faster than linear degradation models suggest, and seasonal conditions compress that window further.

Why it matters

The representative failure cascade, port throughput degradation, then delayed cargo, then inventory drawdown, then aviation surge, then fuel constraints, then medical evacuation limitations, then workforce degradation, then mission impact, is not a worst-case scenario. It is the operationally expected trajectory under the dual-theater disruption modeled in this research. Each link accelerates the next; the failure dynamics are non-linear once threshold conditions are reached, and the self-reinforcing character of the cascade takes over from that point. Planning frameworks built around linear, additive degradation models underestimate the speed of failure and overestimate the window during which intervention remains operationally feasible. The seasonal dimension is decisive: the same disruption produces materially different operational pictures in October–April versus June–July.

What the evidence shows

Commercial inventory buffers in Alaska are measured in days, not weeks. Just-in-time supply chain architecture, pervasive across food, pharmaceuticals, fuel, and consumer goods, produces a structural fragility under disruption: community-level shortages appear within days, not the multi-week adjustment periods that planning frameworks often assume. A West Coast port disruption occurring between October and April operates against already-depleted reserves with no barge resupply option available; depleted reserves cannot be replenished until thaw, compressing the time from port degradation to critical shortage from weeks to days. The same disruption in June presents

a different operational picture, with the barge season active and reserves replenishing. Medical evacuation, which depends on the same constrained aviation, fuel, and weather-window resources as logistics aviation, faces competitive pressure for those resources from the moment the cascade begins. The Days 0–14 window must be planned for as a period of near-complete supply constraint, not as a transitional period before backup arrangements stabilize. Any intervention designed to take effect after Day 14 arrives too late for the populations whose sustainment determines whether the installation workforce can report.

This finding consolidates three of the twenty-two underlying findings, non-linear cascading dynamics, inventory buffers measured in days, and seasonal asymmetry. The implication for planning is structural: the appropriate planning unit for Arctic mission assurance under the dual-theater scenario is not the 30-day or 90-day horizon common in CONUS contingency planning. It is the 14-day window, and the preparatory posture must be in place before that window opens.

Final Report Reference: Key Findings Cluster 3 (Seasonal and Inventory Asymmetry) and Cluster 7 (Cascade Dynamics); Focused Risk and Vulnerability Assessment; Cascading Failure Analysis.

Finding 4: Communications and Workforce Are Cross-Cutting Force Degraders

Headline

Communications degradation and civilian workforce degradation each operate as force degraders on every other failure mode, not as discrete failures within their own sectors. The cyber-physical convergence point amplifies both.

Why it matters

Communications failure eliminates the visibility into logistics status, infrastructure performance, and resource availability that every other response system depends on. The cascade does not require kinetic action on communications infrastructure; adversary gray-zone tactics, interference with commercial satellite service, exploitation of cyber-physical convergence in lifeline operational technology, or disruption of the terrestrial fiber paths that high-latitude communications depend on, make this an exploitable failure mode, not a passive one. The lifeline systems on which Arctic mission assurance depends are increasingly cyber-physical: supervisory control, fuel terminal automation, refinery process control, electrical grid management, and logistics dispatching all depend on networked control systems with varying defensive hardening, and a kinetic disruption to a fuel terminal and a cyber disruption to its control system are convergent vectors against the same operational outcome. Civilian and contractor workforce, the people who actually run the systems, live in the same communities experiencing the disruption; their household-level access to food, fuel, and services degrades within the first week, and their availability falls with it. Intact infrastructure that no one can operate is not operational infrastructure. This is the failure mode that mission assurance frameworks most consistently miss.

What the evidence shows

Commercial satellite operators may alter service availability during a crisis under their commercial terms of service, with no governance framework currently requiring sustained service to defense or emergency management nodes. High-latitude communications depend on a small number of satellite constellations, with limited bandwidth, intermittent reliability above certain latitudes, and meaningful weather sensitivity. The contractor workforce that sustains base operations, fuel distribution, medical services, and lifeline systems is not modeled in most mission assurance frameworks as a constraint, yet its availability is directly tied to the household-level access that supply chain disruption degrades, and that thinness is amplified in specific sectors (seafood processing, maritime crewing, aviation maintenance, fuel terminal operations) where the labor force is both numerically thin and seasonally migratory. A disruption that occurs during a labor turnover window, or that prevents the inbound movement of seasonal workers, compounds the household-level workforce constraint with a labor-supply shock that has no rapid substitute. The cyber-physical convergence point is increasingly important: supervisory control, fuel terminal automation, refinery process control, satellite ground stations, and electrical grid management are now cyber-physical systems whose attack surface extends beyond the defense sector.

This finding consolidates four of the twenty-two underlying findings, communications as force degrader, cyber-physical convergence, workforce as under-modeled dependency, and workforce thinness in seasonal-labor sectors. The recommendation implications run through both the near-term (anti-jam capability, redundant satellite pathways, service-continuity commitments) and the mid-term (sustained workforce-availability modeling, distributed and dual-use communications architecture).

Final Report Reference: Key Findings Cluster 4 (Communications and Cyber-Physical Convergence) and Cluster 5 (Workforce as Critical Dependency); Targeted Dependency Mapping (Layers 3–5); Focused Risk and Vulnerability Assessment.

Finding 5: Medical Capacity Becomes a Day-Zero Adjudication Problem

Headline

Medical evacuation and pharmaceutical cold-chain are not separable from the broader cascade; under the dual-theater scenario, both produce population-level consequences within the first week and create resource-allocation conflicts that require pre-defined adjudication before the disruption begins.

Why it matters

Arctic medical evacuation depends on the same constrained airfield, fuel, weather-window, and crew resources that logistics aviation requires. Under normal conditions, these systems coexist without significant competition. Under the dual-theater scenario, they become directly competitive: every airframe and airfield slot allocated to logistics surge is one not available for medical evacuation, and the resulting allocation decisions are made without a pre-defined adjudication mechanism. The civil–military adjudication gap in medical evacuation prioritization is a Day 0 problem, not a Day 14 problem. The

pharmaceutical cold-chain failure mode runs on an even shorter clinical timeline: a substantial fraction of the population in Alaska depends on temperature-controlled medications whose supply chain depends on continuous power, refrigerated transport, and timely resupply, and disruption to any one of those components produces a population-level health consequence within days, not weeks. The cold-chain failure concentrates risk in populations already disproportionately remote from acute-care capacity.

What the evidence shows

Medical evacuation in Alaska is weather-, fuel-, and crew-limited under normal operations; under the scenario, those constraints become immediate and zero-sum competition with logistics surge. The temperature-controlled pharmaceutical chain (insulin, biologics, refrigerated antibiotics, dialysis-related medications) depends on continuous power, refrigerated transport, and timely resupply; the cold-chain failure mode is structurally similar to the food cold-chain but operates on a shorter clinical timeline. Pharmacy and clinic-level inventory buffers in Alaska are thin, particularly in rural and tribal communities served by the Indian Health Service and regional health corporations. Receiving-facility capacity at Anchorage and Seattle tertiary care centers is finite, and pre-defined coordination across those receiving facilities, the rural sending facilities, and the aviation assets that connect them does not currently exist at the maturity the scenario requires. The medical cascade is not a downstream consequence of the lifeline cascade; it begins simultaneously and competes for the same scarce resources.

This finding consolidates two of the twenty-two underlying findings, medical evacuation as zero-sum competitor with logistics aviation, and pharmaceutical cold-chain as a first-week population-level failure mode. The recommendation architecture identifies medevac surge capacity in the mid-term and pharmaceutical cold-chain pre-positioning in the near-term.

Final Report Reference: Key Findings Cluster 6 (Medical and Pharmaceutical Cascades); Focused Risk and Vulnerability Assessment (Medical Evacuation and Pharmaceutical Cold Chain subsections).

Finding 6: Paper Redundancy and Coordination Gaps Are the Organizing Risk

Headline

Across every lifeline sector and every coordination architecture examined, the redundancies embedded in planning documents do not function under Arctic stress. The absence of pre-defined civil–military adjudication, a sustained sector-neutral intermediary, and exercised coordination doctrine is a structural risk amplifier, not a passive backdrop.

Why it matters

Alternate routes are constrained by weather, ice, crew availability, and throughput limits. Backup fuel supplies exist on paper but are often already committed or seasonally inaccessible. The ALCAN Highway, cited in nearly every contingency planning document

as the overland alternative to maritime resupply, is limited by Canadian sovereignty, by infrastructure chokepoints along the corridor, and by the realistic throughput of overland trucking under winter conditions, a genuine redundancy at low intensity and a paper redundancy at the volumes contemplated under a dual-theater disruption. Satellite communications alternatives provide limited bandwidth with intermittent reliability and no governance framework requiring commercial providers to maintain service during a crisis. The most-likely and most-dangerous threat scenarios diverge in the Arctic operating environment, requiring distinct planning postures; current frameworks treat them as a single continuum, which is itself an analytical risk. Coordination architectures designed for predictable, single-incident response degrade under simultaneous, multi-theater stress, and most Arctic jurisdictions relevant to this research currently operate at the Basic or low-Intermediate tier of the FEMA P3 maturity model, where engagement is event-driven, charters are absent, and exercises against the dual-theater scenario have not occurred. The connective infrastructure Arctic mission assurance requires, non-attribution, sector-neutral, operationally credible, sustained across years rather than activated during incidents, does not currently exist within the federal architecture at the maturity the mission requires.

What the evidence shows

The Substitution Inventory developed in the Final Report documents the gap at the lifeline level: across maritime, aviation, overland, pre-positioned, and cross-modal substitution, each substitution path is bound by a constraint that either eliminates it as a Days 0–14 option or reduces its capacity to a fraction of what mission requirements demand. The distributed microgrid architecture that serves western, northern, and interior Alaskan communities, approximately 150 standalone microgrids, provides partial structural protection against statewide grid failure, but the same architecture concentrates fuel dependency at the community level, and the failure mode that would collapse multiple microgrids simultaneously is the fuel supply chain disruption identified in Finding 1.

The FEMA Public-Private Partnership maturity model documents the gap at the governance layer: most Arctic jurisdictions operate at Basic or low-Intermediate tier, characterized by episodic, transactional, incident-driven engagement insufficient for the sustained coordination Arctic mission assurance requires. The Jones Act waiver framework, the FEMA P3 maturity model, and the contested-logistics literature converge on the same conclusion: coordination architectures designed for predictable, single-incident response degrade under simultaneous, multi-theater stress. Across every lifeline sector examined, the same structural weakness recurs, there is no operationally-tested, sector-neutral intermediary positioned to translate between defense mission requirements and the privately operated systems that sustain them. Existing partnership frameworks are episodic, transactional, or limited to incident response.

This finding consolidates seven of the twenty-two underlying findings, paper redundancy across lifelines, ALCAN as paper redundancy at scale, distributed microgrid architecture as partial-but-fuel-bounded protection, most-likely/most-dangerous divergence, institutional coordination as risk amplifier, the missing trust-based intermediary, and coordination architecture as itself paper redundancy when unexercised. The

recommendation architecture in Section 8 organizes the response: tabletop validation of existing architectures in the near term, advancement to Advanced tier P3 maturity and a permanent cross-sector supply chain intelligence function in the mid-term, and codified DoD–private sector–nongovernmental coordination doctrine in the long term.

Final Report Reference: Key Findings Cluster 7 (Cascade Dynamics and Paper Redundancies) and Cluster 8 (Threat Scenarios and Governance Architecture); Arctic Sustainment Modes and Lifeline Constraints; From Recommendations to Implementation.

Bottom Line

A single page that distills the analytical record into the three structural conditions Arctic mission assurance requires, and that do not currently exist at adequate scale.

Arctic homeland defense cannot be sustained against the dual-theater disruption scenario this research models without three things:

1. Pre-positioned reserves of fuel, food, pharmaceuticals, and critical materiel sufficient to bridge the Days 0–14 window. Inventory buffers in days, seasonal compression in winter, and the operational impossibility of large-scale aviation substitution for maritime resupply combine to make pre-positioning the only mechanism that closes the early-phase supply gap inside the analytical window.

2. Multi-layer communications redundancy that does not depend on any single commercial provider or terrestrial pathway. Communications failure is a force degrader on every other failure mode; closing that gap requires architecture, not procurement of any single technology.

3. A pre-defined civilian-military coordination architecture, exercised against the dual-theater scenario before disruption requires it. Coordination architectures designed for single-incident response, and architectures that exist on paper but have not been exercised, do not function under simultaneous multi-theater stress.

None of the three is a capital investment problem in isolation. All three are systems problems requiring layered, multi-partner, multi-year investment, operating concurrently rather than sequentially. The recommendations in Section 8 organize the path forward.

Final Report Reference: Executive Summary (Bottom Line)

Recommendations at a Glance

This section presents the recommendation set in compact form. Recommendations are layered, not sequenced; all three tiers must run concurrently. Each is summarized in two to three sentences here; full development, partner architecture, cost-benefit framing, and implementation parameters appear in the Final Report's Recommendations section and Appendix H.

Near-Term Recommendations (0–12 Months)

Investments activatable within the current planning and budget cycle that materially compress the activation timeline of early-phase response. Return is realized before disruption occurs.

Pre-Position Strategic Stockpiles for the Days 0–14 Window.

Establish pre-positioned reserves of fuel, food, pharmaceuticals, cold-chain capacity, and critical materiel at Anchorage, Fairbanks, JBER, and rural community hubs whose seasonal resupply windows close with the fall barge season. This is the only mechanism that closes the inventory-buffer gap within the analytical window. Reserves should be sized against the dual-theater scenario, distributed against the geographic and seasonal stress points the dependency mapping surfaces, and managed through a sustained operations-and-rotation framework rather than as a one-time procurement.

Pre-Stage Jones Act Waiver and Regulatory Authority Documentation.

Develop and maintain pre-staged documentation packages supporting Jones Act waivers, transportation weight-limit waivers, hours-of-service waivers, CDL endorsement waivers, and Canadian provincial permitting for ALCAN trucking surge. Convert the waiver process from a capability that may exceed the operational window into a regulatory mechanism activatable within hours of a triggering event.

Initiate Pre-Defined Civil–Military Adjudication Protocols.

Develop protocols for adjudicating civilian and military claims on scarce fuel, food, transportation, and medical capacity. Address allocation across Title 10 authority, state emergency management authority, tribal sovereignty, and commercial cargo prioritization with clearly identified decision authorities, documented prioritization criteria, and pre-coordinated activation triggers.

Conduct Tabletop Exercise–Based Validation of Existing Coordination Architectures.

Test the Alaska Partnership for Infrastructure Protection (APIP), the Mass Care Group and Feeding Task Force, the Alaska Food Value Chain Coordination Council, and the federal-state-tribal-military coordination architecture against the dual-theater scenario. After-action findings convert paper-redundancy coordination architectures into operationally validated capacity, identify the specific decision points at which adjudication mechanisms must exist, and surface the relationships that must be built before they are needed.

Establish Communications Hardening Priorities for Multi-Layer Architecture.

Identify anti-jam capability for primary defense and emergency management nodes; redundant satellite pathways across LEO and GEO providers; high-frequency radio backup; and subsea cable protection. Establish service-continuity commitments with commercial satellite providers operating in Arctic latitudes, recognizing that current commercial terms do not require sustained service to defense or emergency management users during a crisis.

Final Report Reference: Near-Term Recommendations (0–12 Months). Additional near-term items including pharmaceutical cold-chain pre-positioning and sustained stakeholder engagement architecture are developed in full in the Final Report.

Mid-Term Recommendations (12-36 Months)

Sustained investments substantially completable within a three-year horizon. They build resilience above the early-phase floor that near-term recommendations establish.

Distribute Microgrid Capacity to Reduce Single-Point Energy Dependency.

Expand distributed microgrid capacity at rural Alaska communities, regional installations, and dual-use civilian-military sites where energy concentration risk is highest. Investment focuses on reliability: diesel-plus-renewable-plus-storage configurations sized for community and installation demand under disruption conditions, building on the existing baseline of approximately 150 standalone microgrids in Alaska.

Begin Alternate Port Resilience Investments.

Initiate the capital programs required to convert the six identified contingency alternatives to the Port of Alaska, Seward, Valdez, Kodiak, Homer, Whittier, and Mackenzie, into functional substitutes for containerized cargo handling. The return is strategic resilience over multi-year horizons rather than Days 0–14 benefit, but the investment is necessary for long-term reduction of the single-point-of-failure exposure that the dependency mapping surfaces.

Build Medical Evacuation Surge Capacity.

Expand medevac surge through pre-positioned aviation fuel reserves at key medevac airfields; redundant flight-following communications; pre-defined receiving-facility coordination at Anchorage and Seattle tertiary care centers; and crew availability programs that build depth in the thin specialty workforce on which Arctic medical evacuation depends.

Advance FEMA P3 Maturity Beyond Basic and Low-Intermediate Tier.

Progress Alaska's Partnership for Infrastructure Protection and the federal-state-tribal-military coordination mechanisms toward Advanced tier on the FEMA Public-Private Partnership maturity model. Investment includes formalized charters, sustained pre-incident engagement, regular exercise activity, and integration into commercial lifeline operator business processes.

Establish a Permanent Cross-Sector Supply Chain Intelligence Function.

Build, in coordination with the open-source Supply Chain Intelligence Center proposal in the contested-logistics literature, a permanent coordinating capability for anticipating and responding to cross-sector supply chain disruption. Leverages FEMA Defense Production Act authorities, CISA Critical Infrastructure Partnership frameworks, and peer-level coordination with DOT, Commerce, and ODNI.

Final Report Reference: Mid-Term Recommendations (12-36 Months). Additional mid-term items including tribal partner protocols and cold-chain infrastructure at tribal and community health nodes are developed in full in the Final Report.

Long-Term Recommendations (36 Months or More)

Multi-year construction, sustained policy commitment, or sustained research engagement beyond the current grant window. They address the structural conditions of the Arctic operating environment that no near- or mid-term investment alone can resolve.

Extended Sustainment Modeling Beyond the Days 0–14 Window.

Conduct extended sustainment modeling for the recovery dynamics, structural reconstitution, and multi-month sustainment trajectories that lie beyond the Days 0–14 analytical window of this research. Identified as the single most consequential follow-on research priority and the natural extension of the current engagement.

Operational Piloting of Recommended Coordination Architectures.

Pilot the coordination architectures identified in the near-term and mid-term recommendations under live operational conditions, federally declared emergencies, large-scale exercises, and Arctic-specific TTX programs. Converts coordination from validated-on-paper to validated-in-operations, with after-action records that update the doctrine base.

Integrated Arctic Logistics Testbed.

Establish, in partnership with DoD, DOT, MARAD, FEMA, the State of Alaska, tribal governments, and private-sector lifeline operators, an integrated testbed for operational evaluation of mitigation investments, coordination architectures, and emerging logistics technologies under realistic Arctic conditions.

Codified DoD–Private Sector–Nongovernmental Coordination Doctrine.

Develop, in coordination with NORAD, USNORTHCOM, OSD, FEMA, and the broader emergency management and private-sector partnership community, codified doctrine for DoD–private sector–nongovernmental coordination under contested-logistics conditions. Builds on NSM-22, the FEMA P3 maturity model, the Jones Act waiver framework, and the contested-logistics literature.

Long-Horizon Infrastructure Investments, Polar Icebreakers, Processing Capacity, Refining.
Sustain investments in polar icebreaker fleet recapitalization through the Polar Security Cutter program and the April 2025 RFI for Arctic Security Cutters; in-state food processing capacity expansion; and in-state refining capacity assessment. None of these is a Days 0–14 mitigation; all are necessary for the resilience baseline against which future disruption will be measured.

Final Report Reference: Long-Term Recommendations (36 or More Months) and Continuation and Extension.

Strategic Implications for NORAD/USNORTHCOM and Mission Partners

This section translates the findings and recommendations into the language of mission assurance, planning cycles, and partner equities. Each subsection identifies what the analytical record implies for a specific mission partner and where in the Final Report the partner-specific material is developed.

For NORAD / USNORTHCOM

The dual-theater scenario is structurally consistent with the simultaneity problem the 2025 National Defense Strategy identifies as a primary planning challenge. Arctic mission assurance planning should integrate the Days 0–14 decision window at the operational specificity that command-level decisions require, not as a generalized resilience concept but as a defined planning horizon with named dependencies, identified failure thresholds, and pre-defined adjudication authorities. The connective infrastructure framing in Section 10 of this Executive Summary, developed at length in the Final Report, identifies the institutional architecture under which sustained Arctic mission assurance coordination becomes operationally executable.

For the Homeland Defense Institute

This research provides the analytical foundation for follow-on commissioned work in extended sustainment modeling, operational piloting, and the integrated Arctic logistics testbed identified in the long-term recommendations. The connective infrastructure framing identifies the institutional architecture under which subsequent research commissioned by HDI can be executed at the cross-sector specificity Arctic mission assurance requires. The companion study *Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense*, extends the analytical record into the human-systems dimension.

For FEMA, CISA, and DOT

The findings reinforce the relevance of the FEMA Public-Private Partnership maturity model, the CISA Critical Infrastructure Partnership framework, and the Jones Act waiver process, and identify where each currently sits relative to Arctic operational

requirements. Advancement from Basic and low-Intermediate maturity toward Advanced tier is the mid-term institutional priority. Pre-staging of waiver documentation, identified in the near-term recommendations, converts a regulatory mechanism that may otherwise activate too late into an operationally usable tool within the Days 0–14 window.

For Alaska DHS&EM and the State of Alaska

Alaska's Partnership for Infrastructure Protection (APIP), the Mass Care Group and Feeding Task Force, the Alaska Food Value Chain Coordination Council, and the State's coordination architecture with federal, tribal, and military partners are the operational platforms on which Days 0–14 response will run. Tabletop validation against the dual-theater scenario is the near-term priority. The mid-term investment is in advancing P3 maturity across those platforms; the long-term commitment is in sustaining the coordination architecture so that it remains exercised, current, and operationally credible.

For Tribal Governments and Alaska Native Organizations

Engagement is governed by sovereignty, consent, and reciprocity protocols, consistent with the framework PSEMA applied throughout the research. This report identifies the absence of pre-existing relationships between non-tribal institutions and tribal governments as a structural coordination gap and a primary risk-amplifier condition. The implementation parameters, protocols, timelines, and the role of sustained relationship over event-driven engagement, are addressed in detail in the companion study *Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense*, which treats tribal and Alaska Native resilience under sovereignty, consent, and reciprocity framing.

For Private-Sector Lifeline Operators

Engagement throughout this research has been non-attributed, aggregated, and operationally credible. The connective infrastructure framing identifies the architecture under which lifeline operators can engage Arctic mission assurance planning without disclosure of proprietary information, competitive data, or operational specifics that fall outside the scope of mission assurance coordination. Sustained engagement architecture is identified as a near-term recommendation; lifeline operator participation in mid-term P3 maturity advancement and long-term doctrine development is the natural extension.

Final Report Reference: Strategic Implications and From Recommendations to Implementation.

Connective Infrastructure: From Recommendations to Implementation

This section names the institutional finding the research surfaces and the role PSEMA is positioned to play. The compact form here is supported by the full institutional case developed in the Final Report.

Every recommendation in this report identifies multiple external partners: HDI, NORAD/USNORTHCOM, Alaska DHS&EM, FEMA, CISA, tribal governments and Alaska Native organizations, academic institutions, and the private-sector operators across the seven lifeline sectors examined. The recommendations are operationally executable. What requires explicit articulation is the connective infrastructure under which they become operationally executable at scale and across years.

The most analytically significant institutional finding of this research is that the architecture necessary to translate between defense mission requirements and the privately operated systems that sustain them, non-attribution, sector-neutral, operationally credible, sustained across years rather than activated during incidents, does not currently exist within the federal architecture at the maturity Arctic mission assurance requires. Federal entities, by virtue of their statutory mandates and regulatory roles, cannot occupy the sector-neutral intermediary position; commercial operators cannot occupy it across competitors; nongovernmental and academic partners can occupy parts of it but do not have the cross-sector reach across all seven lifeline categories that Arctic mission assurance requires.

PSEMA is positioned to fill that gap. The institutional case is developed in full in the Final Report. The summary points:

- What the gap is, episodic, transactional partnership frameworks insufficient for sustained Arctic mission assurance coordination across the seven lifeline sectors and the multi-year horizons the recommendation architecture requires.
- Why a sector-neutral, non-attribution intermediary is the operational requirement, and why federal entities cannot fill that role within their statutory mandates.
- How PSEMA's membership architecture, engagement protocols, and connective relationships across the seven lifeline sectors align with the requirement, with operational credibility established through the research engagement itself.
- Where PSEMA connects to external partners across near-term, mid-term, and long-term recommendations, including the institutional architecture under which subsequent research commissioned by HDI can be executed at the cross-sector specificity Arctic mission assurance requires.
- What continuation availability looks beyond the May 2026 final submission, sustained engagement, follow-on research, operational piloting, and the integrated testbed identified in the long-term recommendations.

Final Report Reference: From Recommendations to Implementation and The PSEMA Network as Connective Infrastructure.

Continuation, Limitations, and Forward Linkage

An honest accounting of what this research is, what it is not, and where it goes next.

What This Research Is

An analytical foundation for operational planning, organized around five integrated deliverables and validated through structured stakeholder engagement across the seven lifeline sectors examined. The recommendations are operationally specific, the partner architecture is identified, and the time horizons are realistic. The findings are corroborated against an independent body of published, validated research; the analytical posture is operational, not theoretical.

What This Research Is Not

It is not operational planning. It is not classified analysis. It is not a comprehensive solution set for every gap identified. It does not assign blame or attribution to any individual organization. The Days 0–14 analytical window is bounded; extended sustainment beyond Day 14, recovery dynamics, structural reconstitution, multi-month sustainment trajectories, is identified as a follow-on research priority, not delivered within this engagement.

Forward Linkage to the June 2026 HDAS Briefing

The findings, recommendations, and connective infrastructure framing carry forward to the Homeland Defense Awareness Symposium briefing in Colorado Springs in June 2026. The presentation will be positioned against the operational planning cycles of the audience institutions, with the cross-cutting institutional coordination question, the connective infrastructure gap and the PSEMA network framing, surfaced for direct discussion.

Companion Study

Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense addresses the human-systems dimensions of the same dual-theater disruption environment, military families, civilian and contractor workforce, tribal and Alaska Native communities under sovereignty, consent, and reciprocity framing, and governance architecture. The integrated picture across the two studies positions the homeland defense research community to engage Arctic resilience as the systems-design and institutional-coordination challenge the analytical record establishes it to be.

Continuation Availability

PSEMA views this engagement as a foundational exploration that establishes the analytical baseline for the follow-on work the long-term recommendations identify, extended sustainment modeling, operational piloting, the integrated Arctic logistics testbed, and codified DoD–private sector–nongovernmental coordination doctrine. PSEMA is committed to that work and welcomes the opportunity to discuss with HDI, NORAD/USNORTHCOM, and mission partners how this research can be extended and

deepened in support of Arctic homeland defense and the commercial and community systems that make it possible.

Final Report Reference: Conclusion

Reader's Guide to The Full Final Report

A navigation aid for readers moving from the Executive Summary into the Final Report. Section references correspond to the May 2026 Final Report.

List of Acronyms and Glossary.

Recommended first stop for readers new to Arctic homeland defense terminology.

Strategic Implications.

Introduction, strategic context, scope and boundaries, and methodology overview. The starting point for readers who want the full strategic frame before engaging the analytical chapters.

Research Foundation.

Analytical framework, posture, methodological validation anchors, and limits. Recommended for readers evaluating the analytical robustness of the research.

Scenario Development.

Dual-theater disruption framework, West Coast port degradation, Arctic escalation, sustainment modes, stressors, and the Operation Vittles historical analog. Recommended for readers focused on the scenario calibration and historical baseline.

Key Findings.

All twenty-two findings developed in full, in their eight-cluster groupings (Fuel as the Master Dependency; Maritime and Aviation Sustainment Limits; Seasonal and Inventory Asymmetry; Communications and Cyber-Physical Convergence; Workforce as Critical Dependency; Medical and Pharmaceutical Cascades; Cascade Dynamics and Paper Redundancies; Threat Scenarios and Governance Architecture). Recommended for readers who need the detailed analytical record underlying the six consolidated findings in this Executive Summary.

Targeted Dependency Mapping.

Five-layer dependency framework, critical dependencies, cross-sector chokepoints, and coordination architecture. The longest analytical section of the Final Report and the empirical foundation for the fuel, communications, and workforce findings.

Focused Risk and Vulnerability Assessment.

Time-phased degradation, cascading failure analysis, cross-border risk amplifiers, tribal community risk amplifiers, and mission impact. The empirical foundation for the cascade, seasonal asymmetry, and medical findings.

Arctic Sustainment Modes and Lifeline Constraints.

Substitution Inventory, paper vs. operational redundancy, institutional coordination, Jones Act waiver framework, FEMA P3 maturity model, and contested logistics. The empirical foundation for the sustainment substitution and paper-redundancy findings.

Preliminary Mitigation and Cost–Benefit Prioritization.

Prioritization framework, seven priority mitigation categories, and integrated final ranking. The empirical foundation for the recommendation architecture in Section 8 of this Executive Summary.

Stakeholder Engagement.

Engagement methodology, sector coverage, and synthesized findings carried forward from non-attributed stakeholder input.

Recommendations.

Near-term, mid-term, and long-term recommendation architecture in full, with partner mapping and implementation parameters.

From Recommendations to Implementation.

Institutional coordination architecture, the connective infrastructure finding, and the PSEMA network framing.

Conclusion.

End-state framing, continuation availability, and the companion study linkage.

Final Report Reference: Table of Contents

Closing Source Note

This Executive Summary is a synthesis of material developed in full in the *Private-Sector Lifelines in the Arctic: Predicting and Assessing Risk to Critical Infrastructure* (May 2026), Final Report. No new analysis, no new findings, and no new recommendations are introduced in this document. The Final Report is the authoritative source for the substantive material summarized here, including the full reference list, the citation-supported development of each finding, and the partner-specific implementation parameters of the recommendation architecture.

P S E M A · H O M E L A N D D E F E N S E I N S T I T U T E

Private-Sector Lifelines in the Arctic

Predicting and Assessing Risk to Critical Infrastructure

Final Report Date: May 2026

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EXECUTIVE SUMMARY

This report examines the privately operated physical lifeline systems that sustain Arctic homeland defense and concludes that the resilience of NORAD and USNORTHCOM's mission in the North is structurally dependent on commercial fuel, food, transportation, communications, and medical supply networks that lie largely outside Department of Defense control. Through analysis of a dual-theater disruption scenario involving West Coast port degradation and concurrent Arctic escalation, the research demonstrates that these lifeline systems operate with limited inventory buffers, narrow substitution margins, and concentrated chokepoints whose disruption cascades non-linearly into installation-level mission impact. The findings show that failure pathways move faster than existing planning frameworks assume, that paper redundancy frequently does not translate into operational redundancy under Arctic conditions, and that the Days 0–14 window of a crisis is decisive for whether private-sector lifeline degradation can be arrested before it becomes mission-limiting.

The central finding of this research is both straightforward and consequential: Arctic homeland defense depends on a web of privately operated lifeline systems that are not designed for the stress conditions a dual-theater disruption would impose. When they fail, they fail faster than planners assume, cascade more broadly than frameworks anticipate, and leave decision-makers operating without adequate situational awareness during the window when choices are most consequential.

This final report presents the completed findings of *Private-Sector Lifelines in the Arctic: Predicting and Assessing Risk to Critical Infrastructure*, a research effort commissioned by the Homeland Defense Institute in support of NORAD and USNORTHCOM's Arctic mission assurance mission. The research examines privately operated physical lifeline systems, fuel and energy, food and cold-chain distribution, multi-modal transportation, communications, and medical and pharmaceutical supply, as defense critical infrastructure dependencies whose failure cascades directly degrade force generation, sustainment, and mobility in the Arctic theater. The analysis is grounded in DoD Directive 3020.40 (Mission Assurance) and NSM-22 (Critical Infrastructure Security and Resilience), which recognize that military readiness is inseparable from civilian critical infrastructure, a dependence that intensifies in the Arctic where lifeline systems are sparse, fragile, and geographically constrained [69; 80].

The research is analytically distinct from its companion study, C*ontinuity of Society in the North: Multi-Modal Resiliency in Arctic Defense*, which examines the human-systems dimension of Arctic mission assurance, military families, the civilian and contractor workforce, tribal and Alaska Native communities, and the governance architecture that connects them. The two projects share a common dual-theater scenario, West Coast port degradation across Days 0–14 concurrent with Arctic escalation across Days 10–45, but maintain distinct analytical lenses. Where the companion study traces what happens to people because of system failures, this research traces what happens to the systems themselves, how failure pathways move through interdependent

infrastructure, and what those system effects mean for the installations and missions that depend on them.

Five completed deliverables form the analytical backbone of this final report: Scenario Development; Targeted Dependency Mapping; Focused Risk and Vulnerability Assessment; Preliminary Mitigation and Cost–Benefit Prioritization; and Stakeholder Engagement. Together with the Interim Report submitted in March 2026, these deliverables establish the dual-theater operating environment, map the structural chokepoints and no-fail nodes across national supply pathways and in-theater lifeline systems, identify the failure cascades most likely to become mission-limiting within the first fourteen days, evaluate high-impact resilience options against realistic throughput parameters, and validate findings through structured engagement with private-sector operators, government partners, tribal entities, and academic institutions across the lifeline sectors examined.

Throughout, the analytical posture is consistent. Arguments lead; research corroborates. Vulnerabilities are framed in operational terms, what fails first, how fast, with what cascading effect, and at what point intervention is no longer feasible. The objective is not to catalogue Arctic problems. It is to give NORAD, USNORTHCOM, the Homeland Defense Institute, and their interagency and private-sector partners an evidence-based foundation for the investment, coordination, and policy decisions that close the gap between paper redundancy and operational redundancy in Arctic mission assurance.

KEY INTERIM FINDINGS

The findings below are organized into eight analytical clusters that correspond to the structure of the Final Report. Each cluster groups related failure modes, dependencies, or governance conditions that share an underlying logic, allowing the reader to move from the master dependency (fuel) through the lifeline architecture, into the cascade dynamics that connect them, and finally to the governance and coordination conditions that determine whether intact infrastructure can actually be coordinated under stress. Findings are intentionally additive: each contributes a distinctive analytical observation that is not reducible to the others in its cluster.

Cluster 1: Fuel as the Master Dependency

Fuel is the master dependency in the Arctic operating environment.

Fuel disruption does not stay in the energy sector. It simultaneously disables transportation, heating, power generation, and aviation, all of which depend on imported fuel products with limited regional storage and no rapid resupply alternative when maritime supply is degraded. Every other system failure mode identified in this research is accelerated when fuel availability is constrained. Fuel is not one vulnerability among several. It is the foundational dependency whose degradation initiates cascades across the entire lifeline architecture.

In-state refining capacity is structurally concentrated and operationally singular.

Alaska's in-state refining capacity is distributed across a small number of facilities operating under thin margins and tightly coupled to a single crude source. The structural significance is not absolute volume but concentration: there is no parallel refining architecture in-state that can absorb the loss of a single facility, and any disruption to refined product output cascades immediately into aviation, ground transportation, heating, and power generation. Refining concentration converts what elsewhere would be a sectoral disruption into a statewide lifeline event. [71]

Maritime fuel resupply depends on the small number of named commercial carriers.

The maritime fuel chain that feeds western and northern Alaskan communities depends on a sole or near-sole commercial avgas importer and a small number of fuel and freight carriers that together constitute the operational backbone of seasonal resupply. This is not a redundancy gap that can be closed through additional federal coordination; the underlying market structure is consolidated, the assets are specialized (ice-class, shallow-draft, fuel-rated), and there is no substitution path that does not require the same vessels under a different flag. Sole-source maritime fuel dependency is a structural feature of the Arctic operating environment, not a planning gap.

Cluster 2: Maritime and Aviation Sustainment Limits

West Coast maritime gateways are no-fail nodes with no rapid substitute.

The Port of Alaska is the single intermodal hub through which most of Alaska's freight, fuel, and population-supporting supply flows from the Pacific Northwest gateway complex. Peer-reviewed computable general equilibrium modeling of a compound disruption at the twin ports of Los Angeles and Long Beach, the same port complex degraded in this research's dual-theater scenario, finds GDP losses propagating well beyond the port region through commercial supply-chain transmission, with no rerouting alternative available at sufficient scale within a fourteen-day window. Those figures exclude Alaska entirely; Arctic mission-assurance consequences are additive, not captured within existing national loss models.

Aviation cannot replace maritime sustainment at scale.

Airlift is routinely cited in contingency planning as the alternative when maritime routes are disrupted. The RAND Plan Blue 21 wargame documents directly that this assumption is operationally wrong at scale. The historical record (Operation Vittles, 1948–1949) confirms that air-only sustainment requires conditions that do not apply in the Arctic dual-theater scenario, and that even under those conditions falls short of full substitution. Arctic airfields are simultaneously constrained by weather, fuel, maintenance capacity, and crew availability. Replacing one week of Port of Alaska cargo throughput by air would require the equivalent of seven hundred wide-body 747 cargo landings, a figure that does not account for fuel, de-icing, crew, or runway capacity, and that makes the substitution argument operationally implausible. Competition between operational aviation, medical evacuation, and logistics use of limited airfield capacity is immediate and zero-sum. [26]

The Jones Act framework adds friction to maritime substitution under stress.

Cabotage requirements under the Jones Act constrain the universe of vessels available for emergency Alaska resupply to U.S.-flagged, U.S.-built, U.S.-crewed bottoms. Waivers are available but are administratively scoped, time-limited, and require demonstration of national defense necessity through a process that was not designed for the compressed timelines of an Arctic-Pacific dual-theater event. The Jones Act is not the cause of the maritime substitution gap, but its waiver architecture is a friction point that adds time to a window in which time is the binding constraint.

Cluster 3: Seasonal and Inventory Asymmetry

Seasonal asymmetry compresses winter failure timelines from weeks to days.

The vulnerability is not uniform across the calendar year. Alaska's downstream distribution network serving western and northern communities is seasonal and ice-constrained. A West Coast port disruption occurring between October and April operates against already-depleted reserve margins with no barge resupply option available, compressing the time from port degradation to critical shortage from weeks to days because depleted reserves cannot be replenished until thaw. The scenario examined in

this research must be assessed against both dimensions: a national chokepoint failure and a seasonal logistics closure occurring simultaneously.

Commercial inventory buffers in Alaska are measured in days, not weeks.

Just-in-time architecture produces a structural fragility under disruption: community-level shortages appear within days. Planning frameworks built around multi-week adjustment periods before shortages become visible are working from an incorrect baseline. The Days 0–14 window must be planned for as a period of near-complete supply constraint at the community level, with the implication that any intervention designed to take effect after Day 14 arrives too late for the populations whose sustainment determines whether the installation workforce can report.

Cluster 4: Communications and Cyber-Physical Convergence

Communications failure is a force degrader for every other failure mode.

Communications degradation does not produce a single system failure. It eliminates the visibility into logistics status, infrastructure performance, and resource availability that every other response system depends on. Communications degradation is adversarial-exploitable: documented gray-zone tactics targeting both terrestrial and space-segment communications make this a force degrader, not a passive failure mode. Commercial satellite operators may alter service availability during crises, with no governance framework currently requiring them to maintain service. When communications degrade, intact physical infrastructure cannot be coordinated, supply decisions cannot be made, and the entire lifeline architecture operates blind.

Cyber-physical convergence in lifeline systems extends the attack surface beyond traditional defense.

The privately operated lifeline systems on which Arctic mission assurance depends are increasingly cyber-physical: supervisory control, fuel terminal automation, refinery process control, electrical grid management, satellite ground stations, and logistics dispatching all depend on networked control systems with varying levels of defensive hardening. A kinetic disruption to a fuel terminal and a cyber disruption to its control system are not separate scenarios; they are convergent vectors against the same operational outcome. Mission assurance planning that addresses only physical infrastructure or only cyber infrastructure underestimates the actual threat surface.

Cluster 5: Workforce as Critical Dependency

The civilian and contractor workforce is a critical but under-modeled dependency.

Installation sustainment depends on a civilian workforce that lives in communities experiencing the same disruptions as the installation. That workforce is not modeled in most mission assurance frameworks as a constraint, yet its availability is directly tied to the household-level food, fuel, and service access that supply chain disruption degrades within the first seven days. Workforce degradation reduces the functionality of otherwise

intact infrastructure. This is the failure mode that planning frameworks most consistently miss.

Workforce thinness compounds with seasonal labor dependency in specific sectors.

Several lifeline sectors central to the Arctic operating environment, seafood processing, maritime crewing, aviation maintenance, fuel terminal operations, depend on a workforce that is both numerically small and seasonally mobile. A disruption that occurs during a labor turnover window, or that prevents the inbound movement of seasonal workers, compounds the workforce constraint identified in Finding 11 with a labor-supply shock that has no rapid substitute. The implication is that workforce sustainment is not only a household-level problem but also a labor-market sequencing problem that planning frameworks rarely surface.

Cluster 6: Medical and Pharmaceutical Cascades

Medical evacuation capacity competes directly with logistics aviation under stress.

Arctic medical evacuation depends on the same constrained airfield, fuel, weather-window, and crew resources that logistics aviation requires under a dual-theater disruption. Under normal conditions, these systems coexist without significant competition. Under the scenario examined in this research, they become directly competitive: every airframe and airfield slot allocated to logistics surge is one not available for medical evacuation, and the resulting allocation decisions are made without a pre-defined adjudication mechanism. The civil–military adjudication gap in medical evacuation prioritization is a Day 0 problem, not a Day 14 problem.

Pharmaceutical cold-chain dependency creates a population-level failure mode within the first week.

A substantial fraction of the population in Alaska depends on temperature-controlled pharmaceuticals (insulin, biologics, refrigerated antibiotics, dialysis-related medications) whose supply chain depends on continuous power, refrigerated transport, and timely resupply. Disruption to any one of these components produces a population-level health consequence within days, not weeks. The cold-chain failure mode is structurally similar to the food cold-chain but operates on a shorter clinical timeline, and it concentrates risk in populations that are already disproportionately remote from acute-care capacity.

Cluster 7: Cascade Dynamics and Paper Redundancies

Cascading failures occur faster than planning assumes, and non-linearly.

The representative failure cascade, port throughput degradation, then delayed cargo, then inventory drawdown, then aviation surge, then fuel constraints, then medical evacuation limitations, then workforce degradation, is not a worst-case scenario. It is the operationally expected trajectory under the dual-theater disruption conditions the scenario models. Each link accelerates the next. Non-linear escalation occurs once key thresholds are reached, points at which sustainment assumptions fail, and the self-

reinforcing character of the cascade takes over. The critical planning implication is that the window for effective intervention closes faster than linear degradation models suggest.

Paper redundancies consistently fail to materialize under Arctic stress conditions.

Alternate routes are constrained by weather, ice, crew availability, and throughput limits. Backup fuel supplies exist on paper but are often already committed or seasonally inaccessible. Satellite communications alternatives provide limited bandwidth with intermittent reliability. The most analytically significant finding of this research is the gap between the redundancies embedded in planning documents and the redundancies that actually function under Arctic disruption conditions. Closing that gap is the organizing challenge that the recommendations and the PSEMA network framing are designed to address.

Canadian sovereignty makes the ALCAN Highway a paper redundancy at scale.

The Alaska–Canada Highway is the principal overland surge route into Alaska and is cited in nearly every contingency planning document as the alternative when maritime routes are degraded. In practice, ALCAN surge capacity is constrained by Canadian sovereignty over the corridor, by weather and winter road conditions that limit heavy commercial movement for a substantial portion of the year, and by infrastructure chokepoints (fuel availability, driver hours-of-service, customs throughput) that have no rapid surge mechanism. ALCAN is a genuine redundancy at low intensity and a paper redundancy at the volumes contemplated under a dual-theater disruption.

Distributed microgrid architecture is a partial structural protection against statewide cascade.

The approximately 150 standalone microgrids that serve western, northern, and interior Alaskan communities represent a distributed electrical architecture that is not vulnerable to a single statewide grid failure of the kind that would affect a comparably populated CONUS region. The microgrid architecture is a structural asset for resilience. The same architecture, however, concentrates fuel dependency: each microgrid is reliant on imported diesel or aviation fuel, and the failure mode that would collapse multiple microgrids simultaneously is the fuel supply chain disruption identified in Cluster 1. Distributed power architecture is a partial protection, not a comprehensive one, and its protection is fuel-bounded.

Cluster 8: Threat Scenarios and Governance Architecture

Most-likely and most-dangerous threat scenarios diverge, requiring distinct planning postures.

HDI's Research Focus Area framing requests assessment of both most-likely and most-dangerous Arctic threats. These two categories diverge in the Arctic operating environment and require different planning postures. The most-likely disruption profile is a compound natural-hazard event (severe winter weather, seismic activity, volcanic ash) coinciding with an unrelated commercial supply chain disruption, a scenario that is functionally indistinguishable from the dual-theater scenario in its lifeline consequences. The most-dangerous profile is an adversary-initiated dual-theater event with deliberate targeting of no-fail nodes identified in this research. Planning that addresses only one category leaves the other unaddressed.

Institutional coordination is a risk amplifier, not a passive backdrop.

The absence of pre-defined adjudication mechanisms for civilian and military claims on scarce fuel, food, transportation, and medical capacity is the most consequential governance gap surfaced in the research. The Jones Act waiver framework, the FEMA Public-Private Partnership maturity model, and the contested-logistics literature converge on the same conclusion: coordination architectures designed for predictable, single-incident response degrade under simultaneous, multi-theater stress. Building the resolution architecture before the crisis is a high-return, low-capital priority that no infrastructure investment can substitute for.

A sustained, trust-based intermediary is missing from the DoD–private sector–nongovernmental coordination architecture.

Across every lifeline sector examined, the same structural weakness recurs: there is no operationally-tested, sector-neutral intermediary positioned to translate between defense mission requirements and the privately operated systems that sustain them. Existing partnership frameworks are episodic, transactional, or limited to incident response. The connective infrastructure that Arctic mission assurance requires, non-attribution, sector-neutral, operationally credible, does not currently exist within the federal architecture. This finding shapes the recommendations in this report and the framing of the PSEMA network as connective infrastructure for the path forward.

Coordination architecture is itself a paper redundancy when it has not been exercised.

The federal Public-Private Partnership maturity model characterizes private-public partnership capacity along a Basic, Intermediate, Advanced continuum. Most Arctic jurisdictions relevant to this research currently operate at the Basic or low-Intermediate tier, with episodic engagement and resource commitments not formalized through charters or tested through exercises. Coordination frameworks that exist on paper but have not been exercised, formalized through charter, or integrated into lifeline operator business processes do not meaningfully reduce risk during Days 0–14. This is a direct application of the paper-redundancy principle to the governance layer.

Bottom Line

Arctic homeland defense cannot be sustained against the dual-theater disruption scenario this research models without three things that do not currently exist at adequate scale: pre-positioned reserves of fuel, food, pharmaceuticals, and critical materiel sufficient to bridge the Days 0–14 window; multi-layer communications redundancy that does not depend on any single commercial provider or terrestrial pathway; and a pre-defined civil–military coordination architecture, exercised against realistic Arctic disruption conditions, with private-sector cargo prioritization agreements in place before the crisis. Each of these is a preparatory investment whose return is realized before disruption occurs. None can be substituted for during the window when the failures they are designed to prevent are already underway.

The recommendations developed in this report are organized to support exactly that preparatory posture, across near-term, mid-term, and long-term horizons, with explicit identification of which external partners, the Homeland Defense Institute, NORAD/USNORTHCOM, Alaska DHS&EM, FEMA, CISA, tribal governments and Alaska Native organizations, academic institutions, and the PSEMA network, carry which lines of effort. The closing PSEMA Network section articulates the connective infrastructure that this research identifies as essential to making any of these recommendations operationally executable.

STRATEGIC IMPLICATIONS

Arctic mission assurance is constrained by lifeline fragility, not force capability. The lifeline systems on which Arctic homeland defense depends were not designed for the disruption scenario this research models; they were designed for normal operating conditions, and they are thin at every margin, in storage capacity, redundancy, workforce, coordination architecture, and the time available before failures cascade across sectors. Under dual-theater disruption, those margins compress from weeks to days, and in some cases to hours.

Closing the gap requires investments that are pre-positioned, pre-coordinated, and pre-authorized to take effect inside the Days 0–14 window. It requires throughput preservation, not infrastructure protection alone. It requires upstream cascade prevention rather than downstream symptom mitigation. And it requires engaging commercial operators as mission partners, not as regulated entities or contracted vendors. The institutional scaffolding exists, the doctrinal rationale is established, and the analytic foundation has been built. What remains is the deliberate alignment of investment, policy, and coordination commitment to the conditions, geographic, climatic, and operational, that make the Arctic both the most demanding and the most consequential environment for homeland defense mission assurance.

Introduction and Strategic Context

The question at the center of this research is not abstract. If an adversary simultaneously degrades major West Coast port capacity and forces rapid military mobilization through constrained Arctic corridors, which privately operated systems fail first, how fast do the effects cascade, and where do the planning assumptions embedded in current defense strategy break down?

That question matters because Arctic homeland defense cannot be separated from the commercial infrastructure that sustains it. Military installations in Alaska depend on fuel that arrives by barge from Pacific Northwest refineries, food that moves through a handful of maritime gateways, communications systems running on commercial satellite and fiber infrastructure, and a civilian workforce whose stability is inseparable from access to basic services. None of that infrastructure was designed for dual-theater disruption, none of it is owned or controlled by DoD, and almost none of it has been systematically mapped against the specific failure conditions this research examines.

Throughout this report, business-critical infrastructure refers to the privately operated physical lifeline systems on which Arctic homeland defense depends: fuel infrastructure, food and cold-chain distribution, transportation networks, communications systems, and medical and pharmaceutical supply. These systems are owned and operated outside Department of Defense authority but are essential to the projection, sustainment, and recovery of military forces in the Arctic theater. The term is used in preference to "private-sector infrastructure" or "commercial infrastructure" because it specifies operational

function, the systems on which mission and community sustainment depend simultaneously, rather than ownership category alone.

Doctrinal and Strategic Grounding

The institutional grounding for treating commercial Arctic systems as defense-relevant is established in current doctrine and policy. DoD Directive 3020.40, *Mission Assurance*, defines roles and responsibilities for the protection of DoD's critical assets and infrastructures and explicitly recognizes that military facilities and missions depend on civilian systems including energy grids and transportation networks [69]. National Security Memorandum 22, issued in April 2024, defines Defense Critical Infrastructure to include both DoD and non-DoD networked assets and facilities essential to project, support, and sustain military forces and operations worldwide; non-DoD-owned defense critical infrastructure consists of assets from relevant critical infrastructure sectors and subsectors, including as defined by statute [80].

The Navy's 2021 Strategic Blueprint for the Arctic establishes that effective Arctic operations are logistics- and access-defined rather than capability-defined, with sea lines of communication, chokepoint security, and self-sustaining force posture treated as the binding constraints on operational viability [73]. The Blueprint does not assume that resilient supporting infrastructure exists; it assumes operations must continue in a degraded environment, a posture directly consistent with the analytical premise of this research.

The contested-logistics literature developed since 2022, Unified Pacific Wargame Series outputs, Army contested-logistics modernization, and Naval Postgraduate School theses on supply chain resilience, establishes that sustainment is the pacing function in modern conflict, not maneuver, with the homeland increasingly understood as a strategic support area rather than a sanctuary [48; 58; 64]. Together, these frames bring privately operated lifeline systems formally within the mission assurance scope and reframe commercial Arctic logistics as adversarially exploitable rather than environmentally constrained.

Scope and Boundaries

This research is bounded in three deliberate ways that should be transparent to the reader. First, the analytical focus is on the Days 0–14 disruption window. Longer-horizon analysis, recovery dynamics, structural reconstitution, and multi-month sustainment modeling, is out of scope for this engagement and identified as a follow-on research priority. Second, the geographic frame is Alaska, the ALCAN corridor, and the West Coast maritime gateways serving Alaska. Greenland and Canadian Arctic interests are referenced where they directly affect Alaska sustainment, but a full North American Arctic treatment is outside this scope. Third, the analysis is unclassified and operates from publicly available sources, supplemented by structured stakeholder engagement conducted under non-attribution protocols. Where classified context would refine specific findings, the report identifies the boundary and recommends the appropriate follow-on framework.

Within those bounds, the research is intended to be operationally usable. The dependency mapping identifies specific nodes. The risk assessment identifies specific failure thresholds. The mitigation prioritization identifies specific investment categories ranked against five operational criteria. The recommendations identify specific actions, specific external partners, and specific time horizons. The intent is not to characterize Arctic vulnerability in general terms; it is to provide the foundation for command-level decisions about where preparatory investment is most consequential, and which coordination architectures must be built before the next disruption requires them to exist.

Analytical Framework

The research employs a five-deliverable architecture, with each deliverable building on the analytical foundation established by the prior step. The architecture is sequential by design: scenario before dependency, dependency before risk, risk before mitigation, and mitigation refined by stakeholder validation. This sequencing ensures that conclusions at each layer rest on the layer below, and that the integrated picture presented in this final report is internally consistent rather than assembled from independent strands.

Research Sponsor and Institutional Context

This research is sponsored by the Private Sector Emergency Management Association (PSEMA) under federal commission from the Homeland Defense Institute. Reviewers of this report and stakeholders engaging with its findings may not be familiar with PSEMA. The institutional architecture, membership composition, and operational posture that produced this research are summarized here to ground the analysis that follows in the organizational context that generated it.

PSEMA is a 501(c)(6) professional association, with an affiliated 501(c)(3) PSEMA Foundation, founded to close a structural gap in the national emergency management architecture: the absence of a sustained, sector-neutral, non-attribution venue through which private-sector lifeline operators, federal and state emergency management authorities, tribal governments and Alaska Native organizations, military commands, and academic institutions could engage one another on the operational realities that determine whether disaster response, continuity of operations, and homeland defense missions succeed or fail. PSEMA was established to fill that gap, and its institutional architecture is built around the engagement model the gap requires.

Mission and Vision. PSEMA's mission is to advance the integration of private-sector capability into emergency management and homeland defense by convening cross-sector practitioners, producing operationally grounded analytical products, and sustaining the trust-based relationships that crisis coordination depends on. PSEMA's vision is a national resilience architecture in which the privately operated systems that sustain American communities and defense operations, energy, fuel, food, transportation, communications, medical supply, warehousing, and the workforce that operates them, are integrated into preparedness, response, and recovery as full partners rather than as transactional vendors engaged at the point of disruption.

Membership Composition. PSEMA's membership and affiliated partner network span the lifeline sectors examined in this research and the partner categories that surround them. Member and partner organizations include energy and fuel providers; transportation, logistics, and supply chain operators across maritime, rail, trucking, and aviation modes; telecommunications and technology firms; food distribution, retail, and humanitarian feeding organizations; medical and pharmaceutical providers; engineering and infrastructure resilience consultancies; insurance and reinsurance partners; tribal and Alaska Native organizations; humanitarian and disaster-relief nonprofits; and emergency management consultancies and continuity practitioners. Federal, state, and academic partners participate alongside private-sector members through structured engagement protocols. The cross-sector composition is not incidental; it is the structural condition that makes the cross-sector translation function operationally feasible.

Operational Posture. Three features define PSEMA's operational posture and distinguish it from sector-specific trade associations, single-domain consultancies, and federal advisory committees. First, PSEMA operates under non-attribution protocols that enable candid practitioner engagement on system vulnerabilities, operational limits, and decision realities that attributed forums constrain. Second, PSEMA is sector-neutral, not anchored in any single lifeline category, which positions it to facilitate the cross-sector coordination that cascading disruption requires. Third, PSEMA's engagement is sustained across pre-incident, incident, and post-incident periods rather than activated transactionally during a disruption, producing the institutional memory and relationship continuity that operational coordination under stress depends on.

Relationship to This Research. PSEMA's research engagement with the Homeland Defense Institute on this project and its companion study draws directly on the cross-sector membership and engagement architecture described above. The dependency mapping, cascading risk pathways, and mitigation analysis presented in this report reflect practitioner insight gathered through PSEMA's standing engagement protocols, supplemented by literature review, federal policy analysis, and structured stakeholder validation. The non-attribution methodology applied throughout this research is the same methodology under which PSEMA's broader practitioner engagement operates, and it is the methodology that makes the operational realities documented in this report accessible to the analytical record at all. The recommendations developed in this report identify multiple external partners across every line of effort, and the connective infrastructure under which those recommendations become operationally executable is the subject of the From Recommendations to Implementation section. The institutional, methodological, and relational foundations developed here are the foundations that section builds on.

RESEARCH FOUNDATION

The deliverables in this report were developed through three lines of work that together give the analysis its operational and evidentiary grounding. Literature review was the primary source of information for this research, providing the analytical depth that the compressed performance window did not permit through stakeholder engagement alone. Literature review was supplemented by a limited set of structured stakeholder calls with private-sector operators and government partners, a SurveyMonkey-distributed Stakeholder Input Survey, and strategic conversations with current and former NORAD and USNORTHCOM leadership. The methodological objective was not comprehensive coverage of every Arctic infrastructure question; it was the construction of a coherent analytical foundation capable of supporting specific, decision-grade recommendations within the time and resource constraints of the engagement.

Methodology

The research was conducted along three coordinated lines of work, each documented separately and integrated by analytical synthesis. The full methodological foundation, source inventory, mental-models foundation, strategic conversations protocol, and stakeholder engagement instrument detail, is documented in Appendix D. The Stakeholder Input Survey instrument is documented in Appendix E.

The first and primary line of work was an extensive literature review spanning Arctic infrastructure and supply chain vulnerability, DoD mission assurance doctrine and critical infrastructure interdependency frameworks, Alaska-specific logistics and energy systems, civil–military coordination under disruption conditions, and the governance architecture of tribal, municipal, state, federal, and military authorities in Alaska. Given the three-month performance window, literature review carried the analytical weight of the research, providing the specificity of failure timelines, the precision of the dependency mapping, and the identification of cascading risk pathways that are consistent across multiple independent sources. The full inventory of sources with citation IDs is documented in Appendix B.

The literature record converges on four conditions that ground the scenario architecture. The Arctic has shifted from managed cooperation to active great-power competition, and the U.S. icebreaker fleet is critically under-sized against operational requirement. Adversarial gray-zone tactics, GPS jamming, maritime area closures, and interference with undersea telecommunications cables, are documented active practice rather than hypothetical risk. Alaska's dual-use infrastructure architecture concentrates statewide sustainment at the Port of Alaska as a no-fail node, with no viable substitute available within the Days 0–14 window. The unclassified record on Alaska's commercial supply chain is operationally precise and characterizes the just-in-time architecture, the seasonal asymmetry, and the inventory thresholds that the dependency mapping develops in detail. [8 13]

The second line of work was a series of strategic conversations with current and former NORAD and USNORTHCOM leadership. These conversations informed the analytical scope of the research by validating the dual-theater scenario as operationally relevant, confirming that the dependency gaps under examination are recognized at the command level, and identifying the civilian workforce, commercial cargo prioritization, and civil–military governance gaps as research areas consequential for mission-assurance planning. Input from these conversations is reflected in the analytical framing of the deliverables that follow.

The third line of work was structured stakeholder engagement conducted primarily through a SurveyMonkey-distributed Stakeholder Input Survey, supplemented by limited informal conversations with select members of the PSEMA network. The Stakeholder Input Survey served as the principal instrument and consisted of fifteen items combining closed-ended ranking, multi-select, and open-ended response formats. The survey used a core question architecture calibrated to the dual-theater disruption scenario and oriented toward surfacing governance friction, family, and workforce sustainment thresholds, civil–military coordination gaps, and the early-phase decision points where authority structures are most likely to strain.

Supplemental input was gathered through a small number of informal, non-attributed conversations with select PSEMA network members; these exchanges were not conducted as formal structured interviews but provided contextual validation of assumptions and dependencies surfaced through the survey and literature reviews. The engagement window was reopened with an extended response deadline to encourage adequate representation; survey response was nonetheless limited by the three-month performance window, and the analysis presented in this report reflects responses received through the extended deadline.

Three engagement principles govern the stakeholder methodology and apply throughout. First, the engagement is system-of-systems by design: the question architecture surfaces cross-domain interdependencies, the relationships between governance structures, family readiness, workforce sustainment, and tribal community resilience, rather than domain-specific perspectives. Second, the engagement is structured rather than open-ended: a standardized question framework asked across all participants enables comparative analysis across domains and surfaces practitioner-vs-analytical divergences in operationally consequential terms. Third, all input is aggregated and non-attributed by default, with no proprietary or sensitive data required and qualitative thresholds and operational insights treated as sufficient. Participation does not imply endorsement of findings. Tribal engagement was conducted under explicit sovereignty, consent, and reciprocity protocols, and tribal input is governed by those protocols rather than by the default aggregation framework.

Limits, Boundaries, and What This Methodology Cannot Deliver

The methodology is bounded by the three-month engagement window, the data accessible under non-attribution protocols, and the unclassified frame of the work. Three categories of limits warrant explicit acknowledgement and shape the recommendations developed in the final phase of the report.

First, the research does not produce decision-grade quantification of failure thresholds at the precision that command-level investment and pre-positioning decisions ultimately require. The Days 0–14 analytical frame is established with rigor, and the directional findings, that inventory buffers are measured in days, that cascading failure initiates within the first seventy-two hours, that governance friction compounds degradation, are sound. Translating those directional findings into time-to-failure parameters expressed in hours and days, by commodity, location, and disruption condition, is identified as a follow-on research priority.

Second, the research operates from publicly available sources supplemented by limited structured stakeholder engagement under non-attribution protocols. Where classified context, proprietary commercial data, or installation-specific operational information would refine specific findings, the report identifies the boundary and recommends the appropriate follow-on framework. The intent is not to claim more than the methodology supports; it is to be transparent about what the methodology does and does not deliver.

Third, the research is engagement-window-bounded, and the stakeholder engagement component is bounded further by response rate, sector coverage, and self-selection within that window. The three-month performance period was sufficient to construct the analytical foundation, validate it against literature and stakeholder input, and identify the most consequential mitigation and recommendation priorities. It was not sufficient to fully operationalize the recommendations, conduct exercise-based validation of paper coordination architectures, or pilot the integrated Arctic logistics testbed concept that the long-term recommendations identify. Within the engagement component specifically, structured stakeholder calls were limited to a small select group, and survey response was concentrated in the operations, continuity, and emergency management end of the lifeline architecture rather than in upstream lifeline operators. Direct fuel and energy operator participation, direct maritime and port operator participation, and direct communications and telecommunications provider participation are below the levels that sector-specific quantitative claims would require.

Tribal and Alaska Native organizational engagement did not occur within the engagement window and is identified as a priority for future research, with the sovereignty, consent, and reciprocity framing that institutional participation on terms tribal governments determine requires. Survey participation was self-selected; respondents who chose to engage are systematically more disposed toward cross-sector coordination than the sector population as a whole, which is a relevant consideration when interpreting the strength of the joint government-industry coordination consensus. The non-attributed aggregation approach protects respondent confidentiality and was a precondition for candid engagement, but it limits the specificity with which individual findings can be

sourced. Engagement findings are therefore treated throughout this report as directional validation of the analytical record and as identification of practitioner-vs-analytical divergences worth carrying forward, not as the basis for sector-specific quantitative claims in under-represented categories.

> ***Key Insight:*** These limits are not deficiencies in the research; they are boundaries of what can be accomplished within the engagement frame. The recommendations and the PSEMA Network section of this report are constructed to identify what should happen next, by whom, on what timeline, and with what coordination architecture, so that the foundation laid here can be extended into the operationally executable architecture that Arctic mission assurance requires.

The five deliverables that follow build on this methodological foundation in sequence; the dual-theater disruption scenario that grounds all subsequent analysis is developed in the Scenario Framework section that opens the analytical body of this report.

SCENARIO DEVELOPMENT

The Scenario Development deliverable establishes the operating environment against which all subsequent analysis is conducted. It is designed for a broader audience than most defense research, not only DoD planners and logistics analysts, but installation and garrison leadership, private-sector lifeline operators, state and local emergency management partners, tribal governments and Alaska Native organizations, elected officials and civilian policymakers, and the nonprofit and community organizations that support both installations and the populations surrounding them. Understanding the scenario, and why it was constructed as it was, is essential context for everything that follows.

Framework Overview

This research employs a dual-theater disruption scenario to evaluate the limits of U.S. Arctic sustainment capacity under sustained stress. The scenario reflects the central research question: how do interdependencies between defense installations and privately operated lifeline networks influence mission assurance when both are disrupted at scale, particularly during the first Days 0–14 of disruption?

The dual-theater disruption scenario that grounds all subsequent analysis, West Coast port degradation across Days 0–14 concurrent with Arctic escalation across Days 10–45, is developed in full in the Scenario Framework section that opens the analytical body of this report.

The research focuses on how interdependencies between defense installations and privately operated lifeline networks influence mission assurance and sustainment capacity during the first zero to fourteen days of disruption, when formal mitigation options are limited and the decisions made under incomplete information set the trajectory of everything that follows.

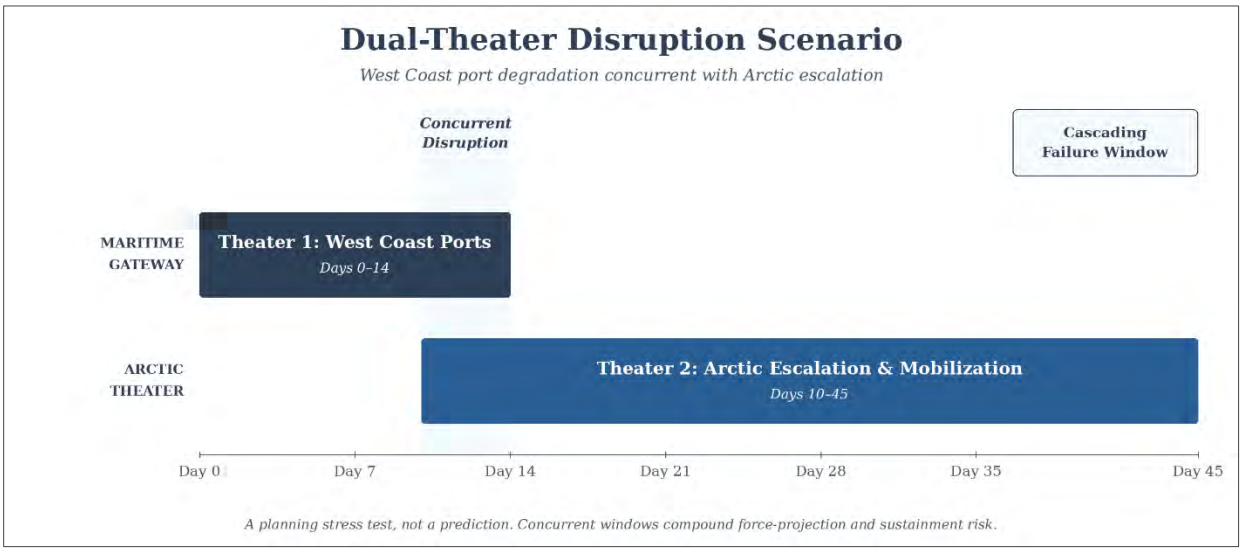
This Scenario Framework is the canonical home for the dual-theater scenario architecture. Subsequent sections, Dependency Mapping, Risk Assessment, Cascading Impact Narratives, and the Mitigation Options framework, reference the operative phase or window only, and rely on this section for the full scenario specification. The two-phase architecture is summarized as: Days 0–14 West Coast port degradation as the early-phase analytical focus, and Days 10–45 concurrent Arctic escalation as the extension window during which mitigation timelines must be evaluated.

The scenario simultaneously stresses two interdependent systems. The first is the CONUS-to-Alaska supply chain supporting fuel and fuel additives, food distribution and cold-chain commodities, pharmaceuticals and medical supplies, telecommunications equipment, and the logistics materiel that defense operations and civilian communities depend on. The second is the in-theater Arctic lifeline system that sustains installations, defense contractors, military families and dependents, and the civilian communities whose stability underwrites operational readiness.

The scenario explicitly incorporates the Arctic-specific realities that make disruptions here qualitatively different from their lower-latitude analogs. Sparse infrastructure and long resupply distances [25]. Severe and prolonged weather conditions. Limited redundancy across fuel, food, power, and communications [26]. Reliance on shared civilian–military systems with no dedicated military alternatives at scale. Jurisdictional complexity involving tribal, municipal, state, federal, and military authorities with different legal bases, different priorities, and no predefined coordination architecture for simultaneous stress [19; 45]. Concentrated dependence on Pacific Northwest maritime gateways with no equivalent substitute [49].

These constraints compress decision timelines and magnify early impacts on supply chains and workforce availability in ways that planning frameworks built around installation self-sufficiency assumptions do not capture. Just-in-time architecture (Glossary; developed in Dependency Mapping) is the structural condition under which the Days 0-14 timeline operates.

> ***Key Insight:*** Early-phase disruption is driven less by catastrophic system failure and more by the rapid degradation of logistics reliability, commercial supply continuity, and the workforce availability that depends on both.



Concurrent West Coast port degradation and Arctic escalation
Scenario framework establishes a planning stress test, not a prediction.

Primary Disruption: West Coast Port Degradation (Days 0-14)

The scenario begins with a coordinated, non-kinetic disruption of major West Coast maritime hubs, the Seattle/Tacoma port complex, which serves as the primary gateway for Alaska-bound containerized cargo, and the Port of Los Angeles/Long Beach. The disruption is compound in character: cyber intrusions targeting terminal operating systems, interference with navigation and cargo-handling systems, and labor safety stand-downs triggered by operational instability. No single failure shuts ports down completely. Throughput degrades. Predictability erodes. Confidence collapses. For Alaska-bound supply chains, the effect is immediate.

The Port of Alaska is the single node through which the majority of Alaska's commercial and defense sustainment flows, handling approximately half of all inbound freight and fuel for nearly the entire state's population. When the Seattle/Tacoma gateway degrades, that concentration means disruption is not distributed gradually across the state. It arrives statewide, immediately, at a single point with no equivalent substitute.

That vulnerability is not uniform across the calendar year. Alaska's downstream distribution network serving western and northern communities is seasonal and ice-constrained. A West Coast port disruption occurring between October and April operates against already-depleted community reserve margins, with no barge resupply option available to offset shortages. For households dependent on bulk seasonal delivery, that compression means the interval between port degradation and critical food or fuel shortage is measured in days, not weeks. The scenario examined in this research must be assessed against both dimensions: a national chokepoint failure and a seasonal logistics closure occurring simultaneously, and landing on communities with the least buffer to absorb either.

The scenario's analytical premise does not require this specific disruption mechanism to hold. Open-source seismic-science analysis of the Cascadia Subduction Zone documents a parallel disruption pathway in which a magnitude 8.0–9.2 full-margin rupture would simultaneously degrade the Ports of Seattle, Tacoma, and Portland; sever Pacific Northwest overland transportation corridors; damage fuel storage and distribution infrastructure including the Puget Sound refineries that process the majority of jet fuel and diesel consumed in Alaska; and generate tsunami inundation across critical intermodal nodes [43]. The non-kinetic disruption modeled here, and the seismic disruption documented in the geological record produce structurally analogous downstream effects on the Alaska supply system; the analytical conclusions of this scenario hold under either pathway.

Immediate Impacts: First 72 Hours

Within the first 72 hours, the effects arrive at the supply chain and workforce level before installation-level impacts become visible. Delays in Alaska-bound shipments of fuel, food, pharmaceuticals, telecommunications equipment, and logistics materiel begin immediately as maritime scheduling reliability erodes. DoD and defense-contractor supply flows routed through West Coast gateways are disrupted. Early backlogs form across container terminals, rail yards, and intermodal transfer points.

Rerouting toward Gulf and East Coast ports stresses rail and trucking capacity in ways that compound the original disruption and extend restoration timelines. Commercial operators, port authorities, and government agencies have limited visibility into recovery sequencing, alternative routing feasibility, or which sectors will be prioritized first in cargo allocation decisions.

The national economic consequence of compound West Coast port disruption, documented in peer-reviewed computable general equilibrium modeling [49], is not primarily a financial indicator in this context. It is a proxy for the speed and breadth of supply chain disruption: food and agricultural flows are among the first affected, and no short-term rerouting exists at the scale Alaska requires.

Early-Phase Stressors: Days 0–14, the Primary Analytical Focus

State emergency management planning in Alaska explicitly recognizes that outside assistance to affected communities may require 24 hours or more to arrive, establishing an early-phase response gap that local and tribal resources alone must absorb. This is not a planning estimate produced by the present research; it is an assumption embedded in Alaska's own operational doctrine, confirming that the Days 0–14 analytical window reflects an operational reality already recognized by state practitioners rather than an externally imposed construct.

Three integrated conditions shape what happens inside that early-phase window, and together they account for why disruption that would degrade gradually in a CONUS context degrades acutely in an Arctic context.

Just-in-Time Fragility Expresses Itself at the Household Level. The 2023 Alaska Supply Chain Baseline Assessment, prepared by Dewberry under the FEMA SCAN Pacific Partnership Project, identifies that commercial inventory buffers for essential goods in Alaska are limited, with food distribution systems commonly estimated at 6–10 days of available supply under normal operating conditions. [26]

Alaska imports upwards of 95 percent of its food, with approximately 95 percent arriving by sea, 4 percent by refrigerated truck up the ALCAN, and 1 percent by air cargo. Food departing the Port of Tacoma arrives at the Port of Alaska in three days and reaches retail shelves on the fourth day; ships arrive on Sundays and Thursdays; the state has no major warehousing capacity. Alaska is, by design, a just-in-time food retail state. What matters for this research is that the 6–10 day inventory window is not an abstract supply chain metric. It is the actual decision horizon within which families, workforce members, and community caregivers must decide about feeding, heating, medical access, and whether to remain in place. In the Arctic, that dependence is not a planning edge case; it is the defining operational condition. [4; 38; 44]

External Dependency with Limited Local Substitution. Fuel, food, medical supplies, and consumer goods are overwhelmingly imported, and the population's capacity to substitute local production for disrupted external supply is bounded. Subsistence systems are resilient and operationally significant, as the Alaska Federation

of Natives' community resilience documentation confirms [[2]], and they extend survival timelines and reduce immediate dependence on external supply flows for the communities that practice them. But subsistence systems are not scalable to offset prolonged commercial disruption at the statewide population level, and the populations most dependent on external commercial supply, the military, the civilian and contractor workforce, and urban populations, have the least access to subsistence substitution. This differential is the analytical basis for the differential vulnerability assessment this research applies across the three populations examined.

Annual petroleum throughput at the Port of Alaska is at a volume for which no alternative routing or delivery mechanism exists within the Days 0–14 window; the infrastructure substitution analysis is examined in the companion study. When that flow is interrupted, the downstream effect is not a gradual reduction in availability; it is an immediate drawdown against fixed reserves with no replacement mechanism on the timescale of the disruption. The dual-theater disruption scenario used in this research is structurally consistent with the simultaneity problem the 2025 National Defense Strategy identifies as a primary planning challenge across the Arctic and Western Hemisphere. The doctrinal frame within which this scenario is analytically significant is developed in the Institutional Coordination section. [70]

Workforce Attrition as Early-Phase Condition. The workforce that sustains transport, processing, logistics, healthcare, and community services in Alaska is heavily dependent on seasonal, imported, and transient labor whose availability is directly affected by the same disruption conditions the scenario produces. Workforce absenteeism driven by childcare, eldercare, household sustainment needs, and subsistence requirements begins within the first week, with simultaneous stress on healthcare, public safety, social services, schools, and emergency response organizations. Workforce attrition under these conditions is therefore not a downstream consequence of the scenario; it is an early-phase condition that propagates into installation support functions, community services, and the civilian sustainment capacity that both populations and installations depend on. The generational digital-dependency vulnerability documented in the Scenario Conditions section compounds this dynamic in the population segment most needed for surge capacity. The contractor workforce that installations rely on for maintenance, logistics, food service, transportation, and support operations is drawn from the same communities experiencing these stressors, which is why installation-level workforce disruption and community-level workforce disruption cannot be analytically separated.

Across households and families, the first fourteen days produce reduced availability of food, heating fuel, medications, and essential household goods. Price volatility, panic purchasing, and rationing behavior emerge in local markets as supply predictability erodes. Families with limited local storage, limited transportation options, or limited income flexibility face these conditions with the least capacity to absorb them. Delays in childcare supplies, medical items, and household necessities destabilize the daily routines that support both family wellbeing and workforce participation.

Across installations and garrisons, civilian workforce disruptions affecting base operations and support services begin within the first week. Increased demand on installation support services from families and dependents intensifies simultaneously. Garrison leaders face growing pressure to balance mission requirements with household and community support concerns, a tension that early-phase uncertainty and limited information make harder to resolve, not easier. Early decisions made under that uncertainty set conditions for prolonged strain even if logistics flows partially recover later.

> ***Key Insight:*** Supply chain and workforce instability emerge before installation-level impacts are fully visible, making early commercial system stress a leading indicator of broader operational degradation.

Concurrent Arctic Escalation (Days 10-45)

As West Coast logistics remain impaired, Arctic operational requirements increase. Elevated air and maritime activity, expanded surveillance and domain-awareness missions, sustained operations at Arctic and sub-Arctic installations, and increased demand for logistics support, fuel, mobility, and installation services all impose greater demands on systems already stressed by the primary disruption. Arctic Edge 2025 is operationally instructive on this point: the exercise was conducted concurrently with an INDOPACOM exercise, generating real competing demand signals across shared sustainment resources and confirming that multi-theater concurrent operational tempo is not a planning hypothesis but a documented exercise condition [67].

Under these conditions, readiness depends directly on the resilience of commercial systems that sustain operations: fuel delivery and distribution, food supply continuity, pharmaceutical and medical resupply, telecommunications infrastructure, and the civilian and contractor workforce sustaining installation operations. RAND Plan Blue 21 is directly applicable: logistics capacity falls below operational demand, coordination speed falls below disruption speed, and civilian system resilience falls below military dependency, creating conditions where the cumulative failure arrives faster and cuts deeper than any individual system failure would predict [46].

The scenario assumes no open kinetic conflict but sustained pressure on systems with limited surge capacity, limited redundancy, and high dependence on commercial infrastructure that was not designed for these conditions. The critical analytical insight is that increased operational demand does not merely add to existing civilian system strain; it amplifies it, accelerating competition for limited resources across military and community systems in ways that governance frameworks are not currently designed to manage.

The Arctic escalation pathway is further shaped by the absence of functional security dialogue mechanisms. Open-source analysis documents that military security dialogue between Russia and other Arctic states has been suspended since 2014, and that no pan-Arctic mechanism exists for managing submarine telecommunications cable security [8; 13; 14; 45]. The Ted Stevens Center's 2025 High North Security Dialogue documents that Arctic security increasingly requires integrated military, civilian, and private-sector

approaches, with infrastructure resilience and domain awareness as foundational enablers and Russia identified as the primary near-term threat layered against Chinese long-term systemic competition [60]. The absence of deconfliction mechanisms means that misperceptions or accidental incidents are harder to de-escalate, compressing the time available before military and civilian systems must adapt to a degraded or contested environment. The Northern Sea Route represents a potential maritime rerouting option that could be closed or degraded by adversarial action under UNCLOS Article 234 claims at the moment of greatest need, eliminating a pathway that otherwise appears as an alternative in benign planning models [49].

> ***Key Insight:*** Increased operational demand amplifies existing commercial system strain, accelerating competition for scarce fuel, cargo capacity, and workforce availability across military and civilian systems simultaneously.

Real-Time Analog: Maritime Chokepoint Disruption and the Compression of Fuel Reserves

The dual-theater scenario this research models is a stress test, not a prediction. Its construction is deliberately conservative: a coordinated non-kinetic disruption of West Coast maritime gateways concurrent with an Arctic escalation requiring rapid northern mobilization. The scenario does not depend on the events of any specific contemporary conflict for its validity. It does, however, gain analytical confidence when the patterns it models are observable in current real-world maritime disruption, and the current Strait of Hormuz situation supplies a structurally analogous case at the time of this report.

Following the outbreak of U.S.–Iran hostilities in late February 2026, Iranian forces declared the Strait of Hormuz closed and asserted regulatory control over transit through the Persian Gulf Strait Authority. Pre-conflict transit averaged approximately 3,000 vessels per month carrying roughly twenty percent of the world's seaborne oil trade and a comparable share of liquefied natural gas. [39] Within weeks, monthly transits had fallen to under ten percent of that level, with daily open transits reaching near zero by early May. War-risk insurance premiums tripled within days of escalation. Importing countries dependent on Persian Gulf throughput, Japan, South Korea, India, and others, began drawing down strategic reserves on timelines compressed against initial planning assumptions. The International Energy Agency coordinated a release of 400 million barrels from member-state reserves. The Federal Reserve Bank of Dallas modeled the disruption as a potential reduction of global GDP growth of 0.2 to 1.3 percentage points depending on duration. These figures are consequential precisely because they describe a system that, in pre-conflict equilibrium, was characterized as resilient and redundant [20; 47].

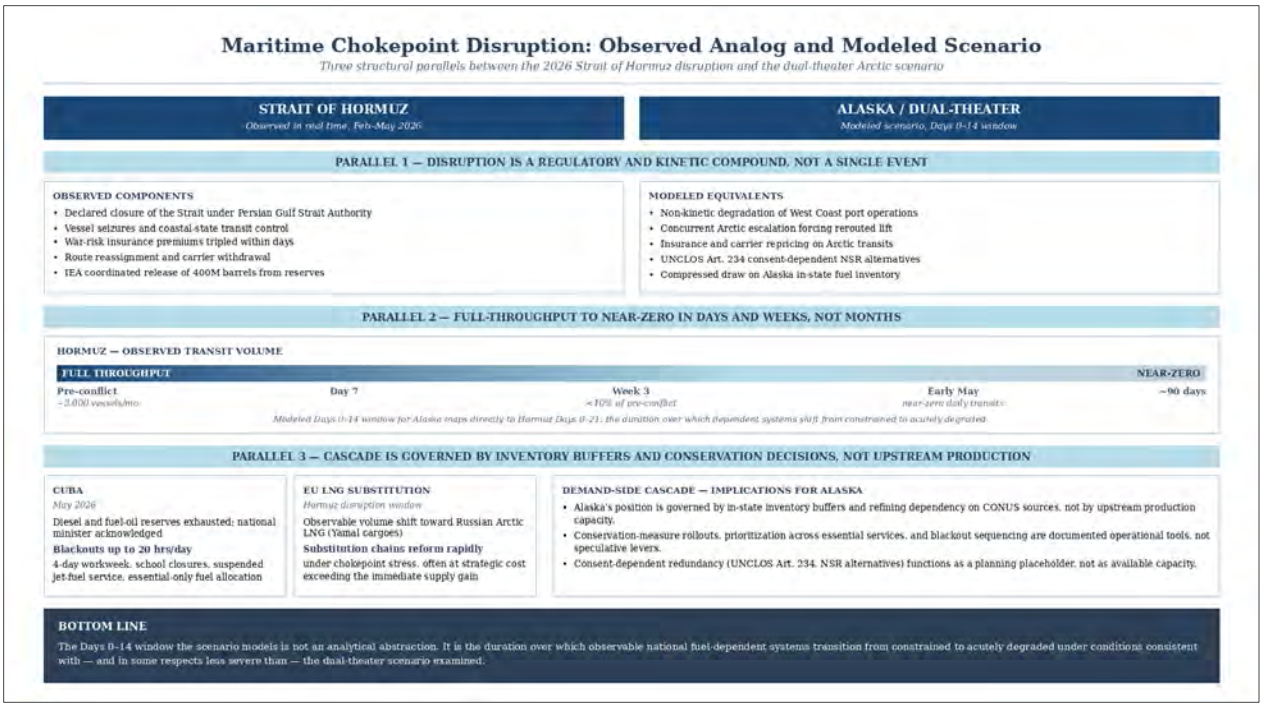
The structural lesson for Arctic mission assurance is not that the Hormuz situation predicts a similar Arctic event. It is that the operational characteristics of maritime chokepoint disruption documented in real time match the operational characteristics this scenario models in three specific ways. First, the disruption is not a single event but a regulatory and kinetic compound, combining declared closures, vessel seizures, insurance-market repricing, and route reassignment under coastal-state authority.

Second, the timeline from full-throughput to near-zero throughput is measured in days and weeks, not months. Third, the cascading impact on dependent national fuel systems is governed by inventory buffers and conservation policy decisions, not by upstream production capacity, which is the same structural condition Alaska faces with respect to its in-state fuel inventory and refining dependency on CONUS sources.

Concurrent with Hormuz, additional fuel-dependent national systems are demonstrating the same pattern at the consuming end. Cuba, whose energy system depends on imported crude and has been subject to U.S. blockade actions since early 2026, has exhausted diesel and fuel-oil reserves to the point that the national energy minister publicly acknowledged in mid-May 2026 that the country has effectively run out, with blackouts of up to twenty hours per day and emergency conservation measures including a four-day workweek, school closures, suspended jet-fuel service, and prioritized fuel allocation to essential services only [52]. Allied LNG dependencies in Europe have shifted observable volume toward Russian Arctic LNG sources, with Yamal cargoes accounting for an outsized share of EU import volume during the Hormuz disruption window, a pattern that demonstrates how rapidly substitution chains reform under chokepoint stress and how those substitutions can carry strategic costs that exceed the immediate supply gain [39].

The relevance of these patterns to the Arctic scenario is direct and bounded. Alaska's fuel position differs from Cuba's in scale, governance, and proximate cause, and the comparison is therefore not equivalent. What the Cuban and broader chokepoint-affected cases establish is that the conditions under which fuel reserves move from operational to critical are not speculative. Conservation-measure rollouts, prioritization decisions across essential services, blackout sequencing, and the political consequences of those choices are being implemented in real time across multiple national systems during the same window in which this research has been conducted. The Days 0–14 window the scenario models is not an analytical abstraction. It is the duration over which observable national fuel-dependent systems transition from constrained to acutely degraded under conditions consistent with, and in some respects less severe than the dual-theater scenario this research examines.

The analog also reinforces the UNCLOS argument developed in the Dependency Mapping section. The Strait of Hormuz crisis is, at its operational core, an exercise of coastal-state regulatory authority over a maritime chokepoint by a state whose interests under disruption do not align with the sustainment requirements of dependent importers. The legal instrument differs from UNCLOS Article 234, Iran has not ratified UNCLOS, and the asserted authority over the Strait is grounded in territorial-waters claims rather than ice-covered-area provisions, but the structural logic is the same. A coastal state asserting unilateral regulatory authority over a transit corridor, in conditions where downstream substitutes are physically constrained or politically unavailable, eliminates redundancy on timelines that substitution-limit modeling does not adequately anticipate. The lesson for Arctic planning is to treat consent-dependent redundancy as a planning placeholder rather than as available capacity, regardless of the specific legal instrument under which the consent operates.



The 2026 Strait of Hormuz disruption mirrors the dual-theater scenario across three structural parallels, confirming the Days 0–14 window as observable operational reality, not analytical abstraction.

Arctic Sustainment Modes and Lifeline Constraints

The Arctic Sustainment Modes framework characterizes how each mode behaves under disruption and surfaces the structural relationships among modes that determine whether substitution is operationally feasible. The underlying infrastructure baseline, throughput, icebreaker capacity, port architecture, and seasonal logistics, is documented in the Scenario Conditions and Constraints section that follows; this section establishes the analytical framework the research applies to that baseline.

Lifeline Sectors and Mission-Control Functions

Stressor domains, onset windows, and household-visible conditions under Arctic dual-theater disruption.

Mode	Capacity & Fuel Profile	Seasonal & Weather Constraints	Cross-Border & Governance Constraint
Maritime: Pacific Northwest to Port of Alaska	**Capacity:** Primary mode, ~50% of all inbound freight and fuel for ~90% of population; >1.47M tons petroleum annually; no operationally available substitute within Days 0–14. **Fuel:** Vessel bunker fuel; low marginal sensitivity at this throughput.	**Seasonality:** Year-round at the Port of Alaska intermodal hub; West Coast gateway congestion is the binding seasonality. **Weather:** Moderate at gateway; high when combined with West Coast port congestion or storm closure.	U.S. domestic. West Coast port and labor disruption is the operative constraint.
Maritime: Western and Northern Alaska barge resupply	**Capacity:** Bulk fuel and bulk goods to rural communities; primary delivery mode for off-road Western and Northern Alaska. **Fuel:** Diesel; depleted reserves cannot be replenished by any other mode until breakup.	**Seasonality:** Late fall delivery cycle, before sea ice and river ice; closed from ice-in to spring breakup. **Weather:** High; ice and seasonal storms are governing constraints.	U.S. domestic.
Aviation: fixed-wing	**Capacity:** Critical-but-capacity-limited tier; replacing one week of Port of Alaska cargo by air would require approximately 700 wide-body sorties (Berlin Airlift Anchor establishes this is not a feasible substitute at scale). **Fuel:** Aviation gasoline for the bush plane network flows through a single importer at the Port of Alaska, single-supplier, single-node dependency.	**Seasonality:** Year-round; airfield availability constrained by weather, fuel, maintenance, and crew. **Weather:** High; Arctic weather, geomagnetic events, and polar route disruption all reduce sortie completion.	Polar overflight permissions where applicable.

Overland: ALCAN Highway	**Capacity:** Constrained-redundancy tier; ~350 vehicles per day operating throughput against 12,000–15,000 vehicles per day theoretical lane capacity (paper-redundancy gap of an order of magnitude). **Fuel:** Diesel for trucking; fuel availability along corridor is itself a constraint under disruption.	**Seasonality:** Year-round in principle; winter closures, weather events, and avalanche conditions produce intermittent unavailability. **Weather:** High at winter; route geometry (1,387 miles) and weather closure cadence are governing constraints.	Approximately 1,200 miles of Canadian transit; sovereign decision points U.S. planning cannot resolve unilaterally.
In-state final-mile: trucking and rail (Alaska Railroad)	**Capacity:** Downstream amplifier mode; throughput determined entirely by upstream maritime, aviation, and ALCAN inputs (not an independent sustainment mode); Alaska Railroad emergency freight activation requires pre-scripted winter-operations pathway and rail-to-port integration at Seward. **Fuel:** Diesel; in-state trucking is workforce-constrained and fuel-constrained simultaneously.	**Seasonality:** Year-round; degraded by snow, ice, and seasonal access at distribution endpoints. **Weather:** Moderate at urban network; high at rural and remote distribution endpoints.	U.S. domestic.

No single mode is independently sufficient under the scenario timeline [73]. The seasonal asymmetry is severe: the modes that carry the greatest peak capacity (maritime through the Port of Alaska intermodal hub and rural barge resupply) are also the modes most compressed by winter conditions and gateway congestion, while the modes available year-round (aviation and ALCAN) carry categorically lower peak capacity.

The Arctic Edge 2025 exercise series, conducted across Alaska under USARAK and USNORTHCOM with joint, interagency, and allied participation, confirms that expeditionary logistics, port operations, and transportation brigade activities are the binding chokepoints on Arctic force flow under exercise conditions, and that infrastructure availability cannot be assumed but must be opened, secured, or restored under degraded conditions [67]. This asymmetry is the operative constraint on Arctic sustainment, not the absolute capacity of any single mode.

Scenario Conditions and Constraints

The scenario incorporates conditions and constraints that define the operating environment for all subsequent deliverables. These are not hypothetical additions; they are the documented realities of Arctic logistics and infrastructure under disruption. The Ted Stevens Center's 2025 High North Security Dialogue convened operators, planners, and Arctic-nation representatives to assess shared concerns, and its proceedings reinforce that infrastructure fragility, communications gaps, and cross-border dependencies are recognized across the Arctic security community as the defining constraints on near-term homeland defense operations [60].

Independent Arctic policy analysis converges on the same conclusion: the operating assumption that the Arctic remains a low-tension cooperative theater is no longer sustainable, and homeland defense planning must increasingly account for strategic competition as the governing condition [8].

Infrastructure and Transportation Constraints

Alaska's transportation architecture is defined by centralization and constraint simultaneously. The Denali Commission establishes that the state operates as a distributed, fragile supply chain system with few redundancies: no continental rail connection, approximately 75 percent of communities inaccessible by road, infrastructure costs two to ten times the Lower 48 baseline, and seasonal logistics cycles with no buffer for sustained disruption [25].

The Port of Alaska's role as the primary consolidation node for approximately 50 percent of all inbound freight and fuel, serving approximately 90 percent of the state's population, means that West Coast gateway disruption is not a partial constraint. It is a statewide supply event concentrated at a single intermodal hub [44]. The macroeconomic stakes are documented: peer-reviewed computable general equilibrium modeling of a compound disruption at the Los Angeles–Long Beach port complex finds GDP losses propagating well beyond the port region through commercial supply-chain transmission to Rest-of-U.S. economic activity, with food and agricultural supply chains among the first and most immediately affected, and no rerouting alternative available at sufficient scale within the early-phase window [49]. The GAO confirms there is no deep-draft port within 800 miles of the Bering Strait, the Nome deepening project remains in design phase. The U.S. icebreaker fleet is critically short of the eight to nine vessels identified as necessary, and GAO has documented that Coast Guard Arctic operations face information gaps and capability shortfalls that further constrain what maritime access can realistically be surged during contested-environment scenarios [34; 37; 59; 68; 75; 77].

The icebreaker capacity constraint that shapes maritime access can be stated specifically. A 2023 Coast Guard fleet mix analysis determined that eight to nine polar icebreakers are needed to meet Arctic mission requirements; the current operational fleet is two vessels, with a third, the converted commercial vessel Storis, anticipated home-ported in Alaska by summer 2026. The Polar Security Cutter program has experienced significant cost growth and schedule delay [21; 68]. For scenario purposes, the operating assumption is

that icebreaker-enabled maritime access during Days 0–14 cannot be surged beyond currently deployed capacity.

Communications and Information Constraints

The Ted Stevens Center's 2023 Arctic Communications Workshop Report documents fragmented and non-interoperable communications systems across agencies and communities, bandwidth limitations constraining real-time coordination, lack of redundancy creating high vulnerability to single-point failures, and insufficient infrastructure in remote communities [58]. For the populations examined in this research, communications degradation is not merely a coordination problem. It eliminates the information access that households depend on to make decisions, increases anxiety and the spread of misinformation, disrupts the digital channels that family readiness programs rely on, and isolates rural and tribal communities from the state and federal coordination structures responsible for their support. Allied partners are pursuing parallel polar satellite communications investments, including Canada's Enhanced Satellite Communications Project, Polar, which is intended to extend assured military communications coverage into the high north, but these investments operate on independent timelines and are not currently integrated into a unified Arctic communications architecture [35].

GAO has separately documented that Coast Guard Arctic communications and domain awareness gaps compound these structural constraints, meaning that the federal entity with the broadest Arctic operational footprint operates with the same information limitations the commercial lifeline systems do [77].

Defense reporting in early 2025 has confirmed that closing the Arctic high north communications gap remains an unresolved priority across DoD and partner-nation discussions, reinforcing that the structural baseline this assessment describes is not a legacy condition awaiting routine modernization but an active capability gap with current operational consequences [11].

Medical and Health System Constraints

Limited healthcare capacity, medevac delays driven by weather and distance, and vulnerable pharmaceutical supply chains create a system operating near its limits before disruption begins. The tribal infrastructure baseline (developed in Dependency Mapping as a risk amplifier) sets entry conditions in which baseline sanitation and water-system stress increase healthcare demand precisely when capacity is most constrained. Pharmaceutical cold-chain disruption is among the earliest consequential failure modes when maritime logistics degrade, with no substitution options and resupply cycles that cannot be compressed [74; 76].

Governance and Coordination Constraints

Overlapping tribal, municipal, state, federal, and military authorities with differing legal bases, different priorities, and no predefined coordination architecture for simultaneous stress create friction at the moments when speed is most operationally critical. The National Academies' report on Indigenous disaster preparedness documents that confusion about tribal authority worsens outcomes, and that tribes exercise real

governmental powers in emergencies that outside agencies frequently misunderstand or ignore. The State of Alaska Emergency Operations Plan (EOP) confirms that no single agency controls all infrastructure or response capabilities, that resource scarcity and competing demands are expected, and that pre-established agreements with private sector and voluntary organizations are critical for effective response, agreements that are rarely in place with the specificity that early-phase conditions require [54]. The governance gap that matters most for the analysis that follows is not the absence of frameworks but the gap between codified authority structures and the operational coordination capacity available under disruption, the paper-vs-real distinction this research applies throughout. Tribal authority within that gap is examined under the sovereignty, consent, and reciprocity framework established in the Research Foundation.

> ***Key Insight:*** No single agency controls all infrastructure or response capabilities, and pre-established agreements with private-sector organizations critical for effective early-phase response, are rarely in place with the specificity those conditions require.

Seasonal Asymmetry and the Berlin Airlift Anchor

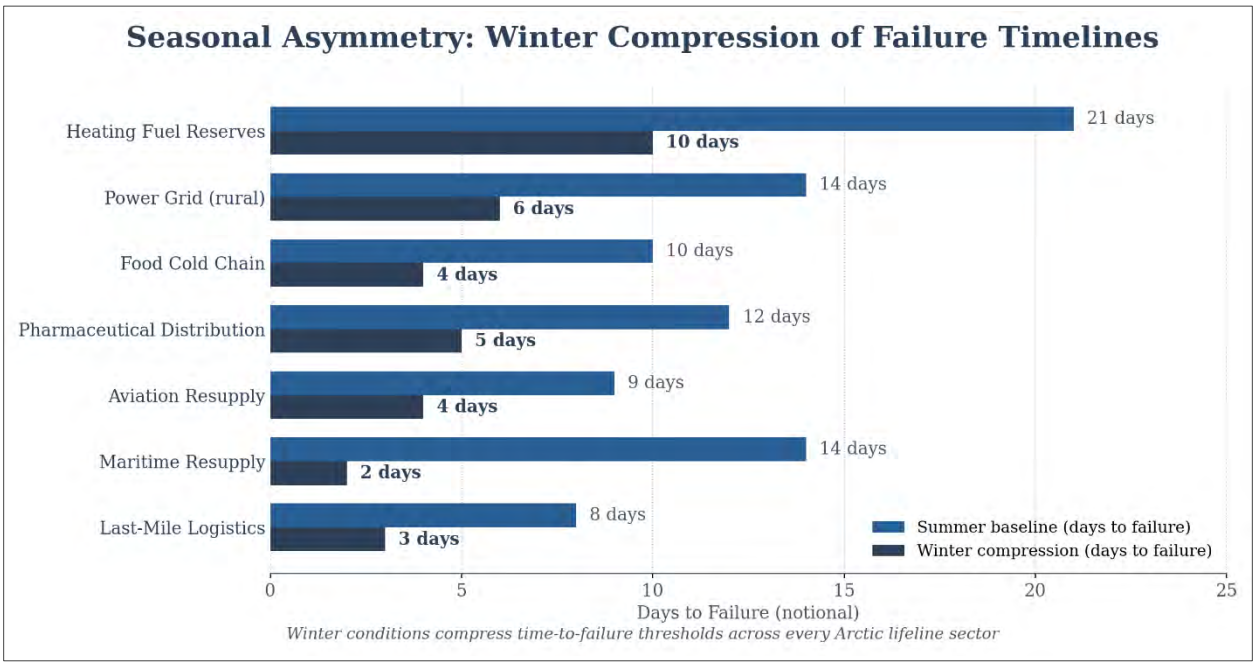
Seasonal asymmetry between summer and winter disruption scenarios is one of the scenario's most consequential and most consistently misunderstood conditions. Bulk fuel delivery to rural Alaska occurs by barge in late fall, before sea ice and river ice set in; any additional fuel required between ice-in and spring breakup must be delivered by bush plane at substantially higher cost and substantially lower volume. A West Coast port disruption occurring during or after the fall barge season therefore operates against communities that cannot receive bulk fuel resupply by the primary delivery mode until spring. Aviation-based fuel resupply under those conditions is not a functional substitute; it is a constrained emergency mechanism operating against depleted reserves with no resupply window. This seasonal asymmetry compresses winter failure timelines from weeks to days because depleted reserves eliminate the barge resupply option.

The rapid-degradation timeline assumed in this scenario is anchored by the historical record. *Operation Vittles*, the Berlin Airlift (1948–1949), is the empirical benchmark for what continuous, highly optimized airlift operations can and cannot deliver under conditions of primary-infrastructure denial. The operation required 321 days of continuous airlift to sustain a defined urban population of approximately two million people, fell short of the approximately 13,500 tons per day requirement during its initial phase, and identified coal (energy) as the dominant sustainment commodity by mass at approximately 2,000 tons per day minimum [66].

The Berlin record is frequently invoked in contingency planning to suggest that aviation can substitute for maritime supply under crisis conditions; on examination it establishes the opposite. The airlift succeeded only because of conditions that do not apply to the Arctic dual-theater scenario: total Allied political commitment sustained over eleven months, a defined and limited urban population, mature undamaged airfield infrastructure, multiple receiving airfields in close proximity, predictable European weather windows, significant international support, and a specific geographic advantage that compressed flight distances. Each of these conditions is favorable rather than

adversarial to airlift feasibility, and yet the operation still fell short of its tonnage requirement during the initial phase. The Berlin record is therefore conservative rather than aspirational when applied to Arctic conditions, and the analytical conclusion the case supports is that airlift is a constrained supplemental mode rather than a substitute for bulk maritime capacity.

Two governing constraints emerge from the Berlin record that this research applies throughout. First, throughput at the receiving airfield, not supply availability at origin, is the governing constraint on aviation-based sustainment: the operation succeeded or failed at the airfield, not at the staging base. Second, the substitution argument fails categorically rather than at the margins: there is no scaling factor at which airlift becomes a feasible substitute for bulk maritime capacity, only thresholds at which it can serve as a constrained supplemental mode for high-value, low-volume cargo. The framework implications of these constraints are developed in the Sustainment Modes Aviation subsection that follows; the case confirms that rapid-degradation timelines of the type modeled in this scenario are a documented historical pattern under infrastructure denial, not a projection.



Winter conditions sharply compress time-to-failure across sectors; heating becomes a binding survival constraint that is not present in summer.

Workforce Digital Dependency

A dimension of workforce disruption that current planning frameworks do not adequately address is the vulnerability concentrated in the youngest segment of the civilian and contractor workforce, teenagers, college students, and young adults roughly ages 13 through 30. This population is disproportionately represented in the entry-level and gig-economy roles that are most immediately stressed under supply disruption: food service, logistics, ride-share and delivery services, retail, childcare, and construction. Their

operational reliability under disruption conditions is constrained by a specific and well-documented vulnerability: near-total dependence on mobile devices and digital connectivity as their primary tools for communication, navigation, information access, and coordination.

In emergency scenarios characterized by power outages, telecommunications failures, and infrastructure degradation, precisely the conditions the dual-theater scenario produces, this digital dependency becomes a concentrated liability. Without reliable cellular networks or internet access, this population loses situational awareness, cannot navigate without GPS, cannot coordinate with employers or community resources, and cannot comply with emergency directives that reach them only through digital channels.

They are simultaneously the workforce segment most needed for surge capacity and the segment least equipped to function when the infrastructure they depend on fails. In rural and tribal communities where this dynamic is compounded by greater distances between services and more pronounced infrastructure fragility, the gap between their potential contribution and their actual availability under disruption conditions is most acute. This generational workforce vulnerability is examined further in the workforce dependency analysis later in this report; the operative point for the scenario architecture is that the labor segment most needed for surge capacity is structurally the segment least equipped to function under the infrastructure conditions the dual-theater scenario produces.

> ***Key Insight:*** The youngest segment of the civilian workforce, heavily digitally dependent and without analog backup skills, represents a latent vulnerability that emerges rapidly when communications infrastructure degrades, precisely when their labor capacity is most needed.

Scenario Stressors to Be Modeled

The scenario introduces four categories of stressors that shape all downstream analysis across the dependency mapping and risk and vulnerability assessment.

Logistics and supply stressors include reduced and unpredictable flow of Alaska-bound cargo; disrupted last-mile delivery to rural, remote, and tribal communities; delays in fuel, food, pharmaceuticals, and essential materiel; and increased reliance on aviation resupply for high-priority goods despite the limited capacity that RAND Plan Blue 21 confirms cannot substitute for maritime cargo volumes at scale [46].

Infrastructure and utility stressors include fuel shortages affecting heating, power generation, transportation, and aviation; power and communications system strain under peak demand; telecommunications outages affecting commercial operators, logistics coordination, and installation support; and the reduced resilience of commercial utility systems under prolonged stress that the DOE Arctic Strategy documents as the foundational vulnerability across all downstream lifeline systems [26].

Workforce and operational stressors include workforce attrition affecting installation support functions, competition between operational and logistics aviation for constrained

airfield capacity, reduced availability of maintenance and support personnel, and degraded operational capacity over time as compounding effects accumulate (the workforce dependency dynamic is examined in detail in Vulnerability Assessment).

*Governance and prioritizatio*n stressors include competing civilian and military claims on scarce fuel, food, transportation, and medical capacity; the absence of predefined adjudication mechanisms for those competing claims; the friction created by overlapping jurisdictions without coordinated decision authority; and the commercial cargo prioritization decisions made by private operators with no obligation to account for defense sustainment requirements [34; 78].

TARGETED DEPENDENCY MAPPING

If the Scenario Framework establishes the conditions under which Arctic lifeline systems will be stressed, the Targeted Dependency Mapping answers the question of what, exactly, is connected to what, and what breaks when those connections are disrupted. This section moves from operational context to structural analysis, building a systematic picture of how commercial infrastructure sustains Arctic defense operations and where the failure points are concentrated.

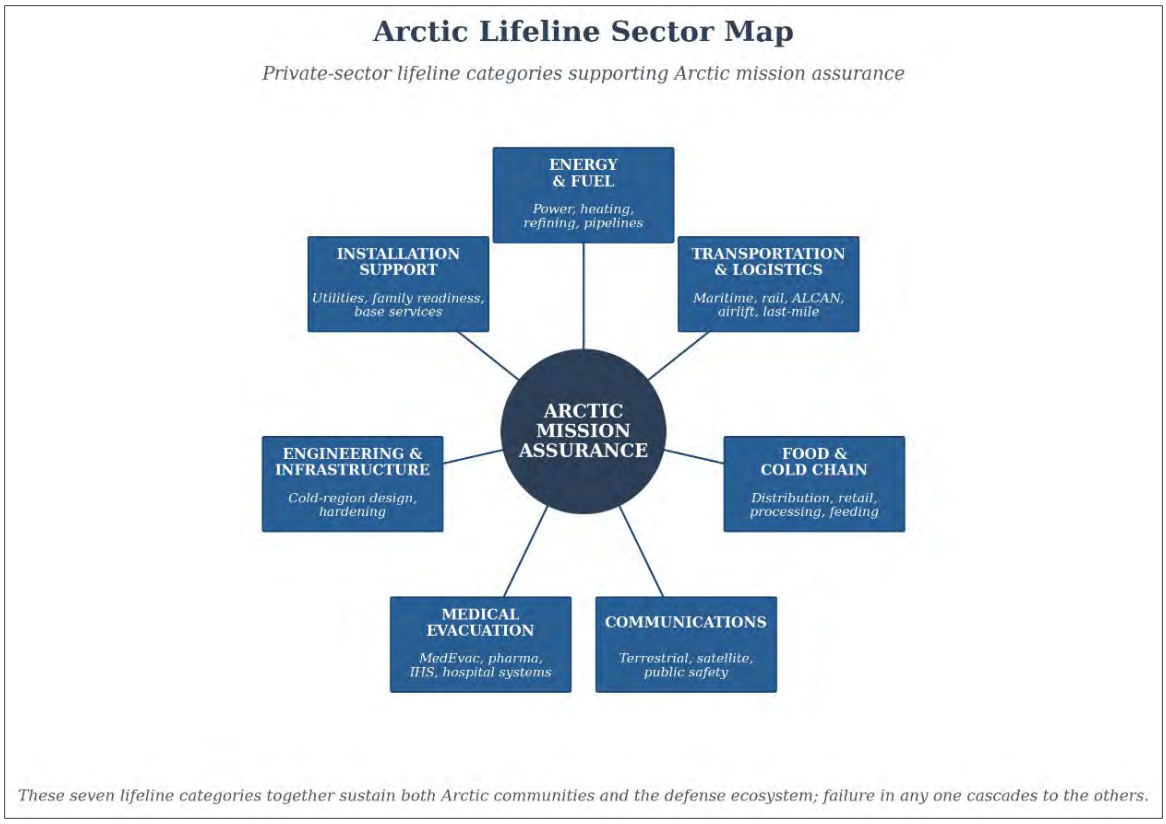
Operative phase for this section: Days 0–14, the early-disruption window when the gap between what planning assumes and what the system can actually deliver is widest and most consequential. Full scenario architecture is developed in the Scenario Framework section. The dependency mapping builds directly on that framework and produces the analytical foundation for the risk and vulnerability assessment that follows.

Framework: Five Layers of Arctic Dependency

Dependencies are assessed across five interconnected layers, each representing a different distance from the point of disruption and a different relationship to defense operations. Together, they describe how a port throughput degradation on the West Coast translates, within days, into mission constraints at Arctic installations. The first three layers identify spatial dependency; the fourth and fifth identify cross-cutting dependencies, workforce, and communications, which propagate across all spatial layers simultaneously.

Each dependency is mapped to identify upstream suppliers and supply pathways; downstream users, defense, contractors, civilian populations; chokepoints and time-sensitive failure modes; storage capacity and inventory duration; and redundancy, substitutability, and recoverability under Arctic conditions.

Critical System Integration: The Master Dependency Chain.
Across these systems, a core dependency pathway is consistently observed: *Fuel → Power Generation → Heat → Water Systems → Communications → Operational Coordination.*



Energy and communications operate as master dependencies; their failure modes propagate across all other lifeline sectors.

Layer 1: National Supply Pathways Feeding the Arctic

The just-in-time architecture (Glossary) that DoD contractors and commercial suppliers rely on begins to erode almost immediately under dual-theater disruption: container throughput drops, congestion drives cargo toward Gulf and East Coast ports, and stressing rail and trucking capacity in ways that compound the original disruption. Food, fuel, and medical inventories intended for northern movement begin drawing down.

The first layer captures the upstream commercial systems whose degradation rapidly invalidates sustainment assumptions during the first fourteen days. This is where the disruption begins: the maritime, rail, and intermodal networks supplying Alaska and northern Canada; the CONUS-origin logistics flows tied to Pacific Northwest ports at Seattle/Tacoma and Los Angeles/Long Beach; the national fuel, pharmaceutical, telecommunications, and equipment supply chains; and the warehouse, cold-chain, and trucking nodes that support northbound movement. Mapping emphasis falls on early loss of throughput and schedule predictability, backlog formation across ports and intermodal systems, limited visibility into cargo prioritization and allocation decisions, and dependency on a small number of maritime gateways with limited surge alternatives. [26; 49]

This layer's maritime chokepoint structure is a specific instance of the exterior-lines vulnerability characterized in the contested-logistics doctrinal frame (developed in the

Institutional Coordination section), in which the same maritime gateway nodes and transit infrastructure that sustain Pacific operations also sustain northern resupply. [30; 50; 58]

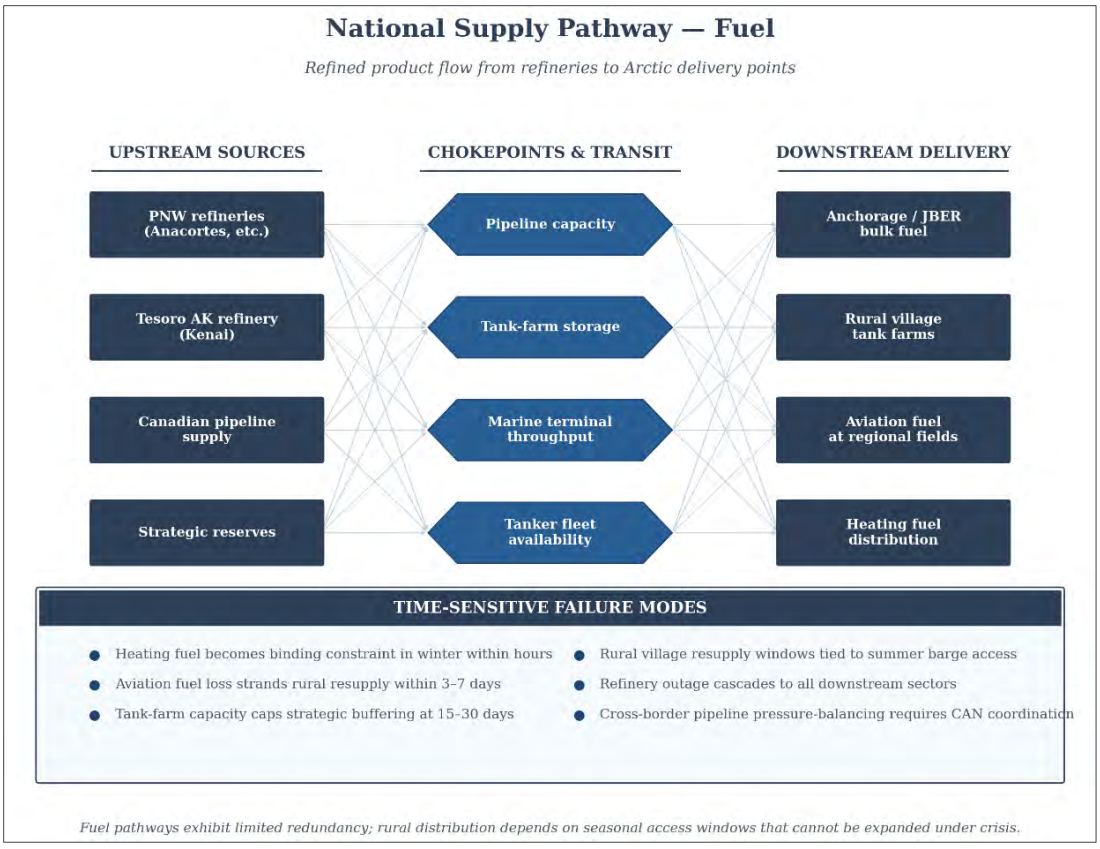
The petroleum volumes the Port of Alaska processes annually, exceeding 1.47 million tons, establish why the fourteen-day window is the critical planning frame. That volume cannot be rerouted, replaced, or approximated by any alternative mode or pathway within the period when sustainment decisions are most consequential. Alaska has no continental rail connection, barge and air cargo serve as last-mile delivery for the majority of western and northern communities, and the entire system was designed for cost efficiency under normal conditions, not for resilience under disruption.

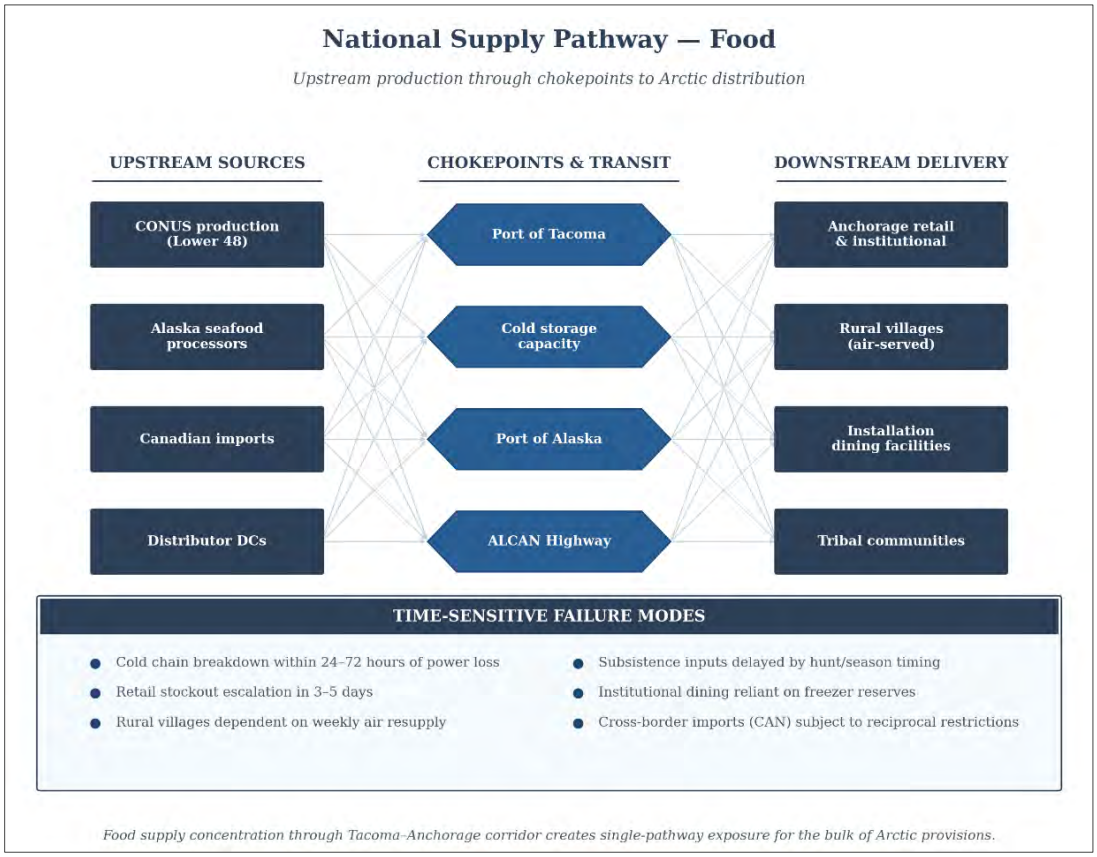
> ***Key Condition:*** *Disruptions at major maritime gateways rapidly constrain the flow of essential goods, including food, fuel components, pharmaceuticals, and equipment, before alternative routing can meaningfully compensate.*

The downstream concentration risk this layer reveals is documented in the Key Findings: a single Anchorage intermodal hub handles the majority of all inbound state freight and fuel. Disruptions at West Coast gateways do not produce distributed degradation; they concentrate failure at that node with immediate statewide effect and no equivalent alternative within the fourteen-day window.

Port disruptions produce nonlinear, cascading impacts across supply chains and regional economies that are not captured by single-event planning models. The key condition this layer reveals is that the United States depends on a small number of maritime gateways for the flow of essential goods into the Arctic, and those gateways have no meaningful surge alternatives available within a fourteen-day window. Disruptions at West Coast gateways do not produce distributed degradation across the state, they concentrate failure at a single node with no equivalent substitute.

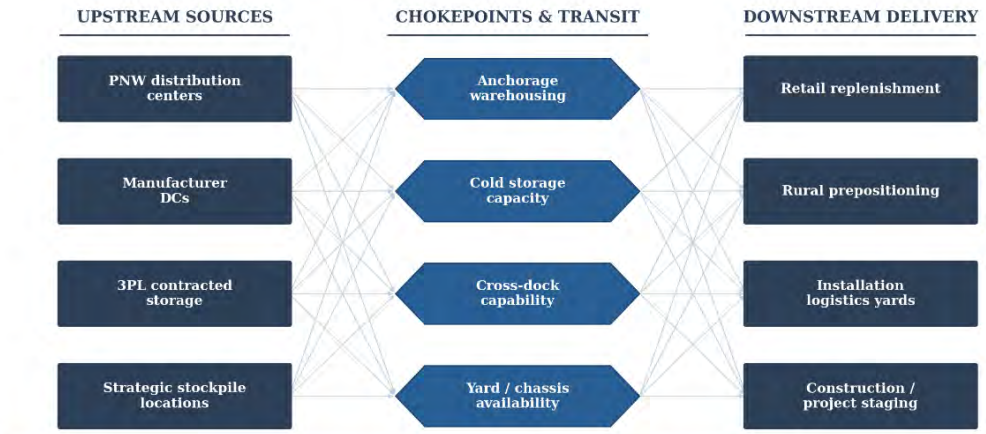
The four pathway diagrams that follow trace the structural geography of how food, fuel, warehousing, and cold-chain goods move from upstream sources, through critical chokepoints, and into Arctic distribution. Each pathway shares the same three-column architecture, upstream sources on the left, the chokepoints and transit infrastructure that actually govern flow in the middle, and the downstream delivery points that depend on them on the right. Read together, the figures expose a recurring pattern: redundancy on paper collapses into a small number of shared chokepoints in practice, and the failure modes that emerge under Arctic stress are tightly coupled across all four sustainment systems. Time-sensitive failure modes are noted at the base of each diagram to indicate where degradation begins, how quickly it propagates, and which downstream populations are exposed first.


National Supply Pathway — Fuel
Refined product flow from refineries to Arctic delivery points
UPSTREAM SOURCES
CHOKEPOINTS & TRANSIT
DOWNSTREAM DELIVERY
PNW refineries (Anacortes, etc.)
Tesoro AK refinery (Kenai)
Canadian pipeline supply
Strategic reserves
Pipeline capacity
Tank-farm storage
Marine terminal throughput
Tanker fleet availability
Anchorage / JBER bulk fuel
Rural village tank farms
Aviation fuel at regional fields
Heating fuel distribution
TIME-SENSITIVE FAILURE MODES
Heating fuel becomes binding constraint in winter within hours
Aviation fuel loss strands rural resupply within 3–7 days
Tank-farm capacity caps strategic buffering at 15–30 days
Rural village resupply windows tied to summer barge access
Refinery outage cascades to all downstream sectors
Cross-border pipeline pressure-balancing requires CAN coordination
Fuel pathways exhibit limited redundancy; rural distribution depends on seasonal access windows that cannot be expanded under crisis.


National Supply Pathway — Food
Upstream production through chokepoints to Arctic distribution
UPSTREAM SOURCES
CHOKEPOINTS & TRANSIT
DOWNSTREAM DELIVERY
CONUS production (Lower 48)
Alaska seafood processors
Canadian imports
Distributor DCs
Port of Tacoma
Cold storage capacity
Port of Alaska
ALCAN Highway
Anchorage retail & institutional
Rural villages (air-served)
Installation dining facilities
Tribal communities
TIME-SENSITIVE FAILURE MODES
Cold chain breakdown within 24–72 hours of power loss
Retail stockout escalation in 3–5 days
Rural villages dependent on weekly air resupply
Subsistence inputs delayed by hunt/season timing
Institutional dining reliant on freezer reserves
Cross-border imports (CAN) subject to reciprocal restrictions
Food supply concentration through Tacoma–Anchorage corridor creates single-pathway exposure for the bulk of Arctic provisions.

National Supply Pathway — Warehouse Capacity

Storage and staging capacity feeding Arctic distribution



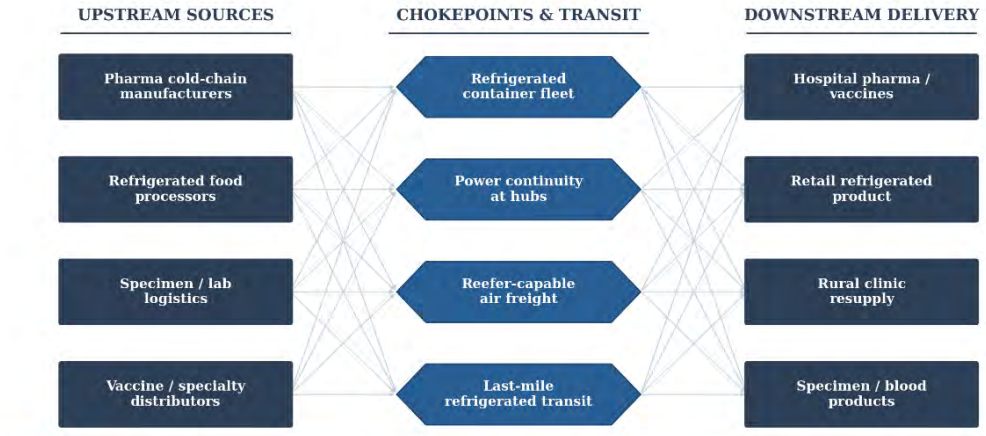
TIME-SENSITIVE FAILURE MODES

- Anchorage warehouse capacity below national throughput norms
- Cold storage saturation forces product diversion or loss
- Cross-dock bottlenecks delay onward movement by 2–5 days
- Yard space competition between commercial and military cargo
- Insufficient buffer for 72-hour to 14-day disruptions
- Limited 3PL surge capacity in-region

Arctic warehousing operates near steady-state ceilings; surge events expose insufficient buffer relative to throughput needs.

National Supply Pathway — Cold Chain

Temperature-controlled flow for food, pharma, and medical supplies



TIME-SENSITIVE FAILURE MODES

- Single power outage breaks cold chain end-to-end
- Reefer container shortage during high-demand windows
- Air freight reefer capacity constrained at regional fields
- Temperature excursions render product unusable
- Vaccine cold chain breaks force re-supply from CONUS
- Last-mile refrigerated transit fails first in rural settings

Cold-chain failure is binary not gradual; a single break can require re-shipment from CONUS at multi-day cost.

Layer 2: In-Theater Arctic Lifeline Systems

This layer captures the regional systems that immediately experience upstream disruption and directly sustain installations, missions, and civilian populations. In-theater Arctic lifeline systems span six interdependent sectors: energy and fuel systems including bulk fuel imports, storage terminals, aviation fuel distribution, local power generation, and heating fuel systems; food supply chains relying on barge lines, air cargo resupply, grocery distribution centers, cold storage, and limited warehouse capacity; medical logistics encompassing pharmaceutical cold-chain systems, medevac capacity, regional hospitals, and limited specialty care; communications and data infrastructure including satellite systems, fiber, microwave, and public-safety communications; transportation networks relying on the ALCAN Highway, regional and remote airfields, limited deep-water ports, and seasonal routes; and water and wastewater systems comprising community water utilities and washeteria-based distribution.

Transportation networks rely on the ALCAN Highway, regional and remote airfields, limited deep-water ports, and seasonal routes. Water and wastewater systems operate under baseline conditions with no parallel in the contiguous United States: twenty-two percent of Alaska households, roughly one in five, are unserved by water utilities, and energy costs for water systems in rural Alaska are up to 260 times higher than in the Lower 48, producing chronic fiscal stress that limits community capacity to maintain resilience reserves [26].

Fuel enables power generation. Power systems support heating, water treatment, and communications infrastructure. Heating systems are critical to preventing infrastructure failure in Arctic conditions. Water and wastewater systems depend on both power and heat. Communications systems depend on stable power and environmental conditions.

Fuel is simultaneously a cross-sector enabler for the emergency response infrastructure that would be activated to address a disruption: Alaska's feeding support operations, emergency response vehicles, and generator-powered fixed sites all depend on the same fuel supply chain whose degradation would trigger plan activation [6; 7]. Fuel disruption does not remain contained within the energy sector; it simultaneously disables the response capacity that would otherwise compensate for the initiating disruption.

The dependency chain is empirically confirmed by the Alaska energy resilience case literature, which documents an energy architecture characterized as fragmented and non-redundant. The centralized Railbelt grid carries approximately 79 percent of Alaska's electricity through a single interconnected corridor concentrated in the urban Southcentral and Interior regions. Outside that corridor, more than 150 isolated microgrids serve communities scattered across geographically remote locations [26]. This bimodal architecture concentrates failure risk in urban corridors served by the Railbelt while limiting recovery options in rural communities served by standalone microgrids; the two populations face structurally distinct vulnerabilities under the same disruption scenario.

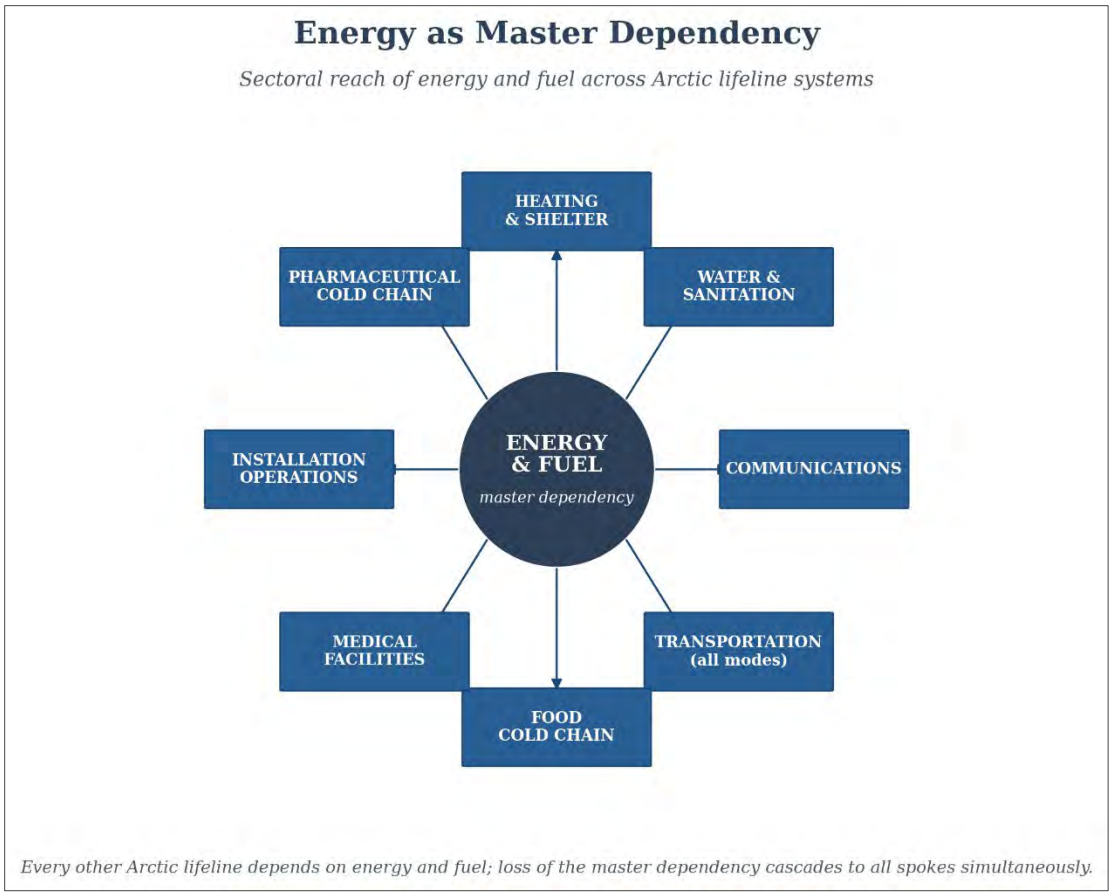
The 150 standalone microgrids are the analytically operative figure for repair logistics, workforce dependencies, and recovery options under disruption. These systems rely

primarily on diesel generation, with renewable penetration typically capped at 10–20 percent before integration complexity yields diminishing returns and at approximately 80 percent before storage requirements become binding [7; 26]. Diesel generators in these communities supply both electricity and recovered heat; renewable substitution reduces fuel consumption but simultaneously reduces heat output, creating secondary failure modes, facility freezing, water system damage, delayed generator startup, that pure-energy substitution models do not capture.

Standalone microgrids do not have access to the mutual aid agreements and shared lineman crews that Railbelt-interconnected utilities can mobilize. When a standalone microgrid fails in a remote community, repair is contingent on a workforce that must be flown in, using fuel that is itself constrained, to address equipment that may be unavailable due to the documented worldwide transformer shortage and the broader workforce constraints affecting Alaska's electric utilities [26]. Documented wind-system failures at Nikolski and St. George illustrate that renewable integration introduces operational complexity that can reduce, rather than increase, reliability in the absence of sustained workforce and maintenance capacity.

Microgrids and distributed energy resources are nonetheless already operational across Alaska and represent realistic resilience strategies to reduce single-point-of-failure exposure and improve local energy security [26]. The mitigation pathway is therefore not theoretical; the analytical question is whether expansion is paired with the workforce capacity, spare-parts logistics, and Arctic-conditioned operational support that determine whether installed capacity translates into operational reliability under disruption.

> ***Key Condition:*** Failure in upstream fuel or energy systems propagates rapidly across all dependent lifelines, creating multi-system degradation rather than isolated outages. Many Arctic systems operate with minimal buffer capacity, meaning disruptions in upstream supply chains translate into localized shortages within days, particularly food, fuel, and medical supplies.



Energy underpins virtually every other lifeline; loss propagates across heating, water, food, transport, medical, communications, workforce, and warehousing.

Layer 3: Mission-Dependent Community Infrastructure

The third layer reflects civilian systems whose degradation directly affects defense readiness, workforce availability, and operational continuity, and which are often absent from defense infrastructure planning because they are neither military-owned nor formally classified as critical infrastructure.

This layer includes local utilities supporting both installations and communities, workforce populations whose access to food, fuel, healthcare, and housing determines their availability for defense-enabling roles, commercial services supporting dependents and contractors, tribal governance systems that shape prioritization and coordination decisions in ways no federal framework fully anticipates, and community-level distribution systems with no meaningful surge capacity.

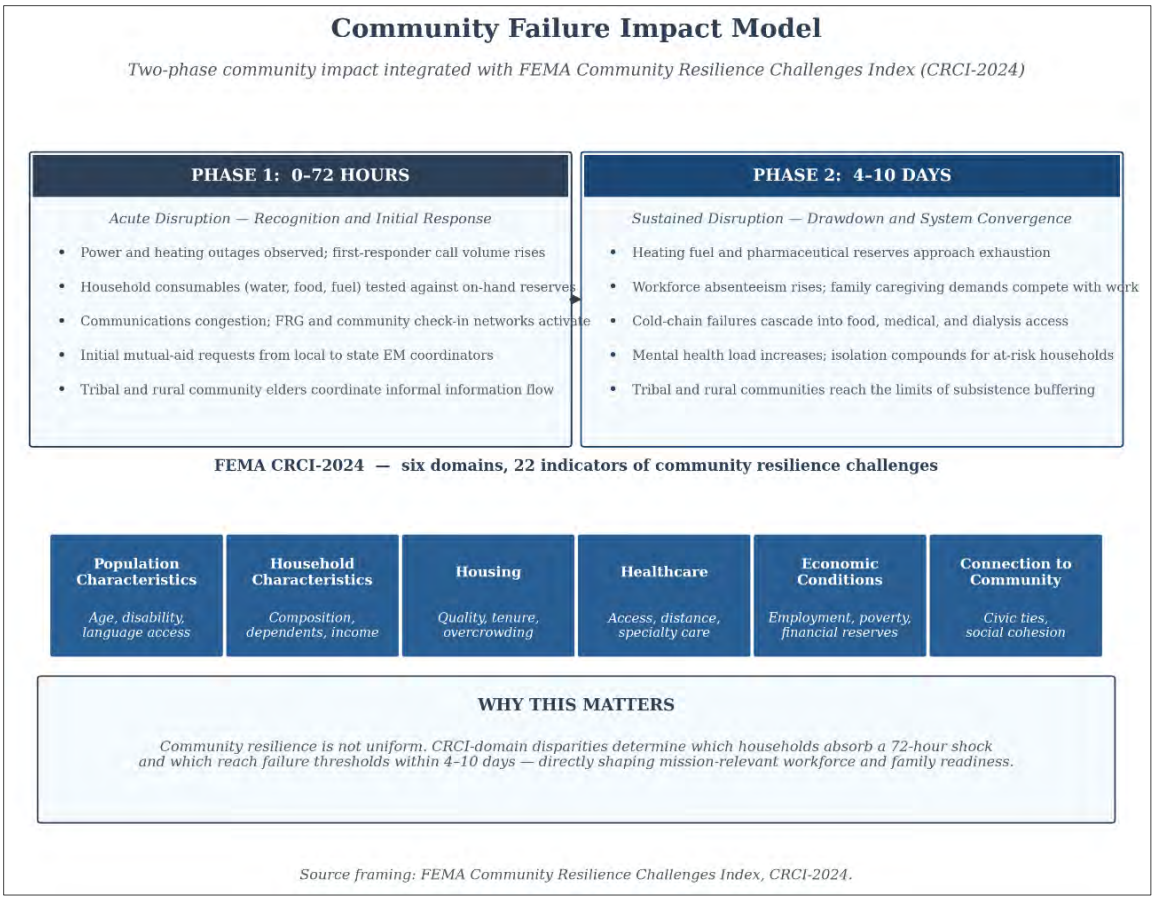
Mapping emphasis falls on workforce fragility driven by supply shortages and service disruption; household-level access constraints that determine whether workforce members can reach installations or receive emergency information during disruption; economic reserve depth that determines how quickly households begin making tradeoffs between work attendance and household survival when food and fuel access becomes unreliable; civil–military competition for scarce resources; governance complexity

affecting prioritization and allocation decisions; and community-level instability as a multiplier of operational risk.

Community-level vulnerability is not uniformly distributed across Alaska, and the pre-existing resilience challenges that shape cascade trajectory can be characterized systematically. The FEMA Community Resilience Challenges Index identifies 22 peer-reviewed indicators across six domains, population characteristics, household characteristics, housing, healthcare, economic conditions, and connection-to-community, which predict where the Days 0–14 failure cascade will move fastest because reserve capacity is thinnest [22 ; 65]. Communities with compounding challenges across multiple domains, high unemployment combined with limited vehicle access; thin healthcare capacity combined with low civic organization density, are the communities where each link in the failure cascade accelerates most quickly.

Peer-reviewed disaster-resilience research from the University of Washington's Disaster Resilience Research Network corroborates that pre-incident community resilience, measured through validated indicators of household composition, civic infrastructure, healthcare access, and economic reserve depth, is a stronger predictor of post-disruption cascade trajectory than disaster magnitude alone [65]. The risk-amplifier framing this section applies to community vulnerability is consistent with that research tradition.

> ***Key Condition:*** Operational readiness directly tied to community sustainment. When workforce availability, family stability, and local services degrade under prolonged disruption, the effects are not confined to the civilian population; they propagate directly into mission capability. This is the layer of the dependency picture most systematically absent from current planning.



CRCI domains shape acute-phase impacts and amplify cumulative stress; Arctic-specific factors compress resilience reserves further.

Layer 4: Workforce as a Cross-Cutting Dependency

Across all three layers, workforce functions as a critical enabling condition, not a standalone sector. Workforce availability directly affects fuel distribution and handling, power generation and system maintenance, transportation and logistics operations, medical care and emergency response, and communications system operation and repair. Under the dual-theater scenario, this dependency becomes more acute rather than more elastic. Deploying additional personnel or materiel into the region to compensate for shortfalls would itself be compromised by Arctic weather conditions, runway closures, ceiling and visibility limits, sea-state and icing constraints on maritime resupply, and ground transportation degradation along the ALCAN corridor all narrow the deployment window precisely when demand is highest. To the extent that resources could be moved by alternate methods, barge timing windows, overland convoy, rotary-wing shuttle, or ad hoc commercial charter, each alternate pathway introduces its own workforce requirement to receive, stage, handle, distribute, and sustain those resources at the point of need. Deployable equipment without deployable people to operate it is not a substitute for in-region workforce capacity; it is a deferred dependency. This compounding effect, that the very act of surging into the region creates new workforce demand at the receiving end, is not consistently captured in current Arctic mission assurance planning, which tends to treat deployment as a workforce solution rather than a workforce multiplier.

Workforce fragility is driven by food availability, fuel and heating access, healthcare capacity, housing and community stability, and pre-existing community resilience challenges measured by validated peer-reviewed indicators, including single-parent household prevalence, households without smartphones, and civic organization density, each of which correlates with workforce disruption risk under stress conditions [22; 65]. Availability also depends on specialized technical personnel, rotational or non-local labor, and the absence of surge capacity under extreme conditions.

Workforce concentration risk is illustrated by the seafood processing sector, which is both a significant Alaska employer and an input to the sustainment supply chain. Approximately 27,000 shoreside seafood processing workers in Alaska are seasonal, including substantial migrant workforce under H-2B visa arrangements [5; 17]. Travel restrictions, immigration policy changes, or transportation disruptions that prevent seasonal workforce arrival would degrade processing capacity, affect fisheries buyers and distribution, and reduce local food supply simultaneously. This is a worked example of workforce fragility acting as a force degrader across multiple lifeline categories from a single disruption pathway.

Layer 5: Communications as the Coordination Spine

Communications systems function as the coordination and decision-making backbone across all lifelines, depending on power and backup power systems, physical infrastructure (fiber, satellite, microwave), and environmental stability. Communications degradation reduces situational awareness during disruption, limits visibility into logistics, infrastructure status, and resource availability, degrades coordination across sectors and agencies, and increases the time required for prioritization and allocation decisions. The dependency picture is defined by two compounding realities: a fragile baseline that exists before any disruption begins, and a degradation surface that converts that baseline into an actively exploitable vulnerability under contested conditions.
[10; 14; 28]

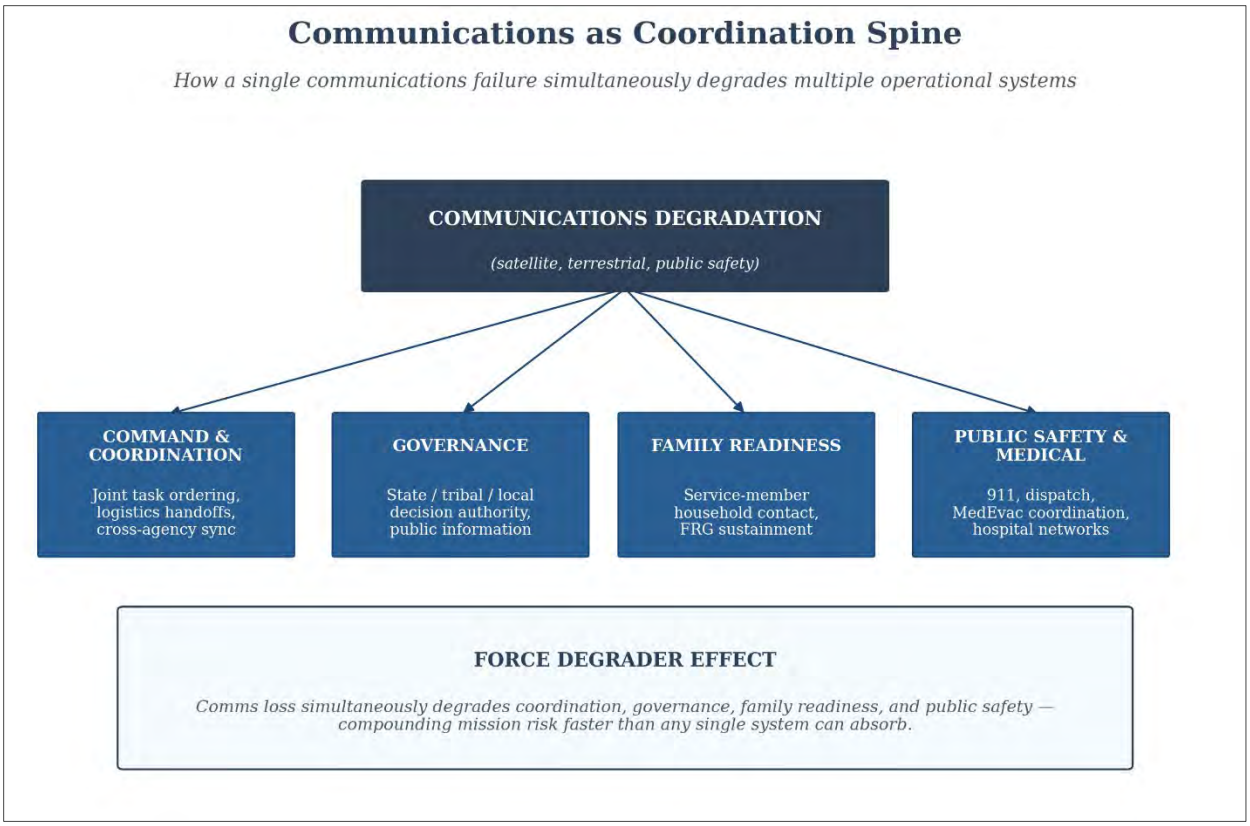
The fragile baseline is structural. Arctic communications operate against fragmented systems, bandwidth limitations, single-point failure exposure, and insufficient remote infrastructure as baseline conditions, not disruption consequences [10; 28; 56]. High-latitude geometry adds a binding technical constraint: geostationary satellites lose effective coverage above approximately seventy degrees north, making low Earth orbit and highly elliptical orbit constellations the primary space-based options for sustained Arctic connectivity. No single satellite architecture provides full Arctic coverage. Terrestrial backup is similarly exposed: the Denali Commission documents ice scour severing subsea fiber cables and producing months-long outages, confirming that the terrestrial layer cannot be assumed to remain available when the space-based layer degrades [25].

The degradation surface compounds that baseline through four overlapping vectors. First, weather, and environmental degradation, including ice scour, geomagnetic events, and Arctic-specific climate stressors, produce baseline reliability degradation that planning frameworks built for temperate-latitude operations consistently underweight. Second, adversarial degradation operates as ongoing capability rather than hypothetical risk:

Russian GPS jamming and potential undersea cable interference are documented gray-zone tactics in the Arctic theater [13; 14; 25; 45]. Third, cyber degradation extends the adversarial vector into the digital layer, with Arctic ground stations, including Svalbard-based facilities supporting commercial, scientific, and U.S. government missions, attractive targets due to remote locations, limited staffing, and internet-dependent operations, susceptible to cyberattack alongside jamming and physical disruption. Fourth, commercial provider autonomy introduces a degradation vector that no other lifeline category exhibits at this magnitude: over half of U.S. satellites are commercially owned and operated, and Arctic operations depend on commercial providers for positioning, navigation, timing, communications, and Earth observation [15; 16]. Commercial operators may alter service availability during conflicts based on financial or political considerations, with no governance framework currently requiring service continuity or military traffic prioritization. The United States has not formally designated space as a critical infrastructure sector, meaning the coordination, information-sharing, liability, and incident response frameworks developed for the sixteen designated sectors do not cleanly apply to the satellite systems those sectors depend on [12].

Cybersecurity within this dependency picture is therefore not a parallel analytical domain but one degradation vector among four operating against the fragile communications baseline. The cyber vector compounds the weather, adversarial-jamming, and commercial-autonomy vectors rather than substituting for them. A communications mitigation portfolio that addresses cybersecurity without addressing the other three operates against an unmitigated dependency on the systems whose failure it is designed to absorb.

> ***Key Condition:*** Communications degradation does not create isolated failure but instead amplifies and accelerates cascading impacts across all other lifelines. Communications degradation operates as a force degrader, and commercial provider autonomy introduces a paper-redundancy pattern analogous to the one documented in maritime and aviation substitution: contractually available capacity is not necessarily operationally available under stress.



A single communications failure simultaneously degrades command, governance, family readiness, and public safety, illustrating how comms loss functions as a force degrader across multiple operational systems rather than a single isolated outage.

Critical Dependencies Being Mapped

Fuel and Energy Systems

Arctic fuel dependency is singular in its operational significance. The entire system, aviation, transportation, heating, power generation, runs on imported fuel products with limited regional refining capacity and constrained storage [26]. Rural Alaska is heavily dependent on diesel-based microgrids with limited redundancy, fuel supply chains dependent on seasonal maritime delivery creating narrow resupply windows, and no grid interconnection that would permit load sharing or rapid recovery across regions. During the first fourteen days of disruption, rapid drawdown of stored fuel inventories competes directly with civilian heating demand and military and aviation requirements. Fuel disruption produces cascading cross-sector impacts affecting transportation, power, heating, and aviation simultaneously, not sequentially.

Two single-source fuel dependencies sit at the heart of the cascade structure. First, Crowley is the sole importer of aviation gasoline in Alaska, and all of it flows through the Port of Alaska [26]. AV Gas powers the small bush aircraft that deliver food and supplies to isolated communities through the Bypass Mail program, meaning that Port of Alaska disruption does not merely affect jet fuel for military and commercial aviation; it severs

the last-mile air delivery system that remote communities depend on for basic sustenance.

A disruption at the Port of Alaska therefore simultaneously disables the bush plane network that delivers food, fuel, and medical supplies to off-road communities, the same air cargo infrastructure on which last-mile distribution depends when maritime alternatives are constrained. This is a convergence failure in which a single upstream node disables both the primary maritime lifeline and its principal substitution pathway simultaneously.

Second, the upstream refining dependency compounds the single-node aviation fuel risk. Approximately eighty percent of crude oil produced in Alaska is refined in Washington and California rather than in-state; the remaining twenty percent is processed across five in-state facilities with a combined capacity of approximately 164,200 barrels per calendar day. Under the scenario, degradation of West Coast port operations simultaneously constrains the delivery of refined petroleum products flowing back into Alaska from CONUS refineries, while in-state refining capacity is insufficient to compensate for the loss. Alaska is therefore in the structurally anomalous position of being a major crude producer that depends on CONUS refining and maritime resupply to receive its own petroleum products in usable form, a dependency chain that collapses at the same maritime chokepoint that governs food and medical supply.

Cook Inlet 2030 depletion creates a structural convergence between the electric power system and the maritime fuel-import chokepoint (full treatment in the Cross-Sector Chokepoints subsection on the next page). The Fuel and Energy Systems implication is that the same disruption that interrupts maritime delivery of refined petroleum simultaneously stresses the gas-fired generation capacity that depends on imported fuel as Cook Inlet declines. [26]

Seasonal fuel delivery structure for rural communities defines a hard constraint on substitution timelines that is distinct from the urban distribution system. Bulk fuel delivery to rural Alaska occurs in late fall by barge, before sea ice and river ice set in; any additional fuel required before spring breakup must be delivered by bush plane at substantially higher cost. The seasonal lock-in dynamic (developed in the Scenario Development's, Seasonal Asymmetry) is the binding scenario constraint: a winter disruption forecloses bulk delivery to rural communities until spring.

> ***Key Condition:*** Aviation-based fuel resupply under those conditions is not a functional substitute for barge delivery; it is a constrained emergency mechanism operating against depleted reserves with no resupply window. This seasonal asymmetry compresses failure timelines from weeks to days in winter disruption scenarios relative to summer.

Food and Essential Commodities

Alaska's food system is structurally bimodal. The commercial food distribution system supplies approximately 95 percent of Alaska's food through a just-in-time architecture optimized for cost rather than resilience, while the subsistence harvest system supplies the parallel pathway through which rural communities sustain themselves outside commercial channels [1; 26; 38]. Both systems are currently under independent stress, and the risk-amplifier condition for the dual-theater scenario is that disruption affecting both pathways simultaneously produces food security outcomes more severe than either system's fragility alone would indicate.

The commercial pathway operates on maritime resupply cycles that underpin the entire system. Food departing the Port of Tacoma arrives at the Port of Alaska in three days and reaches retail shelves on the fourth day; ships arrive on Sundays and Thursdays; the state has no meaningful warehousing capacity; and inventory buffers are commonly estimated at 6–10 days under normal operating conditions. [38] When those cycles are disrupted, the limited commercial inventory begin drawing down immediately. Cold-chain fragility compounds the problem: the perishable goods and medical commodities that require temperature-controlled logistics are among the first to be affected when the systems supporting them are stressed.

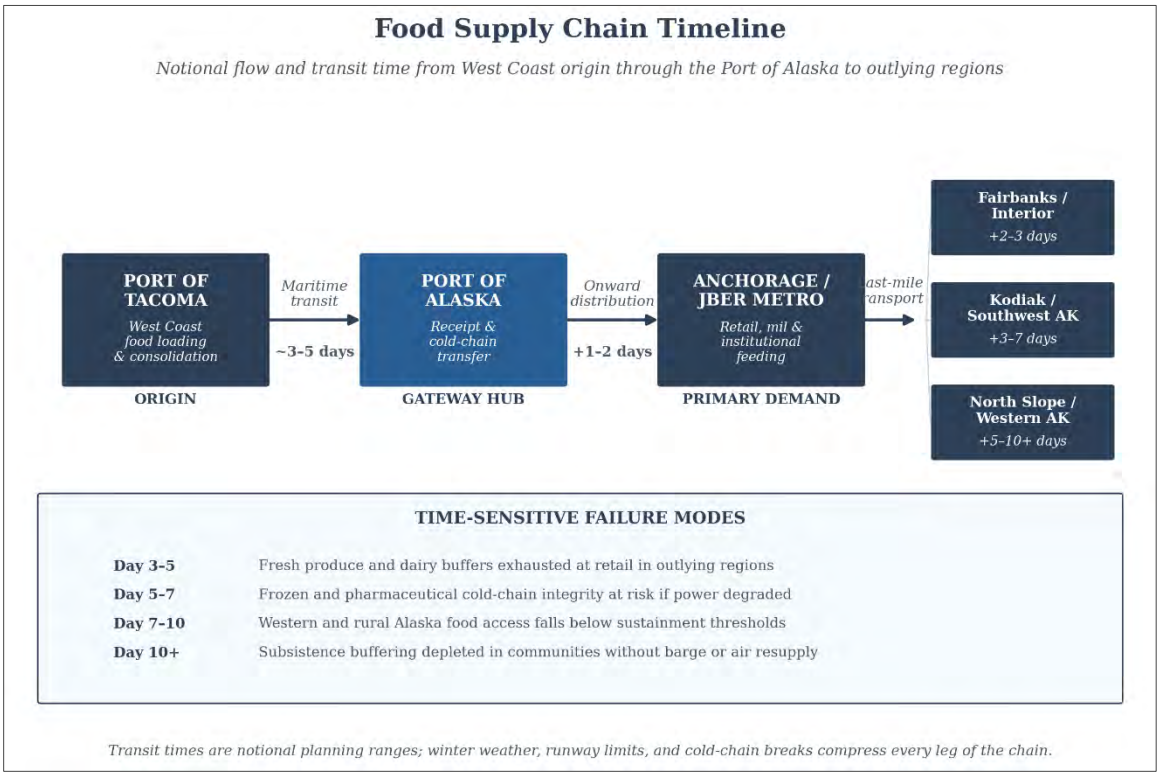
The structural thinness of Alaska's commercial food architecture is characterized by single-source concentration at both the wholesale and retail tiers. At the wholesale tier, a single produce distributor (Charlie's Produce) operates from one Anchorage location; beverage distribution is concentrated across three to seven locations per carrier; and general-line distribution is similarly limited to single locations outside Anchorage. At the retail tier, major chains include Alaska Commercial Company locations, Carrs/Safeway, Fred Meyer, and Walmart. The geographic concentration means that disruption at or upstream of the Port of Alaska simultaneously affects all downstream retail channels, with no independent distribution tier capable of absorbing the disruption. State feeding operations depend on the same commercial supply chain whose disruption would trigger plan activation, including Sysco, Carrs/Safeway, Fred Meyer, Walmart, and Costco, reinforcing that the named commercial partners in Alaska's emergency feeding doctrine are not redundant to the disrupted system but embedded within it [6; 7]. This is a clear instance of paper redundancy without operational redundancy: the backup feeding architecture and the primary commercial food system depend on the same upstream nodes.

In-state processing capacity is similarly thin. Alaska has three USDA-certified meat processing facilities and two mobile units statewide, and two FDA-certified dairies, a structural condition that eliminates local production as a meaningful substitute for imported food within the Days 0–14 window even where raw agricultural output exists. [1; 6]

The subsistence harvest pathway carries the parallel-system load. Wild food harvest of approximately 18,000 tons annually, with replacement value approaching one billion dollars and supporting up to 98 percent participation in some rural communities, is sustained by fish and wildlife populations that are currently in decline, including the

Western Arctic Caribou Herd down 23 percent over two years and 50 percent over ten years, and fishery closures driven by fish population collapse [7]. Subsistence harvest activities, where seasonally available, depend on fuel for boats, snowmachines, and aircraft, meaning that subsistence resilience is itself coupled to the commercial fuel supply chain whose disruption is the initiating condition of the scenario. The bimodal architecture does not provide independent redundancy under disruption; both pathways degrade through partially shared dependencies on fuel, transportation, and workforce.

> ***Key Condition:*** Alaska has no major warehousing capacity. When maritime supply is disrupted, there is no buffer to draw down. Shortages appear at the retail and community level within days not weeks, because the system was designed to function with no slack.



End-to-end food path is sequential: a chokepoint failure at any stage propagates downstream within days.

Medical and Pharmaceutical Supply Chains

Medical system fragility in the Arctic is a function of distance, weather, and supply chain structure simultaneously. Limited regional healthcare capacity and specialty care availability mean that any degradation in the ability to evacuate personnel or resupply pharmaceuticals and medical consumables has immediate operational consequences. [4; 18] Pharmaceutical cold-chain systems are among the most vulnerable elements of the entire supply network. They require temperature control throughout transit, have no meaningful substitution options when disrupted, and operate on resupply cycles that

cannot be easily compressed [74; 76]. Medevac constraints driven by weather and airfield limitations compound the pharmaceutical supply picture.

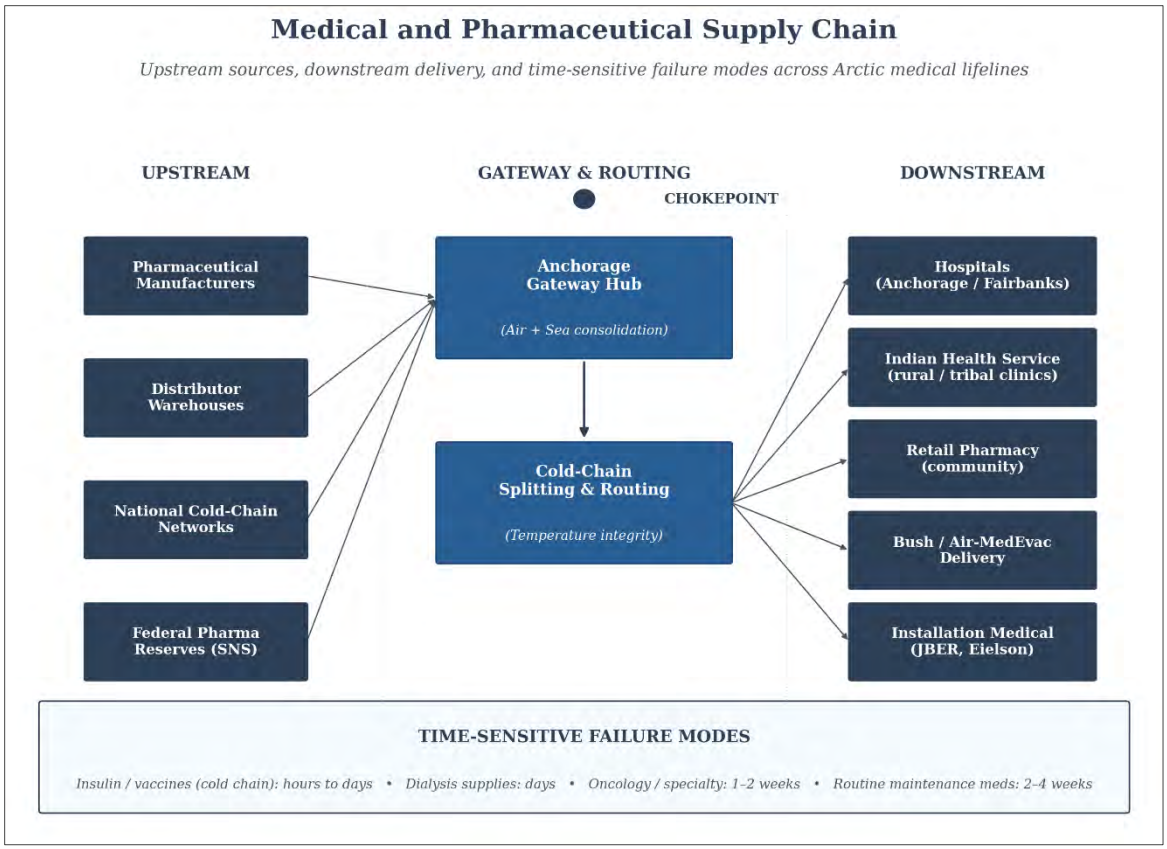
National pharmaceutical distribution operates on just-in-time inventory principles, with retail and hospital pharmacy inventories typically maintained at seven to thirty days of supply [9]. Alaska's position at the end of that pipeline compresses national depletion timelines further, and a disruption that simultaneously affects power, fuel, and communications produces parallel cold-chain failure across distribution centers, pharmacies, and clinics rather than at a single point. Products affected by cold-chain failure are not delayed but destroyed, converting a resupply problem into an inventory-destruction problem that must be resolved through the same disrupted supply chain that produced the failure [27].

Regional healthcare capacity imposes a hard ceiling on surge absorption. Alaska's hospital infrastructure is sized for baseline demand with limited margin, consistent with the broader pattern in which rural and frontier healthcare systems operate near effective capacity under normal conditions [4; 9]. When civilian, military dependent, and contractor populations draw on the same civilian system, the binding constraint is specialty staff rather than beds, and specialty staff cannot be flown in under the fuel and aviation constraints the scenario imposes.

Indian Health Service and tribal health system facilities, operated under tribal compact and co-management arrangements with the Alaska Native Tribal Health Consortium, regional tribal health organizations, and village-level Community Health Aide Programs, extend healthcare access into communities that commercial and state healthcare infrastructure does not reach [40]. This system depends on the same pharmaceutical supply chain, cold-chain infrastructure, medevac network, and fuel availability as the rest of the Alaska healthcare system. It is not redundancy against the disruption scenario; it is an additional set of dependent nodes whose degradation compounds the capacity gap.

Medevac operations represent the convergence pathway in compressed form: Guardian Flight, LifeMed Alaska, and tribal and commercial air ambulance operators depend on aviation fuel, weather windows, thin crew availability, and communications for flight following and receiving-facility coordination, placing medevac capacity in direct competition with cargo airlift, installation sustainment, and defense operational requirements for the same constrained fuel pool [7; 36].

> ***Key Condition:*** Aviation-based fuel resupply under those conditions is not a functional substitute for barge delivery; it is a constrained emergency mechanism operating against depleted reserves with no resupply window. This seasonal asymmetry compresses failure timelines from weeks to days in winter disruption scenarios relative to summer.



Pharmaceutical pathways are heavily concentrated; cold-chain integrity is the binary constraint that converts delay into total product loss.

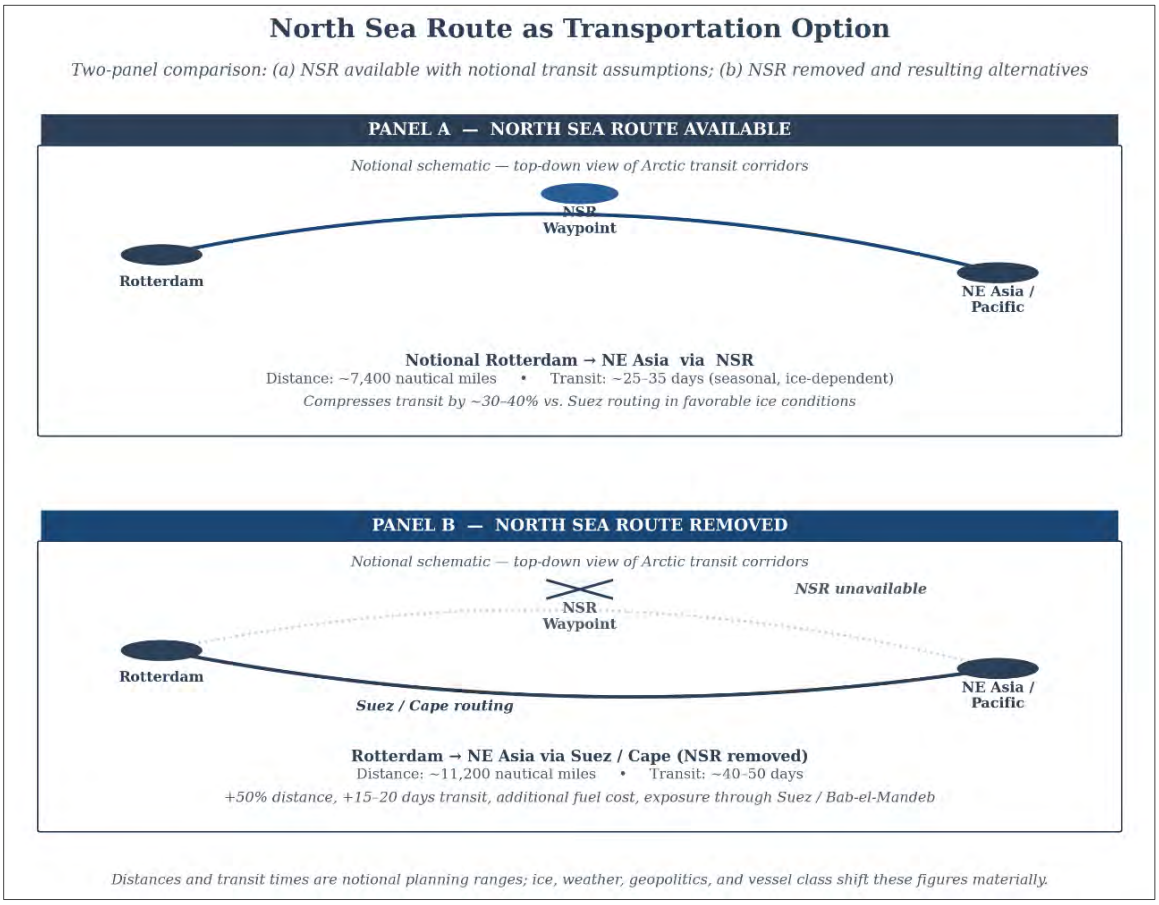
Transportation and Logistics Nodes

The transportation picture in the Arctic is defined by constraint at every level, and the literature is unusually specific about those constraints. There is no continental rail connection to Alaska. The Coast Guard search-and-rescue station nearest to the Arctic is in Kodiak, approximately 820 nautical miles south of Utqiagvik. The Nome port deepening project, the only active effort to create a northern deep-draft capability, remains in design phase with no permanent facility yet constructed [19; 34; 37].

The Port of Alaska handles five times more inbound marine cargo than all other Southcentral Alaska ports combined, with 1.66 million tons inbound and 1.47 million tons of refined petroleum products received in 2019. Its infrastructure includes 3,500 feet of dock frontage, gantry crane capacity, and 3.1 million barrels of liquid fuel storage capacity. No alternative port in Southcentral Alaska replicates this combination of throughput capacity, fuel storage, and land connectivity. The six identified contingency alternatives, Seward, Valdez, Kodiak, Homer, Whittier, and MacKenzie, each require capital investment before they could function as container cargo substitutes [26; 44].

Rerouting feasibility analysis must also account for Northern Sea Route availability under disruption conditions. The NSR represents a potential maritime rerouting option in benign planning models but could be closed or degraded by adversarial action under

UNCLOS Article 234 claims at the moment of greatest need. Russian infrastructure in the Arctic is characterized by a dual-use structure, facilities capable of both civilian search-and-rescue and military use, meaning escalation can convert cooperative infrastructure into contested access points with little warning [14]. This eliminates a rerouting pathway that otherwise appears as an alternative in substitution-limit analyses.



NSR availability is a conditional asset; loss of access materially extends transit times and exposes alternate routes to congestion.

Cross-Sector Chokepoints Identified

The dependency mapping identifies nodes where multiple lifeline systems converge, creating disproportionate mission risk when those nodes are degraded. These are not simply vulnerable points; they are the locations where disruption in one system immediately propagates into multiple others. The literature is unusually consistent in identifying the same nodes across independent sources.

Communications Systems: SATCOM and Commercial Telecom with Limited Redundancy

Communications is treated as a cross-sector chokepoint at the Layer 5 dependency treatment, where the fragile baseline, the four overlapping degradation vectors (weather,

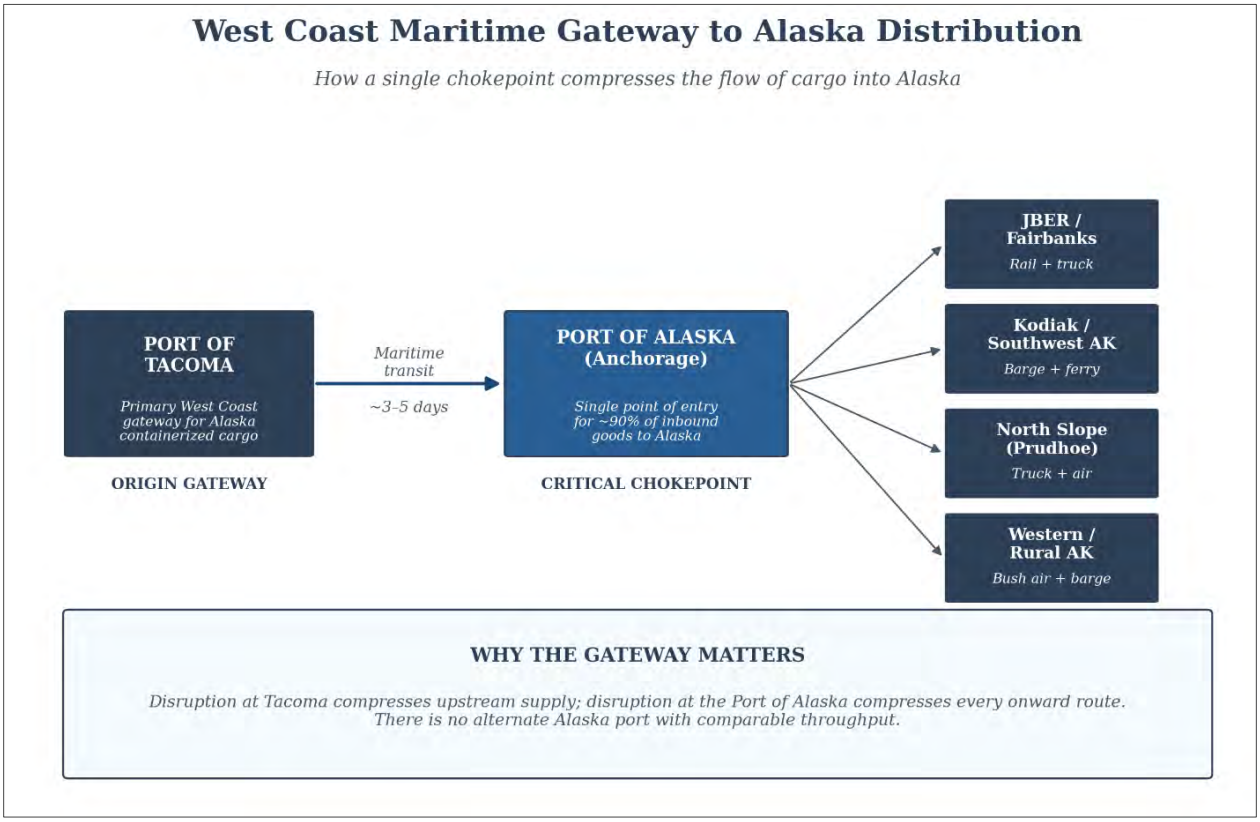
adversarial jamming, cyber, commercial provider autonomy), and the space-as-uncategorized-critical-infrastructure governance gap are developed in full.

Maritime Gateways: West Coast Ports as Primary Entry Point

Documented loss thresholds: peer-reviewed computable general equilibrium modeling of a compound disruption at the Los Angeles–Long Beach port complex finds GDP losses propagating well beyond the port region through commercial supply-chain transmission, with food and agricultural supply chains disrupted immediately and no rerouting alternative available at sufficient scale [49; 73]. That documented impact is not an abstraction for Arctic planning; it is the economic expression of a supply architecture whose failure propagates directly into Alaska sustainment within the first seventy-two hours.

Port of Alaska / Anchorage Hub: Central Consolidation Node

Single facility handling approximately 50 percent of all inbound freight and fuel into Alaska, supplying approximately 90 percent of the state's population through downstream distribution, and functioning as a DoD-designated commercial strategic seaport projecting power across Alaska, the Pacific Rim, and the Arctic [26; 44]. Its 1.47 million tons of annual refined petroleum throughput makes it the primary fuel entry point for a state with no regional refining capacity. Disruption here does not slow one mode; it degrades the entire downstream distribution architecture simultaneously.



Two-port concentration creates single-pathway exposure for Arctic sustainment; redundancy options are limited and seasonal.

ALCAN Highway: Key Overland Redundancy with Limited Surge Capacity

The Alaska–Canada Highway is the category of redundancy this research identifies as most consequential to examine: one that exists in planning documents, moves real cargo under normal conditions, and consistently fails to perform at the scale disruption response requires when conditions are most demanding [19; 26; 46].

Current average daily traffic on the highway is approximately 350 vehicles per day, of which commercial trucks constitute approximately six percent, or 20–25 trucks per day. Road classification can theoretically accommodate 12,000–15,000 vehicles per day under good conditions, suggesting that up to thirty times current traffic volume may be feasible at nominal capacity [55; 62]. The gap between the 350-per-day operating reality and the 12,000–15,000-per-day theoretical ceiling is the empirical expression of the paper-redundancy condition applied to the transportation layer.

Three structural conditions govern whether the gap can be closed under disruption. First, the 1,387-mile route requires pre-staged rest stops, fueling resources, and logistics support infrastructure that does not currently exist; the longest stretch without services is 100 miles, and not all available services operate year-round or on a 24-hour basis. [55; 62] Second, weather exposure across significant route segments produces winter closures and dangerous driving conditions, with closure cadence concentrated precisely in the months when demand for heating fuel and essential goods peaks; this seasonal compression converts the corridor from a marginal redundancy in summer to an unreliable one in winter. Third, the corridor requires transiting approximately 1,200 miles of Canadian sovereign territory, introducing customs, weight-restriction waiver, hours-of-service waiver, and provincial permitting decision points that U.S. planning cannot resolve unilaterally [45].

ALCAN optimization therefore divides analytically into two distinct investment categories with different time horizons. The first, pre-scripted regulatory pathway activation, including weight-restriction waivers, hours-of-service waivers, CDL endorsement coordination, and Canadian provincial permitting, can be achieved within the Days 0–14 window because its cost is administrative rather than capital. The second, physical pre-staging of driver support, lodging, and fuel infrastructure along the route, is a pre-event investment that determines whether the theoretical surge capacity is operationally accessible at all. The corridor is a real redundancy that moves real cargo, but absent both investment categories it cannot deliver the volume, speed, or reliability that maritime disruption response requires within the analytical window.

Airfields: Critical but Constrained for High-Priority Sustainment

The wargame and live-exercise literature converge on this chokepoint. Arctic Edge 2025 documents that multi-domain integration requirements, HIMARS rapid insertion, missile defense, maritime security operations, and expeditionary aviation, compete simultaneously for the same constrained airfield capacity under exercise conditions [67]; the RAND Plan Blue 21 wargame establishes that airlift cannot substitute for maritime supply at the scale Arctic operations require under contested conditions [46].

When the dual-theater scenario introduces fuel constraints simultaneously, the competition between operational aviation requirements and logistics and medevac needs becomes acute in ways that no aviation surge assumption can adequately buffer. Aviation does not substitute for the Port of Alaska's throughput volume under disruption; the operational arithmetic is documented in the Substitution Inventory. The aviation substitution argument does not fail at the margins. It fails categorically.

Electric Power: The Cook Inlet Depletion Trajectory

The electric power system introduces an additional cross-sector chokepoint with a documented long-horizon trajectory. The majority of Alaska's electricity is generated by natural gas, and the principal natural gas source for the Railbelt, the Cook Inlet basin, faces near-term supply decline. The Railbelt grid, which runs from Fairbanks through Anchorage to the Kenai Peninsula, produces most of Alaska's electricity and serves Joint Base Elmendorf-Richardson, Elmendorf's power load, and the majority of the state's population. [26]

As Cook Inlet gas reserves decline, the Railbelt becomes increasingly dependent on fuel imports to sustain generation, an energy dependency that converges directly with the fuel import chokepoint already identified as the primary cross-sector cascade trigger. Infrastructure stakeholder interviews additionally documented a worldwide transformer shortage, aging infrastructure with no rehabilitation plans, and a shortage of trained linemen as compounding conditions that reduce the Railbelt's operational buffer.

Coordination Architecture as a Cross-Sector Chokepoint

Where Business Emergency Operations Centers (BEOCs) and formalized private-public partnership charters exist and have been exercised, coordination moves at the pace disruption requires [33]. The Nebraska Preparedness Partnership's twelve-hour logistics coordination during 2019 flooding and the Houston Supply Chain Group's COVID-19 response pivot illustrate the model. Where these mechanisms do not exist, coordination must be improvised under time pressure. Alaska's Partnership for Infrastructure Protection provides a state-level precedent for formalized cross-sector coordination, though its scope and activation parameters under dual-theater disruption conditions remain a subject for follow-on analysis.

Cross-Border Interdependencies and Sovereignty Constraints

The cross-border sovereignty constraints documented here represent the governance dimension of the paper-redundancy problem identified in the Key Findings. ALCAN, the Northern Sea Route, and Canadian corridor access each appear in planning documents as available alternatives. Each is subject to conditions, sovereign decision-making, adversarial control, pre-defined coordination mechanisms that do not currently exist, that can make them unavailable at the moment they are most needed [13; 14; 19; 45]. The gap between planned redundancy and operational redundancy is widest precisely at the cross-border nodes where U.S. planning authority ends.

The RAND governance gaps analysis provides the most analytically precise framing of why this matters: the risk is not adversarial Canadian action but the absence of pre-

defined coordination mechanisms adequate to resolve competing priorities between two allied nations under simultaneous stress [45]. Existing cooperative frameworks, including Arctic Council mechanisms and bilateral agreements, are not designed for the pace and nature of change now underway, and security dialogue mechanisms have atrophied since 2014. Cross-border dependencies introduce delay and uncertainty precisely when decision timelines are most compressed, converting the ALCAN redundancy from a planning given into an unresolved coordination problem under the scenario conditions this assessment models.

UNCLOS Article 234 and the Conversion of Cooperative Redundancy into Contested Access

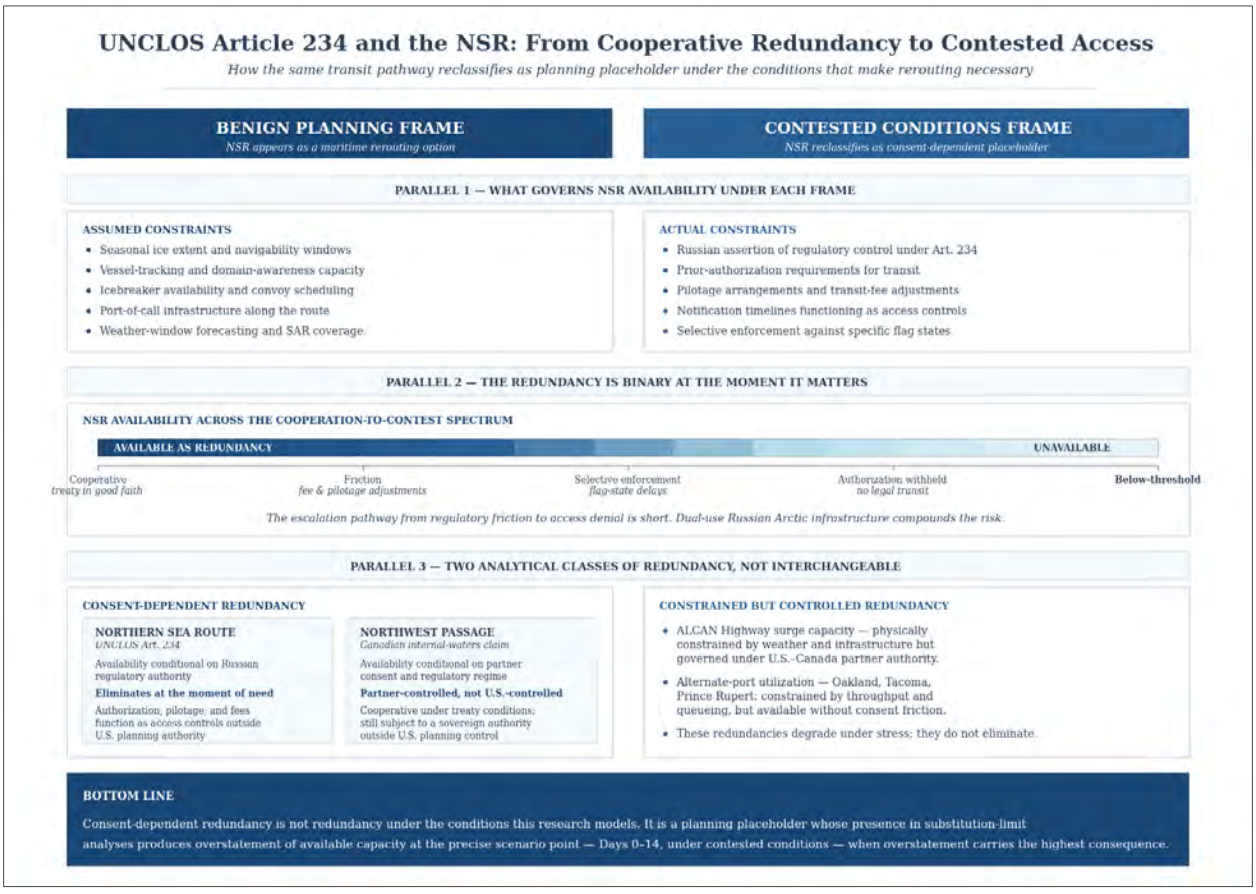
The Northern Sea Route is treated in most contingency planning literature as a maritime rerouting option that becomes available when Pacific gateway access is degraded. Under benign assumptions, the NSR appears as a redundancy whose seasonality is the principal constraint on use. Under the dual-theater scenario, that framing is incomplete in a way that matters analytically. The NSR is not a redundancy whose availability is determined by ice conditions and vessel-tracking capacity. It is a transit pathway whose availability is determined by Russian assertion of regulatory control under UNCLOS Article 234, a provision that authorizes coastal states to adopt and enforce non-discriminatory laws and regulations for the prevention, reduction, and control of marine pollution from vessels in ice-covered areas within the limits of the exclusive economic zone [14; 63].

Article 234 was drafted with environmental protection in view. Russian practice has extended its operational interpretation to require prior authorization for transit, pilotage arrangements, fee payments, and notification timelines that function in effect as access controls [13; 14; 49]. The risk-assessment implication for Arctic mission assurance is straightforward and severe. The NSR is a redundancy whose availability is conditional on conditions outside U.S. planning authority and outside the conditions that would make rerouting most necessary. Under adversarial action below the threshold of overt armed conflict, pollution-prevention enforcement asserted against specific flag states, transit-fee adjustments, pilotage suspensions, or the application of Article 234 standards in ways that selectively delay non-Russian shipping, the NSR becomes operationally inaccessible at exactly the moment West Coast gateway disruption makes it most operationally necessary. The redundancy does not fail because of weather or ice. It fails because the regulatory authority that governs it can be exercised as a disruption tool.

This pattern is consistent with the paper-redundancy condition documented throughout the lifeline architecture of this research. The NSR appears in benign planning models as a maritime option. The same NSR, examined against the conditions under which rerouting would actually be required, eliminates as a redundancy at the moment of greatest need. The dual-use character of Russian Arctic infrastructure, facilities capable of both civilian search-and-rescue and military application, compounds the risk by ensuring that an escalation pathway from regulatory friction to access denial exists and is short [13; 14]. There is no analytical equivalent of soft Article 234 enforcement against a vessel that cannot legally proceed once authorization is withheld; the redundancy is binary at the moment it matters.

The framework consequence is that Arctic rerouting alternatives must be evaluated against a feasibility test that accounts for the sovereign authority that governs them, not only the physical and seasonal conditions under which they would be exercised. The Northern Sea Route, the Northwest Passage, and any other transit pathway whose availability depends on third-state regulatory consent should be classified analytically as consent-dependent redundancy, distinct from constrained but U.S.- or partner-controlled redundancy such as ALCAN surge capacity or alternate-port utilization. Consent-dependent redundancy is not redundancy under the conditions this research models; it is a planning placeholder whose presence in substitution-limit analyses produces overstatement of available capacity at the precise scenario point, Days 0–14, under contested conditions, when overstatement carries the highest consequence.

The same logic applies, with different actors and different legal instruments, to any maritime chokepoint where transit is governed by a coastal state whose interests under disruption do not align with U.S. sustainment requirements. The Strait of Hormuz example documented in the Scenario Development section below illustrates the broader pattern: when a single coastal state asserts regulatory authority over a transit corridor and the conditions under which that authority is asserted shift from cooperative to contested, the operational availability of the corridor collapses faster than substitution-limit modeling typically anticipates. The lesson for Arctic planning is to treat consent-dependent redundancy as a planning placeholder rather than as available capacity, regardless of the specific legal instrument under which the consent operates [20; 49].



Alaska Native and Tribal Community Considerations as Risk Amplifier

Alaska Native communities occupy a distinct position in the Arctic vulnerability picture that this research treats as a risk-amplifier dimension. The CRS Arctic report documents that 144 Alaska Native communities face infrastructure damage from erosion, flooding, and permafrost degradation, that more than one-third face compounding threats, and that more than one-third of highly threatened villages received no relevant federal assistance during the five years reviewed. Twenty federal programs each have at least one characteristic that could obstruct access for small remote communities, including competitive grants, cost-share requirements, cost-benefit analysis rules, and national competition that disadvantages communities with limited administrative capacity. [19; 24]

The Alaska Native governance and economic ecosystem, 12 regional corporations, more than 200 village corporations, tribal governments, regional nonprofits, and statewide Native organizations, collectively controls land, workforce, and local infrastructure that defense operations depend on. [7] Barge routes, airstrip operations, fuel distribution, and local logistics functions that defense supply chains rely on run through or are managed by entities within this ecosystem. Where these infrastructure systems are degraded simultaneously with the broader supply chain, the cascade accelerates rather than stabilizes, because the entities that operate critical last-mile functions are absorbing the same disruption the larger system is experiencing. This is the risk-amplifier framing: tribal community vulnerability does not sit adjacent to the lifeline cascade; it sits inside it.

Functional coordination capacity between non-tribal institutions and tribal governments depends on pre-existing relationships that cannot be improvised under disruption conditions [21; 61]. Statutory consultation mechanisms do not substitute for substantive operational engagement built over time. The risk-amplifier finding does not assign causation to tribal communities; it identifies the structural conditions under which disruption impacts will be most severe and where pre-incident coordination investment yields the highest resilience return.

FOCUSED RISK AND VULNERABILITY ASSESSMENT

The Focused Risk and Vulnerability Assessment is where the scenario conditions and the dependency map converge into operational consequence. Operative phase for this section: Days 0–14, with extension into Days 10–45 where mitigation timelines require it. Full scenario architecture is developed in the Scenario Framework section. The risk and vulnerability assessment below evaluates how disruptions propagate under the scenario conditions documented in the Scenario Framework section.

The central insight this assessment is designed to surface is one that does not emerge from single-system analysis: Arctic mission assurance does not fail because one system breaks. It fails because multiple systems degrade simultaneously, their failure modes interact, and the cumulative effect arrives faster, and cuts deeper, than any individual system failure would predict. The value of this assessment is in mapping those interaction effects with enough specificity that decision-makers can see where the critical thresholds are, and where investment and coordination attention will have the greatest return.

Analytical Approach

The assessment evaluates how disruption propagates across interconnected lifeline systems under realistic Arctic conditions, with emphasis on threshold-driven system degradation rather than steady-state vulnerability. For each lifeline system, fuel, energy, food, communications, medical, transportation, the analysis identifies the initial point of disruption; immediate operational impacts; secondary and tertiary cascading effects across systems; trigger thresholds that shift systems from degraded to failed; and time-to-degradation and time-to-failure under constrained resupply conditions.

Representative Cascade: Port Throughput Degradation

For each lifeline system, fuel and energy, food distribution, communications, medical supply, and transportation, the assessment identifies the initial point of disruption, the immediate operational impacts, the secondary and tertiary cascading effects that propagate across interdependent systems, and the time-to-degradation and time-to-failure under constrained resupply conditions.

Ports and interconnected logistics networks produce nonlinear, compounding effects in which small disruptions generate large consequences [41; 49]. Fuel disruption simultaneously affects transportation, power, heating, and aviation operations, not as sequential effects but as concurrent system failures that share common infrastructure dependencies. In Arctic scenarios, logistics capacity falls below operational demand, coordination speed falls below disruption speed, and civilian system resilience falls below military dependency, creating conditions where the cumulative failure arrives faster and cuts deeper than any individual system failure would predict.

This cascade structure is consistent with FEMA's community lifelines framework and its explicit finding that failure in one lifeline propagates into others through shared

infrastructure dependencies, generating secondary supply chain problems not visible in single-system analysis [32; 41; 45]. Analytical emphasis falls on the throughput-versus-capacity distinction: systems may remain operational, capacity exists, but fail due to insufficient throughput, as in ports that are open, but whose cargo velocity is too low to sustain demand. Threshold-based degradation produces non-linear collapse once inventory falls below minimum operating levels, fuel allocation decisions begin, or workforce availability drops below sustainment requirements.

Risk is not driven by a single failure, but by the convergence of fuel limitations, logistics delays, communications degradation, and workforce instability that together create system-wide degradation. This convergence pattern is the operational expression of the tight-coupling/loose-coupling condition characterized in the contested-logistics doctrinal frame (Institutional Coordination section): tight coupling at the Pacific Northwest maritime gateway concentrates risk at a single node; loose coupling across in-state distribution prevents that risk from being absorbed within the Days 0–14 window. [30; 51]

> ***Key Condition:*** Port throughput degradation delays northbound cargo, which reduces warehouse replenishment rates, which triggers rapid inventory drawdown, which increases reliance on aviation resupply, which drives a surge in fuel consumption, which introduces aviation fuel constraints, which limits medevac

Time-Phased Degradation: Days 0-14

The cascade evolves across three sub-windows in the Days 0–14 frame, each describing both the immediate operational impacts and the threshold indicators that signal transition into the next phase.

Days 0–3: Disruption Recognition and System Strain

The dominant conditions in this window are disruption recognition, backlog formation, and limited situational awareness. Even a single Arctic maritime incident triggers simultaneous demands that exceed local capacity, with notification-system delays and inconsistencies, no common operating picture, and authority-transition friction operating as persistent early-phase constraints. The structural reason is governance: existing coordination mechanisms were not designed for the pace and nature of Arctic change now underway, and post-2014 security-dialogue gaps mean that early-phase uncertainty cannot be resolved through established deconfliction channels.

Days 3–7: Inventory Drawdown and Early Allocation

Warehouse replenishment falls below demand, food and medical inventories begin measurable decline, and fuel consumption increases due to operational adjustments. Threshold indicators include inventory levels falling below reorder thresholds, fuel resupply uncertainty triggering allocation planning, and early prioritization decisions between civilian and defense demand [7; 26]. This is the window in which the Days 0–14 analytical frame transitions from a planning concern to an operational reality, and in

which the absence of pre-defined adjudication mechanisms for civilian–military resource competition begins to produce friction that compounds across systems.

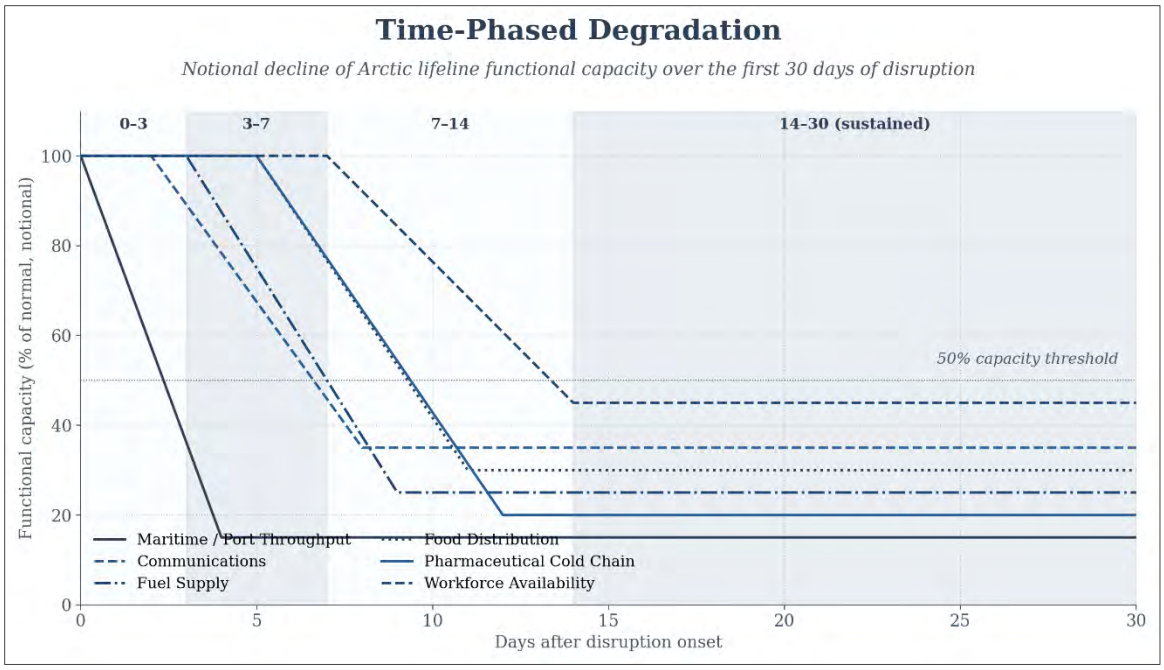
Days 7–14: Resource Scarcity and System Convergence

Sustained lack of resupply creates structural shortages, fuel availability becomes a limiting factor across sectors, and workforce degradation begins affecting operations. Threshold indicators include fuel allocation or rationing initiated, aviation fuel constraints limiting airlift and medevac, food availability impacting workforce stability, and communications degradation reducing coordination effectiveness. The Arctic Climate Change and Homeland Security 2024 report establishes that the environmental conditions of this period are not neutral: the Arctic is warming at approximately four times the global average, wave heights are projected to increase two to three meters, and more frequent and intense cyclones and polar lows are creating compound maritime risk environments [57; 72].

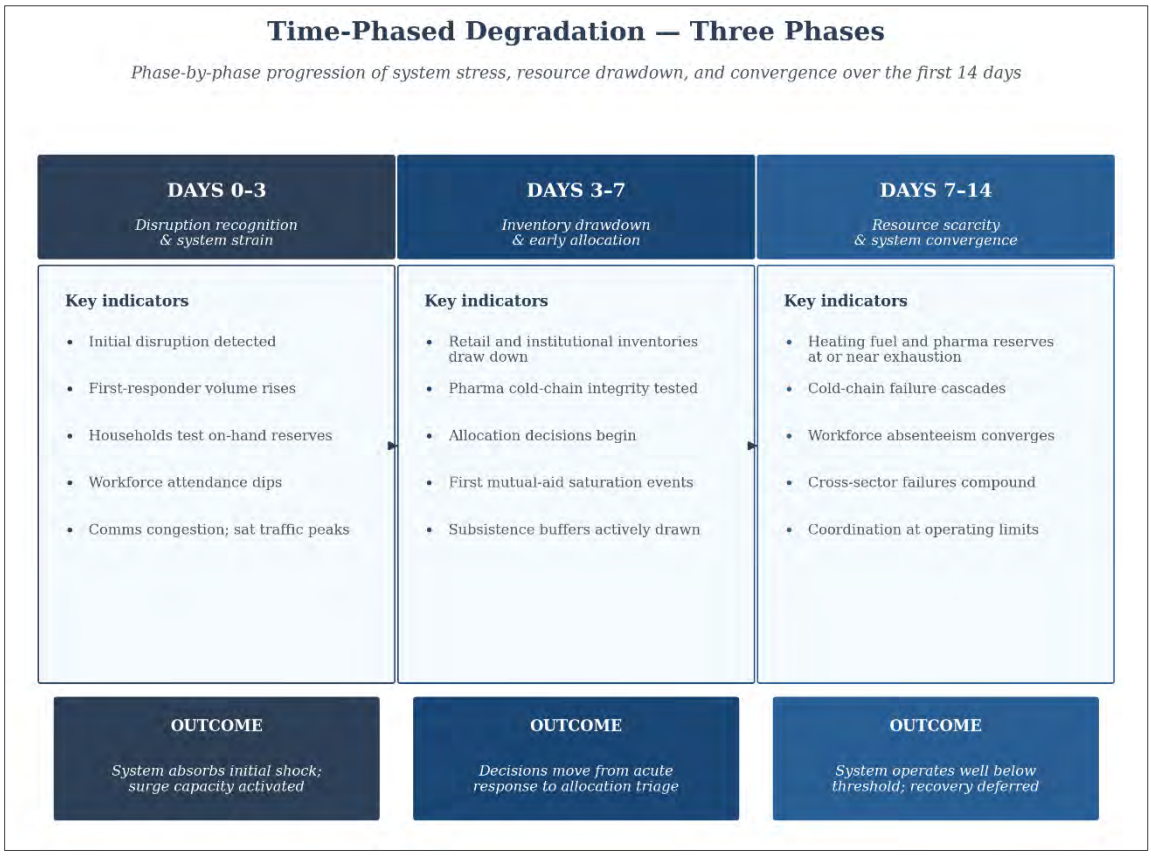
Workforce degradation begins affecting operations as household-level access constraints translate into absenteeism, particularly in workforce categories that lack remote-work options. Threshold indicators include fuel allocation or rationing initiated, aviation fuel constraints limiting airlift and medevac, food availability impacting workforce stability, and communications degradation reducing coordination effectiveness. By Day 14, the system has either stabilized through external intervention or entered a non-linear degradation trajectory that the post-disruption period cannot easily reverse.

Time-to-Failure Thresholds by Sector, Days 0–14

Sector	Initial Impact	Degraded Operations	Failure Threshold	Winter Compression
Heating	Day 0 (winter)	Days 0–1	Within hours	Binding constraint absent in summer
Fuel	Day 2	Days 2–7	Day 12 (summer); Day 3.5 (winter)	–71% time-to-failure
Food	Day 3	Days 3–8	Day 10 (summer); Day 4 (winter)	–60% time-to-failure
Medical	Day 4	Days 4–10	Day 8 (summer); Day 3 (winter)	–62% time-to-failure
Transport	Day 1	Days 1–11	Day 11 (summer); Day 4.5 (winter)	–59% time-to-failure
Workforce	Day 5	Days 5–11	Day 14	Stress compounds in winter
Communications	Day 6	Days 6–12	Day 9 (summer); Day 5.5 (winter)	–39% time-to-failure



Composite curve illustrates aggregated lifeline degradation; component sectors degrade on differing timelines.



Each phase represents a distinct operational regime; transitions reflect cumulative system stress, not discrete events.

Cascading Failure Analysis: From Trigger to Convergence

The assessment identifies six categories of Arctic-specific vulnerabilities driven by geography, infrastructure constraints, and supply chain structure. Each category is presented in operational terms , what fails, on what timeline, with what cascading effect, rather than as a generic vulnerability inventory.

Transportation and Logistics Vulnerabilities

As established in the Dependency Mapping section and the Substitution and Redundancy Reality Check, Arctic transportation alternatives exist in planning documents but cannot deliver required throughput within Days 0–14, leaving the cascading risk pathway without an effective substitution release valve [19; 34; 37].

The scale of the maritime substitution problem can be stated quantitatively. Replacing Port of Alaska cargo volume through aviation alone would require approximately seven hundred wide-body cargo jet landings per week, a throughput requirement that no Arctic airfield network, fuel supply chain, crew pool, or scheduling system can meet within the Days 0–14 window. The 2023 Baseline Assessment evaluated six candidate alternatives for containerized cargo (Seward, Valdez, Kodiak, Homer, Whittier, MacKenzie) and concluded that no viable alternative to the Port of Alaska currently exists; specific infrastructure investments required at each candidate are developed in the Port and Maritime Throughput Resilience mitigation category [26].

> ***Key Risk Condition:*** Alternative routing options exist but are insufficient to replace disrupted maritime throughput within the first 14 days, creating a sustainment gap. This is a paper-redundancy condition applied to the transportation layer.

The substitution alternatives identified in practitioner-level coordination forums underscore the fragility rather than the robustness of these pathways. Alaska's Value Chain Coordination Council, convened in December 2025, examined USPS Bypass Mail and in-state backhaul opportunities for rural supply, confirming that even the alternatives actively under examination depend on federal postal infrastructure and ad hoc logistics arrangements rather than purpose-built redundant systems.

Fuel and Energy Vulnerabilities

Fuel disruption follows a clear cascade sequence: declining bulk delivery, storage drawdown below operational thresholds, allocation decisions, and simultaneous propagation into power generation, heating, water and wastewater, transportation and aviation, and communications. The cascade is amplified by an upstream refining dependency: approximately 80 percent of crude oil produced in Alaska is refined in Washington and California, and in-state refining capacity is structurally insufficient to compensate for disrupted maritime delivery (the operational figures are documented in the Dependency Mapping, Fuel and Energy Systems). West Coast port degradation therefore simultaneously constrains both the maritime delivery of refined product and the in-state refining of locally produced crude, a two-sided constraint that the

vulnerability assessment treats as compound rather than alternative. West Coast port degradation therefore simultaneously constrains refined-fuel delivery into Alaska and the in-state processing capacity that depends on imported additives and blending components [26]. Seasonal asymmetry compounds the cascade: a winter disruption depletes reserves against communities that cannot receive bulk resupply until spring breakup, collapsing the substitution window.

Within the contested-logistics doctrinal frame (Institutional Coordination section), fuel is identified as a non-substitutable foundation whose disruption cascades immediately into transportation, power generation, heating, and operational tempo, and as the ultimate last-mile challenge in contested environments [30; 51]. In the Arctic, fuel is simultaneously the foundation of commercial lifeline continuity and of household survival, where loss of heat becomes dangerous within hours.

> ***Key Risk Condition:*** Fuel system disruption acts as a primary cross-sector trigger initiating cascading failure across all lifelines.

Food and Essential Commodity Vulnerabilities

Urban Alaska operates on approximately seven to ten days of commercial inventory under normal conditions as a just-in-time retail system with no meaningful warehousing buffer *[26]*. Rural systems are shorter and more variable, with most communities served by a single retail food source and no local fallback. Alaska state emergency doctrine formally divides disaster feeding response into three operational phases, with the first phase defined by the absence of outside assistance and the structural constraint that assistance may require twenty-four hours or more to arrive in affected communities. Survivors will not typically arrive at shelters with a 72-hour household food and water supply, directly contradicting a planning assumption embedded in national-average emergency frameworks [6; 7].

As established in the Dependency Mapping section, Alaska's bimodal food system places the highest concentrated risk on rural and Western Alaska communities under the dual-theater scenario, with parallel degradation likely across both the commercial and subsistence pathways. The 2025 termination of the USDA Islands and Remote Areas Regional Food Business Center, which had been standing up technical assistance and coordination infrastructure for Alaska food businesses, demonstrates that the programmatic infrastructure supporting food system resilience is itself subject to cascade dynamics: when federal funding was withdrawn, the relationship capacity required to operationalize supply chain improvements was lost before it could be established [3]. Food system resilience depends on institutional continuity as well as physical inventory, and both are currently thin.

> ***Key Risk Condition:*** Food system disruption rapidly translates into community instability and workforce degradation, amplifying operational risk.

Communications Vulnerabilities

As established in the Layer 5 Communications dependency treatment, the Arctic communications baseline is structurally fragile (high-latitude geometry, bandwidth limitations, single-point failure exposure, insufficient remote infrastructure) and is acted on by four overlapping degradation vectors: weather and environmental, adversarial jamming and physical interference, cyber, and commercial provider autonomy. The risk-assessment implication is that communications failure does not arrive through a single failure pathway but through compounding effects across these vectors operating concurrently. GPS jamming and potential or actual interference with submarine telecommunications cables are active adversarial tactics in the Arctic theater rather than hypothetical failure modes [14; 25]. Russia's 2022 Viasat cyberattack degraded European civilian services as collateral to its military targeting, illustrating the cascade pathway in which the cyber vector compounds the adversarial-jamming and commercial-autonomy vectors and produces simultaneous defense and civilian impacts [28]. The risk this assessment surfaces is not the failure of any single vector but the convergence of all four against a baseline that has no functional redundancy at the Arctic latitude where defense operations depend on the system most heavily.

> ***Key Risk Condition:*** Communications degradation functions as a force degrader, accelerating cascading impacts by delaying coordination and response.

Medical and Pharmaceutical Vulnerabilities

Urban Alaska operates on approximately seven to ten days of commercial inventory under normal conditions as a just-in-time retail system with no meaningful warehousing buffer [26]. Rural systems are shorter and more variable, with most communities served by a single retail food source and no local fallback. Alaska state emergency doctrine formally divides disaster feeding response into three operational phases, with the first phase defined by the absence of outside assistance and the structural constraint that assistance may require twenty-four hours or more to arrive in affected communities.

Survivors will Arctic medical and pharmaceutical sustainment depends on systems that share the same fragile-baseline-plus-degradation-surface architecture documented across the other lifelines, with two domain-specific intensifiers: cold-chain pharmaceutical integrity is non-recoverable once broken within the Days 0–14 window, and medevac access is governed by weather and infrastructure constraints that cannot be substituted by airlift surge in the way bulk cargo theoretically can. Pharmaceutical distribution into Alaska depends on a small number of national wholesalers operating temperature-controlled supply chains across the same maritime and aviation pathways constrained by the dual-theater scenario; once the cold chain is broken, affected inventory is lost, not delayed.

NASEM observations from Hurricanes Harvey, Irma, and Maria document the same cold-chain and pharmaceutical-distribution cascade pattern across U.S. disaster experience: maritime-import-dependent populations experience non-recoverable pharmaceutical loss within compressed timelines when port and aviation logistics degrade concurrently with power and fuel disruption [41]. The Arctic case examined here is structurally

analogous to the Puerto Rico under Maria case but operates under more severe geographic isolation and seasonal compression constraints.

As established in the Dependency Mapping section, the Alaska Native Medical Center hub-and-spoke care model concentrates specialty care, surgical capacity, and pharmaceutical distribution at a small number of urban nodes, with weather-limited evacuation as the connection between those nodes and the rural and Western Alaska communities they serve. The risk-assessment implication is that medevac delay is not a logistics inconvenience but the failure mode that converts a survivable medical event into a non-survivable one, and that medevac and pharmaceutical-shortage degradation are not parallel failures but a convergent failure: when both occur concurrently, the affected population loses access to acute intervention and to the maintenance medications (insulin, anticoagulants, antiepileptics, antibiotics) whose interruption produces clinical deterioration on the same Days 0–14 timeline as the disruption itself.

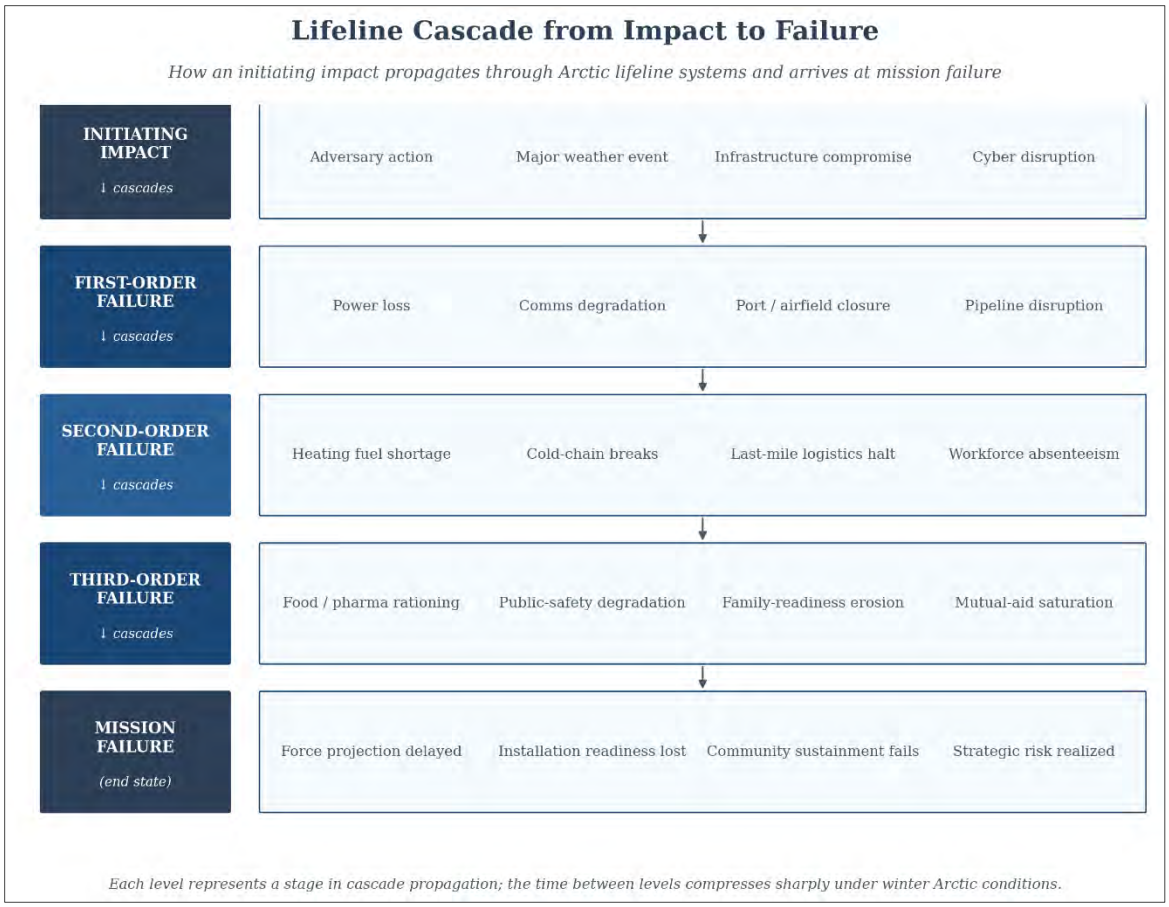
> ***Key Condition:*** Pharmaceutical cold-chain failure is non-recoverable within the Days 0-14 window. Medical system strain becomes mission-liming when medevac delays and pharmaceutical shortages converge

Workforce and Community Dependency Vulnerabilities

The workforce dimension is the most systematically underweighted element of current Arctic defense planning, and the literature supports treating it as an operational dependency rather than a quality-of-life consideration. RAND Plan Blue 21 confirms that workforce and institutional capacity become limiting factors as systems degrade, reducing the functionality of otherwise intact infrastructure [22; 45]. Key logistics, aviation, and processing sectors in Alaska depend on transient, seasonal, and external workforce populations whose availability is directly affected by the same supply disruptions the research is examining. The electric power sector illustrates the failure mechanism in concrete terms: a documented shortage of trained linemen, a worldwide transformer shortage affecting repair lead times, and aging infrastructure with no rehabilitation plans [26]. As established in the Dependency Mapping section, this fragility concentrates at the standalone microgrid layer, where repair under scenario conditions requires airlifting personnel and components using fuel that is itself constrained.

Workforce fragility is not evenly distributed across the communities from which Alaska's civilian and contractor workforce is drawn. Communities with compounding pre-existing resilience challenges (economic sector concentration, limited healthcare access, constrained transportation and communications, thin civic organization density) experience workforce degradation on accelerated timelines because they have the least reserve capacity to absorb early shocks. The FEMA Community Resilience Challenges Index provides a federally validated, annually updated empirical framework for identifying these communities before disruption, enabling workforce vulnerability assessment grounded in peer-reviewed indicators rather than general inference [29; 65].

> ***Key Risk Condition:*** Workforce degradation is not a secondary effect; it is a primary failure pathway, rendering otherwise intact infrastructure non-functional.



Cascade is rarely linear in time; later stages often compress as compounding failures reinforce one another.

Cross-Border Interdependencies as Risk Amplifiers

Cross-border dependencies implicate legal and regulatory chokepoints that informal coordination does not address. The Jones Act can be waived only by the Secretary of Homeland Security upon a finding that the transportation is necessary in the interest of national defense and that no coastwise-qualified vessels are available, a high evidentiary bar requiring advance planning and documentation. The waiver mechanism is recognized in federal supply-chain guidance as a key enabler and chokepoint for OCONUS supply-chain flows.

As established in the Cross-Border Interdependencies and Sovereignty Constraints subsection of the Dependency Mapping section, the Northern Sea Route, UNCLOS Article 234 claims, and the suspension of military security dialogue with Russia since 2014 collectively define a cross-border risk surface beyond Canada. The risk-assessment implication for the dual-theater scenario is that the absence of functional deconfliction mechanisms compresses the time available before military and civilian systems must adapt to a degraded or contested environment, eliminating the rerouting redundancies that benign planning models treat as available at the precise moment they are needed.

> ***Key Condition:*** Coordination architectures that exist on paper but have not been exercised, formalized through charter, or integrated into lifeline operator business processes do not meaningfully reduce risk during Days 0–14. This is the paper-redundancy condition applied at the governance layer.

Cross-Border Dependency Summary (US–Canada)

Domain	Bilateral Dependency	Failure Implication
Maritime corridor	Seasonal Arctic shipping windows shared between US and Canadian operators	Loss of access constrains both nations' Arctic resupply timing
Energy / pipeline	Canadian pipeline supply contributes to PNW refined product flow	Pressure-balancing requires bilateral coordination during disruption
Communications	Cross-border coordination protocols rely on integrated networks	Force-degrader effects extend across both forces simultaneously
ALCAN Highway	Single road link between Lower 48 and Alaska, traverses Canadian territory	Closure or restriction halves operational mobility options
Search & rescue	Joint US–CAN agreements for Arctic SAR coverage	Capacity gaps emerge if either national capability is reduced
Workforce	Specialty labor and contractors cross border for rotational work	Cross-border restrictions compound workforce attrition

Alaska Native and Tribal Community Dependency as a Risk Amplifier

The tribal-community vulnerability picture identified in the Dependency Mapping section becomes a specific risk-amplifier condition in this assessment. As that section establishes, the Alaska Native governance and economic ecosystem operates barge routes, airstrip operations, and fuel distribution channels that defense supply chains depend on, and the 144 communities documented to face baseline infrastructure stress represent fragility that disruption accelerates rather than creates. The risk-assessment implication for the dual-theater scenario is that workforce availability, last-mile logistics, and local coordination capacity in the regions surrounding installations can degrade faster than the dependency mapping alone would suggest, because the entities that operate those functions are simultaneously absorbing the same supply disruption the assessment is modeling. The amplifier effect is concentrated at the same communities whose pre-existing federal-assistance gaps and limited administrative capacity reduce reserve absorption, producing a cascade timeline that compresses for the populations least able to respond.

Mission Impact Assessment

Mission impact in this assessment is evaluated across five operational categories: force mobilization through constrained northern corridors, sustainment of Arctic and sub-Arctic installations, aviation operations dependent on fuel and maintenance, medical support to personnel and dependents, and operational coordination dependent on communications systems. Each category is evaluated against severity (degraded, constrained, or failed capability), likelihood under scenario conditions, time-to-impact during Days 0–14, and dependency convergence at the time of impact. The convergence dynamic established across the Arctic-Specific Vulnerabilities subsections is the operative analytical input; the cascade narrative that follows develops the operational and community-level consequences that convergence produces.

Cascading Impact Narratives

The cascading failure pathway documented in the Focused Risk and Vulnerability Assessment is not abstract. Under the operative phase conditions of Days 0–14, with cascade propagation extending into Days 10–45 as compounding effects materialize, the cascade produces operational and community-level consequences that arrive on a compressed timeline, in coordinated rather than independent fashion, with reinforcement effects that make each subsequent failure mode harder to address than the one before it.

The narrative that follows is organized by time window rather than by sector because the defense and civilian cascades are not parallel processes operating through separate pathways: they are the same process producing simultaneous, mutually reinforcing degradation across military mission assurance and the communities on which mission assurance depends. Within each window, the defense and civilian lanes describe how the same upstream disruption produces concurrent effects on installations and communities that cannot be analytically separated because the underlying systems are not operationally separable.

Days 0–3: Throughput Degradation and First-Cycle Retail Signal

West Coast port throughput drops below normal operating levels through cyber intrusions, navigation interference, and labor stand-downs that produce compound rather than catastrophic disruption. Container backlogs form at terminals and intermodal transfer points, and maritime carrier scheduling reliability collapses for Alaska routes.

Defense lane. DoD and defense-contractor supply flows routed through the Seattle/Tacoma gateway lose schedule predictability. Defense logistics operators have limited visibility into commercial cargo prioritization decisions, and the absence of pre-defined arrangements with private-sector lifeline providers means that DoD cannot exercise meaningful influence over allocation outcomes during the window when those decisions are being made. Planning assumptions embedded in standing logistics frameworks are invalidated before alternative arrangements are operationally viable. The Days 0–3 window is the period during which the disruption is happening but is not yet

visible to most decision-makers, because supply chain effects propagate ahead of the infrastructure-level signals on which planning frameworks typically rely [26; 49].

Civilian lane. The earliest civilian-visible signal is at the retail and healthcare layer. The four-day Tacoma-to-shelf cycle established in the Dependency Mapping (Layer 1 dependency mapping) means that when West Coast throughput degrades, the retail effect appears within the first cycle [26]. Hospital pharmacy inventories, typically maintained at seven to thirty days of supply, begin showing reorder pressure for high-velocity products including insulin, antibiotics, and oxygen [9]. Most rural communities operate with a single retail food source, meaning that disruption upstream of the Port of Alaska eliminates local fallback capacity entirely rather than shifting demand to secondary providers [6; 38]. Communities served by Bypass Mail and the bush plane network see disruption immediately because the distribution architecture has no buffer; communities served by the Anchorage retail and pharmaceutical pipeline see it within days for the same reason.

Days 3–7: Inventory Drawdown, Allocation Friction, and Household Decisions

Warehouse replenishment falls below demand. Food, fuel, and pharmaceutical inventories intended for northern movement begin drawing down against reserves that were never designed to absorb sustained throughput loss. Multiple lifeline systems begin degrading concurrently rather than in sequence.

Defense lane. Defense-relevant cargo competes with commercial cargo for the same constrained transportation capacity, and the absence of a predefined cargo prioritization mechanism means that allocation outcomes are produced by commercial operators making decisions on commercial logic that does not formally account for defense sustainment requirements. Workforce absenteeism begins climbing as household sustainment needs draw workforce toward home rather than installation. Aviation fuel allocation conversations begin between commercial operators, airfield managers, and DoD logistics staff, and the throughput-versus-capacity distinction becomes operationally salient: airfield capacity exists, but aviation fuel availability and competing medevac and cargo demand begin compressing usable throughput [46]. The institutional gap visible in this window is the absence of pre-established adjudication mechanisms for civilian–military resource competition: installation commanders hold Title 10 authority over on-base resource allocation but have no control over commercial cargo prioritization or civilian workforce availability; state authority can reallocate scarcity across competing populations but cannot create supply; commercial operators make prioritization decisions based on commercial logic. The result is coordination friction that is not a failure of any individual actor but a structural absence of agreed authority and prioritization criteria for decisions being made under time pressure.

By the end of this window, the substitution mechanism most often invoked when maritime supply degrades, aviation surge, fails categorically rather than at the margins. Arctic airfields are simultaneously constrained by weather, fuel, maintenance capacity, and crew availability in ways that compound under scenario stress; space weather adds a non-terrestrial constraint as geomagnetic events disrupt navigation systems and force

rerouting or cancellation of polar flights during the conditions Arctic escalation is most likely to produce. As established in the Berlin Airlift Anchor, air-only sustainment requires conditions that do not apply to the Arctic dual-theater scenario; the cascade in this window runs forward without aviation as an effective substitution release valve.

Civilian lane. As inventory drawdown becomes visible, household-level decision-making about food, fuel, and healthcare access begins. Communities with compounding pre-existing resilience challenges, high unemployment combined with limited vehicle access; thin healthcare capacity combined with low civic organization density, experience these decisions earlier and with less reserve capacity to absorb the consequences. Rural and Western Alaska communities, where dependency on a single retail source is highest, experience the first acute shortages. Households with limited economic reserves face tradeoffs between work attendance and household sustainment. Subsistence harvest activities, where seasonally available, become more critical, but those activities themselves depend on fuel for boats, snowmachines, and aircraft, and the fuel supply chain is itself disrupted. Healthcare clinics and the Community Health Aide Program serving villages with no other clinical presence experience pharmaceutical resupply delays for chronic-disease management; insulin, antibiotics, oxygen, and dialysis supplies are the categories where depletion produces life-threatening outcomes within days [76].

State emergency management planning recognizes this dynamic explicitly: outside assistance to affected communities may require twenty-four hours or more to arrive, establishing an early-phase response gap that local and tribal resources alone must absorb. Survivors will not typically arrive at shelters with a 72-hour household food and water supply, directly contradicting a common planning assumption embedded in national-average emergency frameworks.

In this same window, the pre-existing fragility documented in the Dependency Mapping section converts into the early collapse of last-mile logistics functions managed within the Alaska Native governance and economic ecosystem, producing supply shortfalls in precisely the communities whose institutional resilience options are fewest. Barge routes, airstrip operations, fuel distribution, and local logistics functions that defense supply chains rely on run through or are managed by entities within the Alaska Native governance and economic ecosystem; when maritime logistics degrade, these are among the first systems to experience supply shortfalls and among the last to receive resupply prioritization in frameworks that do not formally account for them. Functional tribal coordination capacity depends on pre-existing relationships between non-tribal institutions and tribal governments that cannot be improvised under disruption conditions [21; 61]. Statutory consultation mechanisms do not substitute for substantive operational engagement.

Days 7–14: Convergence, Workforce Degradation, and Mission Impact

Fuel availability becomes a limiting factor across multiple sectors simultaneously, the medical pathway emerges as the fastest-cascading lifeline failure in the scenario, and the workforce dependency that was forming across the prior windows is now operationally realized.

Defense lane. Aviation fuel allocation decisions force tradeoffs between defense operational aviation, cargo airlift, and medical evacuation. Aviation gasoline for the small bush aircraft network that delivers food and supplies to off-road communities, supplied by a single importer through the Port of Alaska, is constrained at the same time and through the same upstream node that constrains defense fuel availability *[26]*. Medevac operations face simultaneous fuel, weather, and crew constraints, competing for the same fuel pool as cargo airlift and operational aviation. Cold-chain failure across distribution centers, pharmacies, and clinics produces parallel pharmaceutical loss rather than at a single point. Outside transfer to Seattle for definitive care, the relief valve on which the Alaska healthcare system depends, is closed by the same maritime and aviation disruption affecting all other lifelines.

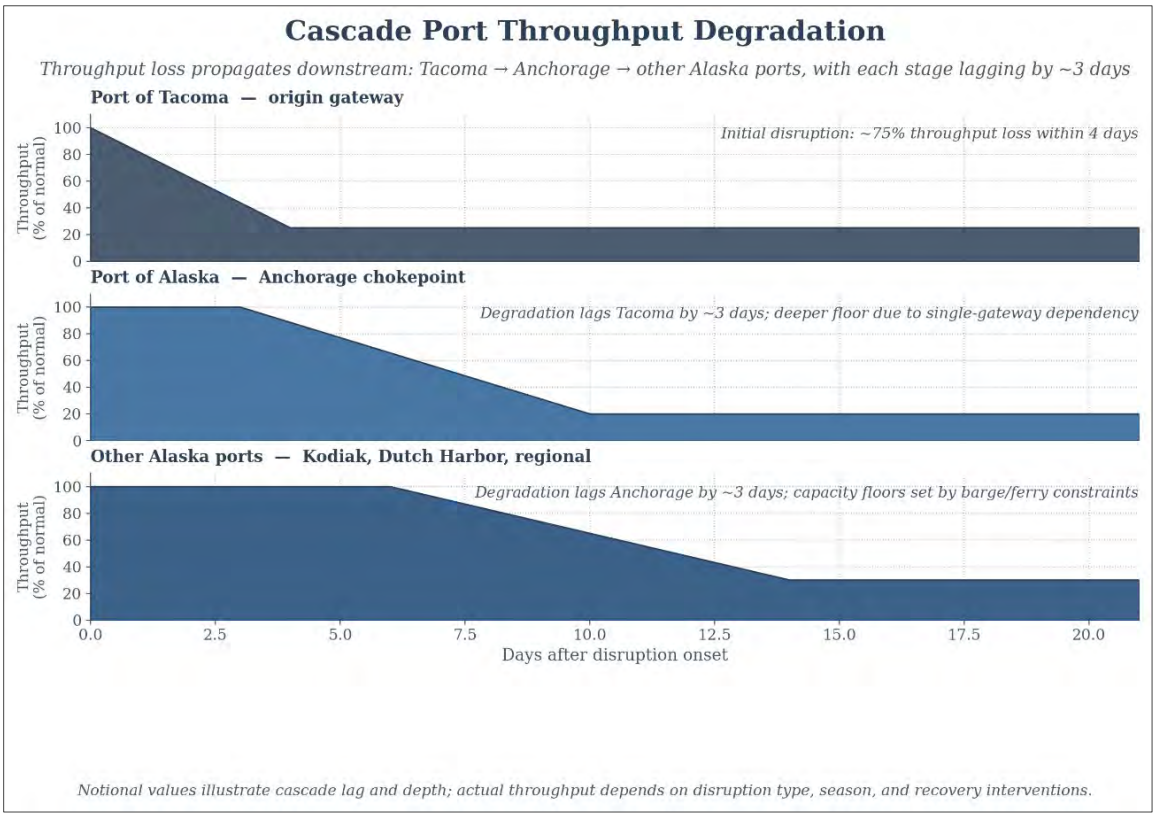
Installation sustainment now depends on a workforce drawn from the same communities the civilian cascade is degrading, and the household decisions of the prior window have now reached the threshold that drives workforce availability. Specialty roles, utility maintenance, medical care, communications operations, fuel handling, are the most consequential because they have the thinnest depth of qualified replacement. The seasonal-workforce concentration risk in the seafood processing sector (the 27,000 H-2B and seasonal-workforce population developed in Vulnerability Assessment, Workforce / Community Dependency Vulnerabilities) is the worked example of workforce fragility acting as a force degrader across multiple lifeline categories from a single disruption pathway.

By Day 14, converging constraints produce mission impacts that are not attributable to any single failure mode. Force projection through northern corridors is constrained by fuel availability, ALCAN throughput limits, and aviation capacity absorbed by competing logistics, operational, and medical evacuation demands. Sustainment of Arctic and sub-Arctic installations is constrained by workforce degradation, fuel allocation tradeoffs, and food and pharmaceutical supply that has not been replenished against initial drawdown. Operational coordination is degraded by communications failures that reduce visibility into logistics status, infrastructure performance, and resource availability. Mission degradation does not arrive as a single recognizable event; it arrives as the accumulation of constraints across systems that were each near baseline before the disruption began.

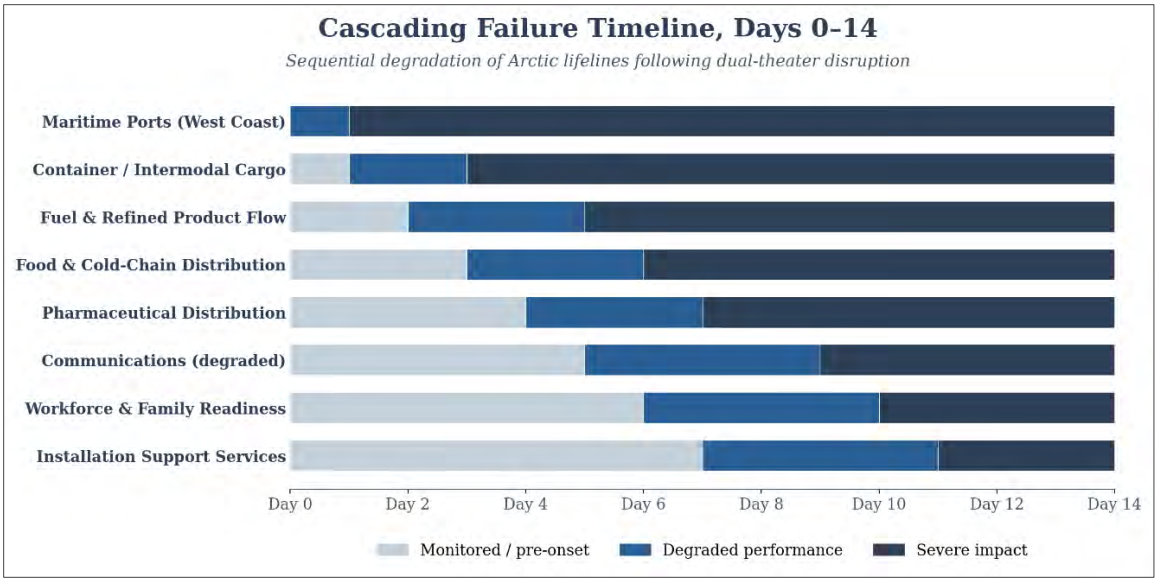
Civilian lane. Civilian healthcare systems experience demand surge from disruption-related injuries, exacerbated chronic conditions, and pharmaceutical access failures. When military dependent and contractor populations draw on the same civilian system because installation medical capacity is exceeded, the binding constraint is specialty staff rather than beds, and specialty staff cannot be flown in under the fuel and aviation constraints the scenario imposes [4]. The Indian Health Service, the Alaska Native Tribal Health Consortium, regional tribal health organizations, and village-level community health aide programs serve communities that commercial and state healthcare infrastructure does not reach; they depend on the same pharmaceutical supply chain, cold-chain infrastructure, medevac network, and fuel availability as the rest of the Alaska healthcare system. Multiple parallel pathways converge on the same constrained upstream resources, with no independent redundancy across them [7; 36].

Heating fuel reserves in rural communities, structured around the fall barge season, are insufficient to bridge to spring resupply. Water and wastewater systems in the 22 percent of Alaska households unserved by utilities face direct degradation as the fuel and power they depend on becomes constrained [26]. The Alaska Mass Care Group and Feeding Task Force [7], which depends on the same commercial supply chain whose disruption triggered the cascade, experiences the paper-redundancy condition directly: the named commercial partners in state feeding doctrine, Sysco, Carrs/Safeway, Fred Meyer, Walmart, Costco, are not redundant to the disrupted system but embedded within it [6].

By the end of the second week, the household-level decisions that began in the prior window produce visible workforce effects across logistics, utilities, healthcare, transportation, retail, and emergency response. The civilian and contractor workforce supporting installation operations experiences absenteeism that intersects with the supply chain degradation that drove household decisions in the first place, a feedback loop in which workforce withdrawal accelerates the supply chain failures that were producing workforce withdrawal.



Throughput losses propagate downstream with damping, but compound at chokepoints; combined effects exceed any single-port loss.



Cross-sector failure propagation across the port degradation window. Cascade timing reflects scenario stress test; actual sector-specific thresholds vary by season and inventory state.

Civilian–Military Convergence

The Days 7–14 window is the point at which the civilian and defense lanes become operationally indistinguishable. Workforce degradation reduces both civilian system functionality and installation sustainment functionality through the same mechanism. Pharmaceutical cold-chain failure destroys both civilian and military-dependent medication supply through the same upstream failure. Aviation fuel allocation forces tradeoffs that simultaneously degrade civilian medical evacuation and defense operational aviation.

The communities, families, and workforce that the civilian lane describes are the same populations on which mission assurance depends. The civilian lane is not a downstream effect of the defense lane; it is a concurrent process operating through the same systems, on the same timeline, with the same driving constraints. Protecting it is not adjacent to mission assurance; it is mission assurance.

The Reinforcement Mechanism

Three specific mechanisms drive the mutual reinforcement that makes the cascade non-linear. First, shared infrastructure means that degradation in any single system simultaneously affects both lanes. Fuel constraints affect both civilian heating and defense operational aviation. Communications degradation affects both retail logistics coordination, command, and control. Pharmaceutical cold-chain failure affects both civilian patients and military dependent populations. The lanes are not analytically separable because the underlying systems are not operationally separable.

Second, workforce dependency means that civilian household-level decisions directly produce defense-side capability degradation: the workforce that installation operations depend on lives in the communities experiencing the civilian cascade, and there is no organizational firewall that separates the two.

Third, governance friction amplifies both lanes. The absence of pre-defined adjudication mechanisms for civilian–military resource competition means that allocation decisions are made under time pressure, with incomplete information, across authorities that have no established shared framework. This friction adds delay to every decision and reduces the effectiveness of any individual mitigation action. The institutional coordination gap is itself a force degrader operating across both lanes simultaneously, and its treatment in the next major section of this report develops the implications for mission assurance planning.

ARCTIC SUSTAINMENT MODES AND REDUNDANCY REALITY CHECK

The most analytically significant finding of this research is the gap between the redundancies embedded in planning documents and the redundancies that actually function under Arctic disruption conditions. Closing that gap is the organizing challenge that the recommendations and the PSEMA network framing address.

This section consolidates the substitution and redundancy analysis distributed across the prior deliverables into a single reality check, organized to make the gap visible at the level of detail that mitigation prioritization requires.

The framework is straightforward. For each redundancy commonly cited in Arctic sustainment planning, the analysis identifies what the redundancy claims to provide on paper, what the redundancy actually delivers under disruption conditions, and what the gap between the two means for the Days 0–14 window. The intent is not to discredit the redundancies, most of them perform real functions under normal conditions, but to clarify the conditions under which they cease to perform the function planning frameworks assume they perform.

The Substitution Inventory: Paper Versus Operational

This section operationalizes the paper-redundancy / operational-redundancy distinction (Glossary) at the level of detail that mitigation prioritization requires. The distinction is not rhetorical; it is the gap between what planning documents assume and what systems actually deliver under Arctic stress, and it is where the highest-consequence planning failures concentrate.

The substitution inventory below evaluates each redundancy commonly cited in Arctic sustainment planning, the modal substitutions (alternate ports, aviation surge, ALCAN overland surge) and the cross-cutting substitutions (backup fuel storage, satellite communications backup, pharmaceutical distribution diversification, state emergency feeding architecture, public-private coordination charters), against the same three-field structure: paper redundancy, operational reality, and Days 0–14 implication. None of the inventoried substitutions function as real redundancies under the conditions this research models.

Maritime Substitution: Alternate West Coast and Southcentral Port

Paper redundancy. Alternate ports appear in planning documents as substitutes for the Port of Alaska under disruption conditions. The 2023 Baseline Assessment identifies six candidates: Seward, Valdez, Kodiak, Homer, Whittier, and MacKenzie [26].

Operational reality. Each candidate requires specific capital investment before functioning as a substitute. Seward requires 200 reefer plug-ins, RO/RO ramp acquisition, and a pre-scripted emergency rail-activation pathway. Valdez requires

ruggedized paired-chassis dollies, weight-restriction exemptions, and crane improvements. Whittier requires construction of a deep-draft cargo port with cranes, ramps, and rail integration. MacKenzie requires conveyor infrastructure, a barge landing, crane pad, fencing, and road improvements. Kodiak and Homer require comparable category-specific investment. Arctic Edge 2025 tested an unimproved Port Mackenzie opening for force-flow reestablishment under exercise conditions, demonstrating that the substitute-port pathway is operationally addressable at the exercise scale but reinforcing the Baseline Assessment finding that container cargo throughput at the scale required by a Port of Alaska disruption is not achievable without the specific infrastructure investments documented above [67]. The documented conclusion of the assessment is explicit: currently, no viable alternative to the Port of Alaska exists, and all options require funding for improvements before they could function as operational redundancy [26].

Days 0–14 implication. Maritime is the primary and dominant sustainment mode on which all other modes depend for their effective volume. Disruption at the West Coast gateways produces immediate, statewide, cross-sector effects. No realistic alternative mode provides equivalent capacity within the analytical window; the alternate-port pathway is a long-horizon investment whose mitigation value lies beyond the scenario window.

Aviation Substitution for Maritime Cargo

Paper redundancy. Airlift is consistently cited in contingency planning as the alternative when maritime routes are disrupted. The wargame literature treats aviation surge as a sustainment option for Arctic operations.

Operational reality. The aviation substitution arithmetic, approximately 700 wide-body cargo sorties to replace one week of Port of Alaska throughput, does not account for fuel, de-icing, crew, runway capacity, turnaround time, or the cold-weather operating constraints that further reduce effective payload per sortie. Arctic airfields are simultaneously constrained by weather, fuel availability, maintenance capacity, and crew availability [26]. Space weather adds a non-terrestrial constraint as geomagnetic events disrupt navigation and force rerouting or cancellation of polar flights [3]. Each sortie also generates a workforce demand at the receiving end that the arithmetic does not capture: ramp crews to marshal and offload aircraft, fuel handlers working in cold-weather PPE at reduced throughput, ground transportation operators to move cargo onward, warehouse and cold-chain personnel to receive and stage materiel, and maintenance technicians to sustain accelerated turn cycles. Surge sortie rates assume surge ground workforce, which Arctic airfields do not have on standby and cannot rapidly import without consuming the same airlift capacity the sorties are meant to deliver. The Berlin Airlift Anchor (Scenario Conditions) establishes the empirical upper bound on aviation-only sustainment under conditions favorable to airlift feasibility and confirms that airlift is a constrained supplemental mode rather than a substitute for bulk maritime capacity; those conditions do not apply to the Arctic dual-theater scenario.

Days 0–14 implication. Aviation is operationally indispensable for high-priority cargo, medical evacuation, personnel movement, and operational lift, but cannot substitute for maritime cargo at the volume sustainment requires. The substitution argument fails

categorically rather than at the margins [46]. Competition between operational aviation, medical evacuation, and logistics use of limited airfield capacity is immediate and zero-sum; every military cargo flight displaces medevac capacity that directly affects civilian and family survivability. Surging additional aircraft into the theater compounds rather than resolves this competition, added sorties draw against the same finite ramp space, fuel stocks, ground crew capacity, and air traffic management bandwidth that medevac and operational lift already depend on, and any deployment of supplemental ground workforce to expand throughput must itself arrive by the same constrained aviation pipeline. The deployment-to-receive workforce dependency identified in Layer 4 applies with particular force here: aviation substitution is not bottlenecked by aircraft availability but by the ground-side human and infrastructure capacity to absorb what the aircraft deliver.

Overland (ALCAN) Substitution: Cross-Border Surge Capacity

Paper redundancy. The Alaska-Canada Highway is cited in planning frameworks as overland surge capacity to substitute for degraded maritime delivery, with theoretical lane capacity of 12,000 to 15,000 vehicles per day across the 1,387-mile corridor.

Operational reality. Operating throughput is approximately 350 vehicles per day, an order-of-magnitude gap below the theoretical ceiling. The corridor crosses approximately 1,200 miles of Canadian sovereign territory, where transit decisions cannot be resolved by U.S. planning authority unilaterally. Optimization requires two distinct investment categories: pre-scripted regulatory pathway activation (weight-restriction waivers, hours-of-service waivers, CDL endorsement coordination, Canadian provincial permitting) achievable within the Days 0–14 window through administrative pre-staging; and physical pre-staging of driver support, lodging, and fuel infrastructure along the route, which is a pre-event capital investment that determines whether theoretical surge capacity is operationally accessible at all [31].

Days 0–14 implication. ALCAN is a constrained-redundancy mode, not a surge alternative. It provides genuine but volume-limited overland capacity subject to Canadian sovereignty, weather closure cadence, and the seasonal-corridor conditions documented in the Dependency Mapping section. The corridor moves real cargo, but absent both investment categories it cannot deliver the volume, speed, or reliability that maritime disruption response requires within the analytical window.

Northern Sea Route: Adversarial-Subject Pathway

Russia's assertion of control over the Northern Sea Route under UNCLOS Article 234 creates a potential rerouting option that could be denied by adversarial action at exactly the moment West Coast disruption makes Arctic routing most necessary [13; 14; 19; 49; 73]. The dual-use nature of Russian Arctic infrastructure means that escalation can convert nominally cooperative transit routes into contested access points with little warning. NSR represents a redundancy whose availability is entirely dependent on conditions outside U.S. planning authority. For the dual-theater scenario, this matters because the NSR could be denied at exactly the moment when West Coast port disruption makes Arctic routing most operationally necessary, eliminating a redundancy at the precise moment it is most needed.

Backup Fuel Supplies and Storage

Paper redundancy. Bulk fuel storage at the Port of Alaska, 3.1 million barrels of liquid fuel storage capacity, and at distribution terminals across the state appears in planning frameworks as a buffer against supply disruption. [26; 44]

Operational reality. Storage levels reflect just-in-time delivery cycles, not strategic reserves. The Days 0-14 implication is that backup fuel storage is a paper redundancy unless pre-positioned product is in place before disruption begins. The Days 0-14 implication is that the substitution argument for backup fuel storage rests not on the existence of in-state refining capacity but on whether pre-positioned product is in place before disruption begins. Aviation gasoline for the bush plane network that delivers food and supplies to off-road communities is supplied by a single importer, with all aviation gasoline flowing through the Port of Alaska, a single-supplier, single-node dependency that makes aviation fuel availability immediately vulnerable to any disruption at the port. Rural communities receive bulk fuel by barge in late fall before ice closes waterways; any additional fuel needed before spring breakup must arrive by bush plane at higher cost and limited volume.

Days 0–14 implication. Storage capacity is not the same as reserve capacity. The fuel architecture is operationally fragile because the storage system functions as a buffer for normal-operations flow rather than as a strategic reserve. Seasonal asymmetry compounds the vulnerability: winter disruption against communities that cannot receive resupply by the primary delivery mode until spring compresses failure timelines from weeks to days.

Satellite Communications as Backup to Terrestrial Infrastructure

Paper redundancy. Multiple commercial satellite systems, Iridium, Starlink, OneWeb, Inmarsat, and traditional GEO providers, are cited in Arctic planning frameworks as redundant communications pathways when terrestrial infrastructure is degraded.

Operational reality. No single satellite architecture provides full Arctic coverage. Bandwidth and latency constraints limit the operational uses each system can support. Commercial provider autonomy means that operators may alter service availability during conflicts based on financial or political considerations, with no governance framework currently requiring them to maintain service or prioritize military traffic. The adversarial-exploitability of communications is documented in the Dependency Mapping. The United States has not formally designated space as a critical infrastructure sector meaning the coordination, liability, and resilience frameworks that apply to other lifeline sectors do not cleanly cover the satellite systems those sectors depend on [12].

Days 0–14 implication. Satellite communications redundancy is a paper redundancy in two distinct senses. First, the technical redundancy assumes that backup systems will function under the conditions that necessitate their use, an assumption directly contradicted by adversarial capability and commercial provider autonomy. Second, the governance redundancy assumes that contractual relationships will produce operational continuity under stress, an assumption directly contradicted by the absence of formal

critical infrastructure designation for the space sector. The Days 0–14 window cannot wait for governance gaps to be closed.

Pharmaceutical Resupply Through Alternative Distribution

Paper redundancy. Pharmaceutical distribution networks include multiple wholesale distributors, mail-order pharmacy services, hospital pharmacy networks, and tribal health system distribution channels, suggesting redundancy through diversified pathways.

Operational reality. The 7-30 day pharmacy inventory threshold (Dependency Mapping) compresses against Arctic delivery cycles, so the operational substitution window for pharmaceutical resupply is shorter than the headline inventory figure suggests. Alaska's position at the end of that pipeline compresses national depletion timelines further. The diverse distribution channels in Alaska, commercial wholesalers, hospital pharmacy networks, the Indian Health Service, the Alaska Native Tribal Health Consortium, regional tribal health organizations, the Community Health Aide Program, depend on the same upstream pharmaceutical supply chain, the same cold-chain infrastructure, the same medevac network, and the same fuel availability. Cold-chain failure across multiple parallel nodes accelerates pharmaceutical loss beyond what single-pathway failure modeling would predict, and products affected by cold-chain failure are not delayed but destroyed.

Days 0–14 implication. The pharmaceutical distribution architecture in Alaska looks redundant in its endpoints but is single-pipeline at the upstream level that disruption affects first. The diverse distribution channels are dependent nodes whose degradation compounds rather than mitigates the capacity gap.

State Emergency Feeding Operations

Paper redundancy. Alaska's Mass Care Group Feeding Support Operations Guide identifies SYSCO, Carrs/Safeway, Fred Meyer, Walmart, and Costco as named commercial partners for emergency feeding response.

Operational reality. The named commercial partners in Alaska's emergency feeding doctrine are not redundant to the disrupted commercial food system; they are embedded within it, drawing from the same upstream supply pathway and reaching the same downstream distribution tier.

Days 0–14 implication. The state feeding system is a paper redundancy in the strict sense: the framework exists, the relationships are documented, the contracts are in place, but the operational capacity the framework describes does not exist independently of the system the framework is supposed to back up. The backup feeding architecture and the primary commercial food system depend on the same upstream nodes. This is a clear instance of paper redundancy without real redundancy at the institutional layer.

Public-Private Coordination Architecture

Paper redundancy. Federal guidance documents formalized private-public partnership frameworks, Business Emergency Operations Centers, executive committees, co-chair structures, formal terms of engagement, as the institutional scaffolding that converts aspirational private-sector cooperation into operationally reliable capacity. Alaska's Partnership for Infrastructure Protection provides a state-level precedent for formalized cross-sector coordination.

Operational reality. Where these mechanisms exist and have been exercised, as documented in the Nebraska Preparedness Partnership's twelve-hour logistics coordination during 2019 flooding and the Houston Supply Chain Group's COVID-19 response pivot, coordination moves at the pace disruption requires. Where they do not exist, coordination must be improvised under time pressure. The presence or absence of pre-established cross-sector coordination architecture directly shapes whether disruptions stabilize or cascade, the paper-redundancy pattern applied at the governance layer. Closing this gap is a pre-incident institutional investment, not a capability that can be improvised during disruption.

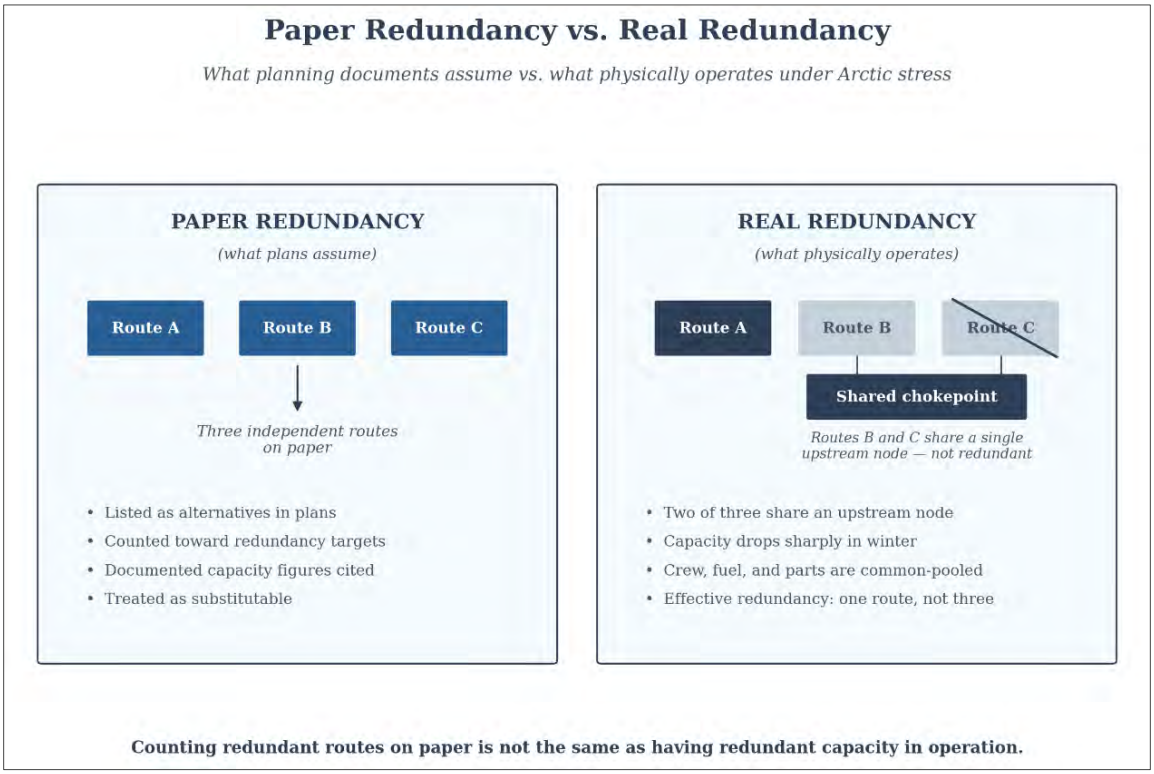
Coordination architectures that exist on paper but have not been exercised, formalized through charter, or integrated into lifeline operator business processes do not meaningfully reduce risk during the disruption window. This is the paper-redundancy condition applied at the governance layer; closing the gap is a pre-incident institutional investment, not a capability that can be improvised during disruption.

Civilian-Versus-Military Competition Across All Modes

The multi-modal picture is further complicated by competition between civilian and military users for the same constrained assets across every mode simultaneously. There is no predefined governance framework for adjudicating this competition. Installation commanders hold Title 10 authority over on-base resource allocation but have no control over commercial cargo prioritization or civilian workforce availability. State authority can reallocate scarcity across competing populations but cannot create supply. Commercial operators make prioritization decisions based on commercial logic that does not account for defense sustainment requirements.

The governance gap this creates is not a coordination problem that better communication solves. It is a structural absence of predefined authority and agreed prioritization criteria for decisions that will be made under time pressure, with incomplete information, across authorities that have no established shared framework. Pre-established governance frameworks and private-sector cargo prioritization agreements are the two highest-return, lowest-capital investments for closing this gap before the next disruption requires those frameworks to exist.

> ***Bottom Line:*** Arctic lifeline sustainment is not a redundancy architecture that provides equivalent alternatives. It is a hierarchy in which maritime is primary and all other modes provide partial, constrained, and competitive supplementary capacity. Planning that treats aviation or ALCAN as viable substitutes for maritime disruption is planning that accepts statewide shortages as the operational baseline.



Paper redundancy is not real redundancy; alternates that share underlying constraints fail concurrently, not independently.

The Pattern: Paper Redundancy as Systemic, Not Incidental

Reading the substitution inventory in aggregate reveals that the paper redundancy problem is not an isolated planning oversight in any single sector. It is a systemic feature of the Arctic sustainment architecture, present at the maritime, aviation, fuel, communications, pharmaceutical, feeding, and governance layers simultaneously. Three structural conditions explain why.

First, planning frameworks consistently substitute capacity figures for throughput figures. A port that remains physically open but processes cargo at a fraction of required velocity, a highway classified to accommodate thousands of vehicles daily but unsupported by fueling and rest infrastructure at emergency volume, an aviation network with airfield capacity but insufficient fuel, crew, and scheduling throughput to substitute for maritime freight, each of these is a system whose capacity remains nominally intact while its throughput collapses below sustainment requirements. The capacity-versus-throughput distinction is the most operationally significant analytical commitment of this

research, and the paper redundancy pattern is what happens when planning frameworks do not maintain it.

Second, redundancies that share upstream dependencies are not actually redundant against disruptions affecting those dependencies. The pharmaceutical distribution architecture in Alaska looks redundant in its endpoints but is single-pipeline at the upstream level. The state feeding architecture is embedded in the commercial food system it is supposed to back up. The diverse satellite communications providers depend on the same regulatory regime that provides no operational guarantees under stress. Architectural redundancy at the visible layer is not the same as substitution capacity at the layer where disruption actually occurs.

Third, the institutional and governance redundancies that should make physical redundancies operationally executable have themselves not been built to the maturity level that disruption response requires. Coordination frameworks exist on paper at maturity levels that have not been exercised. Pre-incident relationship infrastructure between federal, state, tribal, military, and private-sector actors is episodic rather than formalized. The contractual arrangements that should produce private-sector cooperation under stress do not include the prioritization criteria that would convert cooperation into operationally reliable capacity. The paper redundancy pattern at the infrastructure layer is mirrored by paper redundancy at the governance layer, and the two reinforce each other.

> ***Bottom Line:*** Paper redundancy is not a problem of planning quality; it is a structural condition of the Arctic sustainment architecture in which the redundancies that exist on paper consistently do not perform under the specific stress conditions the dual-theater scenario imposes. Risk reduction requires closing both the infrastructure-layer gap and the governance-layer gap simultaneously, and neither can be substituted for the other.

Institutional Coordination as a Risk Amplifier

The paper-redundancy principle does not stop at the infrastructure layer. It applies with equal force, and in some cases with greater consequence, to the institutional and governance architectures on which civil-military coordination depends during disruption. This section examines two specific governance instruments whose operational status determines whether coordination architecture functions as operational redundancy or as paper redundancy: the Jones Act waiver framework for emergency maritime capacity expansion, and the FEMA Private-Public Partnership maturity model for cross-sector coordination.

The Jones Act Waiver Framework: Pre-Staging as the Determining Variable

The Jones Act, Section 27 of the Merchant Marine Act of 1920, restricts coastwise transportation of merchandise to vessels that are U.S.-built, U.S.-flagged, U.S.-owned, and U.S.-crewed. The Secretary of Homeland Security may waive these requirements upon a finding that the transportation is necessary in the interest of national defense and that no coastwise-qualified vessels are available to meet the requirement [31; 22]. The evidentiary bar for the second element is high: documentation of non-availability must be specific, current, and capable of supporting a defensible waiver determination.

The FEMA Restoring Flow guide is explicit about the implication for emergency planners. Jones Act waiver capacity expansion is not an emergency tool that can be activated reactively under disruption conditions; it is a pre-incident planning capability that depends on relationships, documentation, and procedural readiness established before the disruption begins. The FEMA Distribution Management Plan Guide identifies the Jones Act as a key enabler and chokepoint for OCONUS supply chain flows, recognizing that rapid maritime capacity expansion requires regulatory mechanisms that are not pre-positioned for Arctic disruption scenarios [31].

Whether the existing pre-incident relationship infrastructure between FEMA, DHS, and Alaska's maritime stakeholders is sufficient to support a Jones Act waiver request within the Days 0–14 window remains an unresolved question. The research treats this as a paper-redundancy condition at the regulatory layer: the waiver mechanism exists, but its operational accessibility under the scenario's compressed timeline depends on pre-staging that, to the best of this research's ability to determine, has not been completed. Waiver pre-staging is therefore identified as a high-return, low-capital near-term recommendation in the Recommendations section that follows.

The FEMA Public-Private Partnership Maturity Model: Where Arctic Jurisdictions Currently Sit

The federal Building Private-Public Partnerships framework, issued in May 2025 in coordination with the U.S. Chamber of Commerce Foundation, the National Emergency Management Association, the National Association of Counties, Big City Emergency Managers, and the International Association of Emergency Managers, formalizes the institutional scaffolding through which commercial lifeline operators and government entities coordinate under disruption conditions [33]. The framework characterizes P3 capacity along a Basic, Intermediate, Advanced continuum.

Basic-tier P3 programs operate through episodic engagement, ad hoc relationships, and resource commitments not formalized through charter or tested through exercises. Intermediate programs operate through formalized relationships with documented terms of engagement, established Business Emergency Operations Center structures, and exercise-based validation of coordination capacity at moderate intensity. Advanced programs operate through institutionalized cross-sector partnerships with co-chair

executive committees, defined membership, exercised procedures, and integration of P3 stabilization targets into joint scenario planning.

Most Arctic jurisdictions relevant to this assessment currently operate at the Basic or low-Intermediate tier. Episodic engagement during named events. Resource commitments not formalized through charter. Coordination procedures not tested through exercise against the dual-theater scenario or its analogs. Where formalized P3 architecture has been built and exercised, the Nebraska Preparedness Partnership's twelve-hour logistics coordination during 2019 flooding, the Houston Supply Chain Group's COVID-19 response pivot, coordination moves at the pace disruption requires [33; 41]. Where it has not, coordination must be improvised under time pressure with predictably degraded results.

Alaska's Partnership for Infrastructure Protection provides a state-level precedent for formalized cross-sector coordination, though its scope and activation parameters under dual-theater disruption conditions remain a subject for follow-on analysis. The Alaska Mass Care Group and Feeding Task Force, led by the Salvation Army under the Alaska Division of Homeland Security and Emergency Management, already operates a functioning multi-agency coordination mechanism integrating state agencies, nongovernmental organizations, tribal contacts, and private-sector food vendors [55]. The Alaska Food Value Chain Coordination Council, convened in October 2025 under the USDA Regional Food Systems Partnership framework, demonstrates that cross-sector coordination infrastructure can be built when given institutional structure [3].

The Permanent Coordinating-Entity Gap

Open-source analysis identifies a structural gap at the federal level: the federal enterprise currently lacks a permanent coordinating entity for anticipating and responding to cross-sector supply chain disruption; existing responses rely on temporary, ad hoc task forces assembled after crises are underway [12]. A pivot-model Supply Chain Intelligence Center housed at DHS Headquarters that would leverage FEMA's Defense Production Act authorities, CISA's Critical Infrastructure Partnership frameworks, and peer-level coordination with DOT, DOC, and ODNI is one proposed remedy [12]. The proposal is significant because it precisely identifies the analytical architecture whose absence concentrates risk in Arctic early-phase disruption.

FEMA's own Supply Chain Resilience Guide establishes a five-phase methodology, Research and Mapping, Analysis, Outreach, Action, Assess and Refine, that operationalizes pre-disaster coordination through structured private-sector engagement, geospatial flow mapping, and public-private exercise stress testing [32]. CNA's Supply Chain Operational Engagement methodology provides an analytically validated framework for conducting that characterization and has already been applied in partial form to Alaska through the SCAN baseline assessment [15; 26]. The methodological scaffolding to close the coordination gap exists. What is missing is the deliberate, sustained application of that scaffolding to Arctic conditions.

> ***Bottom Line:*** *T*he presence or absence of pre-established cross-sector coordination architecture directly shapes whether disruptions stabilize or cascade. Coordination architectures that exist on paper but have not been exercised, formalized through charter, or integrated into lifeline operator business processes do not meaningfully reduce risk during Days 0–14. This is a direct application of the paper-redundancy principle to the governance layer: the existence of a framework is not equivalent to the operational capacity the framework describes. Closing this gap is an institutional investment with a pre-incident horizon, not a capability that can be improvised during disruption.

Contested Logistics: The Doctrinal Frame

The contested-logistics literature developed since 2022 provides the doctrinal frame within which institutional coordination becomes operationally significant. The literature is unusually convergent across independent sources on three claims that bear directly on Arctic mission assurance. Senior Army leadership articulated the contested-logistics priority at AUSA 2023, characterizing sustainment under contested conditions as a defining institutional challenge for the service over the coming decade and not a residual logistics concern [79].

First, sustainment is the pacing function in modern conflict. The 2025 National Defense Strategy and the contested-logistics modernization writing developed within the Department of Defense, Army Materiel Command's five sustainment imperatives, the Unified Pacific Wargame Series, the Naval Postgraduate School theses on supply chain resilience and crisis-standards prioritization, converge on the conclusion that the operational tempo of modern conflict is set by the rate at which sustainment can flow through pipelines that adversaries actively contest, not by the rate at which forces can maneuver. The Army's establishment of the Contested Logistics Cross-Functional Team in 2023 and the institutional priority articulated in the five sustainment imperatives, predictive logistics, autonomous reach, reduced logistics tail, alternative fuels, and Army watercraft strategy, establish that the analytical framing employed in this research reflects a recognized category of strategic vulnerability rather than a theoretical construct [34; 48; 64; 70].

Second, the homeland is no longer a sanctuary. Adversary actions can obstruct and disrupt power projection and sustainment operations within the United States, including through cyber intrusion, infrastructure interference, gray-zone tactics, and adversarial influence operations. The 2025 National Defense Strategy explicitly names the simultaneity problem, the possibility that adversaries could act in a coordinated or opportunistic fashion across multiple theaters and identifies adversary influence extending across the Arctic [42]. The dual-theater disruption scenario this research employs is structurally consistent with the simultaneity problem the current governing strategy identifies as a primary planning challenge.

Third, the commercial supply architecture optimized for cost efficiency under normal conditions is structurally vulnerable under contested conditions. Tight coupling produces

operational tempo but concentrates exploitable vulnerability at pipeline nodes; loose coupling creates the gaps between supply nodes where cascading failure most readily takes hold. The Arctic commercial lifeline architecture exhibits both conditions simultaneously, as the dependency mapping section of this research documents in detail. The contested-logistics literature converges on a conclusion that bears directly on the institutional coordination question: the response to contested logistics requires commercial and international partnership solutions that the current architecture does not adequately provide. Fuel delivery, identified as the ultimate last-mile challenge in contested environments, illustrates the convergence: the fuel supply chain depends on commercial operators, commercial vessels, commercial refining capacity, and commercial distribution infrastructure, none of which is under direct DoD control, all of which require coordination architecture that does not yet exist at the maturity level the scenario requires.

> ***Doctrinal Implication:*** The contested-logistics literature establishes that institutional coordination between defense and commercial actors is not adjacent to mission assurance, it is mission assurance. The connective infrastructure that enables coordination at the pace and depth modern conflict requires is the high-priority preparatory investment whose return is realized before disruption rather than during it.

Convergent Conclusion: Why Coordination Is the Risk Amplifier

The three treatments above, the P3 maturity model, the Jones Act waiver framework, and the contested-logistics literature, converge on the same operational conclusion. The capacity to adjudicate civilian–military resource competition, route emergency authorities through interagency review, and translate between defense mission requirements and commercial operating logic does not exist at the maturity level the dual-theater scenario requires. Each instance of paper redundancy at the infrastructure layer is mirrored by a corresponding paper redundancy at the institutional layer.

Three specific institutional gaps deserve explicit identification. First, the absence of pre-defined adjudication mechanisms for civilian–military resource competition means that allocation decisions are made under time pressure, with incomplete information, across authorities that have no established shared framework. Installation commanders hold Title 10 authority over on-base resource allocation but have no control over commercial cargo prioritization or civilian workforce availability. State authority can reallocate scarcity across competing populations but cannot create supply. Commercial operators make prioritization decisions based on commercial logic that does not formally account for defense sustainment requirements.

Second, cross-border coordination architecture is structurally underdeveloped. The RAND governance gaps analysis finds existing mechanisms inadequate for the pace of change now underway. Security dialogue mechanisms have atrophied since 2014. The ALCAN corridor, the Northern Sea Route, and the bilateral agreements with Canada that the dual-theater scenario implicates each operate within a coordination architecture that was not designed for simultaneous, multi-theater disruption.

Third, the connective infrastructure between defense, commercial, nongovernmental, tribal, and academic actors that contested-logistics doctrine identifies as essential does not currently exist at the maturity level Arctic mission assurance requires. Episodic engagement, transactional contracting, and incident-response-only coordination are different from a sustained, trust-based, operationally tested intermediary architecture that can translate between defense mission requirements and the privately operated systems that sustain them. This is the institutional gap that frames the recommendations and the PSEMA network section of this report.

> ***Bottom Line:*** Institutional coordination is not a passive backdrop to the cascading failure pathways this research documents. It is itself a primary cascade trigger, and its absence operates as a risk amplifier on every other failure mode.

PRELIMINARY MITIGATION AND COST-BENEFIT PRIORITIZATION

The Preliminary Mitigation and Cost–Benefit Prioritization deliverable moves from vulnerability identification to actionable response. It evaluates high-impact resilience options against five operational criteria, Mission Impact, Cascade Risk, Time Sensitivity, Feasibility, and Dual-Use Value, and identifies the limited set of preparatory investments whose return is realized before disruption rather than during it. The framework is intentionally structured to surface the gap between paper redundancies and operational redundancies, and to distinguish investments that close that gap from those that simply formalize a redundancy that does not function under stress.

The mitigation framework below addresses the failure pathways that emerge under the scenario conditions documented in the Scenario Framework section. The framework is operational in posture, not doctrinal: each mitigation category identifies what the investment is, why it ranks where it does, what its limits are, and how it interacts with the other categories. The objective is not a comprehensive Arctic resilience inventory; it is the limited set of investments that the analytical record identifies as most consequential for closing the gap between current capacity and what the dual-theater scenario requires.

Implication for Mitigation Strategy

Operative phase for this section: Days 0–14 as the decision window, with mitigation timelines evaluated against extension into Days 10–45. Full scenario architecture is developed in the Scenario Framework section.

The operating environment laid out in the research reorders mitigation priorities. Mitigation must prioritize early-phase survivability, throughput preservation rather than infrastructure protection alone, and cross-sector cascade prevention, not as separate workstreams but as integrated design criteria applied to each candidate intervention.

Four implications follow directly from the dependency map and scenario architecture:

Days 0–14 is the decision window, not the response window. Sustainment decisions are made under uncertainty, before federal augmentation arrives, and before the cascade has fully propagated. Mitigation investments that come online after Day 14 address conditions that are already established. Survivability investments must be pre-positioned, pre-coordinated, and pre-authorized to take effect inside this window.

Throughput, not capacity, is the operative metric. The dependency mapping establishes that infrastructure can remain nominally intact while throughput collapses below the level required to sustain the mission. Mitigation strategy must therefore evaluate candidate investments on their contribution to sustained operational throughput under degraded conditions, not on their nameplate capacity or theoretical redundancy. Paper redundancy is not redundancy.

Cascade prevention requires upstream intervention. Because fuel and communications sit at the head of the dependency chain, mitigation directed at downstream symptoms, food shortages, workforce attrition, household instability, addresses outputs of a cascade already in motion. Early-phase mitigation must be directed at the cascade triggers (maritime chokepoints, fuel storage and delivery, communications continuity, and the coordination architecture that links them) to prevent the propagation that drives downstream failure.

Private-sector lifelines must bear more of the early-phase sustainment burden. The dependency map confirms that the systems that fail first are commercially operated: ports, fuel distribution, food wholesale, communications carriers, and pharmaceutical distribution. These are the substrate on which mission assurance depends during the operative phase before federal sustainment can arrive. Mitigation strategy must therefore engage commercial operators as mission partners, not as regulated entities or contracted vendors.

The executive framing localizes to the Arctic theater and to its privately operated lifeline systems the simultaneity problem the 2025 National Defense Strategy identifies as a primary planning challenge across the Western Hemisphere (developed in the Institutional Coordination section) [70].

The remainder of this report develops the calibration that follows from this reordering: a prioritization framework, candidate mitigations evaluated against mission impact, cascade risk, time sensitivity, and cost-benefit, and the early-phase investments most likely to preserve throughput across the window in which the cascade is decided.

Prioritization Framework

Five criteria structure the evaluation of candidate mitigations. Mission Impact captures the degree of degradation to NORAD and USNORTHCOM operations when the candidate system fails or, conversely, the degree of mission preservation when the candidate mitigation operates as designed. Cascade Risk captures the cross-sector failure propagation potential of the system the mitigation addresses; ports, energy, and communications carry the highest cascade risk because their degradation initiates failures across multiple downstream lifelines simultaneously.

Time Sensitivity captures the failure timeline in hours and days under disruption conditions; food, fuel, and communications degrade most rapidly. Feasibility captures the implementation complexity in Arctic conditions, where infrastructure and workforce constraints raise the cost and lengthen the timeline of nominally straightforward investments. Dual-Use Value captures the civil and military benefit of the candidate, with the highest-return investments reinforcing whole-of-society resilience rather than serving narrow operational segments.

Extreme concentration risk.

The Port of Alaska handles five times more inbound marine cargo than any other Alaska port and is non-substitutable within the Days 0–14 window. The Seattle/Tacoma–Port of Alaska corridor is the dominant maritime resupply pathway for Southcentral Alaska. The same West Coast gateways are simultaneously the staging arteries for Pacific–Arctic defense logistics, meaning a single chokepoint sits at the convergence of civilian and defense supply chains.

Minimal redundancy and limited substitution capacity

Alternative pathways such as ALCAN, aviation, and Southcentral port substitution exist on paper but cannot deliver required throughput within the Days 0–14 window. This is the operational meaning of the paper-redundancy condition: capacity exists nominally, but throughput collapses below the level needed to sustain the mission.

Energy as the master dependency across all systems

The dependency chain Fuel → Power Generation → Heat → Water Systems → Communications → Operational Coordination governs all downstream lifeline behavior. As established in the Dependency Mapping section, the Alaska energy architecture is bimodal: the centralized Railbelt grid carries the majority of state electricity through a single interconnected corridor, while approximately 150 standalone microgrids serve geographically isolated communities outside that corridor without mutual-aid substitution. Fuel disruption simultaneously disables the response capacity that would otherwise compensate for the initiating event, and the bimodal architecture concentrates that compounding effect at the standalone-microgrid population, which has the fewest substitution options under the dual-theater scenario.

Communications as the coordination spine and failure amplifier

Communications systems depend on stable power and environmental conditions and degrade as a function of upstream energy failure. In contested or degraded conditions, communications failure operates as a force degrader, propagating across multiple lifeline categories from a single point of failure and disabling the coordination architecture on which civil–military response depends.

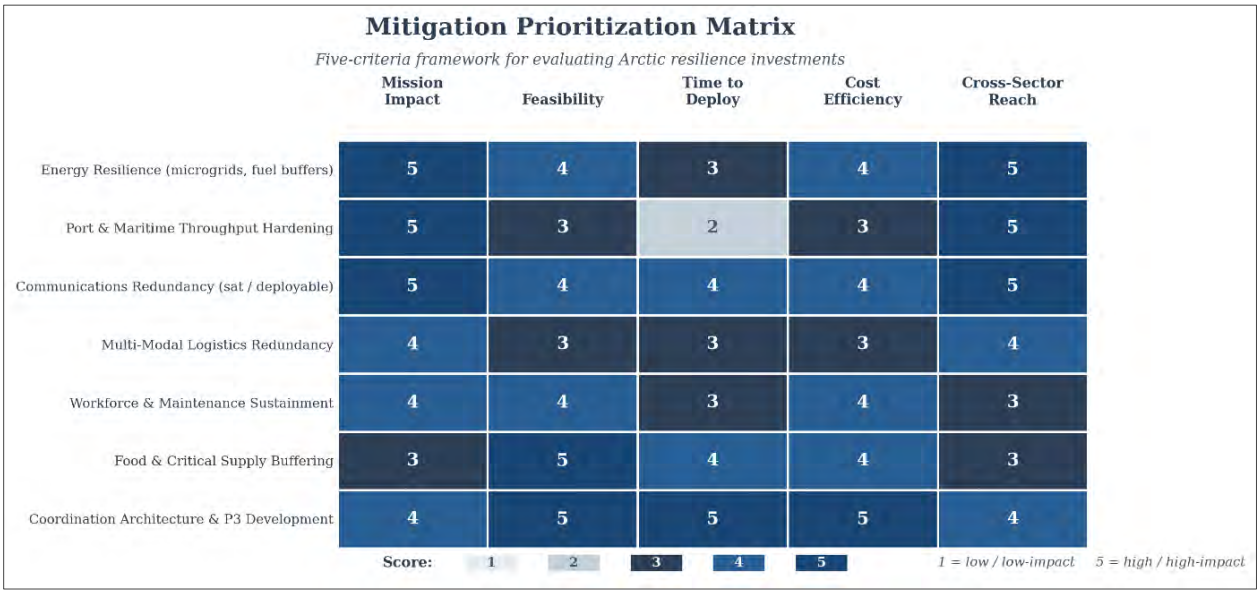
Rapid degradation timelines driven by just-in-time supply chains

Many Arctic systems operate with minimal buffer capacity, meaning upstream supply chain disruptions translate into localized shortages within days, particularly for food, fuel, and medical supplies. Seasonal asymmetry compresses these timelines further: winter disruptions eliminate the barge resupply window entirely, collapsing failure intervals from weeks to days because depleted reserves cannot be replenished until thaw.

These characteristics are not independent. They compound through a common cascade pathway in which a single chokepoint degradation propagates across energy, communications, food, medical supply, and workforce systems within the same compressed window in which sustainment decisions must be made under uncertainty.

Mitigation Prioritization Criteria

The five criteria are not weighted equally. Mission Impact and Cascade Risk are the primary criteria for ranking categories against one another. Time Sensitivity refines the prioritization within high-Mission-Impact categories by distinguishing investments that produce Days 0–14 benefit from those whose return is realized only over longer horizons. Feasibility shapes the recommendation pathway for each category, what can be done now, what requires capital investment with multi-year horizons, and what depends on regulatory or coordination architecture that must be built before it can operate. Dual-Use Value strengthens the case for any category that produces benefit beyond the defense mission, particularly in jurisdictions where investment decisions are shaped by civilian and tribal stakeholder priorities.



Each mitigation option is scored against all five criteria; the composite score informs tier placement in the recommendations roll-up.

The prioritization shown in the above image reflects the interaction of five evaluation criteria, applied consistently across all categories analyzed. Together, these criteria produce the relative ordering displayed in the matrix and form the basis for the tiered recommendations that follow. The criteria are defined below; with the relative weight each carries in the overall prioritization.

Mitigation Prioritization Criteria and Definitions

Criterion	Definition	Scoring Range
Impact magnitude	How much risk does the option reduce, and how broadly?	0–3 points
Time-to-effect	How quickly can the option be in place and operationally effective?	0–3 points
Cost efficiency	Risk reduction achieved per dollar of investment	0–3 points
Implementability	Authority, feasibility, contracting, and partnership requirements	0–3 points
Sustainability	Whether the option creates durable, not one-time, capability	0–3 points
Composite	Sum across all five criteria; informs tier placement	0–15 points

Priority Mitigation Categories

The risk and vulnerability assessment conducted under this research identifies seven mitigation categories that, applied in integrated fashion, materially reduce cascade risk during the Days 0–14 disruption window and beyond. The categories are structurally interdependent; failure in one accelerates degradation across the others. The integrated prioritization that closes this section reflects the strategic value each category contributes to the system as a whole, not a substitution hierarchy in which higher-ranked investments displace lower-ranked ones.

Each category targets a distinct point in the cascade chain, energy at the master-dependency layer, port and maritime throughput at the system's primary intake, communications across the coordination spine, multi-modal logistics in the distribution layer, workforce and maintenance in the operational layer, food and critical supply at the consumption layer, and coordination architecture spanning the entire chain. Together they constitute a layered mitigation architecture in which the value of each category compounds as adjacent categories are addressed in parallel. Investments made in only one or two categories produce limited returns; the resilience benefit emerges from coordinated investment across the set. The categories that follow are presented in order of priority for the Arctic mission-assurance context, but their integrated character means each remains essential regardless of where it falls in the ranking.

Fuel and Energy Resilience

The Dependency Mapping and Risk Assessment sections establish fuel as the master dependency in the Arctic operating environment. Energy resilience therefore ranks first in this prioritization: every other mitigation operates at degraded effectiveness, or fails outright, when the fuel and energy layer is not addressed.

Cost-Benefit Snapshot: Distributed Microgrids (Hybrid Systems)

Cost driver: Capital cost of hybrid generation, storage, and renewable inputs; sustained workforce and maintenance investment is the binding cost factor, not equipment.

Benefit driver: Reduces fuel-import exposure at the standalone microgrid layer where mutual aid is unavailable; reliability-over-penetration design hedges against the workforce-airlift-fuel cascade.

Feasibility: Moderate. Limited by trained-lineman shortage, transformer lead times, and the absence of mutual aid agreements at the standalone-microgrid population.

Time horizon: Medium to long. Hardware deploys on a multi-year horizon; operational benefit requires sustained workforce investment in parallel.

Cost-Benefit Snapshot: Increased Fuel Storage and Pre-Positioning

Cost driver: Capital cost of refined-product storage capacity, distributed across regional and installation locations; calendar timing constraint dominates capital cost as a planning variable.

Benefit driver: Extends the sustainment window beyond the documented 7–10 day commercial inventory threshold; the only mitigation in this category that delivers Days 0–14 benefit reliably.

Feasibility: High when executed before fall barge season closure; categorically infeasible after ice-in until spring breakup if reserves were not pre-positioned.

Time horizon: Short. Pre-positioning effectiveness is determined by calendar timing rather than capital investment and supports Days 0–14 sustainment if executed pre-event.

Cost-Benefit Snapshot: Regional Energy Hubs

Cost driver: High capital cost for shared infrastructure (distributed storage, hybrid generation, interoperable grid capacity) serving communities, installations, and other critical nodes.

Benefit driver: Hedges against two converging risks: scenario disruption and the projected 2030 Cook Inlet natural gas depletion. Dual-use civilian and installation benefit aligns with NSM-22 and DoDD 3020.40 [26; 69;80].

Feasibility: Moderate. Multi-stakeholder coordination across communities, installations, and utilities is the binding feasibility factor; technical engineering is well-established.

Time horizon: Long. Hub investment is a multi-year commitment whose value materializes regardless of whether the scenario occurs.

Cost-Benefit Snapshot: Pre-Scripted Regulatory Waiver Protocols

Cost driver: Administrative effort only; no capital investment, no construction timeline, no workforce expansion. Cost is in pre-event documentation and interagency coordination.

Benefit driver: Compresses Jones Act, weight-limit, hours-of-service, CDL endorsement, and Canadian provincial waiver activation timelines from real-time crisis adjudication to pre-staged activation; return is measured in days saved when days are most expensive.

Feasibility: High. Waiver mechanisms exist in current federal and state frameworks; pre-staging is achievable within current authorities without legislative change.

Time horizon: Short. Activation occurs within the Days 0–14 window if pre-staging is complete; the highest-return-per-dollar option in this category.

Coordination Architecture & Private-Public Partnership Development

The Risk Assessment section and the Substitution and Redundancy Reality Check establish coordination architecture as a risk amplifier in its own right. The four options developed below address that coordination gap and the tempo mismatch stakeholder engagement specifically surfaces: commercial supply chains operate on hours timelines while government activation operates on days' timelines, and pre-staged architecture must include commercial-pace activation triggers.

The disaster-diplomat framing developed in the homeland security practitioner literature characterizes the emergency management function not as a technical or logistical role but as a sustained cross-sector intermediation function whose effectiveness depends on pre-incident relationship investment rather than incident-response activation [53]. The framing is directly applicable to the Arctic civil–military adjudication gap this section addresses.

Cost-Benefit Snapshot: Formalize Cross-Sector Coordination Charters

Cost driver: Relational and institutional investment; modest documentation and convening costs. Building from established precedent (Alaska Partnership for Infrastructure Protection) reduces baseline cost.

Benefit driver: Converts episodic coordination relationships into documented capability that survives personnel transitions, jurisdictional changes, and the operational pace of disruption. Business Emergency Operations Center architecture pre-establishes authority, information flow, and decision sequencing.

Feasibility: High. Builds on existing federal and state frameworks; charter expansion or parallel charter development to cover civil-military adjudication is the principal feasibility step.

Time horizon: Short to medium. Charter formalization is achievable within months; exercised operational capability requires a multi-year exercise cadence.

Cost-Benefit Snapshot: Advance Program Maturity

Cost driver: Sustained relationship and exercise investment over multi-year horizon; funding for personnel continuity, exercise programs, and stabilization-target framework integration.

Benefit driver: Movement from Basic toward Advanced P3 tier produces compounding capability across successive disruption cycles; converts coordination from a personal-relationship dependency into a documented, exercised capability.

Feasibility: Moderate. Capability does not exist at scenario-required level today; advancement requires sustained pre-event investment that current funding cycles do not consistently support.

Time horizon: Long. Multi-year horizon; Days 0–14 benefit depends entirely on pre-event maturation and cannot be reconstituted at pace once disruption is underway.

Cost-Benefit Snapshot: Resource the Disaster Diplomat Function

Cost driver: Personnel cost: emergency management staffing positions and defined authority at state and local levels matched to scenario-conditioned coordination demand rather than to historical disaster experience.

Benefit driver: Operates as the workforce precondition for charter formalization and program maturity; absent staffing, the preceding two options produce documented architecture without personnel capacity to operate it [53].

Feasibility: High administratively (staffing positions and authority definition); moderate politically and fiscally because emergency management capacity is typically resourced against historical disaster experience.

Time horizon: Short to medium. Personnel hiring and authority definition is achievable within current cycles once funding is committed; full coordination capability requires sustained tenure.

Cost-Benefit Snapshot: Leverage Existing Federal Methodology and Data Infrastructure

Cost driver: Application cost rather than development cost; FEMA Supply Chain Resilience Guide five-phase methodology and CNA SCOPE framework are validated outside Arctic context and partially applied within it [33].

Benefit driver: Highest analytical return per dollar in this category because foundational methodology development is complete; requirement is application and adaptation, not invention. Aligned NSM-22 and 2025 NDS policy substrate further reduces speculative risk [70].

Feasibility: High. Federal methodology, partial Arctic application, proposed Supply Chain Intelligence Center architecture, and aligned policy substrate position this option as materially less speculative than equivalent investments in less-developed analytical domains.

Time horizon: Short to medium. Methodology application begins immediately; integrated federal coordinating architecture is a multi-year build operating against established precedent.

Port and Maritime Throughput Resilience

The Dependency Mapping and Risk Assessment sections establish the Port of Alaska as a no-fail node with no rapid substitute. The three options developed below operate against a substitution baseline of zero: redundant port capacity addresses the long-horizon investment gap, pre-positioned stockpiles address the Days 0–14 window where substitution is unavailable, and surge offload capability addresses the intermediate gap between the two.

Cost-Benefit Snapshot: Redundant Port Capacity

Cost driver: Very high capital cost across the six 2023 Baseline Assessment alternatives (Seward, Valdez, Kodiak, Homer, Whittier, MacKenzie); each requires specific infrastructure investment before functioning as substitute.

Benefit driver: Long-horizon strategic resilience against the Port of Alaska single-point-of-failure condition; addresses the vulnerability the Baseline Assessment confirms has no current viable alternative.

Feasibility: Low to moderate. Every identified option requires capital investment before functioning as real redundancy; financing pathway, environmental permitting, and rail/road integration are binding constraints.

Time horizon: Long. Multi-year construction horizon places mitigation value beyond the scenario window; near-term burden falls to pre-positioned stockpiles and surge offload.

Cost-Benefit Snapshot: Pre-Positioned Strategic Stockpiles

Cost driver: Moderate capital cost for stockpile depth across fuel, food, and repair materials; geographic distribution across Seward, Valdez, Fairbanks, and JBER drives deployment cost more than per-unit inventory cost.

Benefit driver: The only option in this category capable of producing operationally meaningful effect within the Days 0–14 window; bridges the gap between disruption onset and substitution availability.

Feasibility: High. Storage infrastructure, inventory rotation, and geographic distribution are well-understood operational requirements; binding constraint is sustained funding for inventory rotation rather than initial deployment.

Time horizon: Short. Pre-positioning delivers Days 0–14 benefit if stocks are in place before disruption occurs; the lowest-capital, shortest-horizon option in this category.

Cost-Benefit Snapshot: Surge Offload Capability

Cost driver: Very high capital cost for polar icebreaker fleet expansion (current operational fleet of two against documented 8–9 vessel requirement) and shore infrastructure investment (more than seven billion dollars in backlog) *[68; 21]*.

Benefit driver: Theoretical intermediate pathway between near-term stockpile reliance and long-horizon port construction; retains marginal value for targeted resupply of installations and high-priority community nodes.

Feasibility: Low to moderate. Polar Security Cutter program cost growth and schedule delay defer fleet capacity; no deep-draft port exists within 800-plus miles of the Bering Strait.

Time horizon: Long. Cannot be assumed available at scenario-required scale and location within the operational window; not a reliable category substitute for Port of Alaska throughput.

Communications and Data Resilience

The Dependency Mapping and Risk Assessment sections establish communications failure as the force degrader operating across every other lifeline category. The three options developed below therefore address a vulnerability whose effects compound across all six other mitigation categories in this section.

Cost-Benefit Snapshot: Multi-Layer Communications Architecture

Cost driver: Moderate to high capital cost for LEO/HEO satellite capacity, HF radio backup, and terrestrial redundancy; contractual and policy reinforcements (liability, segmentation, cybersecurity standards) add policy-development cost.

Benefit driver: Foundational mitigation pathway for Arctic communications resilience; produces real redundancy when each layer carries essential traffic independently of the others rather than sharing upstream nodes.

Feasibility: High technically; moderate at the governance layer. Space is not formally designated critical infrastructure, so coordination, liability, and resilience frameworks do not cleanly apply to the commercial satellite systems Arctic operations depend on.

Time horizon: Medium. Satellite capacity contracting and HF backup deployment achievable within months; correlated-failure architectural review is the analytically time-consuming step.

Cost-Benefit Snapshot: Protected & Resilient Networks

Cost driver: Capital cost for anti-jam capability, redundant pathway design, and subsea cable risk mitigation; correlated-failure analysis adds analytical cost as the most underweighted element of current Arctic communications planning.

Benefit driver: Addresses threat-side and physical-infrastructure-side vulnerabilities multi-layer architecture alone does not resolve; cable repair under Arctic winter conditions cannot be assumed within the Days 10–45 escalation window.

Feasibility: Moderate. Anti-jam capability and pathway redundancy are established technical disciplines; subsea cable risk mitigation is constrained by vessel availability, ice conditions, and seasonal repair envelopes.

Time horizon: Medium to long. Anti-jam capability deployable on shorter horizons; subsea cable redundancy and landing-point diversification are multi-year infrastructure investments.

Cost-Benefit Snapshot: Data & Monitoring Expansion

Cost driver: Moderate capital cost for SCADA expansion across fuel storage, electric generation, water and sanitation, and transportation nodes; real-time logistics visibility platform integration is the principal cost driver.

Benefit driver: Converts the other communications capabilities into decision capabilities; in the Days 0–14 window, materially shortens the interval between failure onset and decision-maker awareness when substitution options are closing fastest.

Feasibility: High. SCADA, remote diagnostics, and logistics-visibility technologies are mature; binding constraint is operator integration and pre-event sensor placement.

Time horizon: Short to medium. Sensor deployment and platform integration achievable within months to a year; full multi-modal logistics visibility requires sustained operator coordination.

Workforce & Maintenance Capacity (Hidden Dependency)

The Dependency Mapping section and the Stakeholder Engagement findings establish workforce degradation as a force degrader and the most binding operational constraint across sectors. Hardware redundancy alone cannot resolve this constraint; an idle pre-positioned generator, a grounded reefer container, or a parked tanker delivers no operational value when the certified operators, mechanics, and dispatchers required to run them are absent or unreachable. The Arctic compounds this dependency through small population centers, narrow specialty pipelines, and seasonal contraction in the available workforce, conditions that mean a localized illness cluster, a single retirement, or a cross-border travel restriction can disable capability that no amount of capital investment can replace on the same timeline. The three options developed below address the operator-and-maintainer dependency that hardware availability cannot resolve on its own, with each option targeting a different layer of the workforce-resilience problem: building local capacity, sustaining the technicians who maintain installed systems, and broadening the skill pool so single-point-of-failure roles do not bring entire sectors to a halt.

Cost-Benefit Snapshot: Local Workforce Development

Cost driver: Sustained multi-year investment in technical trades training across energy, infrastructure maintenance, and logistics; training infrastructure, instructor capacity, and apprenticeship programming.

Benefit driver: Mitigates the binding workforce constraint identified throughout the research; reduces sustained dependence on personnel deployment under exactly the conditions that make personnel deployment most difficult.

Feasibility: Moderate. Training infrastructure exists in partial form; binding constraint is multi-year horizon between investment and operational workforce delivery.

Time horizon: Long. Trade-skilled workforce development operates on multi-year horizons that do not deliver Days 0–14 benefit but determine whether Days 0–14 response continues to depend on flown-in personnel.

Cost-Benefit Snapshot: Regional Maintenance Hubs

Cost driver: Capital cost for pre-positioned equipment, tools, and spare parts at hub locations plus sustained operator presence; geographic catchment design and transportation pathway investment.

Benefit driver: Reduces maintenance response distance and time at the standalone microgrid layer; addresses the geography problem that workforce development alone cannot resolve.

Feasibility: Moderate. Three-component (parts, personnel, transportation) integration is the binding feasibility factor; hubs designed without all three reproduce paper-redundancy condition.

Time horizon: Medium. Hub establishment achievable within a year to two; sustained operator capacity at hub locations requires ongoing workforce investment.

Cost-Benefit Snapshot: Cross-Training Across Lifelines

Cost driver: Moderate cost for multi-lifeline training curriculum development and instructor capacity across power, water, communications, and fuel systems; cost concentrated in remote-community training delivery.

Benefit driver: Expands the available trained-personnel pool in remote communities where the population base does not support specialized full-time positions; produces redundancy against operator unavailability that single-discipline staffing cannot absorb.

Feasibility: High. Multi-skilled-operator training is established practice in remote-community contexts; binding constraint is sustained training delivery to the rural communities where the option produces most benefit.

Time horizon: Short to medium. Initial cross-training certifications achievable within months; full multi-lifeline operator pool requires sustained training cadence.

Multi-Modal Logistics Redundancy

The Dependency Mapping section, the Risk Assessment section, and the Berlin Airlift Anchor establish the Arctic modal hierarchy as maritime-primary with aviation and overland routes in critical-but-capacity-limited and constrained-redundancy tiers. The three options developed below must therefore be evaluated against airfield throughput ceilings, ALCAN seasonal and sovereignty constraints, and the integration vulnerability at handoff points. Multi-modal redundancy is real-redundancy only when the integration mechanism that synchronizes modes is itself pre-staged.

Cost-Benefit Snapshot: Pre-Identified Rerouting Plans

Cost driver: Two distinct cost categories: pre-scripted regulatory pathway activation (administrative effort, low capital) and physical pre-staging of driver support, lodging, and fuel infrastructure along the corridor (high capital, multi-year horizon).

Benefit driver: Activates the constrained-redundancy ALCAN tier as an operationally accessible substitution mechanism; community last-mile delivery (sled teams, snowmobile networks, small-boat distribution, airdrop reception) provides delivery capacity that does not depend on commercial logistics workforce.

Feasibility: High for regulatory pathway activation; low to moderate for physical pre-staging absent multi-year capital commitment to corridor support infrastructure.

Time horizon: Mixed. Regulatory pathway activation deliverable within Days 0–14 window if pre-scripted; physical pre-staging is a long-horizon investment that determines whether theoretical surge capacity is operationally accessible at all.

Cost-Benefit Snapshot: Airlift Prioritization Framework

Cost driver: Administrative cost only; framework development, pre-scripted prioritization protocols, and exercised activation procedures. No capital investment required.

Benefit driver: Triages high-value, low-volume sustainment (pharmaceuticals, repair parts, critical personnel, time-sensitive components); prevents allocation of scarce lift to commodities the mode cannot meaningfully sustain. Berlin Airlift Anchor establishes the empirical benchmark.

Feasibility: High. Framework development is achievable within current authorities; binding constraint is interagency agreement on prioritization criteria across DoD, FEMA, and state authorities.

Time horizon: Short. Framework activatable within Days 0–14 window if pre-scripted; the most analytically constrained but operationally honest option in this category.

Cost-Benefit Snapshot: Intermodal Coordination Systems

Cost driver: Moderate cost for synchronization infrastructure across maritime, rail, road, and air movement; coordination authority, communications infrastructure, and pre-scripted prioritization protocols.

Benefit driver: Converts the multi-modal mode portfolio from a collection of theoretical alternatives into an operational sustainment system; addresses the integration vulnerability that the preceding two options cannot resolve on their own.

Feasibility: Moderate. Analytically dependent on data and monitoring infrastructure addressed in the Communications and Data category; without real-time visibility, intermodal coordination defaults to scheduled handoffs that cannot adapt to ongoing degradation.

Time horizon: Medium. Coordination authority establishment achievable within months; full real-time multi-modal synchronization requires the Communications and Data category investments to be in place.

Food & Critical Supply Buffering (Early-Phase Stability)

The Dependency Mapping and Risk Assessment sections establish Alaska's food system as structurally bimodal across commercial and subsistence pathways, operating against a wholesale chokepoint that emergency feeding doctrine itself depends on. The five options developed below must therefore operate downstream of that chokepoint, address both pathways rather than only the commercial one, and distinguish near-term mitigation horizons (caches, cold chain, prioritization, coordination leverage) from long-horizon investments (processing infrastructure) whose value lies beyond the scenario window.

Cost-Benefit Snapshot: Distributed Food Caches

Cost driver: Capital cost for cache inventory across urban and rural staging locations; geographic distribution downstream of the wholesale chokepoint is the design-driving cost factor.

Benefit driver: The only mitigation in this category capable of inserting real redundancy downstream of the supply chain chokepoint within the Days 0–14 window; rural cache pre-positioning addresses the documented first-phase gap before outside assistance can arrive.

Feasibility: High when caches are positioned downstream of the wholesale chokepoint and use shelf-stable, cold-chain-independent inventory; positioning upstream reproduces the chokepoint condition the mitigation is intended to address.

Time horizon: Short. Pre-positioning delivers Days 0–14 benefit if caches are in place before disruption; the operative condition is downstream geographic positioning rather than capital scale.

Cost-Benefit Snapshot: Cold Chain Resilience

Cost driver: Capital cost for backup power and storage infrastructure protecting perishable inventories; backup generation independence from disrupted fuel and grid restoration is the binding cost factor.

Benefit driver: Addresses the failure mode that converts distributed food caches and existing commercial inventory from real to paper redundancy under scenario conditions; cold chain failure converts inventory into loss faster than any other failure mode.

Feasibility: Moderate. Backup generation that depends on the same fuel supply chain or grid restoration produces paper redundancy at the moment of greatest stress; correlated-failure mitigation is the binding design factor.

Time horizon: Short to medium. Backup power deployment achievable within months; sustained correlated-failure independence requires multi-year fuel-source diversification.

Cost-Benefit Snapshot: Prioritization Protocols

Cost driver: Administrative cost only; protocol development, civilian-military alignment framework, and pre-event interagency agreement. No capital investment required.

Benefit driver: Converts ad-hoc adjudication under crisis conditions into pre-staged decision frameworks activatable within scenario decision cycles; addresses the workforce overlap that makes civilian-military protocols inseparable.

Feasibility: Moderate. Pre-event protocol development is achievable within current authorities; binding constraint is reaching civilian-military and interagency agreement on the workforce-overlap framing.

Time horizon: Short. Protocols activatable within Days 0–14 window if pre-staged; the more time-sensitive of prioritization and rationing because it governs sequence of sustained support.

Cost-Benefit Snapshot: Coordination Architecture as a Leverage Point

Cost driver: Administrative cost only; integration of defense installation feeding dependencies and contractor workforce continuity into the Mass Care Group, Feeding Task Force, and Alaska Food Value Chain Coordination Council. No new institution-building required.

Benefit driver: Low-capital, high-return resilience investment; the underlying coordination mechanisms are already operational and surface transportation, cold storage, and communications gaps in their inaugural sessions.

Feasibility: High. The leverage point is formal integration rather than new institution construction; integration delivers Days 0–14 benefit because the underlying mechanisms are already operational.

Time horizon: Short. Formal integration achievable within months; ongoing benefit accrues across each subsequent disruption cycle.

Cost-Benefit Snapshot: Processing Infrastructure as a Long-Horizon Resilience Investment

> **Cost driver:** Very high capital cost for USDA-certified processing facilities and FDA-certified dairy capacity; long-horizon construction and certification timelines.
>
> **Benefit driver:** Changes the starting position from which future disruptions are absorbed by reducing the baseline import-dependency ratio; not a near-term scenario mitigation.
>
> **Feasibility:** Low to moderate. Cannot substitute for imported food at meaningful scale within the Days 0–14 window or the Days 10–45 escalation window; categorized as recommendation for follow-on funding rather than current-cycle deliverable.
>
> **Time horizon:** Long. Multi-year horizon; mitigation value is realized over decades by changing baseline conditions rather than by responding to the modeled scenario.

Integrated Prioritization (Final Ranking)

Rank	Mitigation Area	Strategic Value
1	Fuel and Energy Resilience	Foundational (enables all systems)
2	Coordination Architecture and P3 Development	Foundational (enables all other mitigation)
3	Port and Maritime Throughput Resilience	Throughput critical
4	Communications and Data Resilience	Coordination backbone
5	Workforce and Maintenance Capacity	Sustains all mitigation
6	Cross-Modal Logistics Redundancy	Enables adaptation
7	Food and Critical Supply Buffering	Stabilizes population

Strategic Takeaways

The priority mitigation categories identified in this research reflect a set of consistent analytical conclusions that extend beyond any single category or investment. Taken together, these conclusions reframe how Arctic resilience must be understood, not as an infrastructure procurement problem, but as a systems design and institutional coordination challenge operating under compressed timelines and extreme operating conditions.

There Is No Single-Point Solution

No individual investment, policy change, or infrastructure upgrade resolves the vulnerabilities identified in this research. Alaska's lifeline architecture fails under dual-

theater disruption because it is structurally concentrated, operationally thin, and institutionally fragmented across multiple sectors and jurisdictions. Resilience sufficient to sustain Arctic mission assurance requires layered mitigation across energy, maritime throughput, communications, coordination architecture, workforce, logistics, and food systems simultaneously. Sequencing matters, energy, and coordination architecture are the foundational categories on which others depend, but the overall architecture is layered, not linear.

Early-Phase (Days 0–14) Is Decisive

The analytical record consistently establishes that mission impact accumulates within the first two weeks of disruption, that the window for intervention closes faster than linear degradation models suggest, and that the convergence of fuel, logistics, communications, and workforce stressors produces non-linear escalation once thresholds are reached. Investment prioritization must therefore reflect Days 0–14 benefit as a primary criterion, not a secondary one. Long-horizon investments, secondary port construction, processing infrastructure expansion, structural workforce capacity-building, are necessary but cannot substitute for near-term investments that close the early-phase gap.

Infrastructure Alone Is Insufficient

The paper redundancy pattern documented throughout this research means that physical investment without complementary institutional investment produces nominal capacity that does not perform under stress. The maturity progression of public-private partnership architecture, the pre-staging of regulatory waiver pathways, and the integration of defense requirements into civilian coordination mechanisms are not adjacent to the infrastructure investments, they are conditions under which the infrastructure investments produce operational benefit. Closing both gaps simultaneously is required; closing either alone leaves the architecture vulnerable to the same failure modes the investments were intended to prevent.

The Highest Return-on-Investment Mitigation Options Are Not Always the Most Capital-Intensive

Coordination architecture and Public-Private Partnership development rank second in cascade reduction value, behind fuel and energy resilience, despite the lowest capital cost in the prioritized framework. Pre-scripted regulatory waiver protocols, pre-positioned strategic stockpiles, and exercised cross-sector coordination charters yield Days 0–14 benefit at orders-of-magnitude lower cost than the secondary port construction or the polar icebreaker fleet expansion that would yield comparable physical resilience benefit only over multi-year horizons. The investment portfolio that produces the greatest near-term cascade reduction is not the most capital-intensive; it is the portfolio that closes the institutional and coordination gaps that paper redundancy has masked.

Arctic Resilience Is a Specific Application of Institutional Coordination Doctrine

The contested-logistics literature, the FEMA Public-Private Partnership maturity model, the Jones Act waiver framework, and the National Security Memorandum 22 framing of defense critical infrastructure converge on the same conclusion: institutional coordination between defense and commercial actors is mission assurance, not adjacent to it. Arctic mission assurance is therefore not a unique problem requiring novel coordination doctrine; it is a specific application of doctrine that the broader institutional record has been developing for years. The Arctic case is distinguished by the geographic concentration, seasonal asymmetry, and cross-border complexity that make the doctrinal failures more consequential, not by requiring different doctrine to fail.

The Paper-Redundancy Gap Is a Planning-Compliance Failure, Not an Engineering Problem

Throughout the research, the gap between documented redundancy and operationally available redundancy is consistently traced to planning frameworks that substitute capacity figures for throughput figures, which treat contractual relationships as operational guarantees, and that assume coordination architectures perform at the maturity level the documents describe rather than the maturity level that has actually been built. Closing the gap requires planning frameworks that maintain the capacity-versus-throughput distinction, integrate stress-condition assumptions into redundancy calculations, and treat institutional coordination maturity as a measurable variable rather than a documented commitment. This is a planning-compliance discipline, not an engineering challenge.

Across every sector represented in stakeholder engagement, practitioners identified workforce availability as the most binding constraint determining whether operations curtail, ration, or stop, a cross-sector convergence that does not alter the master-dependency role of fuel in the integrated prioritization but that elevates workforce and maintenance capacity from a planning consideration to a mission-assurance variable warranting commensurate investment attention.

Integrated Prioritization

Taken in integrated form, the seven mitigation categories reflect a strategic ranking that organizes investment sequencing rather than substitution. Fuel and energy resilience is foundational because it enables all other systems. Coordination architecture and private-public partnership development is foundational because it enables all other mitigation. Port and maritime resilience is throughput-critical. Communications and data is the coordination backbone. Workforce capacity sustains all mitigation. Cross-modal logistics redundancy enables adaptation. Food and supply buffers stabilize the population. The ranking is structural, not exclusive: an investment program that addresses only the top categories without reinforcing the lower-ranked ones produces point improvements that cascade pathways shift around rather than eliminate.

STAKEHOLDER ENGAGEMENT

Stakeholder engagement methodology, instrument design, and the three engagement principles are documented in the Methodology section. Question architecture and the core question set are documented in Appendix E. The substantive engagement findings are presented below, organized first by sector, then synthesized across sectors to identify cross-sector convergences and divergences carried forward into the Mitigation and Recommendations sections.

Sectors Engaged

Engagement across the lifeline sectors examined in this research drew on three complementary inputs: a structured survey instrument distributed across sector contacts, informal conversations with operators and subject-matter experts conducted under the three-month performance window, and a literature review of authoritative public sources, sector reporting, federal and state planning documents, and prior PSEMA engagement records. Findings in this section reflect the convergence of these inputs rather than a structured interview program; where a finding rests primarily on one input, the attribution is identified.

Energy and Fuel

The analysis examined power generation, bulk fuel, aviation fuel, heating fuel, and the single-source aviation gasoline dependency at the Port of Alaska. The analytical scope included the in-state refinery operators representing the approximately twenty percent of Alaska oil refined in-state; the primary Railbelt utility serving Joint Base Elmendorf-Richardson and the Anchorage area, including the Cook Inlet natural gas dependency and its projected 2030 depletion trajectory; and Western Alaska diesel distribution operators representing upstream dependency nodes distinct from the Port of Alaska corridor.

The literature and operator reporting consistently indicate that stated reserve windows do not reliably distinguish between summer and winter operating conditions; reserve figures presented in days-of-supply terms frequently reflect summer or normal-season inventory cadences and do not account for the full winter window between fall barge delivery and spring breakup. The single-source dependency at the Port of Alaska for aviation gasoline is a structural condition documented across operator filings and state planning records, with no immediate mitigation identified. The Cook Inlet natural gas depletion trajectory is a pre-existing concern that compounds maritime fuel-import dependency under any disruption scenario [26]. Informal conversations with sector contacts reinforced the workforce constraint: trained linemen are scarce, transformer lead times have lengthened due to global supply chain conditions, and remote microgrid repair depends on a workforce that must be flown in using the same fuel that is constrained.

Transportation and Logistics

The analysis examined ports, including the Port of Alaska and the Tacoma-to-Alaska maritime corridor, maritime carriers, trucking, and aviation cargo. Analytical scope specifically included the six identified contingency alternatives to the Port of Alaska, Seward, Valdez, Kodiak, Homer, Whittier, and MacKenzie, to surface the specific infrastructure gaps and investment requirements that determine whether each could function as a substitute under disruption conditions. Alaska Railroad analysis focused on the emergency freight activation pathway, including winter-operations constraints, fuel transport capacity, and the rail-to-port integration that would be required for any container cargo rerouting scenario. ALCAN Highway analysis drew on Alaska DOT&PF and Canadian provincial transportation authority documentation to examine the pre-scripted waiver pathways, service-interval conditions, and pre-staging requirements that determine whether the highway's theoretical surge capacity is operationally accessible.

The throughput-versus-capacity distinction was reinforced across both the literature and informal conversations with sector contacts. Port operations reporting consistently establishes that nameplate capacity figures do not reflect operational throughput under crew, equipment, or weather constraints. ALCAN reporting and informal conversations distinguished between theoretical road capacity and operational truck throughput under emergency winter conditions absent pre-staged fueling and rest infrastructure. Aviation sector reporting characterizes airlift as a constrained supplemental mode, not a substitute for maritime cargo at sustainment volumes. The workforce-mobility waiver issue surfaced consistently across the literature: cross-border trucking surge along the ALCAN depends on Canadian provincial permitting, hours-of-service waiver activation, and CDL endorsement coordination that is not pre-positioned [22].

Food and Retail Distribution

The analysis examined wholesale distribution, grocery, cold chain, warehouse operations, and the four-day Tacoma-to-shelf transit cycle. Analytical scope included the primary food service distributors operating from limited Alaska locations; the sole produce distributor operating from a single Anchorage location; and the dominant retail tier whose geographic and organizational concentration creates the single-pipeline architecture identified in the dependency mapping.

Sector engagement and the literature converge on the structural conditions established in Dependency Mapping: the four-day Tacoma-to-shelf cycle and the absence of major in-state warehousing mean that stated reserve figures reflect inventory in transit and on shelves, not warehouse storage, and disruption of the replenishment cycle produces shortfall faster than stated reserve windows suggest. The single-source concentration at the wholesale tier (Charlie's Produce as the primary produce node) is documented in state distribution analyses. [38] The three-figure buffer reconciliation developed in Food Security and Access (seven-to-ten-day commercial pipeline, three-to-seven-day retail warehousing, two-to-three-day effective availability under demand surge) is operationally consistent with what practitioners describe. The single-source concentration at the wholesale tier is documented in state distribution analyses. The continued dependency of state feeding operations on the same commercial pipeline

whose disruption would trigger plan activation is a paper-redundancy condition documented across state planning records [3; 24; 55].

Communications and Data Systems

The analysis examined satellite operators, fiber networks, commercial carriers, and Arctic ground station operators. Analytical focus addressed contractual resilience commitments, network segmentation practices, cyber incident response capacity, and private-sector decision pathways for service continuity during conflict, the conditions under which the paper redundancy pattern documented in the analytical sections is most directly produced.

The literature and operator reporting establish that no single satellite architecture provides full Arctic coverage and that commercial provider service availability during conflict is a discretionary commercial decision, not a contractual guarantee [15; 28]. Cyber incident response capacity for Arctic ground stations is documented as adequate for routine threat profiles but has not been exercised against the integrated cyber-and-jamming scenario the research models. Multi-orbit satellite redundancy is identified across the literature as the most achievable near-term communications resilience investment, with terrestrial backup systems requiring longer pre-incident investment cycles [10].

Medical Supply, Cold Chain, and Medical Evacuation

The analysis examined pharmaceutical distributors, cold-chain logistics, hospital systems, medevac operators, Indian Health Service facilities, and Alaska Native tribal health organizations. The analytical structure treated these as lifeline dependency nodes whose failure produces nonlinear workforce and community destabilization, rather than as healthcare providers secondary to physical infrastructure.

Pharmaceutical cold-chain failure under simultaneous power, fuel, and communications disruption is documented across the literature as a non-recoverable outcome within the Days 0–14 window [74; 76]. Insulin, antibiotics, oxygen, and dialysis supplies are consistently identified across pharmaceutical distribution reporting as the categories where depletion produces life-threatening outcomes within days [76]. Medevac sector reporting establishes that aviation fuel allocation directly constrains medevac capacity and that no pre-established prioritization framework governs fuel allocation between medevac, cargo airlift, installation sustainment, and defense operational requirements. Indian Health Service and tribal health system facilities are additional dependent nodes whose failure would compound rather than mitigate the civilian healthcare capacity gap.

Warehousing and Distribution Nodes

The analysis examined regional hubs, intermodal staging, and pre-positioning operations. Analytical focus addressed capacity for emergency cargo prioritization, surge handling under degraded scheduling, and the relationships between commercial warehousing and the state and federal emergency response architecture that would activate under disruption conditions.

Workforce and Maintenance Operators

The analysis gave specific attention to the transient, seasonal, and locally rooted labor pools on which critical systems depend, and whose availability is itself dependent on community conditions including household-level access constraints and economic reserve depth that determine whether workforce members can sustain work attendance under disruption.

Emergency Management and Coordination Entities

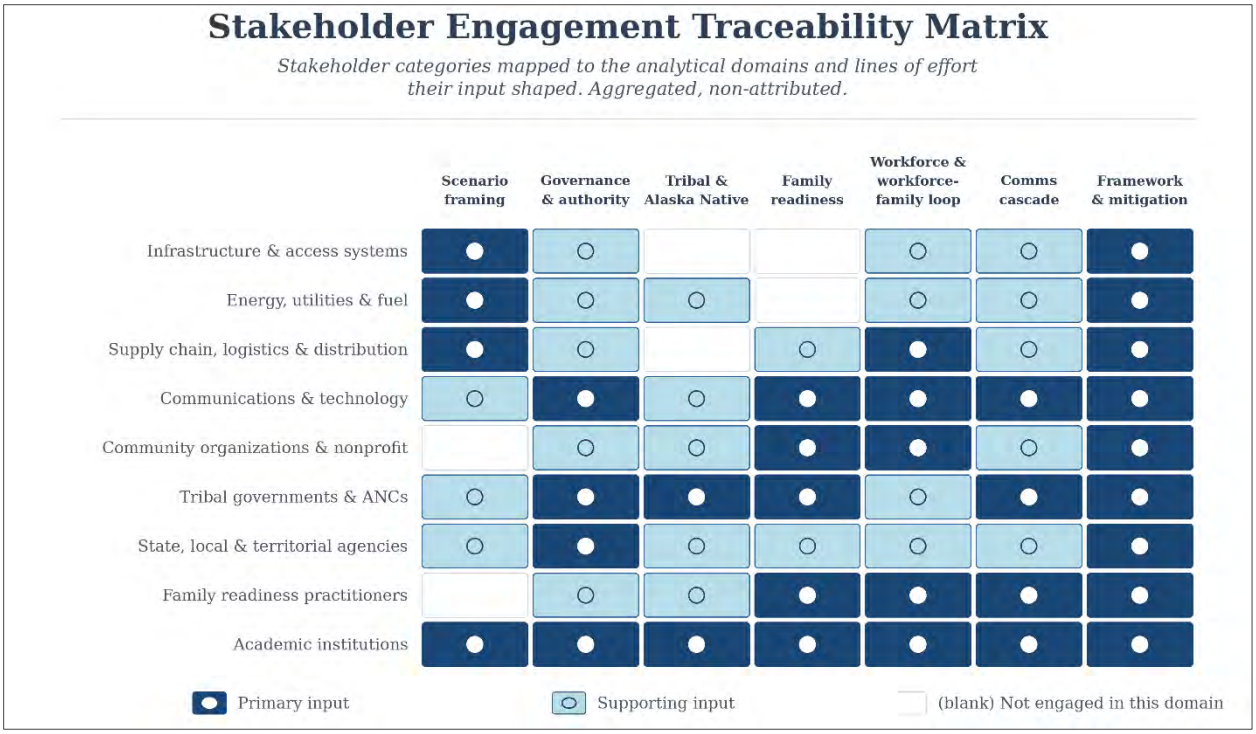
The analysis examined state, federal, and interagency coordination architectures, structured to engage with rather than duplicate existing convening bodies, the Alaska Mass Care Group and Feeding Task Force, the Alaska Food Value Chain Coordination Council stood up in 2025 under the USDA Regional Food Systems Partnership framework, and the partnership architecture that operates under Alaska's Partnership for Infrastructure Protection [3; 55].

The federal P3 maturity finding, Arctic jurisdictions operating predominantly at the Basic or low-Intermediate tier, is documented in the literature and reinforced in informal conversations with state and local emergency management contacts [33]. The absence of pre-established civil-military adjudication frameworks for fuel, airlift, medevac, and communications prioritization is a structural condition documented across federal planning reviews; closure of this gap requires policy and exercise investment, not capital infrastructure. The Mass Care Group, Feeding Task Force, and Alaska Food Value Chain Coordination Council are functioning state-level coordination architectures whose integration with installation and contractor sustainment requirements is incomplete but achievable through deliberate engagement [3; 55].

Findings in this section reflect the three-month performance window and the engagement modalities available within it. Validation through structured stakeholder engagement, including direct interviews with operators across the sectors examined, is identified in the recommendations as a near-term follow-on requirement.

Stakeholder Engagement Sectors and Inputs

Stakeholder Category	Input Domains	Engagement Intensity
Maritime & port operators	Cascade dynamics, sector interdependency, risk mitigation, recommendations	Substantial
Logistics & 3PL providers	Cascade dynamics, sector interdependency, risk mitigation	Substantial
Energy & fuel distributors	Sector interdependency, risk mitigation, recommendations	Substantial
Telecom & satellite providers	Cascade dynamics, risk mitigation	Substantial
Healthcare & EM organizations	Risk mitigation, recommendations shaping	Substantial
Tribal & Alaska Native organizations	Recommendations shaping (risk amplifier framing)	Moderate to substantial
DoD installation liaisons	Sector interdependency, workforce & operations	Substantial
Cross-border (US–CAN) entities	Cascade dynamics, sector interdependency	Moderate



Engagement was distributed across all five analytical domains.

Cross-Sector Assumption Misalignments

The most analytically valuable findings from the engagement were the cross-sector assumption misalignments, the seams at which operators from adjacent sectors held incompatible beliefs about each other's behavior under disruption. These misalignments are operator-level expressions of the paper-vs-real condition this research applies throughout: assumptions documented on paper across sectors fail to align with operational capacity those sectors describe under disruption. Three categories of misalignment recurred consistently across the engagement.

First, downstream operators consistently overestimated the resilience of upstream supply chains. Fuel distributors assumed that maritime delivery cycles would adjust to disruption faster than maritime operators believed they could. Healthcare operators assumed pharmaceutical distribution networks would prioritize Alaska medical supply over other regions; pharmaceutical distributors did not confirm that prioritization would occur absent specific pre-incident agreement. Installation logistics personnel assumed commercial cargo prioritization would account for defense sustainment requirements; commercial carriers confirmed that defense priority is not built into their routine cargo allocation logic.

Second, regulatory and waiver pathway assumptions were inconsistent across operators. Some assumed Jones Act waivers would be available reactively under disruption conditions; the FEMA Restoring Flow guide and the operators with direct waiver experience confirmed that waiver readiness is a pre-incident planning capability. Some assumed Canadian provincial waivers for ALCAN trucking surge would be expedited under emergency conditions; ALCAN operators identified specific procedural and documentation requirements that have not been pre-positioned [32].

Third, communications continuity assumptions diverged significantly across sectors. Operators dependent on commercial satellite communications generally assumed service availability under disruption conditions; satellite operators acknowledged the discretionary character of service continuity decisions during conflicts and the structural absence of governance frameworks requiring service preservation. This misalignment is particularly consequential because it produces a coordination failure mode in which downstream operators continue to operate under invalidated communications continuity assumptions while the system they depend on has already begun degrading service. Communications failure operates as a force degrader across every other lifeline category, and the misalignment between downstream operator assumptions and upstream operator behavior is the mechanism through which that force-degrader effect compounds at the cross-sector level.

Engagement Findings Carried Forward

Three engagement findings are carried forward into the Mitigation and Recommendations sections that follow. First, the paper-redundancy condition is confirmed at the operator level across multiple sectors, validating the substitution-reality-check finding as an operationally grounded conclusion rather than a documentary

inference. Second, the master-dependency role of energy is confirmed by operators across sectors, reinforcing the prioritization of fuel and energy resilience as the highest-return Tier 1 investment category developed in the Cost-Benefit Posture. Third, the absence of pre-established civil–military coordination frameworks is identified by emergency management and lifeline operators as the single most consequential governance gap, validating coordination architecture and P3 development as the highest-return, lowest-capital-cost mitigation category available within the current planning window, the framework finding that operates as a force degrader on mission outcomes when adjudication fails.

Question Architecture

The structured question framework was organized around six analytical emphasis areas, Failure Pathways, Throughput and Capacity Constraints, Resource Competition and Prioritization, Time Sensitivity, Substitution and Operational Reality, and Governance and Authority Gaps, each calibrated to the dual-theater scenario and tied to a canonical home subsection in the body of this report. The full question framework, including the six-area probe lines and the core question set, is in Appendix E.

Synthesized Findings

The synthesis below presents the engagement findings organized by analytical emphasis area. The April 30, 2026, extended response deadline closed the engagement window for the analysis carried into this report; where stakeholder input materially extends, qualifies, or contradicts literature-derived findings, that integration is reflected in the relevant analytical section rather than as a standalone block of practitioner data.

Failure Pathway Findings

This subsection consolidates practitioner perspectives on what fails first under disruption conditions, how impacts propagate across systems, and where cascading effects accelerate. Findings are organized by sector, then synthesized across sectors to identify the convergent failure pathways and the points at which sector-specific practitioner models diverge from the cross-sector analytical model.

The aggregated practitioner record converges with unusual consistency on the cascade pathway the dependency mapping identifies. Across sectors, transportation and logistics interruption is the most-cited early warning sign of Arctic disruption, identified by more than three-quarters of respondents as the first signal that the systems their operations depend on are beginning to degrade. Workforce shortages and employees unable to report to work are the second-most-cited early warning, reported by two-thirds of respondents and confirmed in open-ended responses as the binding constraint that determines whether operations continue, scale down, or stop. Communications and IT system disruptions are third, reported by more than half.

This sequencing matters analytically because it walks the cascade entry from the outside in: external transportation degrades first, internal workforce availability follows, and the coordination layer that would normally compensate for both is itself degraded by

communications loss. The pattern is visible across sectors, defense logistics, healthcare and pharmaceutical, supply chain distribution, engineering, and humanitarian response, with no sector treating its own internal capacity as the cascade entry point. The cascade is overwhelmingly perceived as exogenous to the respondent's operation, originating in shared infrastructure and propagating inward.

The thematic pattern across open-ended cascade descriptions reinforces a finding the analytical record has carried throughout this research: failure is convergent rather than sequential. Practitioners describe fuel, power, and communications as failing together because they share underlying infrastructure, not as a clean sequence in which one system fails before the next. The most operationally distinctive characterization of this convergence comes from the consulting and emergency management sector, which frames communications loss as the cascade amplifier that "force[s] organizations to make choices with bad information," directly validating the analytical position that communications failure operates as a force degrader across every other lifeline category.

A long-horizon cascade dimension surfaced from the engineering / hazard mitigation sector is worth flagging as an extension to the cascade analysis developed elsewhere in this report. Practitioners describe post-event population displacement as a cascading consequence that does not reverse, communities depopulated under prolonged disruption may not return, converting an acute incident into permanent demographic loss. This dimension is not currently developed in the Risk and Vulnerability Assessment, but it is consistent with the analytical posture the research has carried throughout: cascading failure compresses timelines and changes the character of recovery.

Throughput and Capacity Findings

This subsection consolidates practitioner perspectives on maximum operational capacity under degraded conditions, volume limitations, and the gap between theoretical capacity and operational throughput under the specific stress conditions of the dual-theater scenario. Particular attention is given to the substitution-limit findings developed in the Substitution and Redundancy Reality Check section, where practitioner input is most directly evidentiary.

Practitioner input on throughput and capacity converges most strongly on a single proposition: the alternatives most often invoked in contingency planning, particularly aviation surge and alternate-route rerouting, are the alternatives most consistently identified as unrealistic under Arctic conditions. The pattern is visible across sectors. Healthcare and pharmaceutical respondents identify air transportation as structurally unable to absorb ground-mode disruptions at the volume sustained pharmaceutical distribution requires. Defense logistics respondents identify the assumption of single-mode supply continuity as unrealistic, with explicit recognition that maritime supply works in summer and ground transportation works in winter, but neither delivers the full-year throughput planning frameworks assume. Humanitarian response respondents identify air assets as routinely depended on and routinely insufficient when secure stockpile depth is the actual binding constraint.

The throughput-versus-capacity distinction emerges particularly clearly in two operationally specific framings. First, multiple respondents independently identify that alternate routes, alternate airports, and multiple distribution locations are assumed in planning to remain available during disruption, when in practice Arctic events affect wide geographic areas simultaneously and degrade redundant pathways concurrently. The assumption that if Route A fails, Route B is available is itself a paper redundancy when the conditions that fail Route A also fail Route B. Second, the pharmaceutical distribution sector framing characterizes Arctic disruption as a transportation-driven external risk rather than a loss of internal capacity, with operational impact escalating as domestic mobility degrades rather than from internal capacity shortfalls.

A distinctive throughput finding from the commercial pharmaceutical sector concerns the tempo at which commercial supply chains operate. Practitioners describe healthcare distribution as operating on hours-not-days' timelines, with even short transportation, power, or logistics disruptions cascading rapidly into reduced service levels and patient impact. This tempo-mismatch finding is operationally consequential: planning frameworks calibrated to government response timelines, measured in days, do not match the pace at which commercial supply chains absorb disruption or fail.

Resource Competition and Prioritization Findings

This subsection consolidates practitioner perspectives on allocation decisions under constrained supply, interactions between civilian and defense demand, and the internal prioritization criteria that operators apply when external prioritization frameworks have not been pre-defined. Findings here directly inform the institutional coordination analysis and the recommendations for pre-incident adjudication architecture.

Practitioner input on resource competition and prioritization is the single most directionally consistent finding in the engagement record. Roughly nine of ten respondents identify joint coordination between government and industry as the action most likely to strengthen resilience in their sector, the highest-consensus finding across the entire engagement. No single actor, government, industry, military, or nonprofit, is identified by a majority of respondents as the appropriate sole owner of the resilience improvement that matters most. The directional message is unambiguous: the action that practitioners believe is most needed is precisely the action that requires pre-defined cross-sector coordination architecture rather than unilateral capital investment by any single party.

The pattern is reinforced by open-ended responses describing the misunderstanding that decision-makers most often hold about Arctic infrastructure disruption. Practitioners across sectors converge on a small number of recurrent misconceptions: that restoration timelines can be compressed to lower-latitude norms; that civilian commercial operations can absorb delays in the way government systems can; that workforce stability is a secondary concern rather than the binding mission-assurance constraint; that civilian populations and defense installations rely on different lifelines when in fact they share the same energy, transportation, medical, and sustenance pathways. The most concise practitioner statement of this convergence comes from the defense logistics sector: *workforce stability is mission assurance.*

Practitioners also describe the absence of pre-event coordination as the failure mode that converts a disruption into a crisis. An emergency management consulting respondent identifies the assumption that external partners, federal, military, or nonprofit, will arrive with resources ready to deploy as the most consistently unrealistic backup solution in Arctic planning. The pharmaceutical distribution respondent identifies the absence of state-level Emergency Operations Center capacity as a baseline gap that complicates downstream coordination architecture, an observation that warrants verification against the canonical inventory of state EOC postures before further integration but is operationally consistent with the maturity-of-coordination findings developed in the FEMA Public-Private Partnership maturity model.

Substitution and Operational Reality Findings

This subsection consolidates practitioner perspectives on feasible alternatives versus theoretical options, the specific conditions under which a documented redundancy ceases to perform, and constraints driven by weather, workforce, infrastructure, and scale. Findings here directly inform the paper redundancy versus operational redundancy analysis that is central to the report's analytical contribution.

Practitioner input on substitution feasibility is the single most directly evidentiary engagement finding for the analytical position the research has carried throughout: paper redundancy is not redundancy. Across sectors, the alternatives most consistently identified as unrealistic under Arctic conditions are precisely the alternatives the substitution catalog elsewhere in this report identifies as paper redundancies. Aviation surge as a substitute for maritime sustainment is named directly by multiple respondents as an unrealistic backup. Disruptions clearing within days is named as an unrealistic baseline assumption in commercial supply chain planning. External federal partners arriving with resources ready to deploy is named as an unrealistic dependency in the consulting and humanitarian sectors. Alternate airports and multiple distribution locations remaining available during wide-area Arctic events is named as an unrealistic assumption in the pharmaceutical distribution sector. Adequate generator and fuel-storage capacity at the community level is named as an assumed baseline that frequently does not exist in remote Arctic communities.

Two operationally distinctive findings warrant integration into the substitution analysis developed in the Substitution and Redundancy Reality Check section of this report. First, the seasonal lock-in dimension surfaces from defense logistics framing as a binding constraint that the Days 0–14 frame may understate: when fall barge season closes before disruption arrives, the only operational alternative in some communities is to wait for spring breakup, extending the no-substitution window from days into months. This dimension is consistent with the seasonal asymmetry analysis developed elsewhere in this report and reinforces it from the practitioner side. Second, the hub-and-spoke last-mile community delivery pattern surfaces from the humanitarian response sector as a substitution pathway distinct from commercial logistics: where central hubs can receive bulk goods, locally-retrieved last-mile delivery using traditional transport, including sled teams and airdrop reception by community members, provides a delivery mechanism that does not depend on commercial logistics or aviation substitution. This is operationally significant as a community-resilience dimension that the analytical record has not

previously developed and that may warrant inclusion in the cross-modal logistics redundancy mitigation pathway.

The differential between practitioner identification of inputs as essential and identification of those same inputs as hardest to replace is itself an empirical signal of where paper redundancies are most prevalent. Workforce availability and transportation access show no substitution-belief differential, practitioners do not believe these can be replaced. Communications shows the largest substitution-belief differential, suggesting that practitioners may underestimate communications irreplaceability relative to the analytical position developed elsewhere in this report that communications loss is a force degrader across every other lifeline category. This signal is worth flagging because it identifies a practitioner-vs-analytical-record divergence rather than a convergence: the analytical record treats communications as the cascade amplifier; the practitioner record treats it as the most-substitutable of the top-tier essential inputs. Both positions cannot be correct; the divergence is itself a finding worth carrying forward into the Communications and Data Resilience mitigation pathway.

Governance and Authority Gap Findings

This subsection consolidates practitioner perspectives on how scarce resources are allocated in practice, where that authority diverges from formal coordination frameworks, and where civilian, military, tribal, and commercial authority structures operate in parallel without coordination. Findings here directly inform the Institutional Coordination as a Risk Amplifier section and the near-term recommendations for pre-incident coordination architecture.

Practitioner input on governance and authority gaps converges directly with the analytical position developed in the Institutional Coordination as a Risk Amplifier section of this report. Respondents identify the absence of pre-established cross-sector coordination as the failure mode that converts a disruption into a crisis. Practitioners across sectors, defense logistics, supply chain distribution, healthcare, humanitarian response, engineering, identify joint government-industry coordination as the highest-priority resilience action by a margin that is unusually consistent for a multi-sector engagement record.

The practitioner record identifies three specific governance gaps that align with the institutional-coordination analytical work developed elsewhere in this report. First, the gap between commercial supply chain tempo and government response tempo is identified as a coordination failure mode in its own right: pharmaceutical and food distribution operate on hours timelines, while government activation operates on days' timelines, producing a misalignment in which downstream impact develops faster than upstream coordination can adjudicate. Second, the assumption that external partners, federal, military, or nonprofit, will arrive with resources ready to deploy is identified as the most consistently unrealistic baseline assumption in Arctic planning. Third, baseline coordination capacity at the state-EOC level is identified as variable, with some jurisdictions lacking the activation infrastructure that downstream coordination architecture assumes is in place.

These gaps validate the analytical position that pre-established cross-sector coordination architecture is the highest-return, lowest-capital-cost mitigation category available in the current planning window. Coordination capacity that depends on improvisation under time pressure cannot match the pace at which Arctic disruption develops; coordination capacity that has been pre-staged through charters, exercises, and pre-scripted authority transitions can. Practitioner input does not contest this analytical claim; it strengthens it by identifying with specificity which coordination assumptions in current Arctic planning are most consistently invalidated under stress.

Cross-Sector Patterns and Integration with Analytical Findings

Beyond the sector-specific findings consolidated above, the engagement was designed to surface cross-sector patterns, the points at which practitioner observations across multiple sectors converged on the same finding, and the points at which they revealed misalignments in sector-level assumptions about adjacent systems.

Three cross-sector convergences and one cross-sector divergence emerged from the engagement record and warrant integration with the analytical findings developed elsewhere in this report.

The first convergence is the systematic identification of workforce availability as the binding constraint across every sector represented in the engagement. Defense logistics, healthcare and pharmaceutical, supply chain distribution, engineering, and humanitarian response respondents all identify workforce shortages as the curtailment, rationing, or stop-operations trigger under Arctic conditions. Workforce ranks among the top two essential inputs and the top two hardest-to-replace inputs across the dataset, with no substitution-belief differential. This convergence directly validates the analytical position that the civilian and contractor workforce is a critical but under modeled dependency in current planning, and it strengthens the case for the workforce and maintenance capacity mitigation category developed elsewhere in this report.

The second convergence is the systematic identification of paper-redundancy conditions across sectors. The alternatives most consistently named as unrealistic, aviation surge for maritime, short-duration restoration assumptions, alternate airport availability, generator-fuel-volume baselines, external partner arrival on schedule, align directly with the substitution catalog developed in the Substitution and Redundancy Reality Check section. No respondent describes a paper redundancy that the analytical record has not also identified; the cross-sector convergence is in identifying the same paper-redundancy conditions from independent operational vantage points.

The third convergence is the systematic identification of joint government-industry coordination as the highest-priority resilience action. This convergence directly validates the analytical position that pre-established coordination architecture is the highest-return, lowest-capital-cost mitigation category available in the current planning window.

The single cross-sector divergence worth flagging concerns communications. The analytical record treats communications loss as a force degrader across every other lifeline category, the cascade amplifier whose degradation propagates simultaneously into

logistics visibility, coordination capacity, and command awareness. The practitioner record treats communications as the most-substitutable of the top-tier essential inputs, with the largest gap between identification as essential and identification as hardest to replace. This divergence is itself a finding: practitioners may be operating with substitution assumptions about communications that the analytical record does not support. The implication is that the Communications and Data Resilience mitigation category should explicitly address not only the technical substitution gap but also the practitioner-perception gap that converts technical substitutability into operational over-reliance.

Where stakeholder input materially extends, qualifies, or contradicts literature-derived findings in the body of this report, that integration is reflected in the relevant analytical section. Specifically, practitioner input on Days 0–14 inventory thresholds, fuel allocation decision criteria, communications resilience under commercial provider autonomy, and the operational adequacy of existing coordination architecture refines the findings developed in the Targeted Dependency Mapping, Focused Risk and Vulnerability Assessment, Substitution and Redundancy Reality Check, and Institutional Coordination sections, respectively.

The mitigation prioritization framework developed in the next section incorporates stakeholder input directly as a weighting input on the Feasibility criterion. Where practitioner consensus indicated that a proposed mitigation is more or less operationally feasible than literature-derived analysis suggested, the Feasibility scoring is adjusted accordingly, with the basis for adjustment documented in the relevant entry.

Engagement limits are documented in the Methodology section Limits subsection. Follow-on engagement and exercise-based validation are identified in the Recommendations section as near-term priorities, on the basis that the foundation established here is the starting point for the sustained engagement architecture Arctic mission assurance requires.

RECOMMENDATIONS

The recommendations below address the gap between current Arctic mission assurance capacity and the demands the scenario documented in the Scenario Framework section imposes. Each recommendation is tagged against five criteria, Mission Impact, Cascade Risk, Time Sensitivity, Feasibility, and Dual-Use Value, that together identify both the strategic value of the investment and the conditions under which it produces return. Tagging is not a substitute for the integrated analysis that precedes it; it is a planning aid for sequencing investments under resource constraints. The tiering reflects implementation horizon rather than priority: near-term investments are not more important than long-term ones, but they are the investments that close the Days 0–14 gap before the next disruption requires those frameworks to exist.

The recommendations developed in this section translate the analytical findings of the prior sections into specific, actionable lines of effort organized across three time horizons. Each recommendation is tagged by the five-criteria framework introduced in the Mitigation and Cost–Benefit Prioritization section, Mission Impact, Cascade Risk, Time Sensitivity, Feasibility, and Dual-Use Value, and identifies the external partners responsible for the line of effort. The intent is that each recommendation can be read as an executable assignment, not a directional statement of need.

The three-tier structure reflects the analytical conclusion that Arctic mission assurance requires layered investment across overlapping time horizons rather than sequential prioritization. Near-term recommendations close gaps that the Days 0–14 window would expose immediately if disruption occurred today. Mid-term recommendations build the institutional and infrastructure capacity that converts near-term mitigations from improvisation into sustained operational architecture. Long-term recommendations identify the research, infrastructure, and coordination investments whose return is realized over multi-year horizons, but whose initiation cannot wait for those horizons to begin.

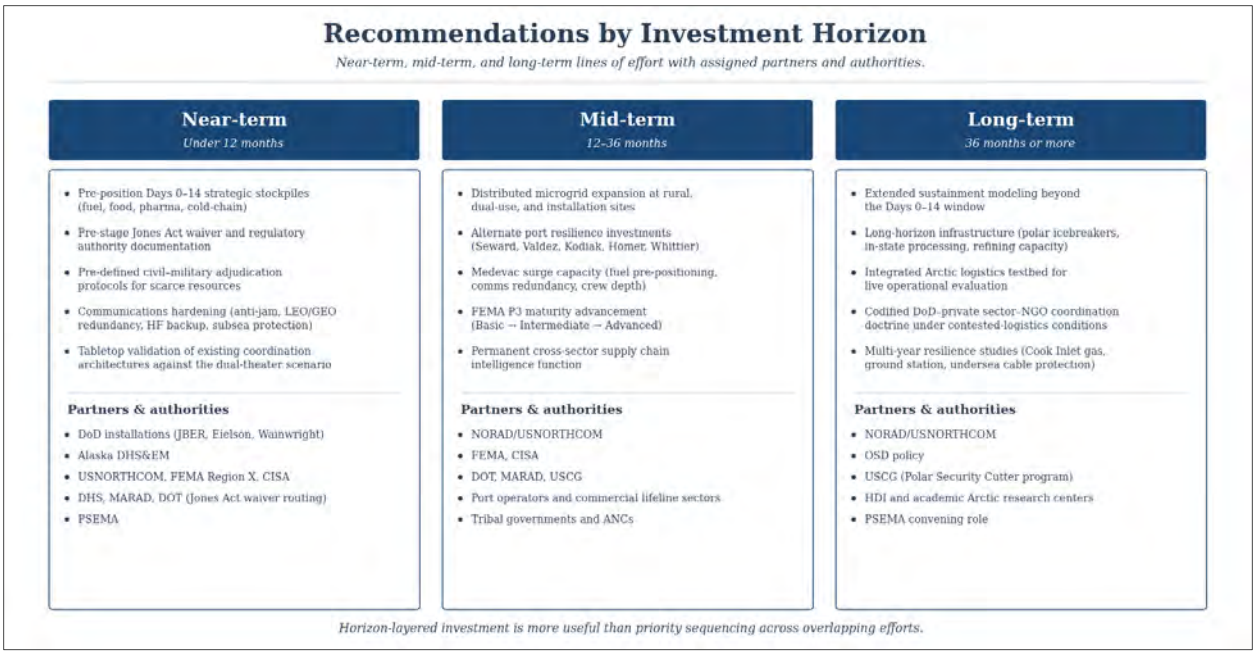
External partner identifiers used throughout this section refer to the following entities. The Homeland Defense Institute is the primary contracting and convening institution for Arctic homeland defense research priorities. NORAD and USNORTHCOM are the operational mission assurance authorities. Alaska DHS&EM is the state-level emergency management authority. FEMA holds federal authority for emergency response coordination, supply chain resilience programs, and the Public-Private Partnership maturity framework. CISA holds federal authority for critical infrastructure protection and cybersecurity coordination. Tribal governments and Alaska Native organizations hold sovereign authority over tribal lands, communities, and citizens, and operate under sovereignty, consent, and reciprocity protocols. [23] Academic institutions, including the Ted Stevens Center for Arctic Security Studies, RAND Corporation, the Naval Postgraduate School, and university research centers with Arctic specialization, provide the analytical capacity for follow-on research and validation. PSEMA functions across these recommendations as the connective infrastructure described in the next section of this report.

Recommendations Roll-Up by Investment Horizon and Decision Authority

Near-term (under 12 months), mid-term (12–36 months), and long-term (36 months or more) lines of effort with assigned partners and authorities.

Horizon	Recommendation Line of Effort	Decision Authority / Partners
Near-term (Under 12 months)	• Pre-position Days 0–14 strategic stockpiles (fuel, food, pharma, cold-chain). • Pre-stage Jones Act waiver and regulatory authority documentation. • Pre-defined civil–military adjudication protocols for scarce resources. • Communications hardening (anti-jam, LEO/GEO redundancy, HF backup, subsea protection). • Tabletop validation of existing coordination architectures against the dual-theater scenario. • Pharmaceutical cold-chain pre-positioning at regional, IHS, and ANTHC nodes. • Sustained stakeholder engagement architecture across DoD, federal, state, tribal, and private-sector lifeline partners.	DoD installations (JBER, Eielson, Wainwright) Alaska DHS&EM USNORTHCOM FEMA Region X CISA DHS MARAD, DOT(Jones Act waiver routing) PSEMA
Mid-term (12–36 months)	• Distributed microgrid expansion at rural, dual-use, and installation sites. • Alternate port resilience investments (Seward, Valdez, Kodiak, Homer, Whittier, MacKenzie). • Medevac surge capacity build (aviation fuel pre-positioning, comms redundancy, crew depth). • FEMA P3 maturity advancement (Basic → Intermediate → Advanced). • Permanent cross-sector supply chain intelligence function. • Tribal partner protocols for sustained engagement under sovereignty, consent, and reciprocity framing. • Cold-chain and pre-positioning infrastructure at tribal and community health nodes.	NORAD/USNORTHCOM FEMA CISA DOT MARAD USCG Port operators and commercial lifeline sectors Tribal governments and Alaska Native Corporations Academic institutions

Long-term (36 months or more)	• Extended sustainment modeling beyond the Days 0–14 window. • Long-horizon infrastructure investments (polar icebreakers, in-state processing, refining capacity). • Integrated Arctic logistics testbed for live operational evaluation. • Codified DoD–private sector–NGO coordination doctrine under contested-logistics conditions. • Multi-year resilience studies (Cook Inlet gas depletion, ground station, undersea cable protection). • Operational piloting of recommended coordination architectures under live operational conditions.	NORAD/USNORTHCOM OSD policy USCG (Polar Security Cutter program) HDI and academic Arctic research centers PSEMA convening role



Tiered visual organizing near-term (under 12 months), mid-term (12–36 months), and long-term (36 months or more) recommendations with assigned partners and authorities.

Near-Term Recommendations (0–12 Months)

Near-term recommendations are investments that can be activated within the current planning and budget cycle and that materially compress the activation timeline of early-phase response. They share a common characteristic: their return is realized before disruption occurs, in the form of pre-positioned stocks, pre-coordinated authorities, exercised relationships, and tested protocols. None of them require new construction; all of them require deliberate institutional commitment and resource allocation.

Pre-Position Strategic Stockpiles for the Days 0–14 Window

Establish pre-positioned reserves of fuel, food, pharmaceuticals, cold-chain capacity, and critical materiel sufficient to bridge the Days 0–14 disruption window at the most consequential nodes, Anchorage, Fairbanks, the Joint Base Elmendorf-Richardson area, and the rural community hubs whose seasonal resupply windows close with the fall barge season. Stockpiles must be sized for both routine demand and disruption-period demand surge, with rotation protocols that preserve operational readiness without producing inventory destruction. The investment is the only mechanism that closes the inventory-buffer gap within the analytical window the dual-theater scenario imposes.

Pre-Stage Jones Act Waiver and Regulatory Authority Documentation

Develop and maintain pre-staged documentation packages supporting Jones Act waiver requests, transportation weight-limit waivers, hours-of-service waivers, CDL endorsement waivers, and Canadian provincial permitting for ALCAN trucking surge. Pre-staged documentation includes vessel availability surveys, shipper demand data, alternative routing analysis, and defense necessity findings updated against current conditions on a rolling basis. Pre-staged interagency coordination architecture exercises waiver routing through DHS, MARAD, the Department of Transportation, and the Department of Defense before disruption requires it. The investment converts the Jones Act waiver process from a capability that may exceed the operational window during a disruption into a regulatory mechanism activatable within hours.

Establish Pre-Defined Civil–Military Adjudication Protocols

Develop pre-defined protocols for adjudicating civilian and military claims on scarce fuel, food, transportation, and medical capacity under the dual-theater scenario. Protocols must address allocation across Title 10 authority, state emergency management authority, tribal sovereignty, and commercial cargo prioritization, with clearly identified decision authorities, documented prioritization criteria, and pre-established communication pathways. Protocols are exercised under the Tabletop Exercise recommendation that follows and integrated into the standing operations of the participating authorities.

The case for elevating this recommendation to near-term priority is the time-compression characteristic of the Days 0–14 window. Adjudication that has not been hammered out before disruption begins will consume the first portion of the operational window in political contestation. Federal, state, and tribal authorities, military commanders, and commercial operators will be making allocation claims against the same constrained pool of fuel, transportation capacity, medical resources, and pharmaceuticals. In the absence of pre-defined criteria and decision authority, those claims will be litigated in real time through media channels, intergovernmental correspondence, and emergency declarations whose legal effect is contested as it is being asserted. The operational consequence is that the degradation pathway the research documents accelerates while the institutional architecture that would slow it is still being constructed. The lost time is not recoverable. Lifeline degradation in the Arctic operates on inventory clocks that do not pause for governance to catch up.

The historical record establishes that the United States has constructed civilian–military resource adjudication architectures under wartime and crisis conditions before, that the architectures have functioned when pre-established and have failed when improvised, and that pre-establishment is feasible within existing authorities without legislative change. Three precedents inform the design.

First, the War Industries Board, established by Executive Order in July 1917 under the Council of National Defense to coordinate U.S. industrial mobilization for World War I, demonstrated both the value of centralized priorities adjudication and the cost of constructing it after the conflict had begun. The WIB operated for sixteen months before achieving substantive operational effectiveness under Bernard Baruch's chairmanship in March 1918, and the period before that operational maturity was characterized by competing military procurement authorities, voluntary compliance regimes that produced inconsistent results, and the inefficiencies that delayed authority-construction always produces. The WIB record is the foundational U.S. case for the proposition that resource-adjudication architecture should be pre-established rather than constructed under stress.

Second, the World War II succession from the Office of Production Management (January 1941) to the Supply Priorities and Allocations Board (August 1941) to the War Production Board (January 1942), all of which adjudicated competing civilian, military, and Allied claims on industrial output and raw materials, demonstrates the same lesson at greater scale. The WPB allocated steel, aluminum, rubber, fuel, and consumer goods across military and civilian demand throughout the war, operating under Executive Order 9024 with priorities and rationing authority over the U.S. industrial base. The WPB's effectiveness rested on the pre-1941 foundation laid by the OPM and SPAB; it did not begin from a standing start in 1942. The lesson is that adjudication architecture builds on prior architecture rather than emerging from improvisation, and that the institutional learning required is not compressible to crisis timelines.

Third, the COVID-19 allocation framework—operating through Defense Production Act authorities, the FEMA temporary final rule on prioritization and allocation of scarce health and medical resources, the CDC Advisory Committee on Immunization Practices guidance on vaccine prioritization, and the National Academies' Framework for Equitable Allocation of COVID-19 Vaccine—provides the closest contemporary analog to the architecture this recommendation calls for. The COVID framework operated under acute time compression to allocate ventilators, personal protective equipment, vaccines, and pharmaceuticals across states with significant inter-state variability, federal-state friction, and contested public health authority claims. The framework worked in important respects and failed in others, and both the successes and failures are operationally instructive. Where pre-existing pandemic planning infrastructure existed, allocation decisions were made faster and with greater institutional defensibility. Where the architecture was constructed during the crisis, allocation decisions were inconsistent across states, litigated in courts and media simultaneously, and produced inequities whose corrective addresses arrived after the most consequential allocation decisions had been made. The Arctic Days 0–14 window will operate under the same time-compression

characteristics with less margin for state-level variability and a smaller pool of authority that can be activated late.

The design parameters for the Arctic adjudication architecture follow from these precedents and from the analytical findings developed throughout this report. Pre-defined criteria for life-safety, mission-critical, and economic-essential categorization, applied consistently across requesting authorities, are required to avoid case-by-case adjudication during the operational window. Pre-identified decision authorities, distinct from the requesting authorities, are required to avoid the conflict of interest that real-time allocation under stress otherwise produces. Pre-established communication pathways between federal coordination (USNORTHCOM, FEMA Region X, CISA), state authority (Alaska Department of Homeland Security and Emergency Management, Alaska Governor's Office), tribal sovereign authorities, and commercial operators are required so that allocation decisions arrive at the operators who execute them within hours rather than days. The PSEMA network is the institutional architecture through which the commercial-operator end of this pathway is operationally addressable; the FEMA Public-Private Partnership maturity framework provides the institutional doctrine; the Jones Act waiver pre-staging recommendation that precedes this item provides the regulatory mechanism through which the adjudicated allocations are implemented at the maritime supply layer [30; 80].

Conduct Tabletop Exercise–Based Validation of Existing Coordination Architectures

Convene structured tabletop exercises that test Alaska's Partnership for Infrastructure Protection, the Mass Care Group and Feeding Task Force, the Alaska Food Value Chain Coordination Council, and the relevant federal-state-tribal-military coordination architecture against the dual-theater disruption scenario this research documents. Exercises must test civilian–military resource adjudication, cross-border coordination with Canadian counterparts, and time-compressed prioritization decisions under the Days 0–14 window. After-action findings convert paper-redundancy coordination architectures into operationally validated capacity.

Establish Communications Hardening Priorities for Multi-Layer Architecture

Identify the specific communications hardening investments, anti-jam capability for primary defense and emergency management nodes, redundant satellite pathways across LEO and GEO providers, high-frequency radio backup for installation-to-installation coordination, and subsea cable protection for Arctic maritime corridors, whose deployment within twelve months produces material reduction in the communications cascade pathway documented in this research. Establish liability and resilience agreements with commercial satellite providers that include service continuity commitments under stress conditions.

Establish Pharmaceutical Cold-Chain Pre-Positioning at Key Nodes

Pre-position the pharmaceutical categories, insulin, antibiotics, oxygen, dialysis supplies, and other categories where inventory depletion produces life-threatening outcomes

within days, at the regional hospital, Indian Health Service facility, and Alaska Native Tribal Health Consortium nodes whose service populations cannot be reached by external resupply within the Days 0–14 window. Investment includes backup power for cold-chain refrigeration, redundant temperature monitoring, and pre-staged distribution agreements with the commercial pharmaceutical distributors operating in Alaska. Cold-chain failure is non-recoverable within the disruption window; pre-positioning is the only mechanism that closes the medical pathway gap.

Initiate Sustained Stakeholder Engagement Architecture

Convert the engagement architecture established for this research into a sustained, recurring engagement mechanism connecting DoD mission assurance planners, NORAD and USNORTHCOM J-staff equivalents, Alaska DHS&EM, FEMA, CISA, tribal governments under sovereignty protocols, and the private-sector lifeline operators across the seven sectors examined in this research. Engagement is structured around the analytical findings of this report, with a recurring cadence that converts episodic stakeholder input into a continuous data architecture supporting future research and operational coordination.

Mid-Term Recommendations (12-36 Months)

Mid-term recommendations require sustained investment beyond a single budget cycle but can be substantially completed within a three-year horizon. They include the institutional and infrastructure investments that build resilience above the early-phase floor that near-term recommendations establish.

Distribute Microgrid Capacity to Reduce Single-Point Energy Dependency

Expand distributed microgrid capacity at the rural Alaska communities, regional installations, and dual-use civilian-military sites where energy concentration risk is highest. Investment focuses on reliability rather than maximum renewable penetration, with diesel-plus-renewable-plus-storage configurations sized for community and installation demand under disruption conditions. Investment includes the workforce capacity and spare-parts logistics required to sustain microgrid operations under Arctic conditions, addressing the workforce-infrastructure-logistics dependency documented in this research.

Begin Alternate Port Resilience Investments

Initiate the capital investment programs required to convert the six identified contingency alternatives to the Port of Alaska, Seward, Valdez, Kodiak, Homer, Whittier, and MacKenzie, into functional substitutes for containerized cargo handling. Specific investments include reefer plug-in installation at Seward, Valdez, and Homer; roll-on/roll-off ramp acquisition; weight-restriction exemption infrastructure; deep-draft cargo port construction at Whittier; conveyor and crane infrastructure at MacKenzie; and the rail and road improvements required to connect alternate ports to Alaska's downstream distribution architecture. Investment yields strategic resilience over multi-year horizons rather than Days 0–14 benefit but is necessary for the long-term reduction of single-point-of-failure exposure.

Build Medical Evacuation Surge Capacity

Expand medical evacuation surge capacity through pre-positioned aviation fuel reserves at key medevac airfields, redundant flight-following and medical consultation communications, pre-defined receiving-facility coordination protocols at Anchorage and Seattle tertiary care centers, and crew availability programs that build depth in the thin specialty workforce that medevac depends on. Investment addresses the convergence pathway in which medevac competition with cargo airlift, operational aviation, and other defense aviation requirements produces the fastest-cascading lifeline failure documented in this research.

Establish Tribal Partner Protocols for Sustained Engagement

Build a standing protocol for tribal and Alaska Native organization participation in Arctic mission assurance research, planning, and operational coordination. Protocols address the absence of pre-existing relationships between non-tribal institutions and tribal governments that the Dependency Mapping section identifies as a structural coordination gap and that the Risk and Vulnerability Assessment establishes as a primary risk-amplifier condition. Implementation parameters appropriate to specific engagement contexts are addressed in the companion study under continuity-of-society framing; this recommendation specifies the institutional investment in coordination architecture required to close the risk-amplifier gap.

Advance FEMA P3 Maturity Beyond Basic and Low-Intermediate Tier

Build the institutional architecture that progresses Alaska's Partnership for Infrastructure Protection, the federal-state-tribal-military coordination mechanisms relevant to Arctic mission assurance, and the cross-sector private-public partnership architecture from current Basic and low-Intermediate maturity toward Advanced tier. Investment includes formalized charters, sustained pre-incident engagement, regular exercise activity, and integration into commercial lifeline operator business processes. Advancement is the institutional analog to the infrastructure investments identified elsewhere in this report.

Establish a Permanent Cross-Sector Supply Chain Intelligence Function

Build, in coordination with the open-source Supply Chain Intelligence Center proposal, a permanent coordinating capability for anticipating and responding to cross-sector supply chain disruption [12]. The capability leverages FEMA's Defense Production Act authorities, CISA's Critical Infrastructure Partnership frameworks, and peer-level coordination with the Department of Transportation, Department of Commerce, and the Office of the Director of National Intelligence. Application of CNA's Supply Chain Operational Engagement methodology and FEMA's Supply Chain Analysis Network framework to Arctic-specific conditions provides the analytical foundation [16; 33].

Build Cold-Chain and Pre-Positioning Infrastructure at Tribal and Community Health Nodes

Expand cold-chain capacity, backup power, and pre-positioned medical supply at the Indian Health Service facilities, Alaska Native Tribal Health Consortium nodes, regional tribal health organizations, and the Community Health Aide Program village sites that

constitute the only line of healthcare access for many Alaska Native communities. Investment is sized to address the parallel cold-chain failure pathway documented in the Risk and Vulnerability Assessment. Implementation parameters appropriate to tribal-health-system context are addressed in the companion study; this recommendation specifies the infrastructure investment required to close the risk-amplifier gap identified in this report.

Long-Term Recommendations (36 Months or More)

Long-term recommendations require multi-year construction, sustained policy commitment, or sustained research engagement beyond the current grant window. They address the structural conditions of the Arctic operating environment that no near- or mid-term investment alone can resolve, and they establish the baseline against which future disruption response will operate.

Extended Sustainment Modeling Beyond the Days 0–14 Window

Conduct extended sustainment modeling for the recovery dynamics, structural reconstitution, and multi-month sustainment trajectories that lie beyond the Days 0–14 analytical window of this research. Extended modeling supports investment prioritization decisions whose horizons exceed the early-phase window and provides the time-to-failure parameter precision that command-level decisions ultimately require. Extended modeling is identified in this research's methodology section as the most consequential follow-on research priority.

Multi-Year Resilience Studies Across Lifeline Sectors

Conduct multi-year resilience studies that build on the dependency mapping and risk assessment foundation of this research. Studies extend the analytical scope to include longer-horizon vulnerabilities, Cook Inlet natural gas depletion, polar icebreaker fleet recapitalization, Arctic ground station resilience, undersea cable protection, which exceed the engagement window of this research but are operationally consequential for Arctic mission assurance over a five-year horizon.

Operational Piloting of Recommended Coordination Architectures

Pilot the coordination architectures identified in the near-term and mid-term recommendations under live operational conditions, including federally declared emergencies, large-scale exercises, and Arctic-specific TTX programs. Operational piloting converts coordination architectures from validated-on-paper to validated-in-operations, addressing the specific concern that paper-redundancy coordination does not perform under the time pressure that disruption response imposes.

Integrated Arctic Logistics Testbed

Establish, in partnership with the Department of Defense, the Department of Transportation, the Maritime Administration, FEMA, the State of Alaska, tribal governments, and the private-sector lifeline operators, an integrated Arctic logistics testbed that supports operational evaluation of mitigation investments, coordination architectures, and emerging logistics technologies under realistic Arctic conditions. The

testbed is the institutional vehicle through which research, operational capability, and private-sector partnership converge for the long-term resilience investment that single-program lines of effort cannot deliver.

Codified DoD–Private Sector–Nongovernmental Coordination Doctrine

Develop, in coordination with NORAD, USNORTHCOM, the Office of the Secretary of Defense, FEMA, and the broader emergency management and private-sector partnership community, codified doctrine for DoD–private sector–nongovernmental coordination under contested-logistics conditions. Doctrine builds on NSM-22's defense critical infrastructure framework, the FEMA Public-Private Partnership maturity model, the Jones Act waiver framework, and the contested-logistics literature, and produces the doctrinal foundation under which the recommendations of this report and the connective infrastructure of the PSEMA network operate [30; 34; 80].

Long-Horizon Infrastructure Investments, Polar Icebreakers, Processing Capacity, Refining

Sustain the long-horizon infrastructure investments identified across this research as necessary for structural Arctic resilience but exceeding the analytical window of this engagement. These include polar icebreaker fleet recapitalization through the Polar Security Cutter program and the April 2025 Request for Information for Arctic Security Cutters [21; 68]; in-state food processing capacity expansion to address the structural processing-capacity ceiling [3]; and in-state refining capacity assessment to address the upstream refining dependency that compounds Alaska's fuel vulnerability. None of these is a Days 0–14 mitigation; all are necessary for the resilience baseline against which future disruption will be measured.

FROM RECOMMENDATIONS TO IMPLEMENTATION

The Coordination Gap This Research Has Identified

The recommendations developed in this report identify multiple external partners across every line of effort: the Homeland Defense Institute, NORAD and USNORTHCOM, Alaska DHS&EM, FEMA, CISA, tribal governments and Alaska Native organizations, academic institutions, and the private-sector operators across the seven lifeline sectors examined in this research. The recommendations themselves are operationally executable. What this section names directly is the connective infrastructure under which they become operationally executable, and the institutional gap that infrastructure exists to fill.

The most analytically significant institutional finding of this research is direct. There is no sustained, trust-based, operationally tested intermediary connecting Department of Defense components to the family readiness organizations, civilian and contractor workforce networks, tribal and Alaska Native governance structures, behavioral health systems, faith-based networks, and community organizations that together sustain Arctic mission assurance at the human-systems layer. Every population in that list is operationally consequential during Days 0–14 disruption. None is currently connected to DoD through a sustained intermediary that combines non-attribution, sector neutrality, and operational credibility. The protocols on paper assume a connector that does not exist with the reach, trust, and operational credibility the Days 0–14 window requires.

The consequence under the dual-theater scenario is direct. Commercial operators, port authorities, maritime carriers, rail operators, trucking companies, telecommunications providers, and pharmaceutical distributors, make cargo prioritization and service-availability decisions based on commercial logic that does not account for defense sustainment requirements. Installation commanders hold authority over on-base resource allocation but have no control over the commercial cargo prioritization or civilian workforce availability on which sustainment depends. State authority can reallocate scarcity across competing populations but cannot create supply. Tribal governments hold sovereign authority over land, workforce, and local infrastructure that defense operations depend on but operate outside formal defense coordination architectures. The result is a structural fragmentation that no individual entity can resolve unilaterally, and that no after-event coordination mechanism can match for speed.

Three structural facts beneath this fragmentation make the gap analytically primary rather than peripheral. First, the privately operated lifelines and human-systems partners that sustain Arctic operations are owned, operated, and led outside DoD authority; mission assurance therefore depends on cross-sector coordination that no single contract, command, or agency is positioned to deliver. Second, the federal partnership architecture has known gaps for the function this research identifies, FEMA's P3 model is calibrated to civilian disaster response, CISA holds critical infrastructure protection authority without functioning as an operational intermediary for Arctic contested-logistics, and DoD itself cannot be sector-neutral by definition because it is one of the parties whose

interests must be coordinated. Third, trust-based intermediation is not a function that contracts can produce. The relationships across commercial operators, federal agencies, state authorities, tribal governments, and academic institutions that the recommendations require depend on sustained engagement under non-attribution protocols, cross-sector translation that respects sectoral confidentiality, and operational credibility built through consistent presence in the spaces where lifeline systems and defense requirements actually intersect. None of those conditions can be created transactionally during a disruption.

The PSEMA Network as Connective Infrastructure

The recommendations developed in the previous section identify what must happen across near-term, mid-term, and long-term horizons to close the gap between current Arctic mission assurance capacity and the demands the dual-theater scenario imposes. Every recommendation identifies multiple external partners: the Homeland Defense Institute, NORAD and USNORTHCOM, Alaska DHS&EM, FEMA, CISA, tribal governments and Alaska Native organizations, academic institutions, and the private-sector operators across the seven lifeline sectors examined in this research. The recommendations are operationally executable. What requires explicit articulation is the connective infrastructure under which they become operationally executable.

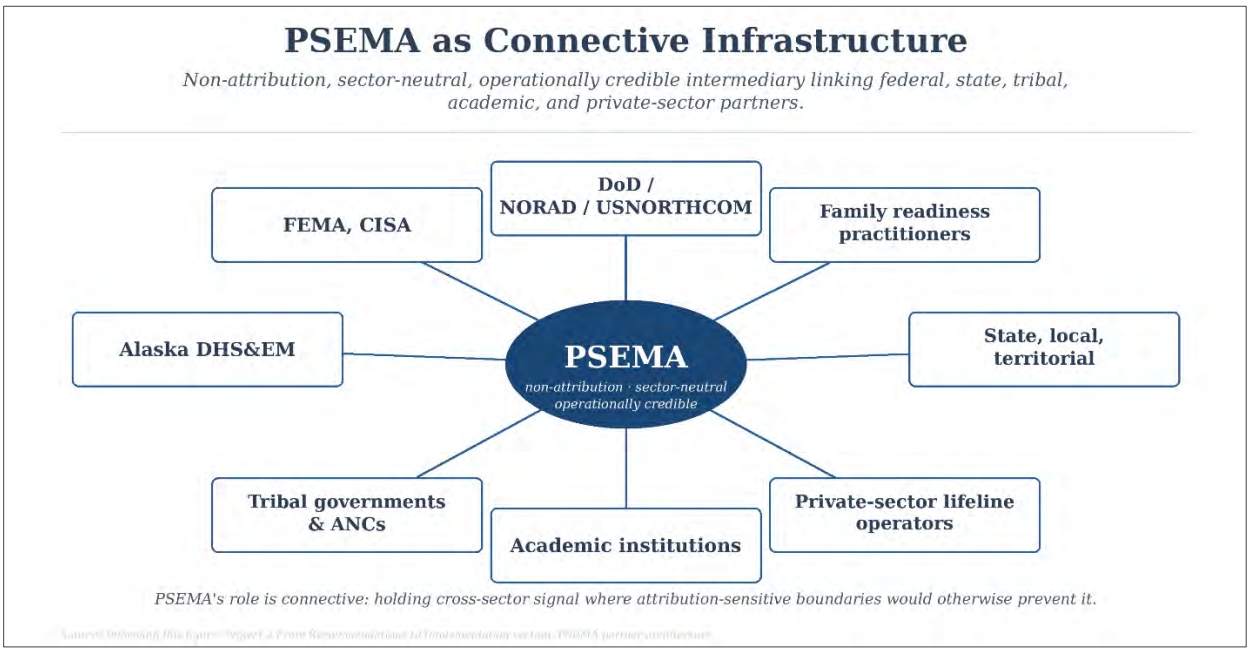
This section names that requirement directly and develops the partnership case that follows from it. The most analytically significant institutional finding of this research is that the architecture necessary to translate between defense mission requirements and the privately operated systems that sustain them, the connective infrastructure that the recommendations and the contested-logistics literature both identify as essential, does not currently exist within the federal architecture at the maturity Arctic mission assurance requires. The Private Sector Emergency Management Association, whose institutional identity, mission, membership composition, and operational posture are described in the Research Sponsor and Institutional Context section earlier in this report, is positioned to fill that gap. This section explains what the gap is, why a sector-neutral, non-attribution intermediary is the architecture required to close it, how PSEMA is structured to operate as that intermediary across the lifeline sectors examined in this research, the specific connection points through which the network engages the external partners the recommendations name, and the continuation availability PSEMA offers to extend this research into the multi-year horizon the analytical record establishes as necessary.

Multi-stakeholder convenings, such as the Ted Stevens Center's High North Security Dialogue, demonstrate that structured cross-sector dialogue is increasingly recognized as a primary mechanism for surfacing operational realities that single-author analysis cannot capture [60].

This research has documented the human-systems coordination weakness that determines whether Arctic continuity of society holds under disruption. The weakness is specific and operationally consequential: there is no sustained, trust-based, operationally tested intermediary connecting DoD and military components to the family readiness organizations, civilian and contractor workforce networks, tribal and Alaska Native

governance structures, behavioral health systems, faith-based networks, and community organizations that together sustain Arctic mission assurance at the human-systems layer. The protocols on paper assume an intermediary that does not exist with the reach, trust, and operational credibility the Days 0–14 window requires.

The Private Sector Emergency Management Association is positioned to fill that role, and this section articulates why and how. The framing is integrated with the companion study's PSEMA Network section to present a unified connective architecture across both research lines: where the companion study positions PSEMA as the intermediary connecting DoD to the privately operated lifeline systems on which Arctic operations depend, this research positions PSEMA as the intermediary connecting DoD to the human-systems partners through whom continuity of society is sustained.



Visual depicting PSEMA's sector-neutral intermediary role linking business, industry, , associations, government, nonprofit, and academic.

Why a Sector-Neutral Intermediary Is The Architecture Required, and Why PSEMA Fits It

The Private Sector Emergency Management Association occupies an institutional position uniquely suited to closing this gap. PSEMA operates as a non-attribution, sector-neutral, operationally credible intermediary across the categories of private-sector lifeline operators that Arctic mission assurance depends on: energy and fuel providers, transportation and logistics firms, telecommunications and technology companies, food distribution and supply chain partners, medical and pharmaceutical providers, warehousing and cold-chain operators, and humanitarian and disaster-relief organizations. The non-attribution posture is operationally consequential: it enables candid practitioner engagement with the operators who would otherwise be constrained

from discussing system vulnerabilities and operational limits in attributed forums, generating the kind of operational data this research has documented as essential to validated mitigation planning.

The sector-neutrality is structurally important because the dependencies this research has mapped are cross-sector. A coordination architecture anchored in any single lifeline category, energy, transportation, communications, food, medical, would replicate the fragmentation that the dependency mapping identifies as the underlying vulnerability. PSEMA's design is the inverse: it is anchored in the cross-sector logic that the cascade pathways require, with sustained relationships across the operator categories whose interactions determine system performance under disruption.

The connective infrastructure Arctic mission assurance requires has a specific institutional shape. It must be non-attribution, because the operational realities that determine system performance under disruption cannot be surfaced in attributed forums. It must be sector-neutral, because the dependencies this research has mapped are cross-sector and a coordination architecture anchored in any single lifeline category would replicate the fragmentation the dependency mapping identifies as the underlying vulnerability. And it must be sustained across pre-incident, incident, and post-incident periods, because the trust and relational continuity that operational coordination depends on cannot be built transactionally once a disruption is underway.

These three characteristics are the operational posture established as PSEMA's institutional foundation in the Research Sponsor and Institutional Context section. This subsection makes the operational case for why they are the right architecture for the gap this research has identified and translates them into three concrete mechanisms through which the PSEMA network closes the paper-redundancy gap at the coordination layer where it originates.

PSEMA's institutional position addresses the paper-redundancy gap at the layer where it originates: the coordination and governance architecture that determines whether documented redundancies translate into operational throughput. PSEMA operates as a sustained intermediary across the categories of private-sector lifeline operators and human-systems partners that Arctic mission assurance depends on, energy and fuel providers; transportation and logistics firms; telecommunications and technology companies; food distribution, supply chain, and humanitarian feeding organizations; medical and pharmaceutical providers; engineering and infrastructure resilience consultancies; tribal and Alaska Native organizations; family readiness and behavioral health networks; and the federal, state, and academic partners whose engagement the recommendations require. The cross-sector reach is not incidental; it is the structural condition that makes the three mechanisms below operationally feasible.

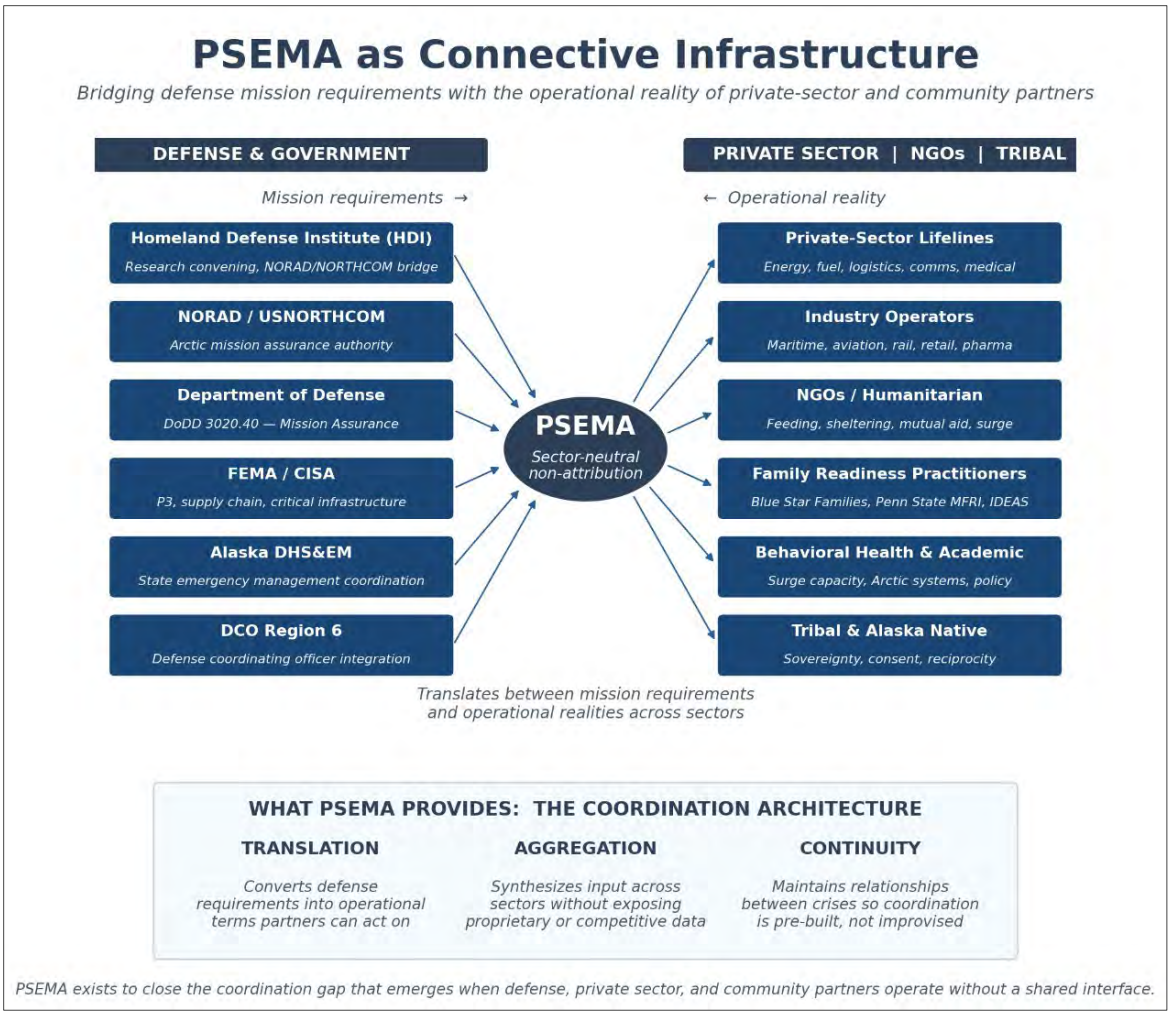
First, PSEMA converts episodic engagement into sustained engagement architecture. The federal Public-Private Partnership maturity model identifies sustained engagement as the precondition for the operational coordination that disruption response requires [28]. Episodic engagement, incident-only, transaction-only, contract-only, produces the paper-redundancy condition that disruption response

exposes. Sustained engagement, pre-incident, through-incident, post-incident, with consistent participants and consistent analytical posture , converts paper relationships into operational architecture. Business Emergency Operations Center architectures connecting commercial lifeline operators, installation commanders, state emergency management, and tribal governments produce the institutional infrastructure through which civil-military resource adjudication can occur under time pressure. PSEMA is structured for the sustained mode rather than the episodic one.

Second, PSEMA functions as the cross-sector translation layer that no single agency or sector is positioned to provide. Defense components understand defense mission requirements; commercial operators understand commercial operating realities; tribal governments understand tribal sovereignty; federal civilian agencies understand federal civilian authority. The translation across these vocabularies, frameworks, and decision-making paradigms is itself the function the contested-logistics literature identifies as missing [44]. Without a sector-neutral translator, each party operates from a partial mental model of how the others will behave under stress, and the assumption misalignments that result are themselves a primary cascade trigger. PSEMA's sector-neutral positioning makes the translation function structurally feasible because PSEMA is not a party to the operational decisions whose coordination it facilitates.

Third, PSEMA produces and disseminates the integrated analytical products that close the information-asymmetry condition that paper redundancy depends on. When defense planners assume that contractual relationships produce operational continuity under stress, when commercial operators assume that defense requirements will be coordinated through standing channels that do not exist at the maturity assumed, and when state and tribal authorities assume that federal-level coordination architecture will deploy at the pace disruption requires, the asymmetry of expectations among the parties is what produces the paper-redundancy outcome. Integrated analytical products, like this research and its companion study, convert asymmetric expectations into shared analytical foundation, against which expectations can be calibrated and operational coordination architecture can be built. The non-attribution posture is what makes the underlying operational data accessible to the analytical record at all; the sector-neutral positioning is what makes the cross-sector synthesis possible; and the sustained engagement is what allows the analytical products to track evolving conditions across the multi-year horizon Arctic mission assurance requires.

> ***Operational Implication:*** Paper redundancy at the governance and coordination layer cannot be closed by infrastructure investment, by additional contracts, or by federal directive alone. It can be closed only by sustained, sector-neutral, operationally credible intermediation that converts episodic engagement into continuous architecture. PSEMA is the connective infrastructure positioned to fill that role across the lifeline sectors examined in this research.



Connection Points: How PSEMA Connects to the External Partners

The recommendations developed in this report identify specific external partners across every line of effort, the full mapping of recommendations to lead actors and supporting partners is consolidated in the Appendix H. The PSEMA network connects to those partners through specific institutional pathways developed over years of sustained engagement. Six connection point categories warrant explicit identification.

HDI and the Homeland Defense Research Architecture

PSEMA's research engagement with the Homeland Defense Institute, including this research effort and its companion study, establishes a direct working relationship with the institutional convening function for Arctic homeland defense research priorities. HDI's partner architecture provides connection to NORAD and USNORTHCOM operational requirements, to interagency research priorities, and to the academic institutions whose analytical capacity supports follow-on research. The connection point is direct, current, and operationally productive; this report itself is a worked example of the architecture in operation.

NORAD/USNORTHCOM Mission Partners

PSEMA's strategic conversations with current and former NORAD and USNORTHCOM leadership establish the operational requirements that the analytical record of this research be designed to support. The connection point provides direct access to the mission assurance authorities that hold Arctic homeland defense responsibility, and to the J-staff equivalents whose operational planning incorporates the findings the research delivers. Sustained engagement converts episodic strategic dialogue into a continuous information architecture supporting both research and operational coordination.

Alaska DHS&EM, FEMA, CISA, and the Federal Emergency Management Architecture

PSEMA's institutional standing within the emergency management community connects directly to Alaska DHS&EM at the state level, FEMA at the federal level, and CISA for critical infrastructure coordination. These connections are particularly consequential because the recommendations developed in this report depend on the public-private partnership maturity progression, the federal supply chain resilience programs, and the critical infrastructure protection coordination that these agencies hold. PSEMA's positioning within the emergency management community, rather than adjacent to it, makes the connection operationally substantive rather than nominal.

Tribal Governments and Alaska Native Organizations

PSEMA's tribal partnerships operate under sovereignty, consent, and reciprocity throughout. Tribal governments are sovereign entities whose engagement with PSEMA proceeds on terms agreed by tribal participants. Reciprocity ensures that PSEMA-mediated coordination returns usable findings to tribal participants rather than extracting data. The connective role is structurally limited: PSEMA does not represent tribal governments or speak for them; it provides the coordination infrastructure through which tribal-DoD-state-federal coordination can occur on tribal terms.

Family Readiness Practitioners and Behavioral Engagement Partners

PSEMA's connection to the family readiness ecosystem combines two complementary capacities. The first is family-research and household-support reach: Blue Star Families, the Penn State Military Families Research Institute, the National Military Family Association, and the broader military family research and support community provide the family readiness coordination capacity that the FRG decline has left open, with bidirectional flow that surfaces practitioner reality from family readiness organizations to DoD planning and surfaces DoD operational requirements to family readiness organizations in operationally usable form.

The second capacity is behavioral engagement and preparedness campaign design, anchored by IDEAS. IDEAS is the creative and engagement firm that designed the original *Kids Get a Plan* preparedness platform in partnership with the Florida Division of Emergency Management, one of the most behaviorally effective youth preparedness platforms in the United States. The platform's assets are now owned by the PSEMA Foundation under an agreement with the Florida Division of Emergency Management that authorizes extension, adaptation, and expansion into new audiences and operational environments. The platform has already been adapted to an Alaskan context through the

Turn Around, Don't Drown expansion developed with the Alaska Division of Homeland Security and Emergency Management. For this research, the IDEAS connection point is operationally consequential: it is the partner through which the *Household-Level Preparedness Extension to Underserved Populations* recommendation and the Arctic Rising Generation Preparedness Initiative move from analytical proposal to deployed campaign capability, without reconstructing methodology, asset library, legal vehicle, or state-agency relationship from the ground up. The combined connection point, family research and support networks paired with a twenty-year behavioral engagement and campaign-design capability, is the architecture through which the household-level and family-readiness recommendations of this report become operationally executable.

Behavioral Health Systems and Academic Institutions

PSEMA's connections to behavioral health systems operationalize the surge capacity gap this research identifies as the workforce-sustainment ceiling, including the trained provider pool, tele-behavioral health infrastructure capable of operating under degraded communications, and tribal-led behavioral health programming under sovereignty protocols. PSEMA's research collaboration with the Ted Stevens Center for Arctic Security Studies, with university research centers specializing in Arctic homeland security, and with research institutions including RAND, the Naval Postgraduate School, and the broader contested-logistics research community provides connection to the analytical capacity required for follow-on research. The connection is operationally productive because academic partners contribute the methodological depth and the multi-year research horizons that single-program engagements cannot sustain, including the longitudinal family readiness studies and multi-year continuity-of-society modeling identified in the long-term recommendations.

Private-Sector Lifeline Operators Across the Seven Sectors

PSEMA's standing engagement spans the seven lifeline sectors examined in this research, and the cross-sectoral reach is the architecture that converts research findings into operationally executable coordination across the recommendations of this report.

Continuation Availability

The recommendations developed in this report exceed the engagement window of this research. Near-term lines of effort begin within twelve months of report submission; mid-term lines require sustained execution across one to three years; long-term and follow-on research lines identify investments whose multi-year horizons begin now but extend well beyond the current research engagement. Each tier requires the connective infrastructure described in this section to operate effectively, and that infrastructure cannot be reconstructed at each transition between research phases.

PSEMA views this research as a foundational exploration of Arctic continuity of society and is positioned to continue the connective infrastructure function across the recommendations identified in this report and across the follow-on research priorities the analytical record supports. Continuation availability is offered to HDI, NORAD/USNORTHCOM, and the broader interagency partnership architecture as a structural commitment, not as a transactional offer.

Specifically, PSEMA is positioned to support the following continuation lines of effort: convening and facilitation of the tabletop exercises identified in the near-term recommendations; engagement architecture development for the sustained stakeholder engagement and the public-private partnership maturity progression identified in the mid-term recommendations; cross-sector facilitation of the operational piloting cross-sector facilitation of the operational piloting and Arctic societal sustainment testbed concepts (paired with the companion study's infrastructure testbed) identified in the long-term recommendations, ongoing coordination with academic partners on the extended sustainment modeling and multi-year resilience studies that exceed the engagement window of this research; and continuing analytical work as new data, new operational conditions, or new research priorities emerge from the integrated picture this report and its companion study deliver.

The closing analytical proposition is straightforward. Arctic homeland defense cannot be sustained against the dual-theater disruption scenario this research models without the layered investment, the institutional coordination, and the connective infrastructure that the recommendations and this section together identify. Closing the gap between current capacity and what the threat environment requires is not a single-program effort; it is a sustained, multi-partner, multi-year line of work whose return is realized over horizons that no single research engagement can encompass. PSEMA is structured for that horizon, positioned within the network the recommendations require, and committed to the continuation availability the analytical record establishes as essential.

CONCLUSION

Arctic mission assurance is a systems problem, not a capital investment problem. The lifeline systems on which Arctic homeland defense depends were not designed for the disruption scenario this research models; they were designed for normal operating conditions, and they are thin at every margin, in storage capacity, in redundancy, in workforce, in coordination architecture, and in the time available before failures cascade across sectors. Under dual-theater disruption, those margins compress from weeks to days, and in some cases to hours.

The five-deliverable architecture this report develops establishes the analytical foundation operational planning requires. The scenario framework defines the Days 0–14 decision window in which sustainment assumptions fail and consequential decisions must be made under uncertainty. The dependency mapping identifies what is connected to what across national supply pathways, in-theater lifeline systems, and mission-dependent community infrastructure. The risk and vulnerability assessment establishes where and when those connections break, how failure cascades non-linearly, and which vulnerabilities become mission-limiting within the first fourteen days. The substitution and redundancy reality check documents the gap between paper redundancy and operational redundancy. The institutional coordination review identifies governance architecture as a risk amplifier in its own right. The mitigation prioritization translates these findings into seven integrated investment categories, and the recommendations organize that pathway across near-term, mid-term, and long-term horizons. The PSEMA Network section identifies the connective infrastructure under which the recommendations become operationally executable.

Five integrating observations close the analytical record.

First, the dual-theater disruption scenario is structurally consistent with the simultaneity problem the 2025 National Defense Strategy identifies as a primary planning challenge, calibrated to Arctic geographic and seasonal realities and grounded in the commercial supply architecture that NSM-22 explicitly recognizes as defense critical infrastructure [70; 80].

Second, failure pathways are non-linear, fast-cascading, and self-reinforcing. The window for effective intervention closes faster than linear degradation models suggest. Inventory buffers measured in days eliminate the multi-week adjustment period that planning frameworks implicitly assume; communications failure operates as a force degrader on every other failure pathway; pharmaceutical cold-chain failure is non-recoverable within the analytical window.

Third, paper redundancy is the systemic condition that makes the failure pathways consequential. Alternate ports, aviation substitution, satellite communications backup, pharmaceutical distribution diversification, state emergency feeding architecture, and public-private coordination frameworks all exist on paper at maturity levels that do not

perform under the stress conditions the dual-theater scenario imposes. The capacity-versus-throughput distinction is the analytical commitment that surfaces the gap.

Fourth, institutional coordination is a primary cascade trigger, not a passive backdrop. The absence of pre-defined civilian–military adjudication mechanisms, underdeveloped cross-border coordination architecture, and the absence of a sustained, trust-based intermediary across defense, commercial, nongovernmental, tribal, and academic actors are structural risk amplifiers. Building the coordination architecture before the crisis is a high-return, low-capital priority that no infrastructure investment can substitute for.

Fifth, the path forward is layered, multi-partner, and multi-year. Near-, mid-, and long-term recommendations operate concurrently rather than sequentially. Energy and coordination architecture are foundational, but the layered architecture across maritime throughput, communications, workforce, logistics, food, and the cross-sector connective infrastructure that holds them together is what produces the resilience baseline Arctic mission assurance requires.

End state. The end state this research is constructed to support is straightforward: an Arctic homeland defense architecture in which privately operated lifeline systems are recognized as defense critical infrastructure under NSM-22 [80]; in which the dual-theater disruption scenario is integrated into mission assurance planning at the operational specificity command-level decisions require; in which preparatory investments, pre-positioned reserves, pre-staged regulatory authority, pre-defined civilian–military adjudication, multi-layer communications redundancy, and exercised coordination architectures, are in place before disruption requires them; and in which the cross-sector connective infrastructure that translates between defense mission requirements and the privately operated systems that sustain them operates at the maturity the threat environment requires. This end state is achievable. The recommendations are operationally specific, the partner architecture is identified, the time horizons are realistic. What remains is the institutional commitment to fund, exercise, and sustain the layered investment the analytical record establishes as required.

Forward linkage. The findings, recommendations, and connective infrastructure framing carry forward to the June 2026 Homeland Defense Awareness Symposium briefing, where they will be positioned against the operational planning cycles of the audience institutions and the cross-cutting institutional coordination question, the connective infrastructure gap and the PSEMA network framing, surfaced for direct discussion. Beyond the symposium, the long-term recommendations identify the follow-on research agenda the analytical foundation supports: extended sustainment modeling, multi-year resilience studies, operational piloting, the integrated Arctic logistics testbed, codified DoD–private-sector–nongovernmental coordination doctrine, and the long-horizon infrastructure investments. The companion study, Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense, addresses the human-systems dimensions of the same disruption environment, and the integrated picture across the two studies positions the homeland defense research community to engage Arctic resilience as the systems-design and institutional-coordination challenge the analytical record establishes it to be.

PSEMA is committed to that work and welcomes the opportunity to discuss with the Homeland Defense Institute, NORAD/USNORTHCOM, and mission partners how this research can be extended and deepened in support of Arctic homeland defense and the commercial and community systems that make it possible.

APPENDIX A: GLOSSARY OF KEY TERMS

The terminology below defines the analytical vocabulary used consistently throughout this research. These terms are used with deliberate specificity; their precise meaning supports the report's evaluative judgments and should not be substituted for general usage.

Arctic Sustainment Modes and Lifeline Constraints. Framework evaluating maritime, aviation, and ALCAN sustainment under Arctic disruption, recognizing maritime as primary, aviation as critical but limited, and overland routes as constrained redundancy.

Business Critical Infrastructure. Privately operated lifeline systems, fuel, food, transportation, communications, and medical supply, that directly support Arctic mission assurance and civilian sustainment.

Cascading Failure. A pattern in which the failure of one lifeline system propagates into other systems on which the first depended or which depended on the first. In this research, cascading failure is the primary mechanism through which the dual-theater scenario produces operational impact within the Days 0–14 window.

Cold-chain. Temperature-controlled logistics systems supporting pharmaceuticals, vaccines, and perishables whose failure under combined fuel, power, and communications disruption creates rapid sustainment risk.

Consent-dependent redundancy. A backup sustainment or transit pathway whose availability depends on approval from another nation or external authority during disruption conditions. This research distinguishes consent-dependent redundancy from constrained but U.S.- or partner-controlled redundancy on the analytical grounds that consent-dependent redundancy can be eliminated at the moment of greatest need without any change in physical conditions. The Northern Sea Route under UNCLOS Article 234 is the canonical Arctic example.

Contested logistics. The operational condition in which adversaries disrupt or degrade the supply chains and transportation systems supporting military operations and sustainment. In this research, contested logistics is treated as the operational condition under which the dual-theater scenario unfolds, distinguishing this analysis from benign-environment sustainment modeling.

Cook Inlet natural gas depletion. The projected decline of Cook Inlet gas reserves that increases long-term energy vulnerability for Anchorage, JBER, and the Railbelt power system.

Days 0–14 Decision Window. T The early disruption phase when inventory depletion, cascading failure, and coordination breakdown occur before alternate sustainment pathways can stabilize operations.

Defense Critical Infrastructure (DCI). Per National Security Memorandum 22, DoD and non-DoD networked assets and facilities essential to project, support, and sustain military forces and operations worldwide. Non-DoD owned DCI consists of assets from relevant critical infrastructure sectors and subsectors, including as defined by statute.

Defense Production Act (DPA). Civilian and military infrastructure essential to sustaining defense operations, including systems not owned or controlled by the Department of Defense.

Dual-Theater Scenario. The stress-test scenario combining West Coast maritime disruption with simultaneous Arctic operational escalation to evaluate sustainment resilience.

Dual-use infrastructure. Infrastructure whose operations serve both civilian and military requirements. In Alaska, the principal example is the Port of Alaska, which simultaneously sustains Anchorage civilian demand and Joint Base Elmendorf-Richardson defense operations. Dual-use infrastructure produces operational efficiency under normal conditions and concentrated risk under disruption.

Fantasy document. A planning document whose contents describe a coordination architecture, redundancy pathway, or operational capacity that does not perform under the conditions it purports to address. In this research, the term is used as a rhetorical anchor for exposing gaps between governance frameworks and actual operational capacity, distinct from documents that describe what should be done but cannot be done within the relevant window.

Force degrader. A condition, such as workforce disruption or communications failure, that reduces operational effectiveness without direct combat engagement.

Inflection point. A specific moment within the disruption timeline at which the operational situation transitions from one phase to the next, typically because an inventory threshold is crossed, a workforce capacity limit is reached, or an institutional decision pathway is activated or fails to activate. Inflection points are identified throughout the time-phased degradation analysis in this research.

Joint Base Elmendorf-Richardson (JBER). The joint Air Force and Army installation in Anchorage, Alaska, headquarters of the Alaskan Command and 11th Air Force, and the principal defense operations node in Southcentral Alaska. In this research, JBER is treated as analytically inseparable from the surrounding civilian infrastructure, especially the Port of Alaska, the Cook Inlet natural gas system, and Anchorage water and sanitation utilities.

Jones Act. U.S. maritime law affecting domestic cargo movement and influencing surge shipping capacity and substitution options during disruption.

Just-in-time architecture. The supply chain configuration in which inventory is sized to consumption rather than to contingency, producing efficiency under normal conditions and rapid shortfall under disruption. In this research, Alaska's commercial supply chain is characterized as a four-day Tacoma-to-shelf just-in-time architecture in which warehouse storage is minimal and stated reserve figures reflect inventory in transit and on shelves rather than in deep storage.

Master Dependency. A foundational system whose disruption simultaneously disables multiple downstream lifelines. Energy is the master dependency in the Arctic operating environment; the dependency chain Fuel → Power → Heat → Water → Communications → Coordination governs all downstream behavior.

Mission assurance. Per DoD Directive 3020.40, a process to ensure that assigned tasks or duties can be performed in accordance with the intended purpose or plan. In this research, mission assurance is treated as the operational outcome that the dependency mapping, risk assessment, and recommendations are constructed to support.

No-Fail Node. A critical infrastructure point whose failure produces major operational or statewide disruption with no viable replacement during the analytical window.

Northern Sea Route (NSR). The maritime transit corridor running along the northern coast of the Russian Federation between the Atlantic and Pacific. In benign-planning literature, the NSR is treated as a potential maritime rerouting option when Pacific gateway access is degraded. In this research, the NSR is treated as consent-dependent redundancy whose availability is determined by Russian assertion of regulatory control under UNCLOS Article 234 rather than by ice conditions or vessel-tracking capacity.

Operation Vittles (Berlin Airlift). The 1948–1949 U.S. and allied operation sustaining West Berlin by air after Soviet ground blockade. In this research, Operation Vittles serves as the historical anchor establishing the upper bound of aviation-only sustainment under conditions of constrained airfield access and weather, against which the feasibility of Arctic airlift substitution for maritime cargo is measured.

Operational Redundancy (or Real Redundancy). Backup capability that has been exercised, resourced, and demonstrated to function under realistic Arctic disruption conditions.

Paper Redundancy. Backup systems or contingency plans that exist in documentation but cannot sustain required throughput during real disruption conditions.

Pre-event planning and relationship preconditions. The standing requirement identified throughout this research, that the planning, coordination, and relationship-building necessary to manage tribal and rural community impacts during Arctic infrastructure disruption must be established before the disruption occurs.

Public-Private Partnership (P3) maturity. The level of coordination and integration between government and private-sector lifeline operators supporting resilience and sustainment planning.

Risk amplifier. A condition or system attribute that increases the severity or speed of cascading impact under disruption, distinct from the proximate cause of disruption itself. In this research, Alaska Native and tribal community vulnerability, cross-border governance gaps, and communications degradation are each identified as risk amplifiers in the dependency and risk-assessment analysis.

Seasonal Asymmetry. The condition in which the operational characteristics of Arctic infrastructure, transportation, communications, and human-systems performance differ substantially between summer and winter, with winter producing the binding constraints on capacity, throughput, and substitution. In this research, seasonal asymmetry is treated as a structural feature of the operating environment rather than as a contingent factor.

Single-Source Dependency. A condition in which one supplier, one node, or one provider is the sole pathway for a critical input. Crowley as the sole source of avgas for Alaska, Charlie's Produce as the sole produce wholesale node, and the Cook Inlet basin as the dominant Railbelt natural gas source are documented single-source dependencies.

Standalone Microgrid. A self-contained electricity generation and distribution system, typically diesel-based with limited renewable integration, that operates without interconnection to a larger transmission grid and without access to the mutual aid agreements available to interconnected utilities.

Substitution-limit modeling. The analytical approach that evaluates contingency alternatives by their operational throughput under realistic constraints rather than by their nameplate capacity. In this research, substitution-limit modeling is applied across maritime, aviation, and overland modes, and consistently establishes that planned substitutes do not perform at the volume, speed, or reliability that benign-planning models assume.

Subsistence Harvest. The traditional fish, wildlife, and wild-plant harvest that sustains rural Alaska communities outside the commercial food distribution system.

Throughput vs. Capacity. Capacity describes what a system can accommodate under optimal conditions; throughput describes what a system can actually move under the specific stress conditions of the disruption environment. The existence of capacity is not evidence of sustainment feasibility; throughput, not capacity, is the analytically operative figure throughout this research.

Time-to-failure threshold. The duration after disruption initiation at which a specific lifeline system, inventory buffer, or coordination architecture transitions from constrained to acutely degraded. Thresholds are evaluated by sector in the Risk and Vulnerability Assessment and are treated as directional rather than as decision-grade quantification.

Tribal and Alaska Native communities as risk amplifiers. The finding that infrastructure disruption disproportionately impacts remote tribal communities, accelerating humanitarian and operational pressures across the broader system.

UNCLOS Article 234. A provision of the 1982 United Nations Convention on the Law of the Sea authorizing coastal states to adopt and enforce non-discriminatory laws and regulations for the prevention, reduction, and control of marine pollution from vessels in ice-covered areas within the limits of the exclusive economic zone. In this research, Article 234 is treated as the legal instrument through which the Northern Sea Route is converted from a benign-planning redundancy into a consent-dependent pathway whose availability depends on conditions outside U.S. planning authority.

APPENDIX B: REFERENCE LIST

This research integrates government, academic, industry, and practitioner sources to assess community stability, workforce resilience, governance dynamics, civil–military coordination, family readiness, tribal and Alaska Native resilience, behavioral health surge capacity, and infrastructure performance affecting societal continuity and mission assurance under Arctic disruption conditions. Entries are listed alphabetically by author or organization. Italicized short-form citation tags in square brackets, used selectively in this report, key to the entries below.

[1] Alaska Department of Labor and Workforce Development, Research and Analysis Section. Processing's Changing Workforce. Alaska Economic Trends, November 2024.

[2] Alaska Federation of Natives. Alaska Native Community Resilience. Anchorage, AK: Alaska Federation of Natives, December 2018.

[3] Alaska Food Policy Council. 2025 Annual Report. Anchorage, AK: Alaska Food Policy Council, 2026.

[4] Alaska Food Security and Independence Task Force. Alaska Food Security and Independence Task Force Report. Fairbanks, AK: University of Alaska Fairbanks and Alaska Food Policy Council, March 2023. Prepared under Administrative Order 334.

[5] Alaska Hospital and Healthcare Association (formerly Alaska State Hospital and Nursing Home Association). Alaska Healthcare Workforce Analysis and Member Facility Data. Anchorage, AK: AHHA, 2023.

[6] American Geophysical Union. Optimizing Polar Air Traffic: Strategies for Mitigating the Effects of Space Weather–Induced Communication Failures Poleward of 82°N. Washington, DC: AGU, August 27, 2024.

[7] ANCSA Regional Association. Alaska Regional One-Pager: Alaska Native Regional Corporations and Community Infrastructure. Anchorage, AK: ANCSA Regional Association, 2024.

[8] The Arctic Institute. Shifting U.S. Arctic Strategy: From Cooperation to Strategic Competition. Washington, DC: The Arctic Institute, 2025.

[9] Behe, Carolina, Kawerak Natural Resources, K. J. Moerlein, and Robert Cameron. "Incorporating Indigenous Voices into Climate Change Adaptation in Rural Alaska." Ecology and Society, 2024.

[10] Bostrom, Ann, Granger Morgan, Baruch Fischhoff, and Cynthia J. Atman. Risk Communication: A Mental Models Approach. Cambridge, UK: Cambridge University Press, 2008.

[11] Breaking Defense. Closing the Arctic High North Communications Gap. January 25, 2025.

[12] Caton, J. J. Uniting Silos of Excellence: Evaluating Alternate Policy Solutions to Enhance Federal Supply Chain Reporting Structures. Master's thesis, Naval Postgraduate School, March 2024.

[13] Center for a New American Security (CNAS), Transatlantic Forum on Russia. Navigating Relations with Russia in the Arctic: A Roadmap for Stability. Washington, DC: CNAS, November 2021.

[14Center for Strategic and International Studies (2026). Northern Defense Strategy: Russian Arctic operations, Northern Sea Route control mechanisms, and dual-use infrastructure analysis.

[15] Center for Strategic and International Studies (CSIS). Commercialization and Dual-Use Space Capabilities in the Arctic. Washington, DC: CSIS, November 21, 2025.

[16] CNA Corporation. Supply Chain Analysis Capabilities and Experience. Arlington, VA: CNA, 2025.

[17] Community Science and Mat-Su Health Foundation. Mat-Su Food Security Assessment. Wasilla, AK: Mat-Su Health Foundation, 2025.

[18] Congressional Research Service. U.S. Space Industry: Launch and Satellite Services. Washington, DC: CRS, December 2016.

[19] Congressional Research Service. Changes in the Arctic: Background and Issues for Congress. Washington, DC: CRS, July 14, 2025.

[20] Congressional Research Service. 2026. Iran Conflict and the Strait of Hormuz: Impacts on Oil, Gas, and Other Commodities. CRS Report R45281. Washington, DC: Library of Congress, March 11, 2026.

[21] Congressional Research Service. Coast Guard Polar Security Cutter (PSC) Program: Background and Issues for Congress. Washington, DC: CRS, January 2026.

[22] Critical Infrastructure Coordination Center / Critical Resilience for Critical Infrastructure (2024). Cross-sector coordination architecture analysis and case-study documentation.

[23] Cybersecurity and Infrastructure Security Agency. National Infrastructure Protection Plan and Designated Critical Infrastructure Sectors Framework. Washington, DC: U.S. Department of Homeland Security, April 2024.

[24] Defense Nuclear Detection / Emergency Support and Coordination Planning Program (2025). Federal program documentation on Alaska Native community vulnerability and federal assistance gaps.

[25] Denali Commission. The Role of Dual-Use Infrastructure in Securing U.S. Arctic Strategic and Economic Interests. Anchorage, AK: Denali Commission, May 2025.

[26] Dewberry Alaska. Alaska Supply Chain Analysis: Baseline Report. Prepared for the Federal Emergency Management Agency Headquarters Logistics Management Directorate under the Supply Chain Analysis Network (SCAN) Pacific Partnership Project. September 2023.

[27] European Space Agency. Arctic Poses Communications Challenges. Paris: ESA, accessed 2025.

[28] European Space Agency. KA-SAT Network Cyber Attack Overview. Paris: ESA, March 2022.

[29] Federal Emergency Management Agency and Argonne National Laboratory. Community Resilience Challenges Index: Methodology for Identifying Commonly Used Indicators from Peer-Reviewed Research Published 2003–2021. Washington, DC: U.S. Department of Homeland Security, August 2024.

[30] Federal Emergency Management Agency. Building Private-Public Partnerships. Washington, DC: U.S. Department of Homeland Security, in coordination with the U.S. Chamber of Commerce Foundation, the National Emergency Management Association, the National Association of Counties, Big City Emergency Managers, and the International Association of Emergency Managers, May 2025.

[31] Federal Emergency Management Agency. Distribution Management Plan Guide. Washington, DC: U.S. Department of Homeland Security, August 2019.

[32] Federal Emergency Management Agency. Restoring Flow of Private Sector Lifelines: Operational Strategies for Emergency Managers. Washington, DC: U.S. Department of Homeland Security, January 2025.

[33] Federal Emergency Management Agency. Supply Chain Resilience Guide. Washington, DC: U.S. Department of Homeland Security, April 2019.

[34] Fox, A. C. Contested Logistics: A Primer. Landpower Essay No. 24-1. Arlington, VA: Association of the United States Army, February 2024.

[35] Government of Canada, Department of National Defence. Arctic Military Communications Partnership: Enhanced Satellite Communications Project — Polar (ESCP-P). Ottawa, ON: Department of National Defence, 2025.

[36] Government of Yukon. Alaska Highway Traffic Statistics and Vehicle Counts. Whitehorse, YT: Government of Yukon, 2023.

[37] Gully, M. "Army Focuses on Contested Logistics — A Threat to Enemy." The United States Army, April 3, 2023.

[38] Hahn, Julia O'Malley, et al. The Messy Reality of Feeding Alaska. Anchorage, AK: Alaska Public Media and Partner Outlets, 2024.

[39] House of Commons Library. 2026. Israel/US-Iran Conflict 2026: Reopening the Strait of Hormuz. Research Briefing CBP-10636. London: UK Parliament.

[40] Journal of Indo-Pacific Affairs. Arctic Strategy. Special issue. Maxwell Air Force Base, AL: Air University Press, 2021.

[41] National Academies of Sciences, Engineering, and Medicine. Strengthening Post-Hurricane Supply Chain Resilience: Observations from Hurricanes Harvey, Irma, and Maria. Washington, DC: The National Academies Press, 2020.

[42] National Oceanic and Atmospheric Administration (NOAA). NOAA Arctic Vision and Strategy. Washington, DC: U.S. Department of Commerce, 2025.

[43] National Science Foundation. Cascadia Region Earthquake Science Center (CRESCENT) Community Risk Work. CRESCENT Research Program, 2023.

[44] Port of Alaska. Port of Alaska Fact Sheet. Anchorage, AK: Municipality of Anchorage, 2020.

[45] RAND Corporation. Insights from the Plan Blue 21 Game. Santa Monica, CA: RAND Corporation, 2021.

[46] RAND Corporation. Exploring Gaps in Arctic Governance: Identifying Potential Sources of Conflict and Mitigating Measures. Santa Monica, CA: RAND Corporation, July 27, 2021.

[47] Richter, Alexander W. 2026. "What the Closure of the Strait of Hormuz Means for the Global Economy." Dallas Fed Economics, March 20, 2026. Federal Reserve Bank of Dallas.

[48] Roque, A. "Army's Contested Logistics Team Close to 'Fully' Operational: General Rainey." Breaking Defense, October 9, 2023.

[49] Rose, Adam, Noah Miller, Zhenhua Chen, Fred Roberts, Andrew Tucci, and Latha Vijayagopal. Economic Impacts of Complex Disasters Affecting Seaports and Their Associated Supply Chains. February 28, 2025.

[50] Scott-Skillern, C., and P. W. Singer. "The Forgotten Part of the Contest: Army Logistics in the Pacific." War on the Rocks, April 29, 2024.

[51] Shaffer, A. R., and W. Miles. "Enabling Logistics in Contested Environments." National Defense, January 5, 2024.

[52] Sherwood, Dave. 2026. "Protests Flare Across Havana as Power Cuts Deepen Amid US Blockade." Reuters, May 13, 2026. Accessed May 18, 2026.

[53] Speranza, C., and D. Taylor. "The Rise of the Disaster Diplomat: A New Era for Domestic Crisis Leadership." HSToday, July 27, 2024.

[54] State of Alaska, Department of Military and Veterans Affairs, Division of Homeland Security and Emergency Management. State of Alaska Emergency Operations Plan. Juneau, AK: State of Alaska, 2024.

[55] State of Alaska, Division of Homeland Security and Emergency Management. Mass Care Group (MCG) Feeding Support Operations Guide. Anchorage, AK: State of Alaska, 2018.

[56] Stockholm Environment Institute and Stockholm Resilience Centre. Arctic Resilience Report. Stockholm, Sweden: Stockholm Environment Institute and Stockholm Resilience Centre, 2016.

[57] Taylor-Catterson, O. "Contested Logistics Industry Week Seeks to Prepare for the Future of Sustainment." The United States Army, September 22, 2023.

[58] Ted Stevens Center for Arctic Security Studies. ICE-PPR Communications Workshop Report. Anchorage, AK: Ted Stevens Center, December 2023.

[59] Ted Stevens Center for Arctic Security Studies. Arctic Climate Change and Its Impact on Homeland Security. Anchorage, AK: Ted Stevens Center, August 2024.

[60] Ted Stevens Center for Arctic Security Studies. High North Security Dialogue 2025: Proceedings and Findings. Anchorage, AK: Ted Stevens Center for Arctic Security Studies, 2025.

[61] Ted Stevens Center for Arctic Security Studies. Arctic Security 2026 Strategy. Anchorage, AK: Ted Stevens Center for Arctic Security Studies, 2026.

[62] Transport Canada. Northern Transportation Infrastructure and Capacity Assessment. Ottawa, ON: Transport Canada, 2022.

[63] United Nations. United Nations Convention on the Law of the Sea (UNCLOS). Concluded at Montego Bay, Jamaica, 10 December 1982; entered into force 16 November 1994. United Nations Treaty Series, Vol. 1833, p. 3. Article 234 (Ice-Covered Areas).

[64] United States Army Pacific. Unified Pacific Wargame Series 2022: Unclassified Overview. Sponsored by the Chief of Staff of the Army. September 2022.

[65] University of Washington Department of Health Systems and Population Health, Disaster Resilience Research Network. Building Disaster Resilient Communities. Seattle, WA: University of Washington.

[66] U.S. Air Force, Headquarters United States Air Forces in Europe. Operation Vittles: The Berlin Airlift, June 26, 1948 – September 30, 1949. Wiesbaden, Germany: U.S. Air Forces in Europe, 1949.

[67] U.S. Army Alaska and U.S. Northern Command. Arctic Edge 2025 Exercise Overview and After-Action Summary. Joint Base Elmendorf-Richardson, AK: USARAK / USNORTHCOM, 2025.

[68] U.S. Coast Guard. Polar Icebreaker Fleet Mix Analysis. Washington, DC: U.S. Coast Guard, 2023.

[69] U.S. Department of Defense. DoD Directive 3020.40: Mission Assurance. Washington, DC: Office of the Under Secretary of Defense for Policy, November 29, 2016.

[70] U.S. Department of Defense. National Defense Strategy of the United States of America. Washington, DC: Office of the Secretary of Defense, 2025.

[71] U.S. Department of Energy. Arctic Strategy: U.S. Department of Energy Strategy to Support Arctic Activities and Priorities. Washington, DC: DOE, October 2022.

[72] U.S. Department of Homeland Security. 2021 Strategic Approach for Arctic Homeland Security. Washington, DC: DHS, January 2021.

[73] U.S. Department of the Navy. A Strategic Blueprint for the Arctic. Washington, DC: Office of the Chief of Naval Operations, January 2021.

[74] U.S. Food and Drug Administration. Good Distribution Practices and Cold Chain Integrity Requirements. Silver Spring, MD: FDA, November 2025.

[75] U.S. Government Accountability Office. Arctic Maritime Infrastructure: Government-wide and Coast Guard Strategies Lack a Means to Measure Progress. GAO-22-104650. Washington, DC: GAO, May 2022.

[76] U.S. Government Accountability Office. Drug Shortages: HHS Should Implement a Mechanism to Coordinate Its Activities. Washington, DC: GAO, July 2024.

[77] U.S. Government Accountability Office. Coast Guard: Arctic Risks Assessed, but Information Gaps and Numerous Challenges Threaten Operations. Washington, DC: GAO, November 2024.

[78] Williams, E. E. FEMA's Contested Logistics: Towards Crisis Standards for Resource Prioritization When Nature Is Your Enemy. Master's thesis, Naval Postgraduate School, March 2025.

[79] Wormuth, C. E. "Secretary of the Army Opening Remarks at AUSA 2023." The United States Army, October 9, 2023.

[80] The White House. National Security Memorandum on Critical Infrastructure Security and Resilience. National Security Memorandum/NSM-22. Washington, DC: Executive Office of the President, April 30, 2024.

APPENDIX C: LIST OF ACRONYMS

The following acronyms appear throughout this report. Entries are grouped by category and listed alphabetically within each group.

Acronym	Full Term
AAC	Alaskan Command
AGU	American Geophysical Union
ALCAN	Alaska-Canada Highway
ANTHC	Alaska Native Tribal Health Consortium
APIP	Alaska Partnership for Infrastructure Protection
ARRC	Alaska Railroad Corporation
ASRC	Arctic Slope Regional Corporation
AUSA	Association of United States Army
AV Gas	Aviation Gasoline
BEOC	Business Emergency Operations Center
CDC	Centers for Disease Control and Prevention
CISA	Cybersecurity and Infrastructure Security Agency
CNA	CAN Corporation (formerly Center for Naval Analyses)
CONUS	Continental United States
COVID-19	Coronavirus Disease 2019
CRCI	Community Resilience Challenges Index (FEMA)
CRS	Congressional Research Service
DCI	Defense Critical Infrastructure
DCIP	Defense Critical Infrastructure Program
DHS	United States Department of Homeland Security
DHS&EM	Alaska Department of Homeland Security and Emergency Management
DOC	United States Department of Commerce

DoD	United States Department of Defense
DoDD 3020.40	DoD Directive 3020.40: Mission Assurance
DOE	United States Department of Energy
DOT	United States Department of Transportation
DOT & PF	Alaska Department of Transportation and Public Facilities
EOC	Emergency Operations Center
EOP	Emergency Operations Plan (State of Alaska)
ESA	European Space Agency
FDA	United States Food and Drug Administration
FEMA	Federal Emergency Management Agency
FEMA P3	FEMA Public-Private Partnership Program
FEMA SCAN	FEMA Supply Chain Analysis Network
GAO	U.S. Government Accountability Office
GEO	Geostationary Earth Orbit
GPS	Global Positioning System
H-2B	Temporary Non-Agricultural Worker Visa Program
HDAS	Homeland Defense Awareness Symposium
HDI	Homeland Defense Institute
HEO	Highly Elliptical Orbit
HF	High Frequency (radio)
HIMARS	High Mobility Artillery Rocket System
IHS	Indian Health Service
INDOPACOM	United States Indo-Pacific Command
JBER	Joint Base Elmendorf-Richardson
LEO	Low Earth Orbit
LNG	Liquefied Natural Gas
MARAD	United States Maritime Administration
MedEvac	Medical Evacuation

NASEM	National Academies of Sciences, Engineering, and Medicine
NDS	National Defense Strategy
NGO	Nongovernmental Organization
NORAD	North American Aerospace Defense Command
NSR	Northern Sea Route
NSM-22	National Security Memorandum 22: Critical Infrastructure Security and Resilience (2024)
OCONUS	Outside the Continental United States
ODNI	Office of the Director of National Intelligence
OPM	Office of Production Management (WWII)
OSD	Office of the Secretary of Defense
P3	Public-Private Partnership
PNW	Pacific Northwest
PPE	Personal Protective Equipment
PSEMA	Private Sector Emergency Management Association
RO/RO	Roll-On / Roll-Off
SAR	Search and Rescue
SATCOM	Satellite Communications
SCADA	Supervisory Control and Data Acquisition
SCOPE	Supply Chain Operational Engagement (CNA methodology)
SCR	Supply Chain Resilience (FEMA Framework)
SPAB	Supply Priorities and Allocations Board (WWII)
TSC	Ted Stevens Center for Arctic Security Studies
TTX	Tabletop Exercise
UNCLOS	United Nations Convention on the Law of the Sea
USARAK	United States Army Alaska
USCG	United States Coast Guard
USDA	United States Department of Agriculture

USNORTHCOM	United States Northern Command
USPS	United States Postal Service
WIB	War Industries Board (WWII)
WPB	War Production Board (WWII)

APPENDIX D: METHODOLOGY AND LIMITATIONS

This appendix documents the methodological detail underlying the analytical lines of work described in the Methodology section of the body of this report. It is organized as a technical reference for HDI, NORAD, and USNORTHCOM reviewers seeking source-level verification of the analytical record, the methodological precedents on which the engagement design rests, and the protocols governing the strategic conversations and stakeholder engagement components. The full Reference List is documented in Appendix B; this appendix references those entries by short-form citation tag rather than reproducing them.

A.1 Literature Inventory by Analytical Strand

The literature review was organized by the five analytical strands the research is structured to address: Arctic lifeline architecture and just-in-time logistics dependency; maritime gateway concentration and West Coast port disruption economics; substitution-mode feasibility (aviation, ALCAN, and Northern Sea Route) under contested conditions; communications fragility and the commercial-provider autonomy gap; and contested-logistics doctrine as it applies to Arctic homeland defense. Each strand is anchored by a small set of authoritative sources, summarized below, and supplemented by domain-specific literature documented in the Reference List.

Arctic lifeline architecture and just-in-time logistics strand. The Dewberry Alaska Supply Chain Analysis Baseline Report, prepared under the FEMA Supply Chain Analysis Network Pacific Partnership Project, provides the most operationally specific characterization of Alaska's just-in-time logistics architecture available in the unclassified literature. [26] It is the primary source for inventory buffer figures, the four-day Tacoma-to-shelf cycle, the absence of meaningful in-state warehousing capacity, single-source wholesale concentration at the Anchorage tier, and the 150 standalone microgrids comprising the rural energy architecture. The Alaska Food Security and Independence Task Force report supplements Dewberry with food-system specificity, particularly the 95-percent import figure and the modal split across maritime, ALCAN, and air-cargo pathways. [4] The State of Alaska Mass Care Group Feeding Support Operations Guide documents the named commercial partners on which state feeding operations depend, establishing the paper-redundancy condition at the operational layer. [55]

Maritime gateway concentration and West Coast port disruption strand. The Port of Alaska Fact Sheet provides the foundational throughput figures: 1.66 million tons inbound, 1.47 million tons of refined petroleum products, 3.1 million barrels of liquid fuel storage capacity, and the dependency picture in which a single intermodal hub serves approximately 90 percent of the state's population. [44] The Rose et al. analysis of economic impacts of complex disasters affecting seaports updates the West Coast disruption economics with current modeling. [49] The Denali Commission dual-use infrastructure assessment provides the cross-state characterization of Alaska as a distributed, fragile supply chain system with few redundancies. [25] The National Academies post-hurricane supply chain resilience study contributes the analytical

framework for cascading port-disruption effects developed across the 2017 hurricane season. [41] The CNA Corporation supply chain analysis capabilities documentation provides the methodological precedent for cross-sector supply chain characterization. [16]

Substitution-mode feasibility strand. The RAND Plan Blue 21 wargame is the primary doctrinal anchor for the substitution-mode analysis, establishing that Arctic operations are logistics-dominated, that sustainment is the pacing function rather than maneuver, and that aviation cannot substitute for maritime cargo volumes at scale. [45] The Operation Vittles operational record from the Berlin Airlift provides the empirical upper bound on aviation-only sustainment under conditions favorable to airlift feasibility, the benchmark against which the seven-hundred-wide-body-sortie figure for replacing one week of Port of Alaska throughput is calibrated. [66] The Government of Yukon Alaska Highway traffic statistics and the Transport Canada Northern Transportation Infrastructure assessment together establish the ALCAN operating throughput (approximately 350 vehicles per day) and the theoretical capacity gap (12,000 to 15,000 vehicles per day) that define the corridor's paper-redundancy condition. [36; 62] The RAND Exploring Gaps in Arctic Governance analysis identifies the cross-border coordination architecture gaps that constrain ALCAN surge under disruption, including Canadian sovereignty, weight-restriction waivers, hours-of-service waivers, and provincial permitting decision points. [46] The U.S. Coast Guard Polar Icebreaker Fleet Mix Analysis and the CRS Polar Security Cutter Program report together document the icebreaker fleet gap that constrains the Northern Sea Route option. [68; 21]

Communications fragility and commercial-provider autonomy strand. The European Space Agency KA-SAT network cyberattack overview documents the 2022 Viasat incident as the canonical cascade pathway in which a cyber vector compounds adversarial-jamming and commercial-autonomy vectors to produce simultaneous defense and civilian impacts. [28] The ESA Arctic communications assessment establishes the high-latitude coverage constraints that make Arctic communications structurally fragile before any disruption begins. [27] The Breaking Defense analysis of the Arctic high-north communications gap and the American Geophysical Union polar air traffic study together document the geostationary coverage limit above approximately seventy degrees north and the space-weather constraints that further degrade polar operations. [6; 11] The CNA supply chain analysis and the CSIS commercialization and dual-use space capabilities assessment document the commercial-provider autonomy vector – the gap in which over half of U.S. satellites are commercially owned and Arctic operations depend on providers whose service availability during conflict is discretionary commercial decision rather than contractual guarantee. [15; 16] The Caton master's thesis at Naval Postgraduate School identifies the federal structural gap that the United States has not formally designated space as a critical infrastructure sector. [12] The Ted Stevens Center ICE-PPR Communications Workshop report and the CSIS analysis of the 2026 National Defense Strategy document the active adversarial degradation vectors: Russian GPS jamming and submarine telecommunications cable interference as ongoing tactics rather than hypothetical risks. [14; 58]

Contested-logistics doctrine strand. The contested-logistics literature developed since 2022 establishes the doctrinal frame within which Arctic mission assurance must now be analyzed. The Shaffer and Miles characterization of logistics in contested environments provides the foundational treatment of fuel as the non-substitutable foundation and the ultimate last-mile challenge in contested theaters. [51] The Fox primer on contested logistics anchors the analytical vocabulary (tight coupling, loose coupling, exterior-lines vulnerability, sustainment as pacing function) used throughout the report. [34] The Scott-Skillern and Singer treatment of Army logistics in the Pacific establishes the exterior-lines vulnerability pattern in which the same maritime gateway nodes sustain both Pacific operations and northern resupply. [50] The Unified Pacific Wargame Series 2022 unclassified overview, the Gully, Roque, Taylor-Catterson, and Wormuth contributions to the Army contested-logistics modernization literature, and the Williams master's thesis on FEMA contested logistics together establish that sustainment is the pacing function in modern conflict and that the homeland is now understood as a strategic support area rather than a sanctuary. [37; 48; 57; 64; 78; 79]

Strategic context anchors. The 2025 National Defense Strategy and the supporting CSIS analysis establish the strategic context in which Arctic mission assurance now operates, including the simultaneity problem the strategy identifies as a primary planning challenge across the Arctic and Western Hemisphere. [14; 70] DoD Directive 3020.40 (Mission Assurance) and National Security Memorandum 22 establish the institutional grounding for treating non-DoD-owned commercial Arctic systems as Defense Critical Infrastructure within mission-assurance scope. [69; 80]

Domain-specific sources supporting fuel and energy systems, food systems, medical and pharmaceutical supply chains, workforce dynamics, and the mitigation and prioritization analysis are documented in the Reference List and are not repeated here.

A.2 Analytical Commitments Embedded in the Methodology

Three analytical commitments are embedded throughout the research and warrant explicit identification.

Throughput, not capacity. The research uses operationally achievable throughput figures rather than theoretical maximum capacity figures whenever the distinction matters. Aviation surge calculations, ALCAN traffic estimates, and port substitution projections are framed against what actually moves under stress conditions, not against nameplate capacity. The capacity-versus-throughput distinction is the single most operationally significant analytical commitment in the research. The seven-hundred-wide-body-sortie aviation substitution figure, the 350-vehicles-per-day ALCAN operating throughput, and the no-viable-alternative finding for the six identified Port of Alaska contingency candidates are each calibrated to throughput rather than capacity. The commitment is methodologically grounded in the contested-logistics literature, which establishes that nameplate capacity figures consistently overstate operational reality under stress. [34; 51]

Operational redundancy, not paper redundancy. The research distinguishes systematically between redundancies documented in plans, contracts, or contingency frameworks and redundancies that perform at the volume, speed, and reliability required under disruption. Mitigation prioritization is shaped by which investments close the gap between the two and which simply formalize a redundancy that does not function. The paper-versus-operational distinction extends to the governance layer: coordination frameworks that exist on paper but have not been exercised, formalized through charter, or integrated into lifeline operator business processes do not meaningfully reduce risk during Days 0–14. The FEMA Public-Private Partnership maturity framework is the source for the Basic / Intermediate / Advanced characterization of coordination capacity that anchors the governance-layer application of the principle. [30]

Arguments lead; research corroborates. The presentation of findings prioritizes the analytical claim and uses the literature to corroborate rather than to construct the argument. Source attribution is precise where it matters operationally, particularly for figures, thresholds, and policy citations, but the report does not foreground source names as a substitute for analytical reasoning. This commitment shapes the citation convention used throughout the body: inline citations are reserved for direct quotes, specific statistics, and named legal or policy authorities, while cross-source synthesis and analytical findings are stated as findings rather than as source-attributed claims. The Reference List in Appendix B is the comprehensive sourcing record; the body of the report is the analytical argument the sources corroborate.

A.3 Cross-Source Validation Patterns

Where individual findings rest on multiple independent sources, the convergence is noted in the body of the report. Particularly consequential examples are documented here as the cross-source validation record for the most analytically significant conclusions.

Maritime gateway concentration risk is cross-validated across the Dewberry Alaska Supply Chain Baseline Assessment, the Denali Commission dual-use infrastructure assessment, the National Academies post-hurricane supply chain resilience study and the Rose et al. economic impacts analysis. The convergent finding across these sources is that a full West Coast port shutdown produces $1.9 to $2.5 billion per day in national economic losses with no rerouting alternative available at sufficient scale within the early-phase window, and that Alaska's downstream concentration at the Port of Alaska means that disruption arrives statewide, immediately, at a single point with no equivalent substitute. [25; 26; 41; 49]

Seasonal asymmetry in fuel delivery is cross-validated across the Dewberry Alaska Supply Chain Baseline Assessment and the Alaska Food Security and Independence Task Force report, both of which document that the October-to-April winter window eliminates barge resupply and concentrates dependence on bush plane aviation. [4; 26] The seasonal asymmetry finding is structurally consistent with the AGU polar air traffic analysis of space-weather and weather-driven aviation constraints. [6]

The paper-redundancy pattern at the maritime, aviation, and overland substitution layers is cross-validated across the contested-logistics literature, the Alaska Supply Chain Baseline Assessment, the Yukon and Transport Canada infrastructure assessments, and the RAND Plan Blue 21 wargame outputs. The convergent finding is that the redundancies invoked in planning documents (alternate ports, aviation surge, ALCAN throughput, Northern Sea Route rerouting) consistently fail to perform at the volume, speed, and reliability the dual-theater scenario requires. [26; 34; 36; 45; 51; 62]

The FEMA Public-Private Partnership maturity model is validated against the operational records of the Nebraska Preparedness Partnership twelve-hour logistics coordination during 2019 flooding and the Houston Supply Chain Group COVID-19 response pivot, both documented in the Building Private-Public Partnerships guidance. [30] These cases establish that where the P3 architecture has been built and exercised, coordination moves at the pace disruption requires; where it has not, coordination must be improvised under time pressure with predictably degraded results.

The commercial-provider autonomy gap in Arctic communications is cross-validated across the CNA supply chain analysis, the CSIS commercialization and dual-use space assessment, the Ted Stevens Center ICE-PPR Communications Workshop report, and the Caton thesis on federal coordination architecture. The convergent finding is that no single satellite architecture provides full Arctic coverage, that commercial provider service availability during conflict is discretionary commercial decision rather than contractual guarantee, and that the absence of a critical-infrastructure-sector designation for space concentrates risk in a coordination architecture that does not currently exist. [12; 15; 16; 58]

A.4 Strategic Conversations Protocol

Strategic conversations with current and former NORAD and USNORTHCOM leadership were conducted on a non-attributed basis. The conversations were not structured interviews and did not follow a fixed question set; they were oriented toward validating the operational relevance of the dual-theater scenario, confirming command-level recognition of the lifeline and infrastructure gaps under examination, and identifying the research areas most consequential for mission-assurance planning within the three-month performance window.

The conversations are not treated in this report as primary data; they are treated as analytical-scope validation. Findings, recommendations, and analytical claims throughout the report rest on the literature record and the structured stakeholder engagement, not on conversation content. This separation is deliberate and protects both the candor of the conversations and the analytical integrity of the report.

A.5 Stakeholder Engagement Instrument and Distribution

Stakeholder engagement was distributed through PSEMA's network and through targeted outreach to the sectors identified in the engagement plan, including infrastructure and access systems; energy, utilities, and fuel; supply chain, logistics, and distribution; communications and technology; community organizations and nonprofit operators; state, local, and territorial agencies; government agencies. The question architecture was calibrated to the dual-theater disruption scenario and oriented toward four analytical objectives: surfacing dependencies and chokepoints across the five-layer Arctic dependency framework; identifying substitution limits and failure thresholds at the operator level; mapping civil–military coordination gaps in the Days 0–14 window; and identifying the early-phase decision points at which lifeline systems are most likely to strain.

Engagement participation was self-selected, and the resulting respondent set is systematically more disposed toward cross-sector coordination than the sector population as a whole. The non-attributed aggregation approach was a precondition for candid engagement and is preserved throughout the analysis. Where stakeholder input refines or extends literature-derived findings, the integration is documented at the section level rather than presented as a separate evidence stream. Where stakeholder input diverges from literature-derived findings, the divergence is treated as an operationally consequential finding to be carried forward rather than as practitioner error to be reconciled. This is the methodological precedent adopted from the mental-models tradition in risk research, which holds that decision-relevant risk information is what practitioners and decision-makers actually use to make choices under uncertainty, not what analysts would prefer that they used.

A.6 Tribal Engagement Protocol

Tribal and Alaska Native communities occupy a distinct position in the Arctic vulnerability picture that this research treats analytically as a risk-amplifier dimension. The risk-amplifier framing identifies the structural conditions under which disruption impacts will be most severe and where pre-incident coordination investment yields the highest resilience return; it does not assign causation to tribal communities, and it does not substitute for the sovereignty, consent, and reciprocity framing under which substantive tribal engagement properly occurs.

Within the engagement window for this research, tribal institutional engagement did not occur at a level sufficient to constitute primary data, and the report does not claim tribal participation it did not have. The risk-amplifier findings rest on the documentary record: the CRS Arctic report on infrastructure damage to Alaska Native communities from erosion, flooding, and permafrost degradation; the GAO Alaska Native Issues report on federal programmatic access barriers; the Behe and related Indigenous voices climate-adaptation literature; and the Alaska Federation of Natives community resilience documentation. [2; 9; 19] Functional coordination capacity between non-tribal institutions and tribal governments is treated throughout the report as dependent on pre-existing relationships that cannot be improvised under disruption conditions, with

statutory consultation mechanisms identified as not substituting for substantive operational engagement built over time.

Substantive tribal engagement on terms tribal governments determine, with sufficient time for the institutional relationships and consent architecture that meaningful participation requires, is identified across the report as a priority for follow-on research and as a precondition for any mitigation pathway involving tribal infrastructure, workforce, or governance contributions. The framing that tribal engagement operates outside the default aggregation framework is itself documented as a methodological commitment of this research, not as a limitation of it. The companion study, *Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense*, addresses tribal and Alaska Native communities under the sovereignty, consent, and reciprocity framing appropriate to its continuity-of-society scope; this research preserves analytical distinctness by holding tribal content within the risk-amplifier frame.

A.7 Bounded Scope and Acknowledged Limits

Methodology limits are documented at section level in the Methodology section of the body and are summarized here for the technical reference record. The directional findings established in this research, that inventory buffers are measured in days rather than weeks, that cascading failure initiates within the first seventy-two hours, that governance friction compounds degradation during Days 0–14, and that paper redundancy at the maritime, aviation, overland, and governance layers consistently fails to perform under Arctic disruption conditions, establish the shape of the problem. They do not yet establish the precise, decision-grade parameters that investment, pre-positioning, and coordination decisions ultimately require. Time-to-failure thresholds expressed in hours and days, by commodity, location, and disruption condition, cannot be derived from published sources alone; they require sustained engagement with operators who have experienced Arctic system performance under real conditions. Producing them is the organizing purpose of the follow-on research and the exercise-based validation identified in the Recommendations section of the Final Report and in Appendix H.

The engagement-window-bounded scope of this research, the unclassified and non-attributed bounds on the data accessible under the engagement, and the absence of decision-grade quantification at precision below the directional findings are documented in the Methodology section. The recommendations section identifies follow-on engagement, exercise-based validation, and the integrated Arctic logistics testbed concept as near-term priorities that address what the current methodology cannot deliver.

A.8 Companion-Study Coordination and Analytical Distinctness

This research is conducted under a shared scenario framework with the companion study, *Continuity of Society in the North: Multi-Modal Resiliency in Arctic Defense*, which addresses human-systems dimensions of the same disruption environment. The two studies are analytically distinct in scope and terminology but mutually reinforcing in conclusion. The shared scenario framework ensures findings are analytically comparable;

the analytical distinctness ensures that each study delivers an integrated picture within its own scope without duplicating the other.

Terminology distinctness is maintained as a standing editorial guardrail. This research uses Arctic Sustainment Modes and Lifeline Constraints as the primary terminology for the four-mode sustainment architecture and the five-layer Arctic dependency framework; the companion study uses Multi-Modal Resiliency as its primary frame. This research uses the paper-redundancy and operational-redundancy distinction as the organizing analytical commitment for substitution-mode feasibility and coordination-architecture maturity; the companion study addresses parallel coordination findings under the continuity-of-society frame. Tribal and Alaska Native engagement in this research is held within the risk-amplifier framing documented in A.6; the companion study addresses tribal and Alaska Native communities under sovereignty, consent, and reciprocity framing. Communications degradation is identified as a force degrader (never as a force multiplier) in both studies, with this research developing the four-vector degradation surface (weather and environmental, adversarial jamming and physical interference, cyber, and commercial provider autonomy) within the lifeline-architecture frame.

Citation conventions are aligned across the two studies. Inline citations are reserved for direct quotes, specific statistics, and named legal or policy authorities. Cross-source synthesis is stated as analytical finding rather than as source-attributed claim. The Reference List in Appendix B is the comprehensive sourcing record for this research; the companion study maintains a separate Reference List under its own analytical scope. Where the two reference lists overlap, the overlap is intentional and reflects the shared scenario framework, not a duplication of analytical scope.

APPENDIX E: STAKEHOLDER ENGAGEMENT DETAIL

This appendix summarizes the stakeholder engagement conducted in support of this research. All individual organizations and individuals are de-identified consistent with the non-attribution principle established in the engagement methodology.

Engagement Principles

Engagement was governed by three principles: (1) system-of-systems by design, with question architecture surfacing cross-sector interdependencies; (2) structured rather than open-ended, with a standardized question framework asked across all participants; and (3) aggregated and non-attributed by default, with no proprietary or sensitive data required and qualitative thresholds and operational insights treated as sufficient. Participation does not imply endorsement of findings.

Sectors Engaged

Engagement was distributed across the lifeline sectors examined in this research:

Energy and Fuel. Power generation; bulk fuel; aviation fuel; heating fuel; the single-source aviation gasoline dependency at the Port of Alaska; in-state refinery operators; the primary Railbelt utility; Western Alaska diesel distribution.

Transportation and Logistics. Ports including the Port of Alaska and the Tacoma-to-Alaska maritime corridor; maritime carriers; trucking; aviation cargo; the six identified contingency alternatives to the Port of Alaska; Alaska Railroad emergency freight activation; ALCAN Highway pre-scripted waiver pathways.

Food and Retail Distribution. Wholesale distribution; grocery; cold chain; warehouse operations; the four-day Tacoma-to-shelf transit cycle; primary food service distributors; the sole produce distributor; the dominant retail tier; state feeding operations.

Communications and Data Systems. Satellite operators; fiber networks; commercial carriers; Arctic ground station operators; contractual resilience commitments; network segmentation practices; cyber incident response capacity.

Medical Supply, Cold Chain, and Medical Evacuation. Pharmaceutical distributors; cold-chain logistics; hospital systems; medevac operators; Indian Health Service facilities; Alaska Native tribal health organizations.

Warehousing and Distribution Nodes. Regional hubs; intermodal staging; pre-positioning operations.

Workforce and Maintenance Operators. Transient, seasonal, and locally rooted labor pools on which critical systems depend.

Emergency Management and Coordination Entities. State, federal, and interagency coordination architectures including the Alaska Mass Care Group and Feeding Task Force, the Alaska Food Value Chain Coordination Council, and the Alaska Partnership for Infrastructure Protection.

Engagement Modalities

Three complementary inputs were used: a structured survey instrument distributed across sector contacts; informal conversations with operators and subject-matter experts; and a literature review of authoritative public sources, sector reporting, federal and state planning documents, and prior PSEMA engagement records. Findings reflect the convergence of these inputs rather than a structured interview program.

Engagement Intensity by Sector

Substantial engagement: Maritime and port operators; logistics and 3PL providers; energy and fuel distributors; food retail and wholesale; emergency management and coordination entities.

Moderate engagement: Communications and data systems; medical supply and medevac; warehousing and distribution; workforce and maintenance.

Engagement gaps: Direct fuel and energy operator participation, direct maritime and port operator participation, and direct communications and telecommunications provider participation were below the levels that sector-specific quantitative claims would require. Tribal and Alaska Native organizational engagement did not occur within the engagement window and is identified as a priority for future research under sovereignty, consent, and reciprocity framing.

Interpretation Considerations

Survey participation was self-selected; respondents who chose to engage are systematically more disposed toward cross-sector coordination than the sector population as a whole, which is a relevant consideration when interpreting the strength of the joint government–industry coordination consensus. The non-attributed aggregation approach protects respondent confidentiality and was a precondition for candid engagement but limits the specificity with which individual findings can be sourced. Engagement findings are treated as directional validation of the analytical record and as identification of practitioner-versus-analytical divergences worth carrying forward, not as the basis for sector-specific quantitative claims in under-represented categories.

Question Architecture

The structured question framework was organized around six analytical emphasis areas derived directly from the dependency, risk, and substitution analyses. Each area was probed through standardized questions designed to surface failure thresholds that operators may not have consciously articulated because the conditions that would trigger them have not yet been experienced.

Failure Pathways
What fails first under disruption conditions; how impacts propagate across interdependent lifeline systems; where cascading effects accelerate from sequential to convergent failure. The canonical home for the cascade analysis is the Cascading Impact Narratives section, with the underlying mechanism developed in the Cascading Failure Analysis: From Trigger to Convergence subsection of the Focused Risk and Vulnerability Assessment.

Throughput and Capacity Constraints
Maximum operational throughput under degraded conditions; volume limitations across maritime, aviation, and overland transportation; the gap between theoretical capacity and operationally achievable throughput under the stress conditions specific to the dual-theater scenario.

Resource Competition and Prioritization
Allocation decisions under constrained supply; interactions between civilian, defense, and tribal demand for fuel, airlift, medical evacuation, and communications capacity; internal prioritization criteria that operators apply when external prioritization frameworks have not been pre-defined. The canonical home is the Civilian–Military Convergence subsection of the Focused Risk and Vulnerability Assessment, with the institutional-architecture dimension developed in the Institutional Coordination as a Risk Amplifier section.

Time Sensitivity
Decision timelines under disruption; points at which delayed action becomes operationally limiting; the Days 0–14 analytical window as practitioners experience it from inside their own operations. Particular attention to the inflection points within the window at which sustainment assumptions fail and the self-reinforcing character of the cascade takes over, and to the tempo mismatch between commercial supply chains operating on hours timelines and government activation operating on days' timelines.

Substitution and Operational Reality
Feasible alternatives versus theoretical options; constraints driven by weather, workforce, infrastructure, and scale; the specific conditions under which a documented redundancy ceases to perform the function planning frameworks assume it performs. This emphasis area operationalizes the paper-versus-operational-redundancy analytical commitment that organizes the substitution analysis throughout the report. The canonical home is the Substitution and Redundancy Reality Check, where the pattern is shown to be systemic rather than incidental across the maritime, aviation, overland, fuel, communications, pharmaceutical, feeding, and governance layers simultaneously.

Governance and Authority Gaps
Decisions on how scarce resources are allocated in practice, and where that authority diverges from what formal coordination frameworks specify; where civilian, military, tribal, and commercial authority structures intersect or operate in parallel without coordination, particularly for fuel, airlift, medical evacuation, and communications prioritization; what pre-established coordination frameworks exist versus where

improvised coordination is the expected mode, and what happens when improvised coordination under time pressure fails. The canonical home is the Institutional Coordination as a Risk Amplifier section, where coordination architecture is treated as itself a primary cascade trigger rather than as a passive backdrop to lifeline failure.

Engagement Conduct

Engagement was conducted under non-attribution and aggregation rules, with no proprietary or sensitive data required and qualitative thresholds and operational insights treated as sufficient. A SurveyMonkey-distributed Stakeholder Input Survey of fifteen items operated in parallel using the same six-area question architecture documented above. Distribution channels included PSEMA member channels, the broader PSEMA distribution list, and direct outreach to non-PSEMA strategic respondents. The engagement window closed April 30, 2026, following an extension to ensure adequate sectoral representation across infrastructure and access systems; energy, utilities, and fuel; supply chain, logistics, and distribution; communications and technology; community organizations and nonprofit operators; state, local, and territorial agencies; and retired military.

Tribal and Alaska Native engagement is governed by sovereignty, consent, and reciprocity protocols rather than by the default aggregation framework applied to other stakeholder input. Within the engagement window for this research, tribal institutional engagement did not occur at a level sufficient to constitute primary data, and the report does not claim tribal participation it did not have. Risk-amplifier findings concerning tribal and Alaska Native communities rest on the documentary record rather than on stakeholder input, and substantive tribal engagement on terms tribal governments determine is identified as a priority for follow-on research. The full tribal engagement protocol is documented in Appendix D, Section D.6.

Integration of Engagement Input into the Analytical Record

Where stakeholder input refines or extends literature-derived findings, the integration is documented at the section level rather than presented as a separate evidence stream. Where stakeholder input diverges from literature-derived findings, the divergence is treated as an operationally consequential finding to be carried forward rather than as practitioner error to be reconciled. This is the methodological precedent adopted from the mental-models tradition in risk research and is documented in Appendix D, Section D.2. Two operationally distinctive findings surfaced through engagement warrant explicit identification here. First, the seasonal lock-in dimension surfaces from defense logistics framing as a binding constraint that the Days 0–14 frame may understate: when fall barge season closes before disruption arrives, the only operational alternative in some communities is to wait for spring breakup, extending the no-substitution window from days into months. This dimension is consistent with the seasonal asymmetry analysis developed in the Scenario Development section and reinforces it from the practitioner side. Second, the hub-and-spoke last-mile community delivery pattern surfaces from the humanitarian response sector as a substitution pathway distinct from commercial logistics: where central hubs can receive bulk goods, locally-retrieved last-mile delivery using traditional transport provides a delivery mechanism that does not depend on commercial logistics or aviation substitution. This is operationally significant as a

community-resilience dimension that the analytical record had not previously developed and that is integrated into the cross-modal logistics redundancy mitigation pathway.

The thematic pattern across open-ended cascade descriptions reinforces the analytical finding that failure is convergent rather than sequential. Practitioners describe fuel, power, and communications as failing together because they share underlying infrastructure, not as a clean sequence in which one system fails before the next. The systematic identification of paper-redundancy conditions across sectors converges with the substitution catalog developed in the Substitution and Redundancy Reality Check, and no respondent described a paper redundancy that the analytical record had not also identified, the cross-sector convergence is in identifying the same paper-redundancy conditions from independent operational vantage points. Practitioner input on Days 0–14 inventory thresholds, fuel allocation decision criteria, communications resilience under commercial provider autonomy, and the operational adequacy of existing coordination architecture refines the findings developed in the Targeted Dependency Mapping, Focused Risk and Vulnerability Assessment, Substitution and Redundancy Reality Check, and Institutional Coordination as a Risk Amplifier sections respectively.

The SurveyMonkey questions and responses are located at the end of the Appendices.

APPENDIX F: HISTORICAL PRECEDENTS REFERENCE

This appendix consolidates the historical precedents referenced or implied in the body of the report. Each precedent is summarized with sufficient detail for follow-on researchers to verify the parallel without re-deriving it.

Operation Vittles (Berlin Airlift), 1948–1949

U.S. and allied operation sustaining the western sectors of Berlin by air after Soviet ground blockade of road and rail access. Operation duration June 1948 through September 1949. Sustainment volume reached approximately 8,000 tons per day at peak operations into Tempelhof, Tegel, and Gatow airfields. Aircraft cycles approached the operational limit of the airfield infrastructure available. Operation analytically significant for Arctic mission assurance because it establishes the upper bound of aviation-only sustainment under conditions of constrained airfield access, weather limitations, and geopolitical pressure. Cited in this report as the analog for evaluating the feasibility of Arctic airlift substitution for maritime cargo.

Galena Flood, May 2013

Yukon River ice-jam flooding of the village of Galena, Alaska. Event significant for Principal Investigator's direct operational experience supporting the State of Alaska response and for the operational lessons drawn about transportation, supply chain continuity, and feeding operations in remote Arctic regions. Cited in the Continuation Statement and as part of the research foundation establishing the operational experience of PSEMA leadership.

Typhoon Mawar, Guam, May 2023

Category 4 typhoon causing extensive infrastructure damage to Guam and Andersen Air Force Base. Cited in the HDI Holistic Defense Awareness research focus area as an example of disasters and adversary actions detrimentally impacting military installations. Relevant to this research as an analog for installation-community interdependency under infrastructure disruption.

War Industries Board (WIB), 1917–1918 (WWI)

Established by Executive Order in July 1917 under the Council of National Defense to coordinate U.S. industrial mobilization for World War I. Operated for approximately sixteen months before achieving substantive operational effectiveness under Bernard Baruch's chairmanship in March 1918. Demonstrates both the value of centralized priorities adjudication and the cost of constructing it after conflict has begun. The foundational U.S. case for the proposition that resource-adjudication architecture should be pre-established rather than constructed under stress.

Office of Production Management (OPM), Supply Priorities and Allocations Board (SPAB), and War Production Board (WPB), 1941–1945 (WWII)

Successive WWII U.S. industrial mobilization agencies. The OPM was established in January 1941. The SPAB was established in August 1941. The WPB was established in January 1942 under Executive Order 9024 with priorities and rationing authority over the U.S. industrial base. Allocated steel, aluminum, rubber, fuel, and consumer goods across military and civilian demand throughout the war. The WPB's effectiveness rested on the pre-1941 foundation laid by the OPM and SPAB. Demonstrates that adjudication architecture builds on prior architecture rather than emerging from improvisation, and that institutional learning required is not compressible to crisis timelines.

COVID-19 Allocation Framework, 2020–2022

Federal framework for allocating ventilators, personal protective equipment, vaccines, and pharmaceuticals across states during the COVID-19 pandemic. The framework operated through multiple integrated components: Defense Production Act authorities; the FEMA temporary final rule on prioritization and allocation of scarce health and medical resources (April 2020); CDC Advisory Committee on Immunization Practices (ACIP) guidance on vaccine prioritization; and the National Academies of Sciences, Engineering, and Medicine's Framework for Equitable Allocation of COVID-19 Vaccine (2020). Demonstrates that pre-existing planning infrastructure (the 2009 H1N1 response architecture, university medical ethics allocation models) produced faster and more institutionally defensible decisions than architecture constructed during the crisis. The closest contemporary analog to the civil–military adjudication architecture this research recommends.

APPENDIX G: LEGAL AND POLICY AUTHORITY INDEX

This appendix indexes the legal and policy authorities referenced in the body of the report, with one-paragraph entries summarizing the relevance of each authority to Arctic mission assurance. Full citations appear in Appendix B (Reference List).

Defense Production Act (DPA)

Statutory authority enabling federal allocation and prioritization of scarce industrial output, raw materials, and other resources under national defense necessity. Identified in this research as the regulatory mechanism through which adjudicated allocations can be implemented during the Days 0–14 window if pre-staged civil–military adjudication protocols exist.

DoD Directive 3020.40, Mission Assurance

Establishes DoD policy framework for mission assurance as the integrating concept for the defense of critical capabilities, assets, infrastructure, and information. Identifies non-DoD-owned infrastructure as a defense dependency. Cited throughout this research as the doctrinal foundation for the proposition that mission assurance depends on civilian critical infrastructure, much of it privately operated.

DoD Instruction 3020.45, Defense Critical Infrastructure Program Management

Implementing instruction for DoDD 3020.40, defining roles and responsibilities for DoD's critical assets and infrastructures. Establishes the Defense Critical Infrastructure Program (DCIP) management framework. Relevant to this research as the framework under which non-DoD-owned Defense Critical Infrastructure (DCI) is identified and protected.

Jones Act (Merchant Marine Act of 1920)

Requires that cargo moved between U.S. ports be carried on vessels built in the United States, owned by U.S. citizens, and crewed by U.S. mariners. Identified in this research as a structural constraint on maritime substitution capacity during disruption. The waiver framework, administered by the Department of Homeland Security with Department of Defense, MARAD, and DOT coordination, is identified as the determining variable in whether maritime surge can be activated within the Days 0–14 window.

National Defense Strategy (NDS-2025)

U.S. national defense policy framework establishing strategic priorities. Cited in this research for its Arctic operations and homeland defense doctrine and as the strategic-level authority connecting Arctic mission assurance to broader national defense objectives.

National Security Memorandum 22, Critical Infrastructure Security and Resilience (30 April 2024)

White House policy framework defining Defense Critical Infrastructure and establishing federal coordination authorities for critical infrastructure resilience. Defines DCI as DoD and non-DoD networked assets and facilities essential to project, support, and sustain military forces and operations worldwide. Cited throughout this research as the strategic-level policy authority for the cross-sector coordination architecture the recommendations call for.

Stafford Act
(Robert T. Stafford Disaster Relief and Emergency Assistance Act)

Federal statutory authority for FEMA-coordinated disaster response, including the Presidential disaster declaration framework, mass care operations, and federal-state cost-share for disaster relief. Relevant to this research as the authority structure under which FEMA Region X would coordinate Arctic disaster response and as one of the legal frameworks across which civil–military adjudication must operate.

Title 10 vs. Title 32 Authority

Title 10 authority governs federal active-duty military forces under the Secretary of Defense; Title 32 authority governs National Guard forces operating under state authority while federally funded. The distinction is operationally consequential in disaster response because Title 32 forces can support law enforcement and civilian authorities in ways Title 10 forces typically cannot. Civil–military adjudication protocols must address allocation decisions across Title 10, Title 32, Stafford Act, DPA, and tribal sovereign authorities.

Tribal Sovereign Authority Frameworks

Alaska Native and tribal governments hold sovereign authority over tribal lands, communities, and citizens. Sovereign authority is exercised under federal trust responsibility, the Alaska Native Claims Settlement Act (1971), tribal consultation policies, and government-to-government coordination protocols. Engagement is governed by sovereignty, consent, and reciprocity principles. Civil–military adjudication protocols must address tribal sovereign authority alongside Title 10, Title 32, Stafford Act, and DPA authorities.

UNCLOS Article 234
(United Nations Convention on the Law of the Sea, 1982)

Authorizes coastal states to adopt and enforce non-discriminatory laws and regulations for the prevention, reduction, and control of marine pollution from vessels in ice-covered areas within the limits of the exclusive economic zone. The legal instrument through which the Northern Sea Route is converted from a benign-planning maritime redundancy into a consent-dependent pathway whose availability depends on Russian regulatory consent. Treated in this research as the canonical case of consent-dependent redundancy.

APPENDIX H: RECOMMENDATIONS BY INVESTMENT HORIZON

The recommendations below consolidate the analytical findings of this report into a single, decision-grade reference, organized into three implementation horizons. Near-Term (0–12 months) covers policy clarifications, planning integration, pre-staged authorities, exercise activity, and low-cost mitigations executable within the current planning cycle. Mid-Term (12-36 Months) covers programmatic build-out, infrastructure hardening at existing nodes, contracting actions, and institutional architecture requiring sustained investment beyond a single budget cycle. Long-Term (36 Months or More) covers new infrastructure construction, doctrinal codification, multi-year research, and capital-intensive resilience investments.

Each recommendation is tagged against the Five Criteria Framework introduced in the Mitigation and Cost–Benefit Prioritization section, Mission Impact, Cascade Risk, Time Sensitivity, Feasibility, and Dual-Use Value, with the level assigned to each criterion shown. Tiering reflects implementation horizon rather than priority; near-term, mid-term, and long-term lines of effort run concurrently, not sequentially. Lead and Supporting Partner attributions reflect the operational accountability and enabling-role architecture identified across the dependency mapping, risk and vulnerability assessment, institutional coordination review, and recommendations findings of this report.

Near Term Recommendations (0–12 Months)

Pre-Position Strategic Stockpiles for the Days 0–14 Window

Description: Establish pre-positioned reserves of fuel, food, pharmaceuticals, cold-chain capacity, and critical materiel at Anchorage, Fairbanks, the JBER area, and rural community hubs sized for both routine and disruption-period demand, the only mechanism that closes the inventory-buffer gap within the dual-theater scenario window.

Five Criteria Framework Tags:

Mission Impact: High
Cascade Risk: High
Time Sensitivity: High (Days 0–14)
Feasibility: Moderate
Dual-Use Value: High

Lead Partner(s): USNORTHCOM; Alaska Department of Homeland Security and Emergency Management (Alaska DHS&EM)

Supporting Partner(s): Defense Logistics Agency; FEMA Region X; commercial fuel and pharmaceutical distributors operating in Alaska; rural community hub operators (barge-season-dependent communities); PSEMA

Pre-Stage Jones Act Waiver and Regulatory Authority Documentation

Description: Develop and maintain pre-staged documentation packages supporting Jones Act, weight-limit, hours-of-service, CDL endorsement, and Canadian provincial waivers, converting the waiver process from a capability that may exceed the operational window into a regulatory mechanism activatable within hours.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: Moderate
Time Sensitivity: High
Feasibility: High
Dual-Use Value: High

Lead Partner(s): Department of Homeland Security (DHS); Department of Defense (DoD) Maritime Administration (MARAD); Department of Transportation (DOT)

Supporting Partner(s): USNORTHCOM; FEMA Region X; Alaska DHS&EM; commercial vessel operators serving Alaska (A.P. Moller–Maersk, Crowley Maritime, Lynden Incorporated); Canadian provincial transportation authorities; PSEMA

Initiate Pre-Defined Civil–Military Adjudication Protocols

Description: Develop pre-defined protocols for adjudicating civilian and military claims on scarce fuel, food, transportation, and medical capacity, addressing allocation across Title 10 authority, state emergency management authority, tribal sovereignty, and commercial cargo prioritization with clearly identified decision authorities and pre-established communication pathways.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: High
Time Sensitivity: High (Days 0–14)
Feasibility: Moderate (political complexity)
Dual-Use Value: High

Lead Partner(s): USNORTHCOM; Alaska DHS&EM; tribal governments and Alaska Native organizations (operating under sovereignty, consent, and reciprocity protocols)

Supporting Partner(s): FEMA Region X; CISA; Alaska Federation of Natives; commercial cargo prioritization authorities; PSEMA

Conduct Tabletop Exercise–Based Validation of Existing Coordination Architectures

Description: Convene structured tabletop exercises that test Alaska's Partnership for Infrastructure Protection, Mass Care Group and Feeding Task Force, Alaska Food Value Chain Coordination Council, and federal-state-tribal-military coordination against the dual-theater scenario, converting paper-redundancy coordination into operationally validated capacity.

Five Criteria Framework Tags:
Mission Impact: Moderate–High
Cascade Risk: Moderate
Time Sensitivity: Moderate
Feasibility: High
Dual-Use Value: High

Lead Partner(s): Alaska DHS&EM (Partnership for Infrastructure Protection convener); USNORTHCOM

Supporting Partner(s): FEMA Region X; Alaska Food Value Chain Coordination Council; Alaska Mass Care Group and Feeding Task Force; Canadian Joint Operations Command counterparts; tribal governments and Alaska Native organizations; PSEMA

Establish Communications Hardening Priorities for Multi-Layer Architecture

Description: Identify the specific communications hardening investments, anti-jam capability for primary nodes, redundant LEO/GEO satellite pathways, HF radio backup, and subsea cable protection, that within twelve months produce material reduction in the communications cascade pathway documented in this research.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: Very High (communications = coordination spine)
Time Sensitivity: High
Feasibility: Moderate (some capital required)
Dual-Use Value: High

Lead Partner(s): USNORTHCOM
Cybersecurity and Infrastructure Security Agency (CISA)

Supporting Partner(s): Alaska DHS&EM; AT&T FirstNet; commercial LEO and GEO satellite providers (Iridium Communications, Starlink, Inmarsat); Alaska Public Safety Communications Services; subsea cable operators; PSEMA

Establish Pharmaceutical Cold-Chain Pre-Positioning at Key Nodes

Description: Pre-position insulin, antibiotics, oxygen, dialysis supplies, and other categories where inventory depletion produces life-threatening outcomes within days at regional hospital, IHS, and ANTHC nodes whose service populations cannot be reached by external resupply within the Days 0–14 window, the only mechanism that closes the medical pathway gap.

Five Criteria Framework Tags:
Mission Impact: High (life safety)
Cascade Risk: Moderate
Time Sensitivity: Very High (non-recoverable cold-chain failure)
Feasibility: High
Dual-Use Value: Very High

Lead Partner(s): Indian Health Service (Alaska Area); Alaska Native Tribal Health Consortium (ANTHC); Alaska Native Medical Center

Supporting Partner(s): Commercial pharmaceutical distributors operating in Alaska (McKesson); regional hospitals; CVS Health; Global Medical Response; Guardian Flight Alaska; FEMA Region X; PSEMA

Initiate Sustained Stakeholder Engagement Architecture

Description: Convert the engagement architecture established for this research into a sustained, recurring mechanism connecting DoD mission assurance planners, NORAD/USNORTHCOM J-staff equivalents, Alaska DHS&EM, FEMA, CISA, tribal governments, and private-sector lifeline operators across the seven sectors examined, converting episodic input into continuous data architecture.

Five Criteria Framework Tags:
Mission Impact: Moderate–High
Cascade Risk: High (coordination architecture)
Time Sensitivity: Moderate
Feasibility: High
Dual-Use Value: High

Lead Partner(s): Homeland Defense Institute (HDI); PSEMA (connective infrastructure)

Supporting Partner(s): NORAD and USNORTHCOM J-staff equivalents; Alaska DHS&EM; FEMA Region X; CISA; tribal governments and Alaska Native organizations; private-sector lifeline operators across the seven sectors examined

Mid Term Recommendations (12–36 Months)

Distribute Microgrid Capacity to Reduce Single-Point Energy Dependency

Description: Expand distributed microgrid capacity at rural Alaska communities, regional installations, and dual-use civilian-military sites where energy concentration risk is highest, with diesel-plus-renewable-plus-storage configurations and the workforce capacity and spare-parts logistics required to sustain operations under Arctic conditions.

Five Criteria Framework Tags:

Mission Impact: High
Cascade Risk: Very High (energy = master dependency)
Time Sensitivity: Moderate
Feasibility: Moderate (Arctic workforce/logistics)
Dual-Use Value: Very High

Lead Partner(s): Alaska Village Electric Cooperative; Chugach Electric Association; USNORTHCOM (installation-side coordination)

Supporting Partner(s): Denali Commission; Department of Energy; National Rural Electric Cooperative Association (NRECA); Cold Regions Research and Engineering Laboratory (CRREL); Alaska Energy Authority; PSEMA

Advance Alternate Port Readiness Investments

Description: Initiate capital investment programs to convert Seward, Valdez, Kodiak, Homer, Whittier, and MacKenzie into functional substitutes for containerized cargo handling, including reefer plug-in installation, RO/RO ramps, weight-restriction infrastructure, deep-draft construction, and the rail and road improvements required to connect alternate ports to downstream distribution.

Five Criteria Framework Tags:

Mission Impact: High
Cascade Risk: Very High (port concentration)
Time Sensitivity: Long (strategic, not Days 0–14)
Feasibility: Moderate (capital intensive)
Dual-Use Value: High

Lead Partner(s): Maritime Administration (MARAD); Alaska Department of Transportation and Public Facilities (DOT&PF)

Supporting Partner(s): Ports of Seward, Valdez, Homer, Kodiak, and Whittier; commercial vessel operators (A.P. Moller–Maersk, Crowley Maritime); USNORTHCOM; FEMA Region X; PSEMA

Build Medical Evacuation Surge Capacity

Description: Expand medical evacuation surge capacity through pre-positioned aviation fuel reserves at key medevac airfields, redundant flight-following and medical consultation communications, pre-defined receiving-facility coordination at Anchorage and Seattle tertiary centers, and crew availability programs that build depth in the thin specialty workforce medevac depends on.

Five Criteria Framework Tags:
Mission Impact: High (life safety)
Cascade Risk: High (medevac competition with cargo airlift)
Time Sensitivity: High
Feasibility: Moderate
Dual-Use Value: Very High

Lead Partner(s): Guardian Flight Alaska; Global Medical Response; Alaska Native Medical Center

Supporting Partner(s): Anchorage tertiary care receiving facilities; Seattle tertiary care receiving facilities; Indian Health Service (Alaska Area); USNORTHCOM (aviation deconfliction); commercial aviation fuel suppliers at medevac airfields; PSEMA

Establish Tribal Partner Protocols for Sustained Engagement

Description: Build a standing protocol for tribal and Alaska Native organization participation in Arctic mission assurance research, planning, and operational coordination, addressing the absence of pre-existing relationships that the Risk and Vulnerability Assessment establishes as a primary risk-amplifier condition.

Five Criteria Framework Tags:
Mission Impact: Moderate–High
Cascade Risk: High (risk amplifier when absent)
Time Sensitivity: Moderate
Feasibility: Moderate (requires sovereign consent and reciprocity)
Dual-Use Value: High

Lead Partner(s): Tribal governments and Alaska Native organizations (operating under sovereignty, consent, and reciprocity protocols); Alaska Federation of Natives

Supporting Partner(s): Native regional corporations (Arctic Slope Regional Corporation, NANA Regional Corporation, Doyon Limited, Cook Inlet Region Inc.); First Alaskans Institute; HDI; USNORTHCOM; Alaska DHS&EM; PSEMA

Advance FEMA P3 Maturity Beyond Basic and Low-Intermediate Tier

Description: Build the institutional architecture that progresses Alaska's Partnership for Infrastructure Protection and the federal-state-tribal-military coordination mechanisms from current Basic and low-Intermediate maturity toward Advanced tier, through formalized charters, sustained pre-incident engagement, regular exercise activity, and integration into commercial lifeline operator business processes.

Five Criteria Framework Tags:
Mission Impact: High (institutional architecture)
Cascade Risk: High
Time Sensitivity: Moderate
Feasibility: High
Dual-Use Value: Very High

Lead Partner(s): FEMA (Region X and Office of Business, Industry, and Infrastructure Integration); Alaska DHS&EM (Partnership for Infrastructure Protection)

Supporting Partner(s): CISA; USNORTHCOM; tribal governments and Alaska Native organizations; private-sector lifeline operators across the seven sectors examined; PSEMA

Establish a Permanent Cross-Sector Supply Chain Intelligence Function

Description: Build a permanent coordinating capability for anticipating and responding to cross-sector supply chain disruption, leveraging FEMA's DPA authorities, CISA's Critical Infrastructure Partnership frameworks, and peer-level coordination with DOT, Commerce, and the ODNI, applying CNA's Supply Chain Operational Engagement methodology to Arctic-specific conditions.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: Very High
Time Sensitivity: Moderate
Feasibility: Moderate
Dual-Use Value: Very High

Lead Partner(s): FEMA (Supply Chain Resilience Office); CISA

Supporting Partner(s): Department of Transportation; Department of Commerce; Office of the Director of National Intelligence; CNA; USNORTHCOM; private-sector lifeline operators across the seven sectors examined; PSEMA

Build Cold-Chain and Pre-Positioning Infrastructure at Tribal and Community Health Nodes

Description: Expand cold-chain capacity, backup power, and pre-positioned medical supply at IHS facilities, ANTHC nodes, regional tribal health organizations, and Community Health Aide Program village sites that constitute the only line of healthcare access for many Alaska Native communities, addressing the parallel cold-chain failure pathway documented in the Risk and Vulnerability Assessment.

Five Criteria Framework Tags:
Mission Impact: High (life safety in isolated communities)
Cascade Risk: High (parallel cold-chain failure pathway)
Time Sensitivity: High
Feasibility: Moderate
Dual-Use Value: Very High

Lead Partner(s): Indian Health Service (Alaska Area); Alaska Native Tribal Health Consortium (ANTHC); regional tribal health organizations; Community Health Aide Program (CHAP)

Supporting Partner(s): Tribal governments and Alaska Native organizations; Alaska Federation of Natives; commercial pharmaceutical distributors operating in Alaska; FEMA Region X; PSEMA

Long Term (36 Months or More)

Extended Sustainment Modeling Beyond the Days 0–14 Window

Description: Conduct extended sustainment modeling for the recovery dynamics, structural reconstitution, and multi-month sustainment trajectories that lie beyond the analytical window of this research, providing the time-to-failure parameter precision that command-level decisions ultimately require.

Five Criteria Framework Tags:
Mission Impact: High (informs all longer-horizon decisions)
Cascade Risk: High
Time Sensitivity: Long
Feasibility: High
Dual-Use Value: High

Lead Partner(s): Homeland Defense Institute (HDI); RAND Corporation

Supporting Partner(s): Ted Stevens Center for Arctic Security Studies; Naval Postgraduate School; CNA; university research centers with Arctic specialization; USNORTHCOM; PSEMA

Multi-Year Resilience Studies Across Lifeline Sectors

Description: Conduct multi-year resilience studies that build on the dependency mapping and risk assessment foundation of this research, extending the analytical scope to Cook Inlet natural gas depletion, polar icebreaker fleet recapitalization, Arctic ground station resilience, and undersea cable protection over a five-year horizon.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: High
Time Sensitivity: Long
Feasibility: High
Dual-Use Value: High

Lead Partner(s): Homeland Defense Institute (HDI); Ted Stevens Center for Arctic Security Studies

Supporting Partner(s): RAND Corporation; Naval Postgraduate School; CRREL; Dewberry; Tetra Tech; AECOM; USNORTHCOM; FEMA Region X; PSEMA

Operational Piloting of Recommended Coordination Architectures

Description: Pilot the coordination architectures identified in the near-term and mid-term recommendations under live operational conditions, federally declared emergencies, large-scale exercises, and Arctic-specific TTX programs, converting coordination from validated-on-paper to validated-in-operations.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: High (addresses paper-redundancy condition)
Time Sensitivity: Moderate
Feasibility: Moderate (requires sustained commitment)
Dual-Use Value: High

Lead Partner(s): USNORTHCOM; FEMA Region X; Alaska DHS&EM

Supporting Partner(s): HDI; tribal governments and Alaska Native organizations; Canadian Joint Operations Command counterparts; private-sector lifeline operators across the seven sectors examined; PSEMA

Establish Integrated Arctic Logistics Testbed

Description: Establish an integrated Arctic logistics testbed supporting operational evaluation of mitigation investments, coordination architectures, and emerging logistics technologies under realistic Arctic conditions, the institutional vehicle through which research, operational capability, and private-sector partnership converge.

Five Criteria Framework Tags:
Mission Impact: High
Cascade Risk: High
Time Sensitivity: Long
Feasibility: Moderate (multi-agency)
Dual-Use Value: Very High

Lead Partner(s): Department of Defense (Office of the Secretary of Defense); Maritime Administration (MARAD); FEMA

Supporting Partner(s): Department of Transportation; State of Alaska; tribal governments and Alaska Native organizations; private-sector lifeline operators across the seven sectors examined; HDI; CRREL; PSEMA

Codify DoD–Private Sector–Nongovernmental Coordination Doctrine

Description: Develop codified doctrine for DoD–private sector–nongovernmental coordination under contested-logistics conditions, building on NSM-22's defense critical infrastructure framework, the FEMA P3 maturity model, the Jones Act waiver framework, and the contested-logistics literature.

Five Criteria Framework Tags:
Mission Impact: Very High (doctrinal foundation)
Cascade Risk: High
Time Sensitivity: Long
Feasibility: Moderate (institutional process)
Dual-Use Value: Very High

Lead Partner(s): NORAD and USNORTHCOM; Office of the Secretary of Defense; FEMA

Supporting Partner(s): CISA; HDI; Joint Chiefs of Staff; emergency management and private-sector partnership community; PSEMA

Sustain Long-Horizon Infrastructure Investments: Polar Icebreakers, Processing Capacity, Refining

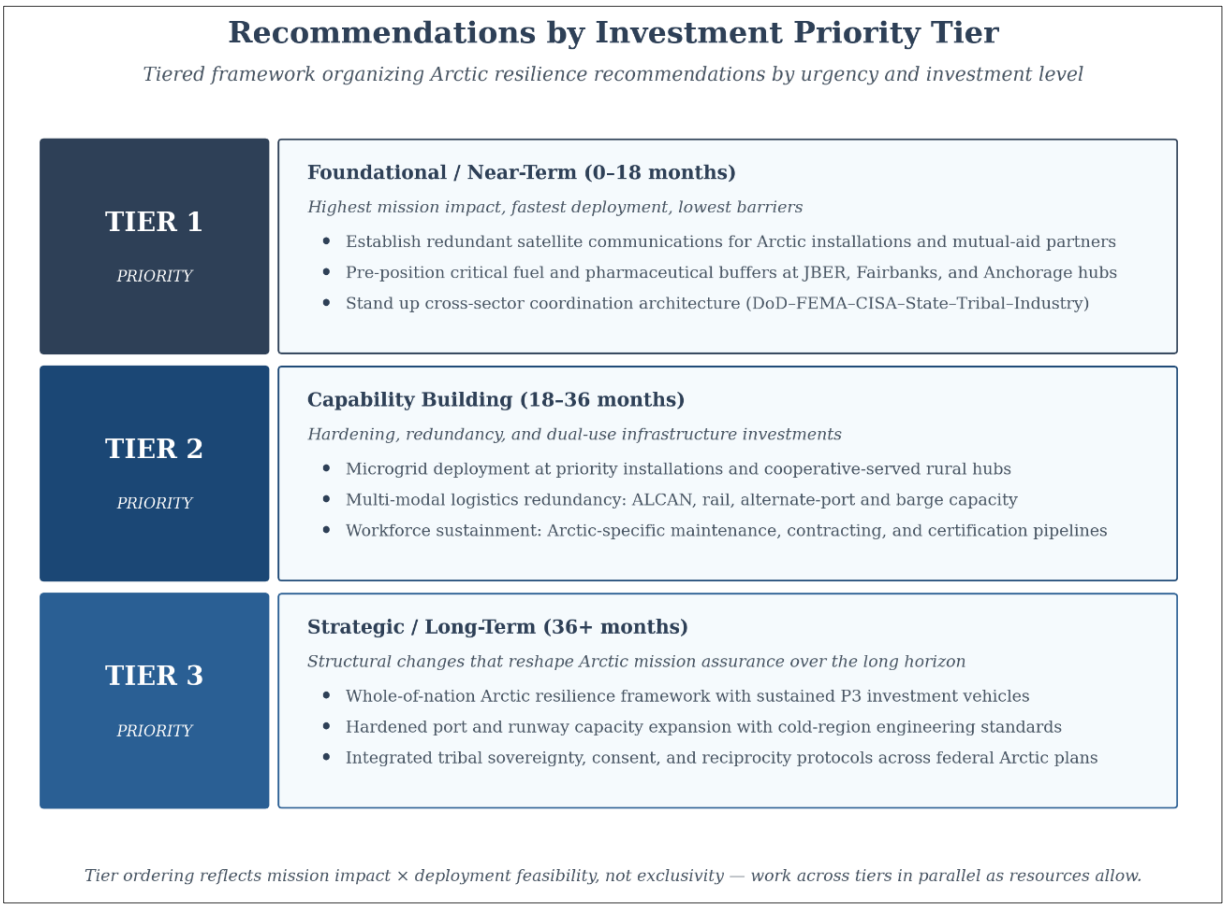
Description: Sustain the long-horizon infrastructure investments necessary for structural Arctic resilience: polar icebreaker fleet recapitalization through the Polar Security Cutter program and Arctic Security Cutter RFI, in-state food processing capacity expansion, and in-state refining capacity assessment.

Five Criteria Framework Tags:

Mission Impact: Very High (structural baseline)
Cascade Risk: Very High
Time Sensitivity: Long (multi-year construction)
Feasibility: Low–Moderate (capital and timeline intensive)
Dual-Use Value: Very High

Lead Partner(s): U.S. Coast Guard (Polar Security Cutter and Arctic Security Cutter programs); Department of Defense; State of Alaska

Supporting Partner(s): Maritime Administration (MARAD); Alaska Federation of Natives (subsistence and in-state food-processing context); in-state refining capacity stakeholders; Department of Energy; Department of Agriculture; PSEMA



Tier assignment reflects composite score and timing; tiers are not strict, and recommendations may move between tiers as conditions change.

Q1

9 responses

Your Perspective1. Which sector best represents your perspective?(Select one)



RESPONSES (0) WORD CLOUD TAGS (0)

Search Responses

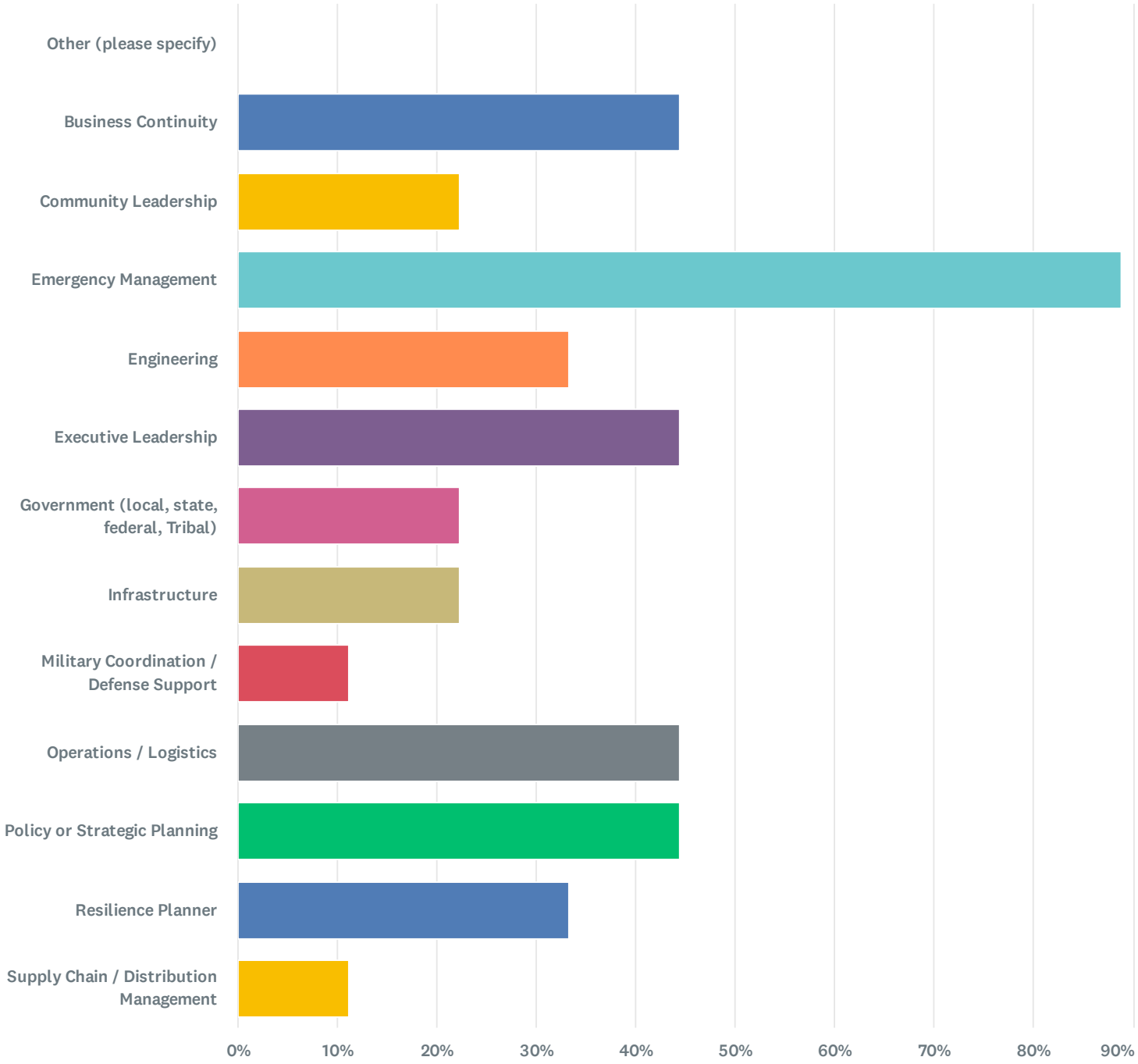
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No responses yet

Q2

9 responses

2. What role best describes your perspective in your organization or professional experience?(Select all that apply)



RESPONSES (0) WORD CLOUD TAGS (0)

Search Responses

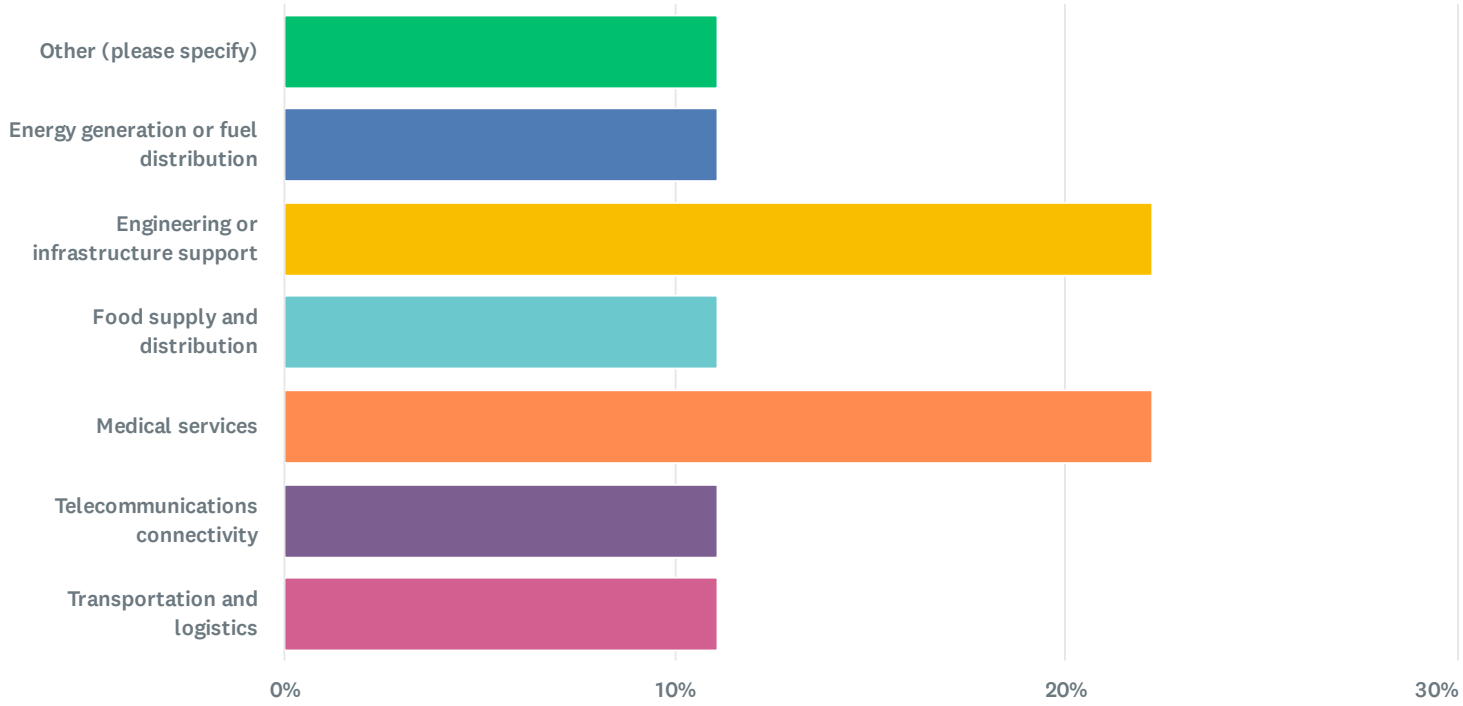
Filter: by tag by sentiment

No responses yet

Q3

9 responses

Essential ServicesFor this survey, an Arctic disruption refers to a prolonged event that limits transportation, energy supply, communications, or resupply to Arctic communities and installations.3. Which core service or operational capability provided by your organization would be most essential during a major Arctic disruption?(Select one)



RESPONSES (0) WORD CLOUD TAGS (0)

Search Responses

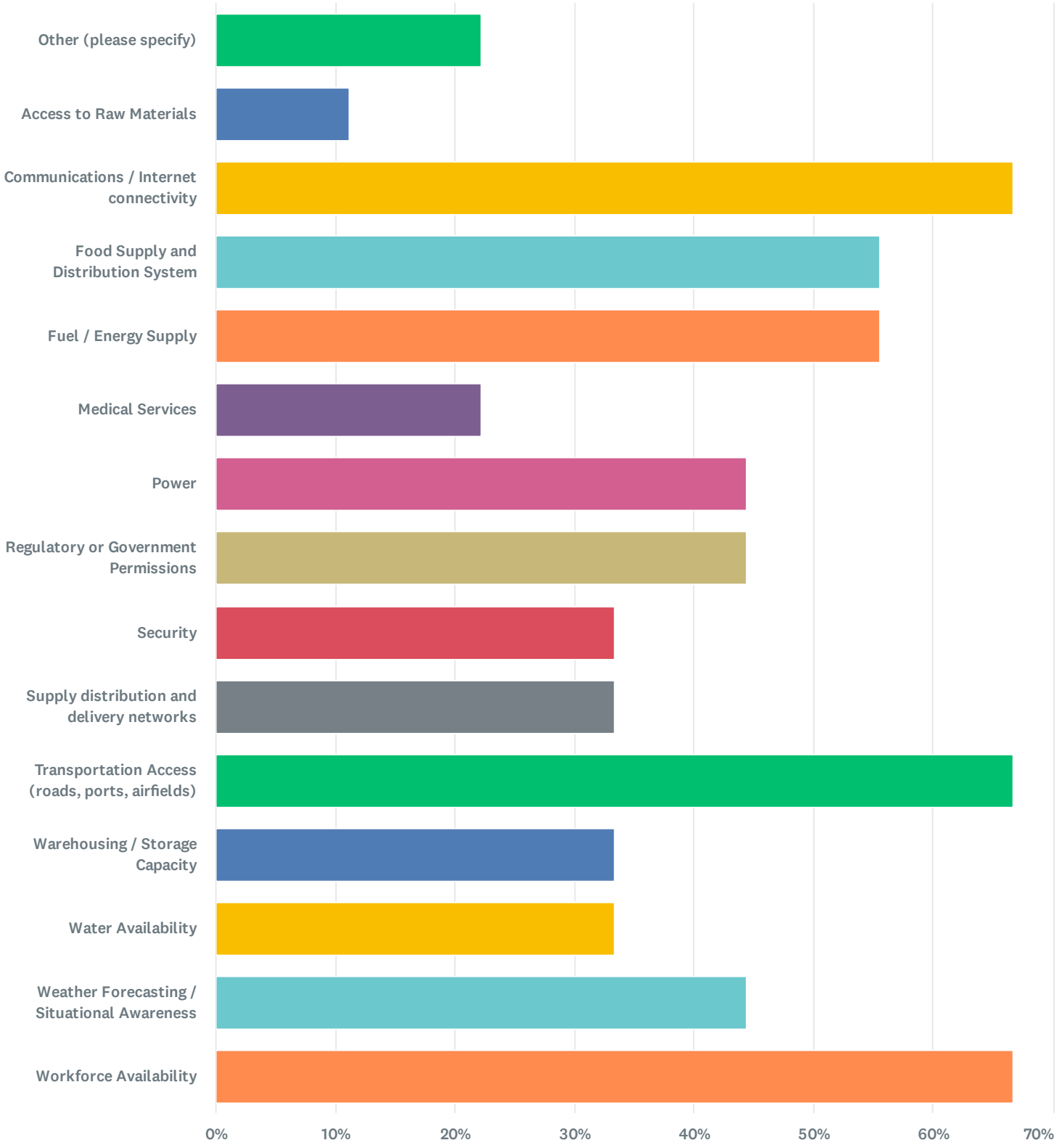
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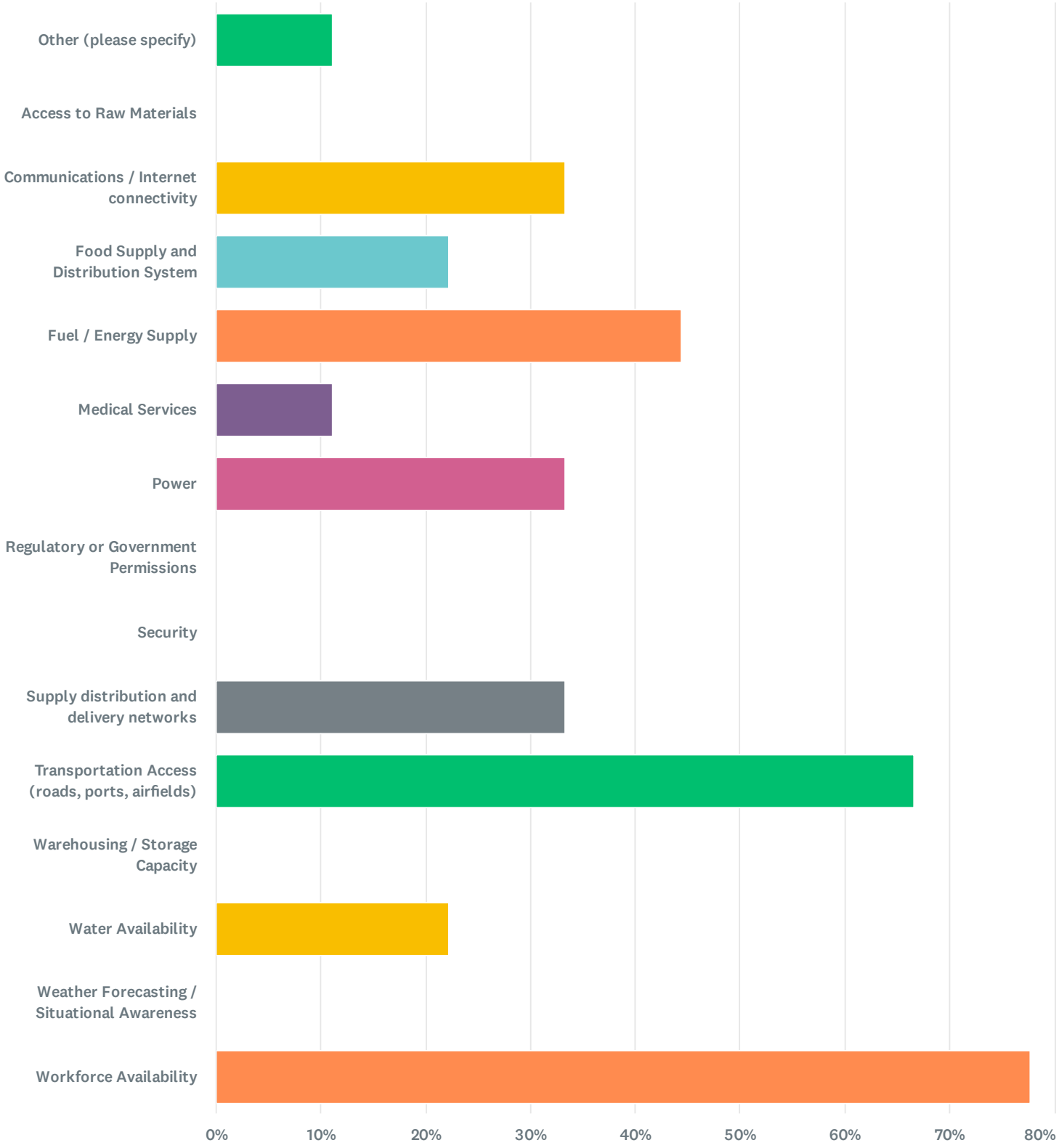
No responses yet

Q5

9 responses

Infrastructure Dependencies5. What inputs are essential for your organization to continue operating?(Select all that apply)





RESPONSES (0) WORD CLOUD TAGS (0)

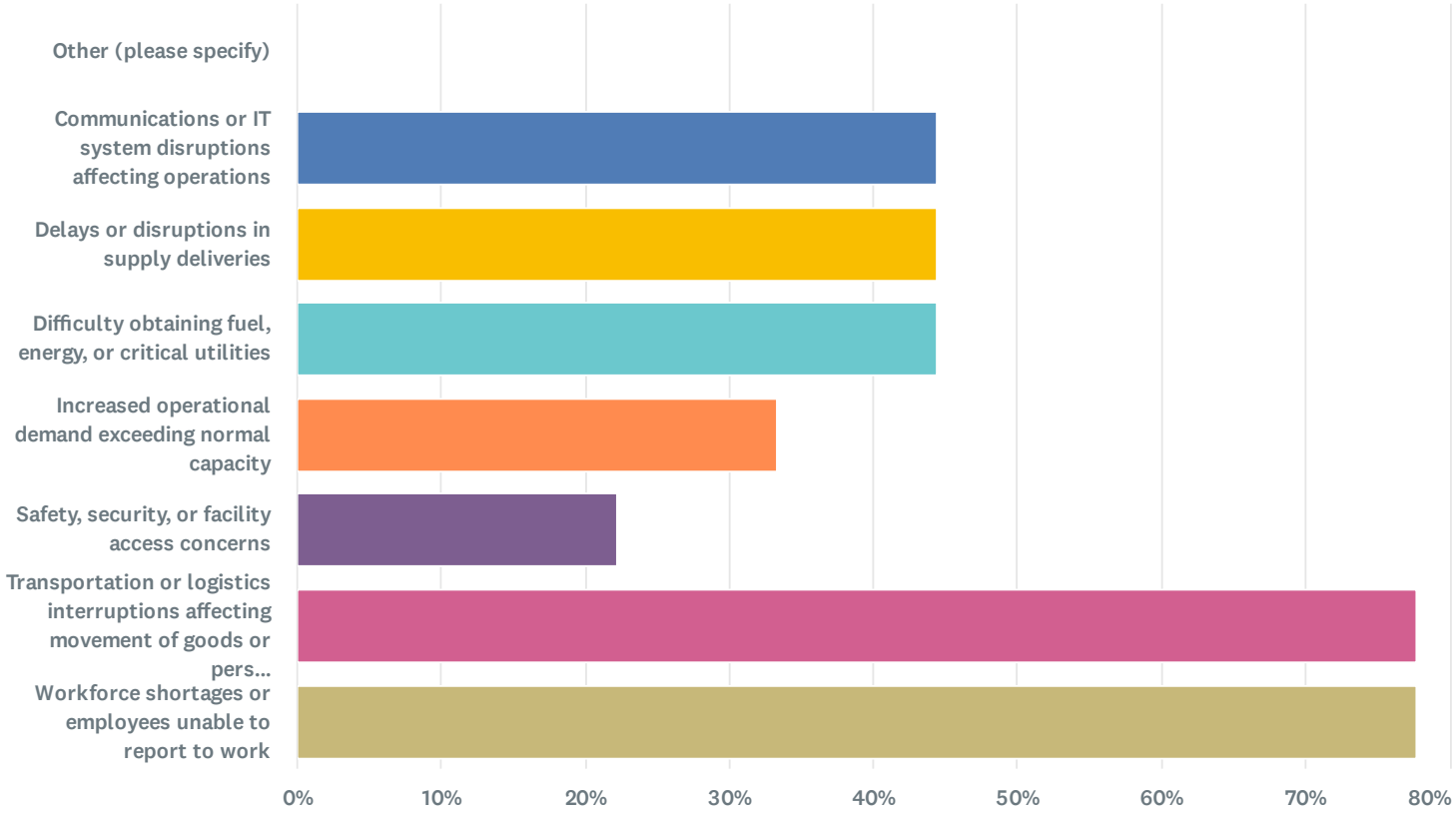
Search Responses

Filter: by tag by sentiment

Q7

9 responses

Early Stress IndicatorsEarly stress indicators are often the first signs that infrastructure systems supporting operations are beginning to degrade during a disruption. 7. During the first days of an Arctic disruption, which warning signs would indicate that the systems your organization depends on are beginning to degrade or fail?(Select all that apply)



RESPONSES (0) WORD CLOUD TAGS (0)

Search Responses

Filter: by tag

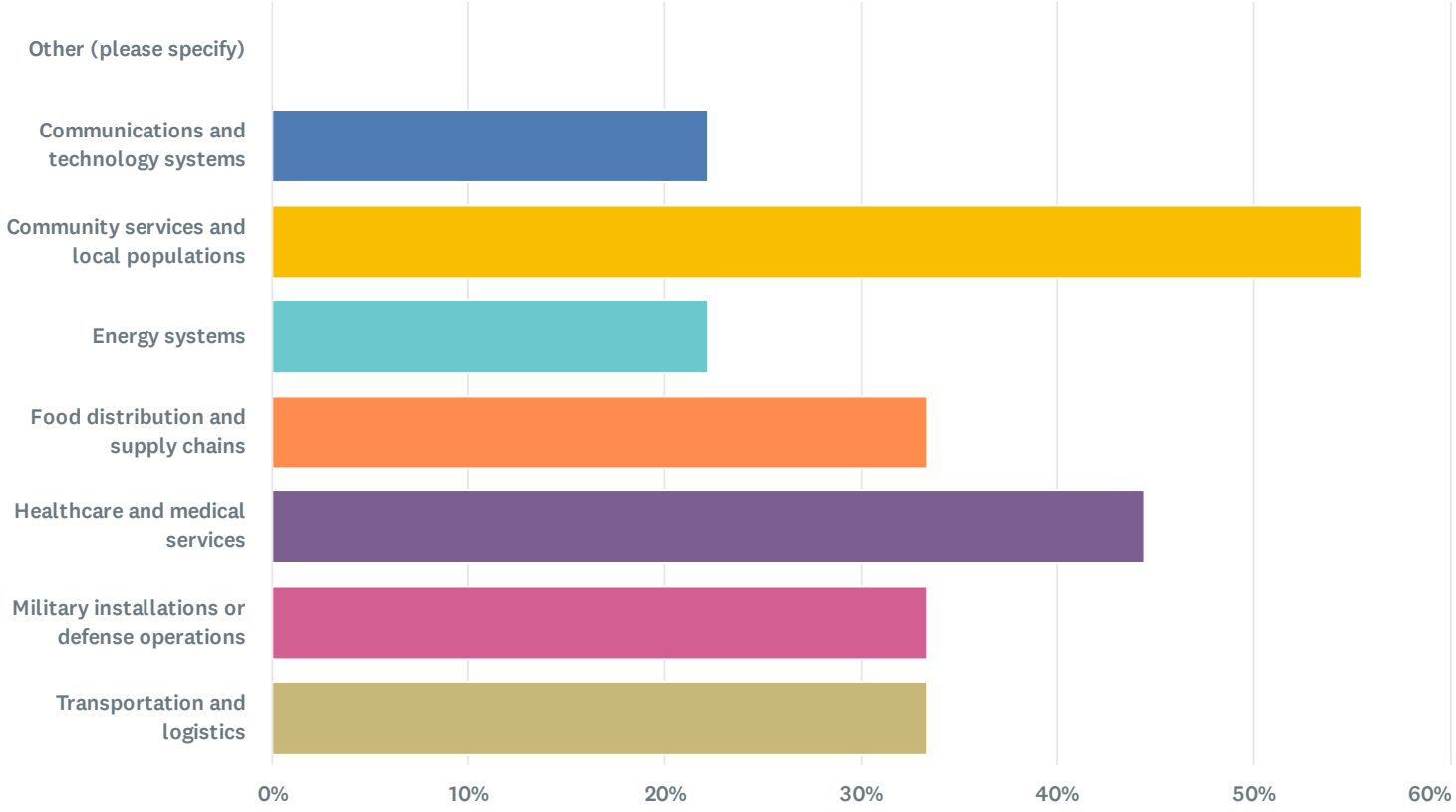
by sentiment

No responses yet

Q9

9 responses

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)



RESPONSES (0) WORD CLOUD TAGS (0)

Search Responses

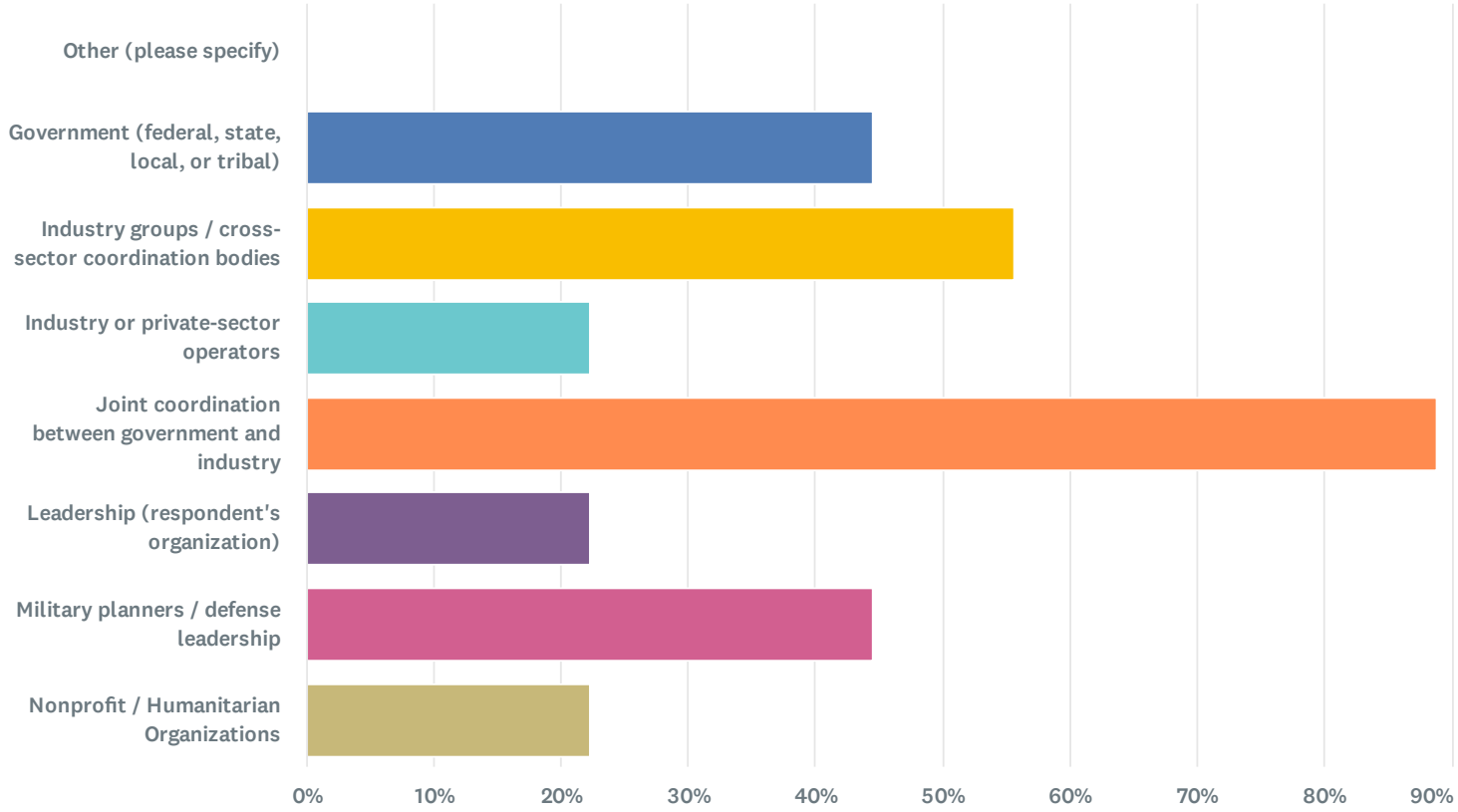
Filter: by tag by sentiment

No responses yet

Q14

9 responses

Strategic Insight14. What one investment, policy change, or coordination improvement would most strengthen resilience in your sector? Who would need to take action?Examples may include infrastructure investment, regulatory change, improved coordination mechanisms, or new communication protocols.(Select all that apply)



RESPONSES (0) WORD CLOUD TAGS (0)

Search Responses

Filter: by tag by sentiment

No responses yet

#1

COMPLETE

Collector:	Project 1 (Web Link)
Started:	Tuesday, March 17, 2026 6:12:54 PM
Last Modified:	Tuesday, March 17, 2026 6:24:14 PM
Time Spent:	00:11:20

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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

Medical services

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Internet connectivity
Workforce availability
Supply chain collapse

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Communications and technology systems

Q10

10. If helpful, briefly describe the dependency relationship.

Inability to provide situation awareness to vulnerable populations

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

Not enough experience to determine

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

Starlink, alternative platforms

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Basic services for food, water, power, shelter and workforce

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

The narrow margins of the civilian communities to sustain themselves around installations.

#2

COMPLETE

Collector:	Project 1 (Web Link)
Started:	Wednesday, March 18, 2026 12:23:26 PM
Last Modified:	Wednesday, March 18, 2026 1:03:59 PM
Time Spent:	00:40:32

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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

If disruption required services to be prioritized, the function that must be preserved first is the continued operation of pharmaceutical distribution and inventory management, including distribution center operations, cold-chain handling, and domestic transportation. Preserving these capabilities ensures that available medications can be allocated, redistributed, and delivered to healthcare providers and patients, sustaining continuity of care even when upstream manufacturing or transportation constraints exist. Maintaining core distribution operations minimizes immediate patient impact, supports critical healthcare infrastructure, and allows the organization to absorb and manage longer-term supply disruptions that originate outside of its direct operational control.

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Under Arctic disruption conditions, service impacts would be driven primarily by the severity and duration of degradation to the domestic transportation network, rather than a lack of internal operational capacity. The organization maintains a robust distribution capability with significant built in redundancy; however, curtailment of service would occur if extreme cold, ice, or snow reduced road accessibility, limited trucking capacity, or constrained air cargo operations, resulting in slower delivery cycles and reduced throughput. Rationing of service would be required if prolonged transportation disruptions materially constrained inbound or outbound movement, necessitating prioritization of critical healthcare deliveries and allocation of limited transportation resources. A full disruption of operations would be unlikely and would only occur under extreme conditions where widespread, sustained failure of both road and air transportation networks prevented safe movement of goods altogether, despite facilities remaining operational and inventory being available. In this context, Arctic conditions represent a transportation driven risk rather than a loss of core distribution capability, with impacts escalating as domestic mobility degrades rather than from internal capacity shortfalls.

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Healthcare and medical services

Q10

10. If helpful, briefly describe the dependency relationship.

We are heavily reliant on the domestic transportation network to deliver products to essential service providers. We are also dependent on product deliveries from the EU, China, and India, which rely on maritime shipping and international air cargo. A disruption to either transportation system can result in product shortages if the disruption is prolonged.

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

When Arctic lifeline systems fail, the most common cascading effects stem from the loss of transportation, power, and logistics connectivity, which rapidly constrains the movement of critical goods and services. Severe cold and infrastructure failures can disrupt road and air transportation, delay or halt deliveries, and limit workforce mobility, leading to reduced distribution capacity and supply imbalances. As these disruptions persist, inventory replenishment becomes increasingly constrained, and patients may not receive the critical medications required for life saving or life sustaining care. While federal initiatives, including efforts led by HHS, are underway to strengthen and expand domestic pharmaceutical production, the vast majority of medications and active pharmaceutical ingredients are still produced outside of the United States, leaving healthcare supply chains vulnerable to prolonged Arctic events that affect both domestic lifelines and global transportation systems.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

If a primary system, route, or supplier failed, several practical alternatives exist, including rerouting through alternate domestic transportation corridors, leveraging multiple distribution locations, engaging secondary or tertiary suppliers, and reallocating available inventory to support critical needs. The organization maintains multiple layers of redundancy across suppliers, transportation modes, and distribution capabilities to absorb localized or short term disruptions. However, if an event were widespread and prolonged, affecting large portions of the transportation network or multiple upstream suppliers simultaneously, these alternatives would become increasingly constrained and would disrupt normal operations, requiring prioritization of services and extended recovery timelines.

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

In Arctic conditions, several backup solutions that are often assumed in planning prove unrealistic due to the scale and severity of impacts. These include the expectation that alternate road routes or nearby airports will remain accessible, when extreme cold, ice, and snow frequently affect wide geographic areas simultaneously. Similarly, leveraging multiple distribution points may be assumed to provide continuity; however, Arctic events can disrupt transportation corridors, utilities, and workforce availability across multiple locations at once, limiting the ability to reposition inventory or shift volume. Assumptions that air transportation can rapidly offset ground disruptions or that suppliers can quickly increase output are also often unrealistic under prolonged Arctic conditions, as aircraft availability, crew safety, and upstream supply constraints are commonly affected in parallel.

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

Decision makers often underestimate how quickly Arctic infrastructure disruptions translate into operational and human impact, particularly in commercial healthcare supply chains that operate on tight margins, compressed timelines, and high velocity. There is a common assumption that private sector operations can absorb delays or “slow down” in the same way government systems often do; however, the speed of business is critical, and even short disruptions to transportation, power, or logistics can rapidly cascade into reduced service levels and patient impact. Arctic events tend to affect large geographic areas simultaneously, limiting practical workarounds and compressing decision timelines to hours—not days or weeks. This research should capture that commercial supply chains are designed for efficiency and continuity of care, not prolonged pauses, and that misalignment between the pace of government response and the tempo required to sustain critical services can exacerbate downstream consequences during Arctic infrastructure failures.

#3

COMPLETE

Collector:	Project 1 (Web Link)
Started:	Friday, March 20, 2026 8:32:24 PM
Last Modified:	Friday, March 20, 2026 8:58:50 PM
Time Spent:	00:26:26

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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

Supply Chain

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Workforce availability
Product availability
Utilities

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Community services and local populations,

Food distribution and supply chains,

Healthcare and medical services

Q10

10. If helpful, briefly describe the dependency relationship.

Many northern communities rely on diesel-based generation and imported fuels, often with no interconnected grid; failures can persist and compound with transport disruption.

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

A resilient population, lack of transportation, workforce & utility would lead to bigger impacts

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

We have a limited footprint in Alaska and Northern Canada. If distribution routes where impacted, alternative routes would be shipping and air. If workforce is not available, there is no alternative option.

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Short term disruptions would be cleared within days. In this case, that can't be assumed.

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

- “Workforce is a secondary concern” — workforce stability is mission assurance
- “Infrastructure failure is binary” — it isn’t
- “Logistics problems are short-term” — Arctic recovery is slow by default
- That DoD assets (airlift, generators, satellite comms, fuel) can readily replace civilian lifelines during disruption. The private sector (food, fuel, retail, logistics) is not “adjacent support”—it is foundational infrastructure for Arctic operations.

#4

COMPLETE

Collector:	Project 1 (Web Link)
Started:	Monday, April 06, 2026 2:54:59 PM
Last Modified:	Monday, April 06, 2026 3:18:36 PM
Time Spent:	00:23:37

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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

Energy/fuel for initial people/life support

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

1. Curtail - supply chain disruptions
2. Ration - Fuel and workforce limitations
3. Stop - extreme weather and lack of transit capability (roads, airfields, etc).

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Community services and local populations,

Energy systems,

Food distribution and supply chains,

Healthcare and medical services,

Military installations or defense operations,

Transportation and logistics

Q10 Respondent skipped this question

10. If helpful, briefly describe the dependency relationship.

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

People/community/medical support becomes a priority vs. supporting security infrastructure or non-life sustaining systems. This can cause significant infrastructure degradation due to severe weather impacts.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

The plan would need to build in the primary, alternate and contingency alternatives up front. A singular system or route is not sufficient. But if industry does not have contingencies, then support would have to come from US or State government agencies (FEMA, National Guard, Army, etc).

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

It's unrealistic to rely on one supply method or route. Routes and logistics will be impacted seasonally, so planning for marine supply routes would work in summer months but must transition to an alternative (e.g. ground transportation) during winter months. This will reduce goods flow and resupply times.

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

Decision makers often misunderstand the weather and size/scale of the Arctic, along with the time it will take to build life support and operational infrastructure. Ice engineering is difficult at best, and the population level is mostly confined to cities and larger towns. You also don't have readily available materials to build with (lumber, concrete, etc.,) and obtaining the goods and civilian support for arctic response will be difficult and time consuming due to distances from the L48.

#5

COMPLETE

Collector:	Project 1 (Web Link)
Started:	Monday, April 06, 2026 4:35:47 PM
Last Modified:	Monday, April 06, 2026 4:53:35 PM
Time Spent:	00:17:47

Page 1

Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

During a major Arctic disruption, the limiting factor is rarely what aid exists—it's whether it can reach isolated communities safely and fast. Team Rubicon's strongest and most consistently deployed capability is operational logistics in austere, disrupted environments, including:
Route clearance and access restoration
Ground and air movement coordination in damaged or weather-constrained areas
Last-mile delivery of critical supplies (fuel, food, medical materials)
Incident management, staging, and integration with government and military partners
Operating when normal transportation networks are degraded or unavailable
In Arctic conditions—where terrain, weather, ice, limited infrastructure, and long distances dominate—logistics is the enabling function that makes every other core service possible.

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Extreme weather has historically affected our ability to provide services. The life safety risk are too high for the roles we typically provide that are not directly life safety task.

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Community services and local populations,

Food distribution and supply chains,

Transportation and logistics

Q10

10. If helpful, briefly describe the dependency relationship.

Respondent skipped this question

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

Food chain and essential services are critical. The Arctic region historically does a good job of food reserves for the winter months but an early or late season arrival disrupts that timing. Access for transportation would make meaningful delivery of last mile logistics difficult. Resource allocation of life sustaining goods in an equitable way becomes very difficult.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

Based on limited population centers in the Arctic the delivery to central hubs and routes is important. A hub and spoke delivery model where locals can use their resources to retrieve goods through traditional transport means they use is the most realistic. From sled teams to airdrops.

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Air assets are depended on. Stockpiles of supplies that can be housed securely is a key factor.

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

The complexity of delivering services and supplies in the Arctic is dependent on transportation heavily. The military is the best equipped to solve this issue. The coordination across the public/private/non-profit sector with the Military is critical. People to execute the mission is not the blocker, the ability to get the people to the mission will be.

#6

COMPLETE

Collector:	Project 1 Email Follow-Up (Email)
Started:	Friday, April 10, 2026 8:52:43 AM
Last Modified:	Friday, April 10, 2026 9:03:20 AM
Time Spent:	00:10:37
First Name:	136.226.2.85
Last Name:	
Email:	
IP Address:	

Page 1

Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

Supply Chain - drug delivery to markets

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Inability to access fuel that power N+1 generators, ingress and egress of couriers to DCs and out to markets due to road conditions or airport access and/or availability of staff.

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Healthcare and medical services

Q10

10. If helpful, briefly describe the dependency relationship.

McKesson delivers 40% of all market sure drugs in North America. If those drugs to not make it to hospitals, clinics and pharmacies, it can cost lives.

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

Lack of fuel for redundant power, reduced capacity of food supplies, power down for extended periods as Alaska likely does not have DOT resources to respond at scale, communications challenges, airport restrictions.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

Transfer order lines to lower 48, work on solution to get supplies to market via air, hot shot drivers, other means. Put employees up in local hotels (essential employees) so they they can man the DCs, engage with EOC on time to clear roads.

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Fuel availability, food supplies, communication and the overall preparedness of the government to respond timely.

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

That it takes both private sector and the government to coordinate successfully and the government does not typically put effort into preparedness coordination in many states with only 28 states having an EOC at all.

#7

COMPLETE

Collector: Project 1 Email Follow-Up (Email)
Started: Wednesday, April 22, 2026 9:14:06 AM
Last Modified: Wednesday, April 22, 2026 10:42:40 AM
Time Spent: 01:28:33
Email:
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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

In my opinion, organizational operations and structure must be preserved first. Organizations can provide no external assistance or participate in any kind of response without first stabilizing their internal structures.

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Our industry is all about the ability to communicate and arrive on client site. If we would need to curtail services if communications are disrupted, we are unable to access clients, or if our workforce is unable to participate.

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Communications and technology systems,

Community services and local populations,

Military installations or defense operations

Q10 Respondent skipped this question

10. If helpful, briefly describe the dependency relationship.

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

The first thing to go in a disruption event are strong communications and shared operating picture. These initial disruptions cascade down into every sector and industry and force organizations to make choices with bad information.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

None

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

The idea that outside or external partners (like government) will show up with manpower and resources ready to be utilized.

Q15 Respondent skipped this question

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture? Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

#8

COMPLETE

Collector:	Project 1 (Web Link)
Started:	Monday, April 27, 2026 3:50:33 PM
Last Modified:	Monday, April 27, 2026 4:16:41 PM
Time Spent:	00:26:08

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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

Power, communications, workforce, and physical restoration.

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Given the remote nature of a 501(3)(c) non-profit, the National Hazard Mitigation Association would be severely impacted by long-term power outages, hampering all communications and potentially dismantling the organization's capacity to re-emerge.

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Community services and local populations

Q10

10. If helpful, briefly describe the dependency relationship.

NHMA's main pillar is training and education of local communities, organizations, and students in emergency management, with specific aim on hazard mitigation and resilience.

Q11

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

A blackout would entail a complete lack of situational awareness. This would leave people blind to hazard risk during or after a catastrophic event. Without communication, most interagency and organizational initiatives would fail to reduce that harm. Post-event restoration may be prolongued, causing people to migrate to non-affected areas, with a majority of the population not returning in the long run.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

For power, electric generators and a few carbon-free options exist. However, these options only serve for limited time frames and are unable to replace the grid system.

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Large generators with thousands of fuel gallons are needed, but in artic conditions, this may not exist. However, artic populations might not be as high as compact urban areas. Therefore, planned assessments often show that it is possible to have sufficient backup for artic communities, but if hazard intensity keeps growing, well begin to see these unrealistic assessments made during planning stages.

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

If would say, "The time required to restore critical services", as this may be highly volatile to extreme weather conditions, further aggravated by below-zero temperatures.

#9

COMPLETE

Collector:	Project 1 Email Follow-Up (Email)
Started:	Wednesday, April 29, 2026 9:00:08 PM
Last Modified:	Wednesday, April 29, 2026 9:06:37 PM
Time Spent:	00:06:29
First Name:	
Last Name:	
Email:	
IP Address:	

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Q4

4. If disruption forced you to prioritize services, which function must be preserved first?(Please answer based on the services or capabilities your organization provides.)

Restoration of the transportation system

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Under Arctic disruption conditions, what circumstances would force your organization to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Inability to maintain supply lines

Q9

Cascading Impacts9. If your system were disrupted, which sectors would be be most immediately affected? Consider organizations, sectors, or communities that depend on your service to continue operating.(Select all that apply)

Energy systems,

Military installations or defense operations,

Transportation and logistics

Q10

10. If helpful, briefly describe the dependency relationship.

Respondent skipped this question

Q11

Respondent skipped this question

11. In your experience, what cascading effects are most likely when Arctic lifeline systems fail?

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

Other commercial suppliers

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Availability of military airlft

Q15

15. What do decision-makers (military / government leadership) most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

Lack of interconnected road system for many communities throughout the state.

#1

COMPLETE

Collector:	project 1 military (Web Link)
Started:	Tuesday, March 17, 2026 1:44:54 PM
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Time Spent:	00:09:52
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Q1

Your Perspective1. Which sector best represents your perspective?(Select one)

Military Operations or Command Leadership

Q2

2. What role best reflects your experience?(Select all that apply)

Command or Senior Leadership

Q3

Essential ServicesFor this survey, an Arctic disruption refers to a prolonged event that limits transportation, energy supply, communications, or resupply to Arctic communities and installations.3. Based on your operational experience, which service or operational capability is most essential to sustaining Arctic operations during a major disruption?(Select one)

Energy generation or fuel distribution

Q4

4. If disruption forced operational prioritization, which function would need to be preserved first?Please answer based on the services or capabilities most critical to sustaining operations or supporting installations.

First, energy and fuel distribution closely followed by transportation and logistics.

Q5

Infrastructure Dependencies5. Based on your experience, what resources or systems are essential for Arctic operations to continue functioning?(Select all that apply)

Access to Raw Materials,

Supply distribution and delivery networks,

Transportation Access (roads, ports, airfields),

Warehousing / Storage Capacity,

Fuel / Energy Supply,

Regulatory or Government Permissions,

Communications / Internet connectivity,

Power,

Security,

Water Availability,

Weather Forecasting / Situational Awareness,

Food Supply and Distribution System,

Medical Services,

Workforce Availability

Q6

6. Which of these dependencies would be most difficult to replace or substitute during Arctic conditions?Arctic conditions refer to environments where extreme weather, long distances, and limited infrastructure reduce redundancy and make transportation, communications, energy supply, and logistics difficult to replace if disrupted. (Select all that apply)

Supply distribution and delivery networks,

Transportation Access (roads, ports, airfields),

Fuel / Energy Supply,

Communications / Internet connectivity,

Security,

Food Supply and Distribution System,

Medical Services,

Workforce Availability

Q7

Early Stress IndicatorsEarly stress indicators are often the first signs that infrastructure systems supporting operations are beginning to degrade during a disruption. 7. During the first days of an Arctic disruption, which warning signs would indicate that the systems supporting Arctic operations are beginning to degrade or fail?(Select all that apply)

Communications or IT system disruptions affecting operations

,

Delays or disruptions in supply deliveries,

Difficulty obtaining fuel, energy, or critical utilities,

Increased operational demand exceeding normal capacity

,

Safety, security, or facility access concerns,

Transportation or logistics interruptions affecting movement of goods or personnel

,

Workforce shortages or employees unable to report to work

Q8

Operational ThresholdsOperational thresholds are conditions where infrastructure systems can no longer operate normally due to Arctic constraints such as extreme weather, transportation disruption, fuel shortages, or workforce limitations.8. Based on your experience, what conditions would force infrastructure systems supporting Arctic operations to:Curtail service (reduce capacity)Ration service (prioritize limited availability)Stop operations

Fuel shortages, logistics supply support, workforce limitations.

Q9

Cascading Impacts9. If a critical infrastructure system supporting Arctic operations were disrupted, which sectors would be most immediately affected?(Select all that apply)

Community services and local populations,

Energy systems,

Food distribution and supply chains,

Transportation and logistics

Q10

10. If helpful, briefly describe the dependency relationship.

Respondent skipped this question

Q11

11. Based on your experience, what cascading effects are most likely when Arctic lifeline systems fail?

Basic public services. Support to security and military operations.

Q12

Backup and Resilience12. If your primary system, route, or supplier failed, what practical alternatives exist?

Not currently applicable but I've seen the only alternative be to wait for spring weather.

Q13

13. What backup solutions are often assumed in planning but are unrealistic in Arctic conditions?

Resupply using commercial air or maritime means.

Q14

Strategic Insight14. What investment, policy change, or coordination improvement would most strengthen resilience in Arctic infrastructure systems? Who would need to take action?Examples may include infrastructure investment, regulatory change, improved coordination mechanisms, or new communication protocols.(Select all that apply)

Government (federal, state, local, or tribal),

Industry or private-sector operators,

Joint coordination between government and industry

Q15

15. What do decision-makers most often misunderstand about Arctic infrastructure disruption that this research should capture?Examples may include misunderstandings about transportation availability, supply chain timelines, workforce capacity, communications reliability, or the time required to restore critical services.

That most locations have a duel support requirement where both the local population and the government or industry rely on the same means for energy, transportation, medical, and sustenance.